Governing Disclosure Risk

Probability, Evidence and Accountable Decisions
for Repeated Statistical Releases

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Preface

This book grew out of a paper presented at the workshop on data security and privacy that Banco de Portugal and the Irving Fisher Committee held in Lisbon in September 2026. The paper argued that a disclosure risk figure only becomes useful to an institution once it is attached to a declared attack, an honest statement of its uncertainty and a written decision rule. In discussion, a participant asked what happens to the risk when an institution gives several data sets to the same researcher. The question went to the centre of the argument, and a conference paper could only answer part of it. Chapter 9 answers it in full, and the rest of the book builds the framework that answer needs.

The book has two audiences, and it is arranged so that each can read it without the other's material getting in the way. The first is the statistician, data protection officer or release committee member at a central bank, statistical office or health registry who has to decide whether a file goes out. For that reader every chapter opens with the problem in institutional terms, states its results in words before it states them in symbols, and closes its main text with a short block headed *In practice* that says what to write down and what to check. The second audience is the researcher in statistics, computer science or economics who wants to know exactly what is claimed and under which assumptions. For that reader every result carries a numbered statement and a full proof, and the proofs sit in a final section of each chapter so that the practitioner can pass over them without losing the thread.

Every number in the book comes from one research pipeline, the Python package `privacy_risk` in the repository that accompanies the book. Where a chapter reports a figure, it names the module and function that produced it, so that a reader can rerun the calculation, change an assumption and see what moves. Most of the evidence is synthetic. That is a limitation and the book says so in the chapters it affects: the credit register in Part V shares the structure of a supervisory collection, but its parameters are chosen, not estimated, and the floors it implies are conditional on that design.

The book also reports what did not work. The chapter on model-proposed attacks includes a pilot in which augmenting the training data did not improve the model, and the audit chapter shows how often an analyst who retries a failed audit certifies an unsafe release. These results belong in the record because a method for governing risk has to be judged on its failures as well as its successes.

Each chapter opens with a belief that a careful reader is likely to hold, states it fairly, and then shows the fact that breaks it, because a reader who thinks the answer is already known has little reason to read on. The chapter then asks the one question it will answer and builds the answer as a connected case. Two threads run through it. One is technical. The other follows people: documented cases from the published record of re-identification, and a small cast at a fictional central bank, called here the Bank. Naledi, its disclosure control officer, runs the assessments; the chair of its statistics release committee signs the decisions; Tomás, a visiting researcher, asks for several files from the credit register; Pieter,

a credit analyst at a reporting lender, is the insider; and Grace, from internal audit, reads the decision records afterwards. They are composites, drawn from the roles that release decisions involve, and none of them is a real person.

This is a first full draft. Every chapter is written, with its proofs, and an appendix maps the software. Several chapters report figures that the repository itself records in more than one version; where that happens the chapter says which run it uses, and the list of such points is kept with the source files for the next revision. Comments on the draft are welcome.

Part I

The problem

Chapter 1

A risk figure is not a decision

Most people who work with confidential statistics hold a belief that seems too plain to need stating: once the disclosure risk of a release has been measured carefully, the decision follows. If the measured risk is low, the release is safe; if it is high, the file needs more protection. The hard work, on this view, is the measurement.

The belief has a strong case behind it. The methods of statistical disclosure control have been developed over four decades, are documented in handbooks, implemented in tested software and taught to the staff who apply them (Willenborg and de Waal, 2001; Hundepool et al., 2012; Dwork and Roth, 2014). Cell suppression, minimum cell counts, global recoding, record swapping, added noise, synthetic data and formal privacy mechanisms are all well understood. An institution that measures risk with these tools, and releases only when the measured figure is small, appears to have done everything a careful institution could do. What could a committee add to a well-measured number except its signature?

Two facts break the belief, and both come from the evidence reported in this book. The first is that one release has no single risk figure. On the simulated release of Chapter 3, three measures that are routinely reported under the one label "re-identification risk" give 0.466, 0.064 and 0.023. Each is correct, and each describes a different experiment. A committee that reads "0.023" has not measured the risk; it has measured the answer to one question, and nothing on the page says which. The second fact is that even a well-defined figure is an estimate. For a release whose true risk sits exactly at a tolerance of 0.12, a simulation of 1,200 attacks produces an estimate at or below the tolerance with probability 0.52. A committee that releases whenever the estimate is below the line therefore releases a boundary file about half the time, and its records, which hold only the estimate, cannot show this.

This chapter asks one question, and the rest of the book answers it in detail: *what must an institution add to a measured risk before the result can be called a decision?*

1.1 A Thursday committee

Consider the statistics release committee of the Bank, the fictional central bank whose staff appear throughout this book. On Thursdays the committee reviews the files proposed for release. Naledi, the disclosure control officer, presents a table for each file: the share of sample-unique records, the smallest cell count, a mean re-identification probability. The chair asks whether anyone objects, and if nobody does the file goes out.

The procedure works much of the time, because the people in the room know the data and the users. Its weakness shows only later. A year after one such meeting, Grace from

internal audit is asked to check a release that a journalist has written about. She finds the minutes and Naledi's table. She cannot find what adversary the table assumed, what probability of harm the committee meant to accept, who accepted it, or when the assessment was meant to lapse. The decision may well have been right. Nobody can show that it was.

1.2 Three objects run together

Grace's difficulty has a precise cause. A release decision involves three different objects, and practice treats them as one. The first is a probability model for an attempted linkage: given what a declared adversary knows and what the release shows, how likely is it that the adversary names the right record. The second is a statistical estimate of that probability, which in all but the simplest cases comes from simulating the attack many times and so carries sampling error. The third is a policy decision: how much residual risk is acceptable, for which harm, on whose authority and until when.

The first is a modelling problem, the second an inference problem and the third a governance problem. Each needs its own discipline. Running them together produces failures that are easy to miss, because each looks reasonable from inside one of the three.

The first failure concerns the model. A risk figure means nothing until it states its adversary: who attacks, with what knowledge, against which targets, and whether the target is assumed to be in the file. Chapter 2 derives the most familiar measure in the field, the reciprocal of the key frequency, from four assumptions, and shows that two of them fail in the ordinary case. A release that keeps ten records in every key class can still hold a record that an adversary with one extra variable identifies with probability 0.74.

The second failure concerns the estimate, and it is the coin toss described above. Chapter 4 shows that replacing the point estimate with a one-sided upper confidence bound controls the chance of a wrong release at a stated level, and Chapter 5 extends the control to the choice among several protection settings.

The third failure concerns the decision. An institution that says "we apply a minimum cell count of ten" has stated a protection setting and called it a policy. It has not stated the probability of harm it accepted, so nobody can check whether the setting achieves the policy, and nobody owns the judgement that the residual risk is acceptable. When the data or the users change there is no record of what the setting was meant to deliver, and so nothing to reassess against.

Naledi met the second failure in a meeting that Chapter 4 recounts. Her estimate for a household survey extract stood just below the tolerance; her upper bound stood just above it. The chair asked, reasonably, why a second and larger number should decide. The answer to that question is the first piece of what an institution must add.

1.3 What a decision rule needs

The book's answer is a written decision rule with three parts, one for each object. The model part declares an adversary: who attacks, what it knows, which targets it attacks and which conditioning event the risk figure assumes. The inference part replaces the point estimate with an upper confidence bound at a stated error probability. The governance part states a tolerance and attaches to it what turns a number into a decision: an owner who accepts the residual risk, an expiry date, the harm class the tolerance is set for, the legal basis and the parties it protects, the access route through which the data will be used, and, for a series of

releases, the horizon over which risk is assessed. Chapter 7 sets out these elements and who should write each one.

A release is then admissible when the upper bound for every target in a declared target set lies at or below the tolerance. The rule is simple, and its simplicity is the point. Two analysts given the same declarations and the same simulation output reach the same decision, and an auditor can check afterwards that the decision followed from the evidence. When Grace reads a record written under this rule, every question she could not answer in the old minutes has a line of its own.

The rule has a price, and the book does not hide it. The evidence needed to certify a tolerance grows as the tolerance shrinks. If an attack is simulated many times and never succeeds, the number of clean attempts needed for a 95 per cent upper bound to reach one in a hundred is 299. For one in a thousand it is 2,995, and for one in a hundred thousand it is 299,572. Observing no successful attack never shows that the risk is zero, and choosing a tolerance and choosing the evidence standard turn out to be one decision. The tolerance itself is not arbitrary either. Chapter 8 shows that a stated weight on harm against the value of the data implies a tolerance for each release, and that one tolerance for every release is usually the wrong choice.

1.4 Several releases from one register

Single releases are not how institutions publish. They publish programmes: a quarterly series, a research file and a public extract, all drawn from one register. An adversary who holds several of these is not obliged to use them one at a time. When records keep the same pseudonym across releases, the adversary can intersect the candidate sets that each release leaves, and the result can identify a target that no single release exposes.

The consequence is stronger than intuition suggests. Chapter 9 proves that no function of the per-release risks can bound the risk of an adversary who pools the releases. On a simulated register, four releases that each keep at least three records in every key class, and that contain no unique record at all, identify the target with probability 0.458 once pooled, against an upper bound of 0.253 implied by the four marginal risks.

At the Bank, this is the problem Tomás brings. He is a visiting researcher who wants a loan-level file from the credit register for six months, a table of lending by sector and a public series, for one study. Each request, judged alone, would pass Naledi's assessment. The committee's question is whether the three together do, and its old procedure cannot answer it, because the procedure only ever looks at one file.

1.5 A running example

Most chapters return to one example: a synthetic credit register built to share the structure of a supervisory collection such as the euro area's AnaCredit (Israël et al., 2017). It holds 20,000 firms as borrowers, 60 lenders with unequal shares, 18 sectors, 12 regions and four size classes, observed over six monthly vintages. About 27,000 loans are live at any time. Each released loan carries a persistent pseudonym, its sector, region and size class, and its exposure.

The register is synthetic because real loan-level credit data are not released for general use, and that choice limits what the example can show. Its parameters are chosen by design, so the protection settings it implies describe the simulation and not an actual institution.

What it can show is how the decision rule responds when the adversary changes. At an illustrative tolerance of 0.12, an assessment of one release at a time passes the file with no suppression. An adversary who pools the six vintages and knows in which of them the target's loan was live requires a minimum cell count of 75. One who also holds a reading of the loan's exposure in each vintage requires 150, and at that floor the file keeps 20 per cent of loans but only 4 per cent of exposure, with no loan to a large firm. Against that adversary a loan-level file is the wrong product for Tomás's study, and Chapter 13 follows him to the secure room where he does his work instead. Pieter, a credit analyst at one of the reporting lenders, is the hardest adversary of all, because he knows his own loans almost exactly; no floor up to 300 stops him.

1.6 How the book is arranged

Part I sets up the problem: this chapter, the definition of risk as the outcome of a declared attack (Chapter 2), and the measures that answer different questions (Chapter 3). Part II treats the risk figure as an estimate: its uncertainty, the choice among settings, and certification while an attack is monitored. Part III turns to the decision: the elements of a governance threshold and the tolerance a stated harm weight implies. Part IV deals with programmes of releases and with formal privacy. Part V reports the evidence from simulation, public microdata, the credit register and a pipeline in which a model proposes attacks. Part VI is for practice: falsification tests, an operational protocol, worksheets and the questions that remain open. Appendix A maps the software that produces every number.

A reader who must decide on a release next month can read this chapter, the *In practice* blocks at the end of each chapter, and Part VI. A reader who wants to check the claims will find each one stated as a proposition with its assumptions and proved in the last section of its chapter.

1.7 The answer in brief

The question was what an institution must add to a measured risk before the result can be called a decision. It must add three things. It must name the attack the figure describes, so that the figure answers a stated question. It must replace the estimate with an upper bound at a stated error probability, so that a boundary file is not released by chance. And it must write down a tolerance with an owner, an expiry date, a harm class, a legal basis, an access route and, for a programme, a horizon, so that someone answers for the residual risk and someone else can check the decision afterwards. Without these the committee has a number; with them it has a decision.

IN PRACTICE

Before the next release decision, write down four things that are usually left implicit: the adversary the assessment assumes, the tolerance the release must meet, the person who owns that tolerance, and the date after which the assessment lapses. If any of the four cannot be written down, the committee is not yet in a position to say whether the release is safe, whatever the risk table shows.

Chapter 2

Disclosure risk as the outcome of a declared attack

The minutes of the Bank's release committee contain one phrase more often than any other: "every cell holds at least ten records, so no record carries a risk above one in ten." The chair wrote it, the committee approved it, and most readers of this book would approve it too. It is the belief that a record's disclosure risk is the reciprocal of the number of released records that share its key, so that a key class of ten means a one in ten chance.

The belief deserves its standing. The formula is old, it appears in every handbook, and it answers a clear experiment: an adversary who knows a person's key, finds the released records with that key, and picks one of them at random will be right one time in ten. The rule that every class must hold at least k records, known as k-anonymity, rests on it, and so do most minimum cell counts in use at statistical institutions. If an adversary cannot tell the records in a class apart, the formula is exact.

The fact that breaks it comes from a simulated register of 200,000 people with a 5 per cent sample. A release that keeps at least ten records in every key class contains a record whose own chance of being identified is 0.74, with a one-sided 95 per cent lower bound of 0.64. The adversary in that simulation knows one thing the key leaves out: a noisy reading of a continuous variable that the release also publishes. That is enough to tell the ten records apart.

This chapter asks one question: *what does a disclosure risk figure measure, and under what conditions does the reciprocal of the key frequency answer it?* The answer starts from the objects every assessment involves, defines risk as the probability of one experiment, and derives the familiar formula from that experiment under four assumptions. Stating the assumptions shows where they fail, and two of them fail in the ordinary case.

2.1 Population, sample, key and mechanism

Let the finite population be $P = \{1, \dots, N\}$. The institution holds a confidential sample, indexed by a random set $I \subseteq P$ of size n, and we write $S_i = \mathbf{1}\{i \in I\}$ for the inclusion indicator of unit i and $\pi_i = \mathbb{P}(S_i = 1)$ for its inclusion probability. A census is the case $I = P$. Each unit has a confidential record X_i.

An adversary does not see all of X_i. It works through a key, a map K_a from records to a set of key values, where the subscript a names the attack scenario. The key collects the variables the adversary can observe about a person or firm from outside: for a household

survey perhaps age band, sex and district, for a credit register perhaps sector, region and size class. For each key value k the population and sample frequencies are

$$F_k = \sum_{i \in P} \mathbf{1}\{K_a(X_i) = k\}, \qquad f_k = \sum_{i \in P} S_i \, \mathbf{1}\{K_a(X_i) = k\}, \tag{2.1}$$

so that $0 \leq f_k \leq F_k$. A record is sample unique when $f_k = 1$ and population unique when $F_k = 1$.

The institution does not publish X_I itself. It applies a protection mechanism, which we write as a probability kernel M_θ indexed by a setting θ, and publishes a release Y_θ drawn from $M_\theta(\cdot \mid X_I)$. The notation covers deterministic transformations, such as recoding or suppressing small cells, as the case in which the kernel puts all its mass on one output. A minimum cell count of five, a coarser age classification and a Laplace mechanism with $\varepsilon = 1$ are all settings in this sense, and nothing so far requires a setting to carry a formal guarantee.

Everything that follows depends on the randomness in the release, because a disclosure risk is a probability over four sources at once: the sampling design, the mechanism, the error in the adversary's own information, and any randomness in the adversary's choice. An assessment that integrates over some of these and conditions on others must say which, since the same file can give risk figures an order of magnitude apart depending on that choice.

2.2 One experiment

Fix a target t in the population. The adversary holds auxiliary information Z_t about the target, observes the release, and applies a linkage rule δ_a that either names a released record as the target's or abstains. Correct identification is the event that the target is in the file and the rule names its record.

Definition 2.1 (Scenario-conditional risk)**.** Given the background information $\mathcal{B}_a$ that scenario a grants the adversary, the unconditional and membership-conditional risks for target t are

$$R_a^{\mathrm{unc}}(t, \theta) = \mathbb{P}\{S_t = 1, \ \delta_a(Z_t, Y_\theta) = t \mid \mathcal{B}_a\}, \qquad R_a^{\mathrm{in}}(t, \theta) = \mathbb{P}\{\delta_a(Z_t, Y_\theta) = t \mid S_t = 1, \ \mathcal{B}_a\}. \tag{2.2}$$

The two differ by the chance that the target is in the file at all. If a correct link is impossible when the target is absent,

$$R_a^{\mathrm{unc}}(t, \theta) = \pi_t \, R_a^{\mathrm{in}}(t, \theta). \tag{2.3}$$

This is the multiplication rule of probability and involves no independence assumption. It has practical weight because a sampling fraction is often applied or omitted without comment, and with a 5 per cent sample that silent choice moves the reported risk by a factor of twenty.

2.3 Where the reciprocal of the key frequency comes from

The most familiar disclosure measure assigns a record with key k the risk $1/f_k$: if three released records share the target's key, the adversary has a one in three chance. The formula is usually presented as self-evident. It is in fact the answer to the experiment above under four assumptions, which are worth stating one by one.

Assumption 2.1. The released key is unperturbed: applied to the release, the key map gives the same values as applied to the confidential sample.

Assumption 2.2. The adversary observes the target's key without error.

Assumption 2.3. All records that share the target's key are exchangeable in the adversary's information: nothing the adversary knows favours one of them over another.

Assumption 2.4. The adversary chooses uniformly among the records compatible with what it knows.

Proposition 2.2 (Reciprocal frequency)**.** *Under Assumptions 2.1 to 2.4, for a target in the file whose key is k,*

$$\mathbb{P}\{\delta_a = t \mid S_t = 1,\ K_a(X_t) = k,\ f_k\} = \frac{1}{f_k}. \tag{2.4}$$

The result is old (Duncan and Lambert, 1989; Skinner and Elliot, 2002), and the value of restating it lies in the list of conditions. Assumption 2.3 fails whenever the adversary holds a variable, correlated with something the release publishes, that distinguishes candidates within a key class. In financial and health data that is the normal case: a lender knows roughly how large its own loan to a firm is, and a neighbour knows roughly how old someone is. Assumption 2.2 fails whenever the adversary's own source contains reporting error, which is also the normal case. When either fails, the right object is no longer a frequency but a posterior probability.

At the Bank this is where the chair's phrase runs into trouble. The credit register publishes each loan's exposure beside its sector, region and size class. Pieter, a credit analyst at one of the reporting lenders, knows the size of every loan his bank has made, give or take a rounding. When he looks for one of his own borrowers in a class of ten, he does not choose at random. He chooses the record whose exposure matches his books, and Assumption 2.3 no longer holds.

2.4 When the assumptions fail

Suppose the adversary's information and the released records are compared through a likelihood, in the sense of record linkage under error (Fellegi and Sunter, 1969). Let L_{tj} be the likelihood of what the adversary observes under the hypothesis that released record j belongs to target t, let q_{tj} be the prior weight on that hypothesis, and let $j = 0$ stand for the hypothesis that the target is not in the file. Bayes' rule gives the posterior

$$p_{tj} = \frac{q_{tj} L_{tj}}{q_{t0} L_{t0} + \sum_{\ell \in I} q_{t\ell} L_{t\ell}}, \tag{2.5}$$

and an adversary that names the most probable candidate succeeds with probability

$$R_a^{\text{in}}(t, \theta) = \mathbb{E}\Big[\max_{j \in I} p_{tj} \;\Big|\; S_t = 1,\ \mathcal{B}_a \Big]. \tag{2.6}$$

When the compatible records have equal likelihood and the no-match prior is zero, (2.5) and (2.6) reduce to the reciprocal frequency. In every other case they can lie above or below it.

Two consequences follow, and both cut against common practice. First, measurement error cannot be assumed to reduce risk. It changes the likelihood ratios, and it may spread

the posterior across candidates or concentrate it on one. In the simulation study of Part V, adding reporting error to the key lowers the adversary's success from 0.466 to 0.381, while giving the adversary one noisy reading of a continuous variable brings it back to 0.464. The reciprocal-frequency figure was close to the truth in that case by cancellation, not by argument.

Second, a release can satisfy a minimum cell count on its declared key and still expose individual records. If the release publishes a continuous variable, such as an exposure or an income, and the adversary holds a noisy reading of it, the posterior concentrates on the record whose value is nearest the reading, whatever the size of the key class.

2.5 k-anonymity is a threshold statement

The rule that every released key class should contain at least k records, k-anonymity in the sense of Sweeney (2002), is the most widely used protection criterion in this field. The experiment above shows exactly what it delivers.

Proposition 2.3 (k-anonymity as a decision rule)**.** *Under Assumptions 2.1 to 2.4, and for a tolerance $\tau \in (0, 1]$, the largest prosecutor risk in the release is at most τ if and only if every occupied key class has at least $\lceil 1/\tau \rceil$ members:*

$$\sup_{t \in I} \frac{1}{f_{K_a(X_t)}} \leq \tau \iff \min_{k:\, f_k > 0} f_k \geq \lceil 1/\tau \rceil. \tag{2.7}$$

If Assumption 2.3 fails, for instance because the release publishes a continuous variable of which the adversary holds a noisy reading, the implication from right to left fails: there are releases that are k-anonymous on the declared key in which some target's risk exceeds $1/k$.

Read this way, k-anonymity is a special case of the decision rule developed in Part III. It is the case in which the risk model is exact exchangeable matching, the estimate carries no uncertainty, and the tolerance is the reciprocal of an integer. Each of these simplifications is a place where an institution can be wrong, and only the third is a policy choice. The failure in the second half of the proposition is not hypothetical. On a simulated register of 200,000 units with a 5 per cent sample, a release that is 10-anonymous on its declared key contains a record whose own re-identification probability is 0.74, with a one-sided 95 per cent lower bound of 0.64. Raising the minimum class size on that key does not establish a bound for the enriched attack, because the variable that separates the records is not part of the key.

The refinements of k-anonymity follow the same pattern. The ℓ-diversity criterion (Machanavajjhala et al., 2007) addresses the case in which all records in a key class share a sensitive value, and t-closeness (Li et al., 2007) limits how far the distribution of a sensitive value within a class may differ from its overall distribution. Both protect against attribute disclosure, which can occur without any record being identified. They change the risk model rather than the decision rule, and they cannot be read as interchangeable tolerances for correct linkage.

Naledi ran Pieter's attack on the register before the next meeting. Against an adversary who reads exposure with small error, the minimum cell count that the committee had treated as a one in ten guarantee bounded nothing at all: Chapter 13 reports that no floor up to 300 brings his success below the committee's tolerance. The chair's phrase was true of a different adversary, one who knows only the key.

2.6 What a risk figure must carry

The argument of this chapter reduces to a requirement. A reported risk figure should specify its scenario: who the adversary is, what it knows, which targets it attacks, and which conditioning event the figure assumes, in particular whether the target is known to be in the file. Chapter 3 shows that three measures commonly reported under one label, all computed on the same simulated release, give 0.466, 0.064 and 0.023. Each is correct for its own experiment, and only a declared scenario says which experiment the institution meant.

2.7 The answer

The question was what a disclosure risk figure measures and when the reciprocal of the key frequency answers it. A risk figure measures the probability that a declared adversary, with declared knowledge, names the right record in one defined experiment, and it means nothing until that experiment is stated. The reciprocal of the key frequency is the exact answer when the released key is unperturbed, the adversary observes it without error, nothing it knows distinguishes the records in a class, and it chooses among them at random. When the adversary holds a variable the release also publishes, or observes the key with error, the answer is a posterior probability that can lie well above or below $1/f_k$, and a minimum cell count on the key alone no longer bounds it. The chair's minutes should have read: "no record carries a risk above one in ten against an adversary who knows only the key."

IN PRACTICE

For every risk figure in a release assessment, record the adversary's auxiliary variables and how precisely it observes them, whether the target is assumed to be in the file, and whether the release publishes any variable outside the key that the adversary might also know. If it does, a minimum cell count on the key alone does not bound the risk, and the assessment needs a simulated attack that uses that variable.

In the software. The measures of this chapter are in `privacy_risk.risk` (`prosecutor_risk`, `marketer_sample_risk`, `marketer_unconditional_risk`), and the posterior linkage of (2.5) and (2.6) is in `privacy_risk.linkage` (`posterior_linkage`, `maximum_posterior_success`, and `attack` for the simulated version).

2.8 Proofs

Proof of Proposition 2.2. Condition on $S_t = 1$ and $K_a(X_t) = k$. By Assumption 2.1 every released record carries its confidential key, and by Assumption 2.2 the adversary observes k. The compatible set is therefore $\{j \in I : K_a(X_j) = k\}$, which has f_k members by (2.1) and contains t. By Assumption 2.3 the adversary's information is invariant to permutations of that set, and by Assumption 2.4 it selects each member with probability $1/f_k$. Exactly one member is the target, so the probability of a correct choice is $\sum_j f_k^{-1}\mathbf{1}\{j = t\} = 1/f_k$. □

Proof of Proposition 2.3. By Proposition 2.2, the largest prosecutor risk in the release is $1/\min_{k:\, f_k>0} f_k$. It is at most τ if and only if $\min_k f_k \geq 1/\tau$, and since frequencies are integers this holds if and only if $\min_k f_k \geq \lceil 1/\tau \rceil$, which is k-anonymity with $k = \lceil 1/\tau \rceil$.

For the failure claim, drop Assumption 2.3. Let the release publish a confidential variable y beside the key, let a key class have exactly k members with distinct values of y, and let the adversary observe $z_t = y_t + \sigma\xi$ with ξ standard normal. Under the Gaussian likelihood the

posterior (2.5) is largest for the member whose y is nearest z_t. Let $d > 0$ be the smallest gap between two values of y in the class. The maximum-posterior rule is correct whenever $|\sigma\xi| < d/2$, an event whose probability tends to one as $\sigma \to 0$. Hence for all sufficiently small σ the target's risk exceeds $1/k$, while the release is k-anonymous on the key for every σ. $\square$

Chapter 3

Measures that answer different questions

Most disclosure assessments end in a table, and most such tables have a row headed "re-identification risk". Readers of these tables, careful ones included, tend to bring one belief to that row. They take the measures in circulation, the prosecutor risk, the journalist risk, the marketer risk, the share of sample-unique records and the reciprocal of a key frequency, to be different summaries of one underlying quantity. On this view the measures should roughly agree, an analyst may report whichever suits the release, and a cautious analyst can quote the largest and be safe.

The belief deserves a fair statement, because it has real support. Every one of these measures is computed from the same key frequencies. Each is a number between zero and one that is read as a probability. Each falls when key classes grow and rises when they shrink, so across two releases they usually rank the files in the same order. The handbooks present them together (Hundepool et al., 2012), and software prints them side by side. Measures that move together and share a vocabulary look like views of one thing.

The first fact that breaks the belief is a single release. On the synthetic register used throughout this book, a population of 200,000 units from which a 5 per cent sample is released, the mean prosecutor risk of the unprotected release is 0.466. The same mean computed with population frequencies in place of sample frequencies is 0.064, and the marketer risk for an adversary who does not know who is in the file is 0.023. These are three figures for one file, one key and one adversary's knowledge, and the largest is twenty times the smallest. None of them is a computing error. A factor of twenty separates a figure that most release committees would accept from one that none would. The rule of quoting the largest fails as well, for a reason this chapter reaches near its end: the largest of these measures is not an upper bound on anything once the adversary knows more than the key.

This chapter therefore asks one question. What experiment does each measure describe, and when can a figure computed for one experiment stand in for a figure computed for another?

3.1 Three records of one release

The question first reached the Bank as an audit finding. Grace, from internal audit, had been asked to check that the Bank's past release decisions could be reproduced from their records. She pulled the files for three products drawn from the same household survey:

a public extract, a scientific-use file for accredited researchers, and a tabular release. All three records reported a "re-identification risk", and all three cited a tolerance. Grace could reproduce none of the figures from the records alone, because none of them said whether the target was assumed to be in the file, whose frequencies had been used, or whether the sampling fraction had been applied.

Naledi, the Bank's disclosure control officer, recomputed each figure from the archived data and found that the three records had used three different quantities under one name. Her note to the chair of the release committee was short. The measures were not three estimates of one risk. They were the answers to three different experiments, and the records had not said which experiment each answered. The rest of this chapter follows the argument her note had to make.

3.2 Three experiments on one file

Chapter 2 defined a disclosure risk as the probability of one experiment: a declared adversary, holding declared information about a target, applies a linkage rule to the release, and succeeds when it names the target's record. The names prosecutor, journalist and marketer, which El Emam and Dankar (2008) gave to three conventional scenarios, are labels for three such experiments. They differ in two choices that the adversary's position fixes before any data are seen. The first choice is whether the adversary knows that the target is in the file. The second is whether the adversary attacks one target or many.

The *prosecutor* knows that the target is in the released file, perhaps because the target told someone that they took part in the survey, or because the file is a census of a population the target is known to belong to. Under the four assumptions of Proposition 2.2, its chance of naming a target whose key is k is $1/f_k$, where f_k is the number of released records with that key. The file-level summaries are the mean of $1/f_k$ over released records, which describes a target drawn uniformly from the file, and the maximum, which describes the worst released record.

The *journalist* does not know whether the target is in the file. It starts from a released record that looks rare and asks whether that record can be tied to one person in the population. The quantity that decides this is the population frequency F_k of the record's key, which the journalist cannot see in the release.

The *marketer* attacks many targets at once and values the share it links correctly. Its risk is an average over a declared population of targets. Whether that population is the released file or the whole population, and whether the marketer knows which units are in the file, are two further choices, and each changes the figure.

Figure 3.1 places the ten summaries that the simulation study computes for the unprotected release on one logarithmic axis and groups them by the experiment they describe. The axis runs over three orders of magnitude. The bars with the same shade and hatch belong to the same experiment, and within each group the figures agree with one another or differ for a stated reason. Between groups they differ by factors from two to more than a hundred. The sections below take the groups in turn.

3.3 The prosecutor and an identity

Start with the group at the top of Figure 3.1, where the target is known to be in the file and the adversary chooses among released candidates. Two of its bars have the same length,

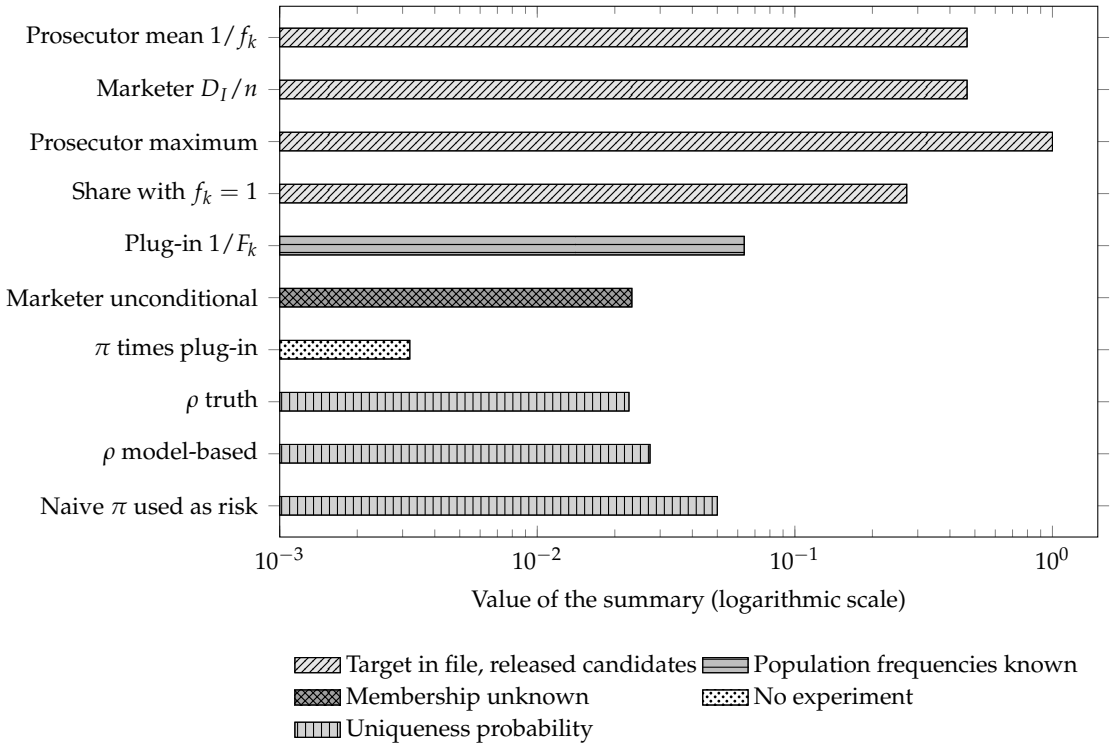


FIGURE 3.1: Ten summaries of one synthetic release ($N = 200{,}000$, $\pi = 0.05$, $n = 10{,}035$), grouped by the experiment each describes. Shade and hatch mark the group. The axis is logarithmic, so equal distances are equal ratios.

0.4663, and that is no coincidence. The mean prosecutor risk over released records and the marketer risk computed as the number of distinct released keys divided by the number of released records are the same quantity.

Proposition 3.1 (The in-file marketer identity)**.** *Let D_I be the number of distinct keys among the n released records. Under Assumptions 2.1 to 2.4,*

$$\frac{1}{n}\sum_{i\in I}\frac{1}{f_{K_a(X_i)}} = \frac{D_I}{n}. \tag{3.1}$$

The identity holds because each occupied key class contributes f_k records, each with risk $1/f_k$, and so contributes exactly one to the sum. It has a plain reading. A marketer who attacks every released record and picks uniformly within each key class gets one correct link per class on average, so its expected haul is the number of classes. On the synthetic release there are 4,679 distinct keys among 10,035 records, which gives the figure in the first two bars. The identity also gives a check on software: a program that reports a mean prosecutor risk different from D_I/n on an unperturbed release has an error somewhere.

The other two bars in the group summarise the same distribution differently. The maximum prosecutor risk is 1, because 27.2 per cent of the released records are unique in the sample. A maximum of 1 says that at least one record is fully exposed to a prosecutor. It says nothing about how many, which is what the share of sample uniques adds. A committee that receives the whole distribution of $1/f_k$ across released records can read both summaries from it, and more besides.

The prosecutor figures carry a condition that the label hides. The value $1/f_k$ is the success probability only under the four assumptions of Proposition 2.2: an unperturbed key,

an adversary that observes the key without error, records that are exchangeable within the key class, and a uniform choice among them. Chapter 2 showed that the third assumption fails whenever the adversary holds a reading of a released variable outside the key, and that in that case the success probability can exceed $1/f_k$. The prosecutor maximum is then the largest value of a formula, not the largest risk. This is why the cautious rule of quoting the largest measure fails. The largest measure in the table is computed under assumptions that an informed adversary breaks, so it bounds only the risk of an uninformed one.

The public record shows how easily an adversary comes to know that a target is in a file. In the case reported by Sweeney (2002), a state agency released hospital records of state employees and their families with names removed. The file covered a defined group, so anyone who knew that a person belonged to the group and had been treated knew that the person's record was in it. Sweeney identified the record of the state governor by his date of birth, sex and postcode. Whoever holds that kind of knowledge is a prosecutor in the sense of this section, and a file in which those three attributes leave most records alone in their key class gives it little protection.

3.4 Sample and population frequencies

The second group in Figure 3.1 holds one bar, the mean of $1/F_k$ over released records, where F_k is the population frequency of the record's key. Its value is 0.064, against 0.466 for the same mean with sample frequencies, a ratio of 7.3. Both figures have been published as the re-identification risk of a file, and an analyst who did not know which had been computed could not tell from the number alone.

The two figures describe different experiments. The mean of $1/f_k$ describes an adversary choosing among released candidates. The mean of $1/F_k$ describes an adversary who starts from a released record and must find its owner among all members of the population with the same key, as one would by searching a full population list for the record's combination of attributes. The second experiment is the one behind the second step of that case, where the released record was matched against the voter list of the governor's town (Sweeney, 2002). Sweeney's estimate that simple demographics identify most of the United States population (Sweeney, 2000), revisited with later census data by Golle (2006), is a statement about population frequencies: it counts how often $F_k = 1$. Neither experiment is more correct than the other. They answer different questions.

What can be said in general is the sign of the gap. Since the released sample is a subset of the population, no key class can have more members in the sample than in the population, and so the sample-frequency measure can never fall below the population-frequency measure.

Proposition 3.2 (Ordering and plug-in bias)**.** *For every key k with $f_k > 0$, $1/f_k \geq 1/F_k$ pathwise, so for every released record i,*

$$\mathbb{E}\Big[\frac{1}{f_{K(i)}}\Big] \;\geq\; \mathbb{E}\Big[\frac{1}{F_{K(i)}}\Big], \tag{3.2}$$

with equality only if every member of every occupied key class is in the sample. If f_k is positive with probability one and has finite mean, then

$$\mathbb{E}\Big[\frac{1}{f_k}\Big] \;\geq\; \frac{1}{\mathbb{E}[f_k]}, \tag{3.3}$$

strictly unless f_k is degenerate.

The first part fixes the direction of the error when one frequency is used in place of the other. An analyst who has population frequencies to hand and reports the mean of $1/F_k$ for an adversary who in fact chooses among released candidates understates that adversary's success, and by a factor that can be large: on the synthetic release it is 7.3. The reverse substitution overstates. The direction is known before any data are seen, and that is useful, because a reviewer can tell from the description alone which way a mislabelled figure errs.

The second part concerns a different shortcut. Some reports compute the average key frequency and quote its reciprocal as the risk, on the reasoning that a typical record shares its key with that many others. By Jensen's inequality the reciprocal of an average frequency is never larger than the average of the reciprocal frequencies. On the synthetic release, where the mean sample frequency of a released record's key class gives a reciprocal of 0.191, the shortcut reports less than half of the mean prosecutor risk of 0.466. The reason is that rare keys carry most of the risk, and an average frequency is dominated by the common ones. A reciprocal-of-average figure is therefore biased downwards for the in-file experiment, with a known sign, and it should not appear in a decision record as a risk.

At the Bank, this was the substance of the first finding. The scientific-use file of the earlier record had been assessed with population frequencies taken from the census, because the census was available and seemed the more complete source. The researchers who would receive the file, however, would be choosing among released records. Naledi's recomputation put the in-file risk at several times the recorded figure, and the record had compared the smaller figure with the tolerance.

3.5 When membership is unknown

The third group in Figure 3.1 holds the marketer who does not know which population units are in the file. Its risk is the unconditional risk of Definition 2.1, averaged over a declared population of targets. By the multiplication rule (2.3), each target's unconditional risk is its inclusion probability times its in-file risk. Under a common inclusion probability π and exact exchangeable matching the file-level figure is π times the mean prosecutor risk, and on the synthetic release that is $0.05 \times 0.4663 = 0.0233$. This figure is correct for its experiment, which is an adversary who draws a target from the population without knowing whether that target was sampled. Its smallness is real: in nineteen of every twenty attempts there is no record to find.

The fourth group holds a figure that answers no experiment, and it appears in the table because reports have used it. It multiplies the population-frequency measure by the sampling fraction, giving 0.0032. The product discounts for non-membership twice. The population frequency F_k already counts all the population members who share the key, sampled or not, so an adversary choosing among them is already paying for not knowing who is in the file. Multiplying by π then charges for the same ignorance a second time. The result is less than one per cent of the in-file figure and about a seventh of the correct unconditional one, and it has no adversary behind it.

The two figures of the third and fourth groups show how a sampling fraction enters a risk figure silently. In Grace's sample of past records, the public extract had been assessed at 0.023, and the record compared that figure with a tolerance of 0.05 without mentioning π. The figure was the right one for an adversary who did not know who was sampled. It was the wrong one for the adversary the release committee had in mind, a neighbour or a journalist who knew that a particular household had been interviewed, for whom the factor

of π does not apply. The chair's question, when Naledi presented the finding, was which adversary the committee had meant. The record did not say.

3.6 The journalist and the uniqueness probability

The last group in Figure 3.1 belongs to the journalist, and it is the one group that needs a model for something no one can observe. The journalist looks at a released record that is unique in the sample, $f_k = 1$, and asks whether the person behind it is also unique in the population. The classical object, studied by Skinner and Holmes (1998) and Skinner and Elliot (2002), is the probability that a sample unique is a population unique,

$$\rho_k = \mathbb{P}(F_k = 1 \mid f_k = 1). \tag{3.4}$$

It is not quite the journalist's success probability. A journalist who finds that the record's key is shared by F_k people in the population and picks one of them at random succeeds with probability $1/F_k$, and averaging over what it does not know gives $\mathbb{E}[1/F_k \mid f_k = 1]$, which is at least ρ_k. It differs again from the journalist risk $1/\min_k F_k$ of El Emam and Dankar (2008), which is a worst case over key classes. The three are close in a large population with many uniques and can be far apart otherwise, which is reason enough to state which one a report contains.

Proposition 3.3 (The uniqueness probability under Bernoulli sampling)**.** *Suppose each population unit enters the sample independently with probability π, and let $g(v) = \mathbb{P}(F_k = v)$ for $v \geq 1$ be the law of the population frequency of a key class. Then*

$$\rho_k = \frac{\pi\, g(1)}{\sum_{v \geq 1} v\pi(1-\pi)^{v-1} g(v)}, \tag{3.5}$$

and the journalist's success probability for a sample unique satisfies

$$\mathbb{E}\Big[\frac{1}{F_k} \;\Big|\; f_k = 1\Big] \;\geq\; \rho_k. \tag{3.6}$$

Equation (3.5) shows what the sampling fraction contributes and what it does not. It enters through the chance that a class of size v leaves exactly one member in the sample, which is $v\pi(1-\pi)^{v-1}$. The law g of the population frequencies enters as well, and π alone does not determine ρ_k. Practice sometimes reports π itself as the journalist risk, on the reasoning that a population unique is caught in the sample with probability π. That reasoning computes the wrong conditional probability: it is the chance that a population unique is sampled, not the chance that a sample unique is a population unique. On the synthetic register the true value of ρ at $\pi = 0.05$ is 0.0227, so using π overstates it by a factor of 2.20, and at $\pi = 0.5$ by a factor of 1.64. The error happens to be conservative here, but it carries no guarantee of that, since the factor depends on g.

The question that (3.5) leaves open is where g comes from. The institution sees the sample, and the population frequencies are exactly what it does not see. Rocher et al. (2019) put the practical side of this problem sharply: they argued that releasing only a small sample of a population gives less protection than it seems, because a model fitted to the sample can estimate how likely a given match is to be correct. Their argument is a journalist argument, and the remedy they point to, a fitted model for what the sample does not show, is also what the institution needs in order to assess its own release. The rest of this section sets out the simplest such model.

3.7 Estimating population uniqueness

A model for g must be estimable from the sample, and one family makes that possible in a single step. If the population frequencies of the key classes follow a negative binomial law, the sample frequencies follow a negative binomial law with the same dispersion. This is the model behind the gamma-Poisson approach of Bethlehem et al. (1990), whose marginal law is the negative binomial, and later work by Rinott and Shlomo (2006) and Shlomo and Skinner (2010) refines it. Write $\mathrm{NB}(r, \mu)$ for the negative binomial law with dispersion $r > 0$ and mean $\mu > 0$, so that

$$\mathbb{P}(F = v) = \frac{\Gamma(v + r)}{\Gamma(r)\, v!} \Big(\frac{r}{r + \mu} \Big)^{r} \Big(\frac{\mu}{r + \mu} \Big)^{v}, \qquad v = 0, 1, 2, \ldots \tag{3.7}$$

Small r means a skewed law with many empty and many small classes beside a few large ones, which is the shape of real key-frequency tables.

Proposition 3.4 (Binomial thinning preserves the negative binomial)**.** *If $F \sim \mathrm{NB}(r, \mu)$ and, given F, $f \sim \mathrm{Binomial}(F, \pi)$, then $f \sim \mathrm{NB}(r, \pi\mu)$. Consequently, if every one of the C cells of the key space has a population frequency drawn independently from $\mathrm{NB}(r, \mu)$ and the sample is a Bernoulli sample with known π, the parameters (r, μ) are identified from the C sample cell counts, empty cells included.*

The proposition turns an unobservable law into an estimable one. The analyst counts the released records in every cell of the key space, including the cells with none, fits a negative binomial to those counts by maximum likelihood, reads off the dispersion r and the sample mean $\pi\mu$, divides the second by the known π, and substitutes the fitted law into (3.5). Every step is mechanical. The model, however, is a model: real key-frequency tables are not draws from one negative binomial, since the cells of a cross-classification have structure that a single law ignores, and the log-linear models of Skinner and Holmes (1998) and Elamir (2004) exist to capture some of it.

On the synthetic register the key space has 13,440 cells, of which 12,596 are occupied in the population. The fit recovers a mean population cell size of 14.83 against a true value of 14.88, within 1 per cent. At $\pi = 0.05$ the model-based estimate of ρ is 0.0274 against the truth of 0.0227. Figure 3.2 repeats the comparison at five sampling fractions from 0.02 to 0.5. The model-based estimate lies above the truth at every fraction, with a relative error between 8 and 25 per cent, and the naive use of π lies above both. For a quantity that cannot be observed, an error of that size in a known direction is a usable result.

The fit also contains a warning, and it is the more instructive half of the result. Proposition 3.4 says that the dispersion should not depend on the sampling fraction, since thinning leaves it unchanged. In the fits it does: the estimated dispersion rises from 0.33 at $\pi = 0.02$ to 0.51 at $\pi = 0.2$ and 0.59 at $\pi = 0.5$. A parameter that the model says is fixed, drifting steadily with the sampling fraction, is evidence that the population law is not negative binomial. The estimate of ρ was still close, because ρ depends mostly on the small classes and the fit gets those roughly right, but the drift shows that its accuracy rests on the shape of this particular population. The synthetic study can see the drift because it draws samples at several fractions from a known population. An institution has one sample, but it can draw Bernoulli subsamples of that sample at smaller fractions, which by the same thinning argument are Bernoulli samples of the population at the product fraction, and fit the model to each. A dispersion that moves across those fits is a reason to report ρ with the model

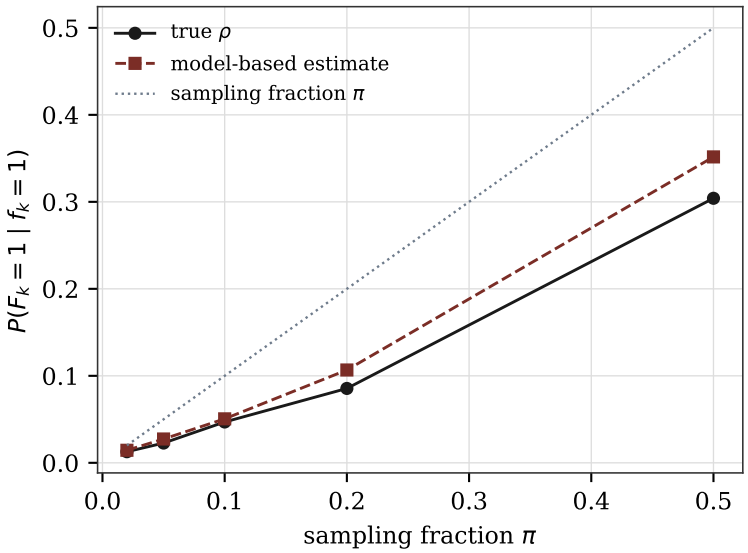


FIGURE 3.2: The uniqueness probability $\rho = \mathbb{P}(F_k = 1 \mid f_k = 1)$ against the sampling fraction on the synthetic register: the truth, computed from the known population; the estimate from a negative binomial law fitted to the sample cell counts; and the sampling fraction itself, which practice sometimes reports in its place. Source: the long-form paper, Figure 5.

stated beside it and to treat it as an estimate, which is how Chapter 15 treats it among its falsification tests.

3.8 The table Naledi sent back

Naledi's revised assessment kept all the numbers and changed the table. Each row now named the experiment it described: who the adversary was, whether it knew the target was in the file, whose frequencies were used and whether π had been applied. The row for the uniqueness probability named the fitted model and its dispersion. The row that multiplied a population-frequency measure by π was removed, with a sentence in the record saying why. Above the table she added the one line the committee would act on: the declared adversary for the scientific-use file was a prosecutor, so the figure to compare with the tolerance was the in-file risk, and the other rows were context.

The chair accepted the table and asked for one change. Since the earlier records could not be reproduced, the committee would treat their tolerances as unowned until each had been restated against a named experiment. Grace recorded that decision as the response to her finding.

3.9 What each measure answers

The question of this chapter was what experiment each measure describes and when one can stand in for another. The answer has three parts.

The measures describe different experiments, and they agree only when the experiments coincide. The mean prosecutor risk and the in-file marketer risk are one quantity by the identity of Proposition 3.1. Every other pair differs by a factor that the experiment fixes: the inclusion probability separates the in-file figure from the unconditional one; the ratio of

sample to population frequencies separates the prosecutor's choice among released records from the search of a population list, with a known sign by Proposition 3.2; and the uniqueness probability needs a model of the population frequencies, without which the sampling fraction is a guess that was off by a factor of two on the synthetic register.

One figure can therefore stand in for another only when the record shows that the two experiments are the same, and never by taking the largest. The largest conventional measure is computed under exact exchangeable matching, and an adversary with information outside the key is not bound by it. A risk figure is complete when it states its adversary, whether the target is known to be in the file, which frequencies it used, whether π was applied and, for the journalist, which population model supplied the frequencies it could not see. With those stated, the figure is no longer one of several measures. It is the answer to one question, and the next chapter asks how precisely that answer is known.

IN PRACTICE

Replace the row headed "re-identification risk" with one row per declared experiment, and in each row state the adversary, whether the target is assumed to be in the file, whether sample or population frequencies were used, and whether the sampling fraction was applied. Check on any unperturbed release that the mean prosecutor risk equals the number of distinct released keys divided by the number of released records. Never multiply a population-frequency measure by the sampling fraction. Do not report the sampling fraction as a journalist risk. If the uniqueness probability is needed, fit a population frequency model, state it, and refit it on subsamples of the release to see whether its parameters hold still.

In the software. The measures are in `privacy_risk.risk`: `prosecutor_risk`, `marketer_sample_risk`, `marketer_population_frequency_risk`, `marketer_unconditional_risk` and `journalist_unique_probability` for a stated prior on F_k. The negative binomial model is `FrequencyFit`, fitted by `fit_frequency_model` to the sample counts of every key cell, zeros included, and evaluated by `journalist_from_fit`; `journalist_truth` computes the target when the population is known, as in a simulation. The ordering of Proposition 3.2 is checked by `privacy_risk.uncertainty.plugin_ordering`, and `privacy_risk.experiments.conventional_measures` produces every value in Figure 3.1. The uniqueness probability under a fitted law is short enough to read in full:

```
def journalist_from_fit(pi: float, fit: FrequencyFit) -> float:
    support, mass = fit.probabilities()
    numerator = pi * float(mass[0])
    denominator = float(np.sum(support * pi * (1 - pi) ** (support - 1) * mass))
    return numerator / denominator if denominator else 0.0
```

The support starts at one, so the mass at zero is left out of both numerator and denominator, and it cancels as (3.5) requires.

3.10 Proofs

Proof of Proposition 3.1. By Proposition 2.2, each released record i has prosecutor risk $1/f_{K_a(X_i)}$. Group the sum by key value:

$$\sum_{i \in I} \frac{1}{f_{K_a(X_i)}} = \sum_{k:\, f_k > 0} \; \sum_{i \in I:\, K_a(X_i) = k} \frac{1}{f_k} = \sum_{k:\, f_k > 0} f_k \cdot \frac{1}{f_k} = \#\{k : f_k > 0\} = D_I.$$

Dividing by n gives (3.1). □

Proof of Proposition 3.2. By (2.1), $f_k = \sum_{i \in P} S_i \mathbf{1}\{K_a(X_i) = k\} \le \sum_{i \in P} \mathbf{1}\{K_a(X_i) = k\} = F_k$ for every k and every realisation of the sample, since $S_i \le 1$. For an occupied class, $0 < f_k \le F_k$, so $1/f_k \ge 1/F_k$ pathwise, and taking expectations preserves the inequality, which is (3.2). Equality in expectation requires equality almost surely, because the difference $1/f_k - 1/F_k$ is non-negative; that is, $f_k = F_k$ for every class the average touches, so every population member of each such class is sampled.

For (3.3), the function $x \mapsto 1/x$ is strictly convex on $(0, \infty)$. Jensen's inequality for a random variable that is positive with probability one and has finite mean gives $\mathbb{E}[1/f_k] \ge 1/\mathbb{E}[f_k]$, with equality if and only if f_k is almost surely constant. □

Proof of Proposition 3.3. Under Bernoulli sampling with common probability π, the sample frequency of a class with $F_k = v$ members is a sum of v independent Bernoulli(π) indicators, so $f_k \mid F_k = v \sim \text{Binomial}(v, \pi)$ and $\mathbb{P}(f_k = 1 \mid F_k = v) = v\pi(1-\pi)^{v-1}$. A class with $f_k = 1$ has $F_k \ge 1$. Bayes' theorem gives

$$\rho_k = \frac{\mathbb{P}(f_k = 1 \mid F_k = 1)\, g(1)}{\sum_{v \ge 1} \mathbb{P}(f_k = 1 \mid F_k = v)\, g(v)} = \frac{\pi\, g(1)}{\sum_{v \ge 1} v\pi(1-\pi)^{v-1} g(v)},$$

which is (3.5). For (3.6), note that $1/F_k \ge \mathbf{1}\{F_k = 1\}$ whenever $F_k \ge 1$, since the left side equals the right side when $F_k = 1$ and is positive otherwise. Taking conditional expectations given $f_k = 1$ gives $\mathbb{E}[1/F_k \mid f_k = 1] \ge \mathbb{P}(F_k = 1 \mid f_k = 1) = \rho_k$, with equality only if $\mathbb{P}(F_k \ge 2 \mid f_k = 1) = 0$. □

Proof of Proposition 3.4. Use probability generating functions. With $p = r/(r+\mu)$, the generating function of $\text{NB}(r, \mu)$ is

$$G_F(s) = \mathbb{E}[s^F] = \Big(\frac{p}{1-(1-p)s}\Big)^r = \Big(1 + \frac{\mu}{r}(1-s)\Big)^{-r}, \qquad |s| \le 1.$$

Given F, f is a sum of F independent Bernoulli(π) variables, whose common generating function is $1 - \pi + \pi s$. Hence

$$G_f(s) = \mathbb{E}\big[\mathbb{E}[s^f \mid F]\big] = \mathbb{E}\big[(1-\pi+\pi s)^F\big] = G_F(1-\pi+\pi s) = \Big(1 + \frac{\mu}{r}\,\pi(1-s)\Big)^{-r},$$

which is the generating function of $\text{NB}(r, \pi\mu)$. A generating function on $|s| \le 1$ determines the law, so $f \sim \text{NB}(r, \pi\mu)$.

For identification, the C sample cell counts are then independent draws from $\text{NB}(r, \pi\mu)$. The negative binomial family is identified in its parameters: its mean is $\pi\mu$ and its variance is $\pi\mu + (\pi\mu)^2/r$, so the first two moments of the count law determine $\pi\mu$ and r uniquely. Since π is known, μ follows, and with it the population law $\text{NB}(r, \mu)$ of every cell. The maximum likelihood estimator of $(r, \pi\mu)$ from the C counts is consistent as $C \to \infty$ under the usual regularity conditions for this family (van der Vaart, 1998). □

Part II

Evidence

Chapter 4

A risk figure is an estimate

A committee that has declared its adversary, as Chapter 2 asks, and settled which experiment its figure describes, as Chapter 3 asks, still has to obtain the figure. For any realistic adversary the figure has no closed form. It comes from simulating the attack many times and counting how often it succeeds. Most committees then apply a rule that seems too plain to question: if the estimated risk is below the tolerance, the release is safe enough to make.

The rule has a strong case. A simulated success rate is an unbiased estimate of the risk it targets. Its standard error falls with the square root of the number of simulated attacks, so a few thousand replicates give an estimate whose error is a small fraction of any tolerance a committee is likely to set. The tolerance itself is a judgement, often rounded to one significant figure, and set beside that judgement the second decimal place of a well-run simulation can seem beside the point. On this view the estimate is the best single guess at the risk, and a decision based on the best guess is the reasonable one.

The fact that breaks the rule concerns the releases it is meant to stop. Take a release whose true risk sits exactly at a tolerance of 0.12 and estimate that risk from 1,200 independent simulated attacks, the replicate count of the simulation study. The estimate falls at or below the tolerance with probability 0.52. At the boundary the rule is a coin toss, and it stays close to one for releases a little above the boundary: at a true risk of 0.125 it still releases about one time in three. A committee using it believes that it keeps risk below the tolerance. What it keeps below the tolerance is a random number whose mean is the risk, and the decision record, which holds only that number, cannot show the difference.

The chapter asks one question. How much evidence must a simulated risk figure carry before a committee may conclude that the risk lies at or below its tolerance?

4.1 A figure just under the line

The question reached the Bank in a committee meeting. Naledi had assessed a household survey release with age in twenty-year bands, three occupation groups and a minimum cell count of ten. Against the stronger of the two declared adversaries her simulation named the target in 127 of 1,200 attacks, a rate of 0.1058, and the committee’s tolerance was 0.12. The chair read the figure, saw that it was below the line, and asked whether anyone objected to release.

Naledi did. Her paper carried a second column, an upper confidence bound at the 95 per cent level, and in that column the same setting stood at 0.1216, above the tolerance. She recommended that the committee not release on this evidence. The chair’s reply was

the natural one: the best estimate of the risk was below the tolerance, so on what grounds should a second, larger number decide? The rest of this chapter is the answer Naledi gave.

4.2 The estimate and its error

Let the declared attack be simulated B times under the declared scenario, each replicate drawing a target and the adversary's information afresh, and let $C^{(b)} = 1$ if replicate b names the target correctly. The Monte Carlo estimator of the risk R is the success rate,

$$\widehat{R} = \frac{1}{B}\sum_{b=1}^{B} C^{(b)}, \qquad \mathbb{E}[\widehat{R}] = R, \qquad \operatorname{Var}(\widehat{R}) = \frac{R(1-R)}{B}. \tag{4.1}$$

The variance formula holds because the $C^{(b)}$ are independent Bernoulli variables with success probability R, and $X = B\widehat{R}$ is then binomial. Both facts depend on how the simulation was built. If the replicates share an auxiliary file, a set of released records or a fitted model, they are not independent, and (4.1) understates the variance. The cure is to resample the unit that is shared, the auxiliary file or the fitted parameters, and recompute the risk within each resampled unit (Efron and Tibshirani, 1993), so that the spread between units carries the dependence. Section 4.7 returns to this as an analysis of variance. For now take the replicates to be independent, which in the simulation study they are, conditional on the realised sample and mechanism output.

The estimate is unbiased, and its standard error at the Bank's figure is 0.0089, the square root of $0.1058 \times 0.8942/1200$. That is less than a tenth of the tolerance, which is the source of the chair's confidence. The gap between the estimate and the tolerance, 0.014, is less than two standard errors, and that is the source of Naledi's doubt.

4.3 What a point-estimate rule does at the boundary

The behaviour of a decision rule is best read from its operating characteristic, the probability that it releases as a function of the true risk. Figure 4.1 plots it for two rules at $B = 1{,}200$ and a tolerance of 0.12. The solid line is the rule that releases when $\widehat{R} \le \tau$. The dashed line is the rule that releases when an exact one-sided 95 per cent upper bound is at or below τ, the rule defined in the next section.

The solid curve crosses one half at the tolerance. That is the general pattern, and it holds for any estimator that is centred on the truth.

Proposition 4.1 (A point estimate at the tolerance)**.** *Suppose a rule releases whenever $\widehat{R}_B \le \tau$, where $\widehat{R}_B$ is a consistent estimator of R with $\sqrt{B}(\widehat{R}_B - R)$ converging in law to a distribution symmetric about zero and continuous at zero. If $R = \tau$, the rule releases with probability tending to $1/2$ as $B \to \infty$. If instead the rule releases only when a one-sided upper bound $\overline{R}_{1-\alpha}$ with $\mathbb{P}\{\overline{R}_{1-\alpha} \ge R\} \ge 1-\alpha$ is at or below τ, then for every $R > \tau$ the rule releases with probability at most α.*

The proposition does not say that point estimates are poor estimates. It says that a rule which compares a point estimate with a line has no control over the error it exists to prevent. More replicates sharpen the solid curve in Figure 4.1 towards a step at τ, but they do not move the half at τ itself, and the releases near the line are the ones a committee most needs its rule to handle. The bound rule fixes the chance of a wrong release for every risk above

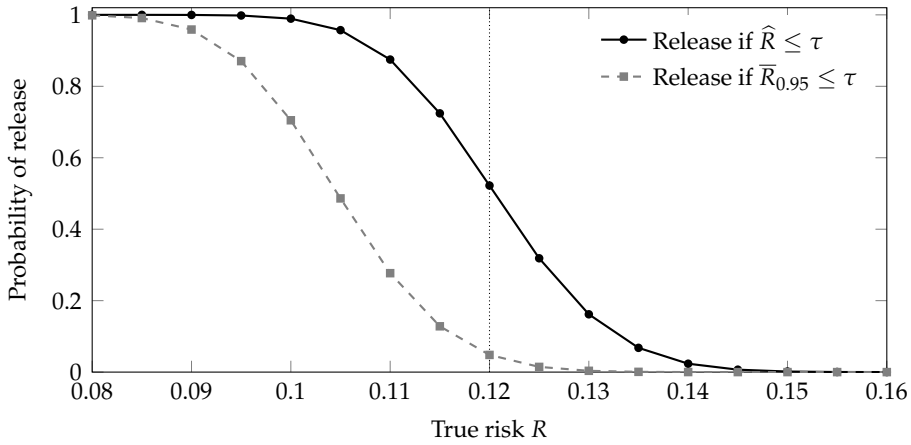


FIGURE 4.1: Probability that each rule releases, against the true risk, with $B = 1{,}200$ independent simulated attacks and a tolerance $\tau = 0.12$ (dotted line). The rules are told apart by line style, shade and marker. At $R = \tau$ the point-estimate rule releases with probability 0.52 and the bound rule with probability 0.048. Exact binomial calculation.

the tolerance at α or less, and the dashed curve shows the price: it releases less often below the line. Section 4.6 weighs that price.

Grace, reading the Bank's archive for the audit of Chapter 3, met the consequence in its plainest form. The past decision records held point estimates only, some of them a few thousandths below their tolerances. For each such record she could see that the committee had released, and she could not tell whether the release had been a sound one or the lucky side of a coin. The record did not hold the number of replicates, so the uncertainty could not be rebuilt. Her finding recommended that every risk figure in a decision record carry its replicate count and its bound.

4.4 Exact one-sided bounds

The bound rule needs a bound that is valid at the risks the committee faces, which are often small. The normal approximation, $\widehat{R} + z_{1-\alpha}\sqrt{\widehat{R}(1-\widehat{R})/B}$, is unreliable there: its coverage falls short of the nominal level for small R and moderate B, and it collapses to zero when no attack succeeds (Brown et al., 2001). The exact bound of Clopper and Pearson (1934) has no such failure. If $X = x$ of B independent attacks succeed, the one-sided $(1-\alpha)$ upper bound is

$$\overline{R}_{1-\alpha}(x, B) = \mathrm{Beta}^{-1}_{1-\alpha}(x+1,\ B-x), \qquad x < B, \tag{4.2}$$

and one when $x = B$, where $\mathrm{Beta}^{-1}_{q}(a, b)$ is the qth quantile of a beta distribution. Equivalently, $\overline{R}_{1-\alpha}(x, B)$ is the risk at which the chance of seeing x or fewer successes is exactly α: a larger risk would make the observed count implausibly low.

Proposition 4.2 (Validity of the exact bound)**.** *If* $X \sim \mathrm{Binomial}(B, R)$, *then* $\mathbb{P}_R\{\overline{R}_{1-\alpha}(X, B) < R\} \le \alpha$ *for every* $R \in [0, 1]$.

The worked example of the long-form paper shows the bound in use. On the confirmation draw for the single-release attack on the synthetic credit register, the attack named the right loan for 337 of 4,000 targets, a rate of 0.084. The 0.95 quantile of the $\mathrm{Beta}(338, 3663)$ distribution is 0.092, below the tolerance of 0.12 with a margin of 23 per cent of the tolerance,

so the file passes for that adversary. The decision rests on the bound and not the rate. Had the rate been 0.115 on the same number of targets, the bound would have been 0.123, and the release would have failed.

The case of no successes deserves its own formula, because it is the case a committee with a small tolerance hopes to see. With $x = 0$ the beta quantile has a closed form, since $\mathbb{P}_U(X = 0) = (1 - U)^B$, and setting that to α gives

$$\overline{R}_{1-\alpha}(0, B) = 1 - \alpha^{1/B} \approx \frac{-\log \alpha}{B}. \tag{4.3}$$

At $\alpha = 0.05$, $-\log \alpha$ is close to three, which is the rule of three that Hanley and Lippman-Hand (1983) set out for clinical work: if none of B patients suffers an adverse event, the event rate may still be as high as about $3/B$. The same arithmetic applies to an attack that never succeeds. No number of failed attacks shows that the risk is zero. What a run of failures shows is an upper bound, and its strength is set by B and nothing else.

4.5 The price of a small tolerance

Equation (4.3) turns a tolerance into a budget. A committee that wants its bound at or below τ from a run with no successes needs enough attacks to push $1 - \alpha^{1/B}$ under τ.

Proposition 4.3 (Certification cost)**.** *Suppose B independent simulated attacks produce no success. The exact one-sided $(1 - \alpha)$ bound satisfies $\overline{R}_{1-\alpha} \leq \tau$ if and only if*

$$B \;\geq\; \frac{\log \alpha}{\log(1 - \tau)}, \tag{4.4}$$

and the right side equals $-\log(\alpha)/\tau + O(1)$ as $\tau \downarrow 0$.

The cost grows in inverse proportion to the tolerance. Figure 4.2 plots the required number of attacks at five tolerances and two error levels. Certifying a tolerance of 0.01 at $\alpha = 0.05$ takes 299 independent attacks with no success. A tolerance of one in a thousand takes 2,995, and one in a hundred thousand takes 299,572; at $\alpha = 0.01$ the last becomes 460,515. At the other end, a tolerance of 0.05 needs only 59 attacks, and the long-form paper's worked example puts the figure for the credit register's tolerance of 0.12 at 24. These counts are lower limits. They assume that every attack fails. If some succeed, more are needed, and inverting (4.2) for the observed count gives the number.

The practical content is that a tolerance and an evidence standard are one decision. A committee that writes a tolerance of one in a hundred thousand into its policy has committed itself to run several hundred thousand independent attacks for every release, and to run them so that they really are independent, or else to replace simulation with an argument from a model. If it cannot do either, the tolerance is not supported by its evidence, and the decision record should say so. Choosing a small tolerance because it sounds prudent, without pricing the evidence, produces a rule that no release can pass on the evidence.

The published record of attacks offers no shortcut. A systematic review of re-identification attacks on health data (El Emam et al., 2011) found that most were small, and that most targeted data which had not been de-identified to any standard. It is tempting to read such a record as evidence that properly protected data are rarely re-identified. The record does not support that reading in the terms of this section. It is not a count of independent attempts against protected releases, so it does not supply the B of (4.3), and an absence of reported successes is not a run of observed failures.

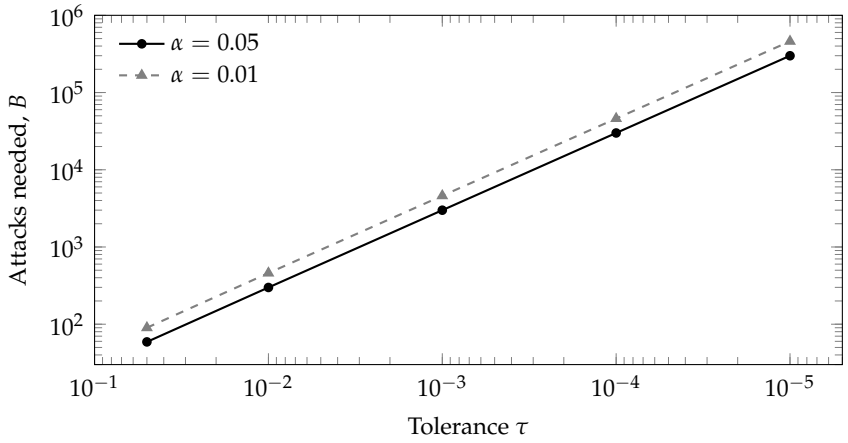


FIGURE 4.2: Independent simulated attacks, all failing, needed for the exact one-sided bound to reach the tolerance, from Proposition 4.3. Both axes are logarithmic and the tolerance falls from left to right. The error levels are told apart by line style, shade and marker. Ten times smaller a tolerance costs ten times as many attacks.

4.6 The cost of the other error

The bound rule controls one error and leaves the other to the sample size. As in the testing framework of Neyman and Pearson (1933), it fixes the more serious error, release when the risk exceeds the tolerance, at a stated level. The other error is suppression of a release whose true risk is below the tolerance, and Figure 4.1 shows how large it can be. At a true risk of 0.105, well inside the tolerance, the bound rule with 1,200 attacks releases less than half the time. The Bank's setting, with an estimate of 0.1058, sits in exactly that region.

The remedy is more attacks, and the calculation is short. The bound exceeds the estimate by roughly $z_{1-\alpha}$ standard errors, about $1.645 \times 0.0089 = 0.015$ at the Bank's figure, and that margin shrinks with $\sqrt{B}$. If the true risk is near 0.106, a margin of 0.014 between risk and tolerance needs the standard error to fall below about 0.0085 for the bound to clear the line half the time, and well below it for the bound to clear the line reliably. Four times the attacks halves the standard error. A committee that sees an estimate below its tolerance and a bound above it has learned that the evidence is too thin to decide, and the right response is to buy more evidence, not to choose the number that agrees with the hoped-for answer.

That was Naledi's recommendation. The chair accepted that the committee could not release on 1,200 attacks, and asked how many would settle the matter. Naledi's answer was that the count depended on where the true risk lay, which was the thing being estimated. If it was near the estimate, about 5,000 attacks would put the bound below 0.12 in nineteen runs of twenty. If it was nearer the tolerance, no affordable number would, and the committee would then be choosing between a stricter setting and a different access route. Chapter 6 shows how to run the attacks in stages and stop as soon as the evidence is sufficient, which an analyst may not do with the fixed-sample bound of this chapter.

4.7 Where the variance comes from

The binomial variance in (4.1) describes one source of error, the simulation, and it describes it conditional on everything the simulation holds fixed. A risk estimate has at least two further sources: the confidential sample, which is one draw from the sampling design, and any

parameters the adversary's model fits from an auxiliary source. Writing $\mathcal{D}$ for the realised sample and $\widehat{\eta}$ for the fitted inputs, the law of total variance separates the two kinds:

$$\operatorname{Var}(\widehat{R}) = \underbrace{\mathbb{E}\{\operatorname{Var}(\widehat{R} \mid \mathcal{D}, \widehat{\eta})\}}_{V_{\mathrm{MC}}} + \underbrace{\operatorname{Var}\{\mathbb{E}(\widehat{R} \mid \mathcal{D}, \widehat{\eta})\}}_{V_{\mathrm{input}}}. \tag{4.5}$$

The binomial expression estimates only V_{MC}. Increasing B drives that term towards zero and leaves V_{input} untouched. An interval built from (4.1) alone can therefore become as narrow as the analyst's computing budget allows while the uncertainty that the institution actually faces stays where it was.

A nested design measures both terms. Draw outer replicates $s = 1, \dots, S$; in each, redraw the confidential sample according to its design, refit any auxiliary model, apply the mechanism and run B attacks, giving an estimate $\widehat{R}_s$. With $\overline{R} = S^{-1} \sum_s \widehat{R}_s$, the analysis of variance is

$$\widehat{V}_{\mathrm{within}} = \frac{1}{S} \sum_{s=1}^{S} \frac{\widehat{R}_s (1 - \widehat{R}_s)}{B}, \qquad \widehat{V}_{\mathrm{between}} = \frac{1}{S-1} \sum_{s=1}^{S} (\widehat{R}_s - \overline{R})^2 - \widehat{V}_{\mathrm{within}}, \tag{4.6}$$

with the second truncated at zero when it falls slightly negative.

Proposition 4.4 (What the nested estimator estimates)**.** *Let the outer replicates be independent, let R_s be the risk conditional on the sample and inputs of replicate s, and let $B\widehat{R}_s \mid R_s \sim \mathrm{Binomial}(B, R_s)$. Write $V_{\mathrm{MC}} = \mathbb{E}[R_s(1 - R_s)]/B$ and $V_{\mathrm{input}} = \operatorname{Var}(R_s)$. Then*

$$\mathbb{E}\big[\widehat{V}_{\mathrm{within}}\big] = \frac{B-1}{B} V_{\mathrm{MC}}, \qquad \mathbb{E}\Big[\frac{1}{S-1} \sum_s (\widehat{R}_s - \overline{R})^2\Big] = V_{\mathrm{input}} + V_{\mathrm{MC}}, \tag{4.7}$$

so that, before truncation, $\mathbb{E}[\widehat{V}_{\mathrm{between}}] = V_{\mathrm{input}} + V_{\mathrm{MC}}/B$.

The bias terms are of order $1/B$ and negligible at any useful replicate count, so the two estimators split the variance as (4.5) does. The simulation study ran this design on one setting of its grid, the same twenty-year, three-group, minimum ten setting as the Bank's example above: 120 outer replicates, each redrawing the 5 per cent sample from the same population and running 500 attacks. The within-replicate standard deviation was 0.0112. The between-replicate standard deviation came out at zero after the correction, and the standard error of the overall mean was 0.0010 whether the design term was included or not. On this design, with 120 outer replicates, the sampling term was not detectable at that number of replicates.

The result does not license ignoring the design, and the reason is in its interval. A chi-square interval for the total variance, carried through to the share of variance that the design contributes, runs from 0 to 0.19. The data would not have distinguished a design contribution of up to a fifth of the variance from none. That interval assumes the outer estimates are roughly normal, which at 500 attacks and a risk near 0.07 is reasonable, though it would not be for a risk near zero. Three features of the synthetic design kept the term small: the population was large and fixed, the sampling fraction was modest, and the adversary's model was not refitted between replicates. None of them is guaranteed at an institution. In particular, an institution that reuses one auxiliary file across all its replicates has made the auxiliary file a shared unit, and (4.5) then has a third term that (4.1) cannot see. An analyst who reports a very tight interval from a very large B should say which term the large B drove down.

Grace put this point to Naledi as a question about the Bank's own practice. The earlier assessments had used one extract of the business register as the adversary's auxiliary file

for every replicate, so every simulated attack had drawn on the same file. Naledi confirmed it, and agreed that the binomial bounds in those records described the simulation and not the file. The new procedure resampled the auxiliary file within the nested design, and the record stated which variance the reported bound covered.

4.8 The worst target

The risk in the preceding sections is an average over a target drawn from a declared population. Rule (4.8) below is stated for a set T of targets, as Chapter 1 introduced it, and the set has to be explicit: a maximum over released records, a maximum over a protected subgroup and an average over a population are three controls, and a release can pass one and fail another. A release is statistically admissible only if

$$\sup_{t \in T} \ \overline{R}_{1-\alpha}(t, \theta) \ \le \ \tau. \tag{4.8}$$

When T is finite, the rule asks every target's bound to meet the tolerance. That is an intersection–union test in the sense of Berger (1982), and it needs no correction for the number of targets.

Proposition 4.5 (Every target must pass, at level α each)**.** *Let T be a finite set of targets, and for each $t \in T$ let $\overline{R}_t$ be a one-sided bound with $\mathbb{P}\{\overline{R}_t < R_t\} \le \alpha$. The rule that releases when $\overline{R}_t \le \tau$ for every $t \in T$ releases with probability at most α whenever some target has $R_t > \tau$, whatever the dependence among the bounds.*

The worst target raises a different question when the committee wants to report it. On the synthetic register the long-form paper examined the targets in the smallest released cells under minimum cell counts from 3 to 20, attacking each target 250 times. At a minimum count of ten, 230 targets were examined, and the worst had an estimated risk of 0.744 against a nominal $1/k$ of 0.1. That figure is selected: it is the largest of 230 estimates, and the largest of many noisy estimates is biased upwards. A lower bound computed for that target as if it had been chosen in advance would inherit the bias. The correction is to take each target's lower bound at level α/q for q examined targets, which makes the statement hold for all targets at once.

Proposition 4.6 (A lower bound for the selected worst target)**.** *Let T hold q targets, and for each let $\underline{R}_t$ be a one-sided lower bound with $\mathbb{P}\{R_t < \underline{R}_t\} \le \alpha/q$. Then*

$$\mathbb{P}\Big\{ \max_{t \in T} R_t \ \ge \ \max_{t \in T} \underline{R}_t \Big\} \ \ge \ 1 - \alpha, \tag{4.9}$$

whatever the dependence among the bounds, and in particular whatever rule selected the target that attains the maximum on the right.

With the correction the worst target's lower bound at a minimum count of ten is 0.638, and the conclusion survives it: some target in a 10-anonymous release has a risk above six times $1/k$, at the 95 per cent level simultaneously over all 230 targets examined. The same table reports the average over uniformly drawn targets at 0.1129, with an upper bound of 0.1178 from 12,000 targets. The average and the worst case differ by a factor of more than six, which is why the committee has to say which it is controlling. Proposition 4.5 covers the release decision, and Proposition 4.6 covers a report of how bad the worst case is. They answer different questions and use different corrections, and confusing them produces either an overly strict decision or an overstated worst case.

4.9 The decision the chair signed

Naledi's revised paper for the household release had three columns where the first had one: the estimate, the number of attacks behind it, and the exact upper bound. A note beside the table said that the replicates were independent conditional on the sample, and gave the nested design's result for the share of variance the sample contributed, with its interval. The chair signed a decision that the setting did not meet the tolerance on the evidence available, and directed a larger simulation before the committee's next meeting. Grace, reading the record later, could check each step: the tolerance, the error level, the count of successes, the bound and the comparison.

The question of this chapter was how much evidence a simulated risk figure must carry before a committee may conclude that the risk lies at or below its tolerance. The answer is that the figure must carry an upper bound at a stated error level, computed exactly from the count of successes and the number of independent attacks, and the committee may conclude only when that bound, not the estimate, is at or below the tolerance. With a point estimate a release at the boundary passes about half the time; with the bound it passes at most α of the time. The bound has a price, which is paid in attacks: in inverse proportion to the tolerance, so that one in a thousand costs 2,995 failed attacks at the 95 per cent level and one in a hundred thousand costs 299,572. It has a second price, paid when the true risk is below the line but the evidence is thin, and the remedy for that is more evidence. Finally, the bound covers only the error it was built to cover. The simulation's binomial error is one term of the variance, and a committee should know whether the sample and the adversary's fitted inputs add others.

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Report every simulated risk with three numbers: the count of successes, the number of independent attacks, and the exact one-sided upper bound at the stated level. Decide on the bound. Before adopting a tolerance, compute the number of attacks it needs from Proposition 4.3 and confirm that the institution can run them. If the estimate is below the tolerance and the bound above it, do not release; run more attacks or move to a stricter setting. If the replicates share an auxiliary file or fitted parameters, resample the shared unit or run a nested design, and report the share of variance it contributes with its interval. When the decision covers a set of targets, require every target's bound to pass at level α; when reporting the worst target, correct its lower bound for the number of targets examined.

In the software. `privacy_risk.uncertainty` holds `binomial_upper` and `binomial_lower` (the exact one-sided bounds), `monte_carlo_summary` (estimate, standard error and bound from a vector of successes), `certification_replicates` (Proposition 4.3), `selected_maximum_bound` (the corrected lower bound of Proposition 4.6) and `nested_variance` ((4.6), with a chi-square interval for the design share that assumes roughly normal outer estimates). `privacy_risk.experiments.design_variance` runs the nested design and `certification` produces Figure 4.2. Two cautions from the repository's review apply: `selected_maximum_bound` rebuilds success counts by rounding rates times replicates, so it should be given rates that came from whole counts, and `nested_variance` should be read with its normality assumption in mind. The two functions a decision rests on are short:

```
def binomial_upper(successes: int, trials: int, alpha: float = 0.05) -> float:
    """One-sided Clopper--Pearson upper confidence bound."""
    if trials <= 0 or not 0 <= successes <= trials or not 0 < alpha < 1:
        raise ValueError("invalid binomial inputs")
    if successes == trials:
```

```
        return 1.0
    return float(beta.ppf(1 - alpha, successes + 1, trials - successes))

def certification_replicates(tolerance: float, alpha: float = 0.05) -> int:
    """Replicates needed to certify the bound at `tolerance` with no observed success."""
    if not 0 < tolerance < 1 or not 0 < alpha < 1:
        raise ValueError("tolerance and alpha must lie in (0, 1)")
    return math.ceil(math.log(alpha) / math.log(1 - tolerance))
```

4.10 Proofs

Proof of Proposition 4.1. Let $Z_B = \sqrt{B}(\widehat{R}_B - R)$ converge in law to Z, symmetric about zero with a distribution function continuous at zero. At $R = \tau$ the rule releases when $\widehat{R}_B \le \tau$, that is when $Z_B \le 0$. Since zero is a continuity point of the law of Z, $\mathbb{P}(Z_B \le 0) \to \mathbb{P}(Z \le 0)$, and symmetry with continuity at zero gives $\mathbb{P}(Z \le 0) = \mathbb{P}(Z \ge 0) = 1/2$.

For the bound rule, let $R > \tau$. The rule releases only if $\overline{R}_{1-\alpha} \le \tau$, and on that event $\overline{R}_{1-\alpha} \le \tau < R$. Hence $\mathbb{P}(\text{release}) \le \mathbb{P}\{\overline{R}_{1-\alpha} < R\} = 1 - \mathbb{P}\{\overline{R}_{1-\alpha} \ge R\} \le \alpha$. □

Proof of Proposition 4.2. For $x < B$ write $G_x(p) = \mathbb{P}_p(X \le x)$. The identity $G_x(p) = \mathbb{P}\{\text{Beta}(x+1, B-x) > p\}$ holds for every $p \in [0,1]$, as can be checked by differentiating both sides in p and comparing values at $p = 0$. Hence G_x is continuous and strictly decreasing on $[0,1]$, from one to zero, and $U_x := \overline{R}_{1-\alpha}(x, B)$ is the unique solution of $G_x(U_x) = \alpha$. For $x = B$ the bound is one and the event $\{\overline{R}_{1-\alpha} < R\}$ is empty.

Fix R. For $x < B$, strict monotonicity gives $U_x < R$ if and only if $G_x(R) < G_x(U_x) = \alpha$. Since $G_x(R)$ is non-decreasing in x, the set of x with $G_x(R) < \alpha$ is $\{0, 1, \dots, x_0\}$ for some x_0, possibly empty. Therefore

$$\mathbb{P}_R\{\overline{R}_{1-\alpha}(X, B) < R\} = \mathbb{P}_R(X \le x_0) = G_{x_0}(R) < \alpha$$

when the set is non-empty, and the probability is zero otherwise. □

Proof of Proposition 4.3. With $x = 0$, $G_0(p) = (1-p)^B$, so the bound solves $(1-U)^B = \alpha$ and equals $1 - \alpha^{1/B}$. Then $\overline{R}_{1-\alpha} \le \tau$ if and only if $\alpha^{1/B} \ge 1 - \tau$. Taking logarithms, $B^{-1} \log \alpha \ge \log(1-\tau)$. Both logarithms are negative, so multiplying by $B/\log(1-\tau) < 0$ reverses the inequality and gives $B \ge \log \alpha / \log(1-\tau)$. For the expansion, $\log(1-\tau) = -\tau(1 + \tau/2 + O(\tau^2))$, so

$$\frac{\log \alpha}{\log(1-\tau)} = \frac{-\log \alpha}{\tau}\big(1 - \tau/2 + O(\tau^2)\big) = \frac{-\log \alpha}{\tau} + \frac{\log \alpha}{2} + O(\tau),$$

which is $-\log(\alpha)/\tau + O(1)$. □

Proof of Proposition 4.4. Condition on R_s. Since $B\widehat{R}_s$ is binomial, $\mathbb{E}[\widehat{R}_s \mid R_s] = R_s$ and $\mathbb{E}[\widehat{R}_s^2 \mid R_s] = R_s^2 + R_s(1-R_s)/B$. Hence

$$\mathbb{E}[\widehat{R}_s(1-\widehat{R}_s) \mid R_s] = R_s - R_s^2 - \frac{R_s(1-R_s)}{B} = \Big(1 - \frac{1}{B}\Big) R_s(1-R_s).$$

Dividing by B, taking expectations and averaging over s gives the first identity in (4.7). By the law of total variance, $\text{Var}(\widehat{R}_s) = \text{Var}(R_s) + \mathbb{E}[R_s(1-R_s)]/B = V_{\text{input}} + V_{\text{MC}}$. The $\widehat{R}_s$ are independent and identically distributed, so their sample variance with divisor $S-1$ is unbiased for $\text{Var}(\widehat{R}_s)$, which is the second identity. Subtracting the first from the second gives $V_{\text{input}} + V_{\text{MC}} - (1 - 1/B)V_{\text{MC}} = V_{\text{input}} + V_{\text{MC}}/B$. □

Proof of Proposition 4.5. Suppose $R_{t^*} > \tau$ for some $t^* \in T$. The rule releases only if every bound is at most τ, so release implies $\overline{R}_{t^*} \leq \tau < R_{t^*}$. Hence $\mathbb{P}(\text{release}) \leq \mathbb{P}\{\overline{R}_{t^*} < R_{t^*}\} \leq \alpha$. The argument uses the bound of the single target t^* and no property of the joint law of the bounds. □

Proof of Proposition 4.6. By Boole's inequality,

$$\mathbb{P}\{R_t < \underline{R}_t \text{ for some } t \in T\} \leq \sum_{t \in T} \mathbb{P}\{R_t < \underline{R}_t\} \leq q \cdot \frac{\alpha}{q} = \alpha.$$

On the complementary event, of probability at least $1 - \alpha$, every t has $R_t \geq \underline{R}_t$. Let t' attain $\max_t \underline{R}_t$. Then $\max_t R_t \geq R_{t'} \geq \underline{R}_{t'} = \max_t \underline{R}_t$. The event holds for all targets at once, so the identity of t', and any rule by which it was chosen, does not affect the conclusion. □

Chapter 5

Choosing among protection settings

Chapter 4 gave a committee a rule for one setting: release only if the exact upper bound on the declared risk is at or below the tolerance. In practice a committee rarely faces one setting. It faces a menu: minimum cell counts of 1, 5, 10 and upwards, coarser and finer classifications, more or less suppression. The procedure that most analysts follow next seems to need no defence. Test every setting on the menu with the bound rule, set aside those that fail, and among those that pass choose the one that costs the users least.

The procedure deserves its strongest statement. Each setting is tested with a valid bound at level α, so each setting that passes has passed an honest test. Choosing among the passing settings is an ordinary constrained optimisation: maximise the value of the release subject to the risk constraint, which is how the risk–utility literature frames mechanism choice (Duncan et al., 2001). An analyst who chose a more protective setting than the least protective one that passes would throw away utility and gain nothing that the test had not already secured. On this view the menu is a convenience, and the guarantee of Chapter 4 carries over to whichever setting is chosen.

The guarantee does not carry over, and simple arithmetic shows why. The setting that is chosen was chosen with the same bounds that certify it. Suppose five settings on the menu are unsafe, each with a true risk just above the tolerance, and each is tested on its own independent draw of targets at $\alpha = 0.05$. Each passes falsely with probability up to 0.05, and the procedure picks the least protective setting that passes, which will be an unsafe one whenever any unsafe setting passes. The chance that at least one of five passes is $1 - 0.95^5 = 0.226$, four and a half times the level the committee believes it is working at. With more unsafe settings near the line it grows further. Each test was honest; the choice among them was not tested at all.

The chapter therefore asks one question. How can a committee choose a protection setting from a menu, using the same simulated evidence that certifies it, and keep the chance of releasing an unsafe setting at its stated level?

5.1 A request for loan-level data

At the Bank the question arose from a request by Tomás, a visiting researcher, for a loan-level extract of the credit register: one record per loan, with the borrower's sector, region and size class and the amount of the exposure, released monthly. The evidence in this chapter comes from the synthetic register that stands in for the Bank's, with 20,000 borrowing firms, 60 lenders and six monthly vintages, of which the first holds 27,167 loans. Its parameters

were chosen, not estimated from a real register, and the floors reported below inherit those choices: exposures that do not change over time and lending activity that does not depend on firm size are two of them.

Naledi's protection lever was a minimum cell count on the key of sector, region and size class, with loans in smaller cells suppressed. Her menu held thirteen floors, from 1, meaning no suppression, through 5, 10, 15, 20, 30, 40, 50, 75, 100, 150 and 200 to 300. The committee's tolerance was 0.12 at an error level of 0.05. Tomás asked, reasonably, for the lowest floor that passed. A lower floor kept more loans, and the loans that a higher floor removed were the large ones, in the thin cells, which were the ones his research was about.

5.2 The constrained choice

Write Θ for the menu of settings, $U(\theta)$ for the utility of the release under setting θ, declared in advance as a loss on the analyses the users will run, and $\overline{R}_{1-\alpha}(t,\theta)$ for the exact bound for target set T. The choice the committee wants to make is

$$\theta^{\star} \in \arg\max_{\theta \in \Theta} U(\theta) \quad \text{subject to} \quad \sup_{t \in T} \overline{R}_{1-\alpha}(t,\theta) \le \tau. \tag{5.1}$$

When the menu is a continuum, as it is for a noise scale or a privacy parameter, it is worth knowing first that (5.1) has a solution at all.

Proposition 5.1 (Existence)**.** *If Θ is a compact metric space, U is upper semicontinuous, the map $\theta \mapsto \sup_{t \in T} \overline{R}_{1-\alpha}(t,\theta)$ is lower semicontinuous, and some θ satisfies the constraint, then* (5.1) *has a solution.*

The conditions fail for the menus institutions actually use, which are finite grids of suppression thresholds and recoding levels. There the maximum exists trivially, because a finite set has one, and the difficulty lies elsewhere. On the household simulation of the long-form paper the grid had 45 settings, of which 11 were nondominated in risk and utility, and the frontier they traced was not concave: successive slopes ran from 6.0 down to 0.006, and not monotonically. A report on such a grid should name the setting chosen and the resolution of the grid, and should not imply a global optimum that the grid cannot show.

Existence, however, is a statement about the problem with its constraint known. The committee does not know its constraint. It knows a random bound for each setting, and it uses the same random bounds both to decide which settings are feasible and to pick the best of them. That double use is where the arithmetic of the opening comes from.

5.3 Why scanning a grid inflates false certification

Call a setting *unsafe* if its true risk exceeds τ, and say it *passes* if its bound is at or below τ. The scanning procedure tests every setting and chooses the least protective one that passes.

Proposition 5.2 (The error of a scan)**.** *Let $H \subseteq \Theta$ be the set of unsafe settings, and let each setting's bound satisfy $\mathbb{P}\{\theta \text{ passes}\} \le \alpha$ for $\theta \in H$. Then the scan chooses an unsafe setting with probability at most $\min(1, |H|\alpha)$. If every unsafe setting is less protective than every safe one, the scan chooses an unsafe setting exactly when some unsafe setting passes, and if the settings are tested on independent draws with pass probabilities p_θ, that probability is*

$$1 - \prod_{\theta \in H} (1 - p_\theta). \tag{5.2}$$

The condition in the second part is the ordinary one. Unsafe settings are the weakly protective ones, at the permissive end of the menu, and safe ones are at the strict end. With p_θ near α for settings whose risk sits just above the tolerance, (5.2) grows with the number of such settings, as the opening example shows. If the settings share one draw of targets, the pass events are correlated and the inflation is smaller, but the first part of the proposition is then the only general bound, and it grows linearly in $|H|$. A committee cannot know $|H|$ in advance, because knowing it would require knowing which settings are unsafe. The longer and finer the menu, the larger the number of settings near the line, and the worse the scan becomes. A finer menu is exactly what a committee builds when it wants to release as much as it safely can.

Clinical trials met this problem decades ago. A sponsor testing several doses of a drug against a placebo, and claiming an effect for whichever doses beat it, multiplies its chances of a false claim. Maurer et al. (1995) described the remedy for the case in which the hypotheses can be put in order before the data are seen, as doses can: test them in that order, each at the full level, and stop at the first that fails. A disclosure grid has the same structure, because its settings can be ordered by how much protection they give before any attack is simulated.

5.4 Fixed-sequence testing

Definition 5.3 (Fixed-sequence selection)**.** Let $\theta_1, \theta_2, \dots, \theta_L$ be the settings in an order fixed before any outcome of the selection draw is seen, from the most protective to the least. Test θ_1 with the level-α bound rule. If it passes, test θ_2, and so on. Stop at the first setting that fails, and choose the last setting that passed. If θ_1 fails, choose none.

Proposition 5.4 (Fixed-sequence selection controls the error)**.** *Under Definition 5.3, with each bound valid at level α, the probability that the chosen setting is unsafe is at most α, whatever the true risks, and whether or not they are monotone in the order. The order may depend on data, provided those data are independent of the selection draw.*

The argument is short enough to give here, and the proof at the end of the chapter only writes it out. The procedure can choose an unsafe setting only if it reaches one and passes it. The first unsafe setting in the order, if there is one, is fixed before the draw, because the order is. To choose any unsafe setting, the procedure must pass that first one, since it cannot get past it otherwise. Passing a fixed unsafe setting has probability at most α. Nothing in this uses the risks of the other settings, which is why monotonicity is not needed.

Two features of the definition carry the guarantee, and both are procedural. The first is the order. It has to be written down before the selection draw is run, and anything that lets the analyst reorder the menu after seeing results reopens the scan. The second is the stop. A setting after the first failure is never tested, even if it would pass, and an analyst who looks at it anyway and argues for it has returned to scanning. Both features are things an auditor can check. Grace's checklist for the loan-level decision began with them: the record had to show the menu and its order, with a time stamp earlier than the first simulated attack, and it had to show that no floor below the first failure appeared anywhere in the analysis presented to the committee.

The price of the guarantee is power, and it is paid only when the bounds are out of order. If a less protective setting would have passed but a more protective one before it failed by chance, the sequence stops early and chooses a stricter setting than necessary.

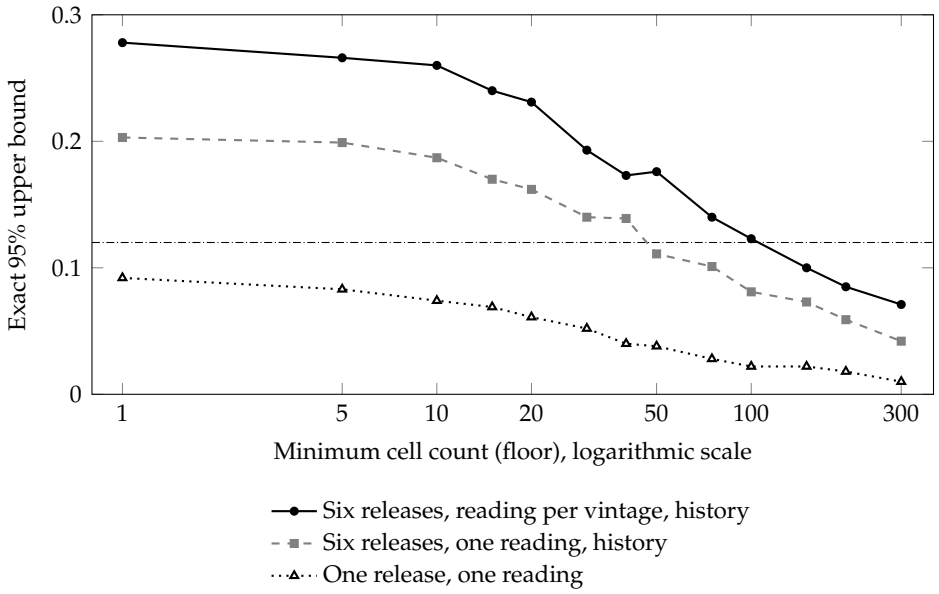


FIGURE 5.1: Selection draw on the synthetic credit register: the exact one-sided bound at each floor for three declared adversaries, 4,000 targets per floor, with the tolerance of 0.12 as a dash-dotted line. Adversaries are told apart by line style, shade and marker. Fixed-sequence selection reads each line from the right and stops at the first point above the line.

Proposition 5.5 (Fixed sequence against the scan)**.** *On every realisation of the selection draw, the setting chosen by fixed-sequence selection is at least as protective as the setting chosen by the scan, and the two coincide unless some setting after the first failure in the order passes. If utility increases strictly along the order and no setting after the first failure passes, the fixed-sequence choice solves* (5.1) *over the grid with the realised bounds.*

The comparison makes the cost concrete. Fixed-sequence selection gives up utility only on the draws in which the bounds are out of order past the first failure, and on those draws the scan's choice rests on a setting that came after a failure, which is the event that breaks the scan's error control in the first place.

5.5 The sequence on the credit register

Figure 5.1 shows the selection draw for three of the adversaries Naledi declared, each attacked with 4,000 targets at every floor. The bounds fall as the floor rises, because higher floors suppress the thin cells in which targets are easy to single out. The sequence reads each line from the right, starting at 300, and stops at the first bound above the tolerance.

For the adversary who attacks one release with one reading of a loan's exposure, every floor passes, the sequence runs to the end, and the chosen floor is 1. For the adversary who pools the six vintages, holds one reading and knows in which vintages the target's loan is live, the bounds are 0.042 at 300, 0.059 at 200, 0.073 at 150, 0.081 at 100, 0.101 at 75 and 0.111 at 50, all below 0.12, and 0.139 at 40, which fails. The sequence stops there and selects 50. For the full pooled adversary, with a reading per live vintage as well as the history, the sequence passes 300, 200 and 150 and stops at 100, where the bound is 0.123. These selections are the starting points of the floors that the long-form paper reports for the credit register, before the confirmation step described in the next section.

The full pooled line is not monotone: the bound at 40 is 0.173 and at 50 is 0.176. Both lie far above the tolerance, so the disorder does not touch the choice here, but it shows that monotone bounds cannot be assumed even when the true risks are expected to fall with the floor. On this register the scan and the fixed sequence chose the same selection floor for every one of the ten adversaries declared, including the variants with exposure rounded to one significant figure. That agreement is not the reason to prefer the fixed sequence. The reason is that its error bound would have held had the two disagreed, and the scan's would not.

5.6 The confirmation draw

The selection step controls the chance of choosing an unsafe floor. It leaves a second problem in the figure the committee reads. The floor that selection chooses is, by construction, the last to pass, and so the one whose bound sat closest to the tolerance among those that passed. Its selection bound is the one most likely to have passed by a small margin of luck, and a decision record that quotes it quotes the most optimistic number the procedure produced. The remedy is to audit the chosen floor again on a second draw of targets, independent of the first, and to carry that confirmation into the record.

Definition 5.6 (Confirmation with fallback)**.** Let $\hat{\theta}$ be the setting chosen by fixed-sequence selection. On a second draw of targets, independent of the selection draw, compute the bound at $\hat{\theta}$. If it passes, report $\hat{\theta}$ with its confirmed bound. If it fails, move to the next more protective setting in the order and compute its bound on the same second draw, and continue towards the most protective setting until one passes. If none passes, choose none.

Proposition 5.7 (Fallback keeps the error bound)**.** *Under Definitions 5.3 and 5.6, the probability that the reported setting is unsafe is at most α.*

The reason is that fallback only moves towards protection. Every setting more protective than $\hat{\theta}$ also passed the selection sequence, so the reported setting is always one that passed selection, and the argument of Proposition 5.4 covers every such setting at once: any unsafe setting can pass selection only if the first unsafe setting in the order passes, which has probability at most α. The guarantee rests on the selection step alone. The confirmation draw adds a fresh figure for the record and guards against reporting a floor that passed by chance near the line. What fallback must never do is move the other way. Choosing a less protective setting after a failed confirmation, or after seeing that a setting below the first failure would have passed on the confirmation draw, is a scan with extra steps.

The adversary who pools six vintages with one reading and the history shows the procedure at work. Selection chose 50. On the confirmation draw the same attack at 50 named the target for 458 of 4,000 targets, a rate of 0.115 with a bound of 0.123, above the tolerance. The procedure moved to the next stricter floor, 75, where 384 of 4,000 targets were named, a rate of 0.096 with a bound of 0.104, and 75 became the floor. The full pooled adversary, selected at 150, confirmed there at 0.101 with a bound of 0.109. The single-release adversary confirmed at a floor of 1 with 0.084 and a bound of 0.092.

When Naledi presented these results the chair asked the question that a committee always asks at this point: if 50 failed by so little, why not 40, which would keep more loans? The answer was in the record. Forty had failed selection at 0.139, and it was the first failure in the declared order, so the procedure never reached the question of whether some later

draw might pass it. Falling back to 75 kept the error bound, because 75 had already passed at selection. Choosing 40 or 30 after seeing these numbers would not have kept it. The chair accepted the floor of 75 for that adversary.

5.7 What the floors cost

Tomás's question was about the full pooled adversary, because a researcher who receives six monthly files and can see which loans appear in which month holds exactly that adversary's information. Its floor was 150. At 150 the release keeps 20 per cent of the loans of the first vintage, 4 per cent of the exposure and no loan to a large firm; at 75 it keeps 45 per cent of loans and 11 per cent of exposure. Large firms carry the largest exposures and sit in the thinnest cells, so they are the first to go. A file that meets the tolerance against an adversary who knows loan histories has lost the borrowers that concentration and supervisory research are about.

The insider fares worse. Pieter, a credit analyst at a reporting lender, knows the exposure of his own institution's loans almost exactly, and the declared insider adversary reads exposure with an error of 0.01 on the log scale. No floor on the menu stops it: its success is 0.772 with no suppression and still 0.416 at a floor of 300, because a precise reading of exposure separates candidates in a cell of any size. A minimum cell count is the wrong lever for that adversary, and the fixed sequence reports this plainly: its first test, at 300, fails, and it chooses none.

The committee's decision followed. Against an adversary who pools vintages and knows the histories, a loan-level public or scientific-use file was the wrong product. Tomás was offered access to the loan-level file in the Bank's secure environment, where the declared adversary no longer holds the file and the Five Safes treat the access route as a control of its own (Desai et al., 2016), together with aggregates for anything he wished to publish. The floors behind that decision are synthetic, and a central bank would have to reproduce them on its own register before relying on their values. The procedure that produced them does not depend on the register.

5.8 Several adversaries, subgroups and tolerances

A committee usually declares more than one adversary for a release, and it may require every subgroup of targets, such as every firm size class, to meet the tolerance. Both requirements fit the fixed sequence without any division of α. At each setting the committee requires every declared attack, or every subgroup, to pass. That is an intersection–union test (Proposition 4.5) and is valid at level α as it stands. Ordering the settings and stopping at the first failure then gives a fixed sequence of intersection–union tests.

Corollary 5.8 (A fixed sequence of intersection–union tests)**.** *Let each setting θ_ℓ carry several coordinates, one for each declared attack or subgroup, each with a bound valid at level α, and let θ_ℓ pass when every coordinate passes. Fixed-sequence selection over $\theta_1, \ldots, \theta_L$ chooses a setting at which some coordinate's true risk exceeds τ with probability at most α.*

The long-form paper runs this choice as a release chain with all three outside attacks as coordinates. Over 20 independent target streams it chose 150 in 19 and 100 once, against an estimated true floor of 100, and no stream chose an unsafe floor. The gap to 100 comes from the full attack's risk at that floor, 0.116, which lies so close to the tolerance that an audit

of practical size certifies it only by chance. Chapter 6 describes how that chain monitors its evidence as it arrives. The subgroup control is stricter again: with every size class required to pass, the single-release floor is 20, pooling with the history needs 200, and for the full attack no floor up to 200 passes. Those subgroup floors were chosen by the same fixed sequence but were not confirmed on a second draw, and a record that used them should say so.

The tolerance belongs to the declaration as well, and repeating the selection at other tolerances, from the same draw, shows how much it moves the answer. At a tolerance of 0.05 the single-release floor is 40, pooling with the history needs 300, and the full attack has no floor up to 300. At 0.20 the three floors are 1, 5 and 30. Across that range the floor moves by an order of magnitude. A committee that is unsure of its tolerance should read the whole table before the selection draw is run, and should fix the tolerance before it is. A tolerance chosen after the floors are known is a scan over tolerances, and it inflates the error in the same way as a scan over settings.

5.9 Menus that are not a line

The credit register's menu was one lever, and its order was obvious. The household grid of the long-form paper had three levers, band width, occupation groups and minimum cell count, and 45 settings that no single ordering of protection ranks without judgement. Proposition 5.4 allows any order fixed independently of the selection draw, and that permission is what makes a multi-lever grid workable. An analyst can run a pilot draw, estimate the frontier from it, and declare the path through the grid that the pilot suggests, from the most protective nondominated setting towards the least. The selection draw and the confirmation draw are then run fresh on that path. The pilot may be as informal as the analyst likes, because nothing it produces is used as evidence. The settings off the path are not tested, and the record should list them so that a reviewer can see what was left out. Chapter 14 uses the same division of labour when a model proposes the order and an independent audit decides.

5.10 A choice that keeps its error

The question of this chapter was how a committee can choose a protection setting from a menu, using the same simulated evidence that certifies it, and keep the chance of releasing an unsafe setting at its stated level. The answer is to fix the order of the menu, from the most protective setting down, before any attack is run; to test the settings in that order with the level-α bound and stop at the first that fails; to choose the last setting that passed; and to confirm that setting on an independent draw, moving only towards more protection if confirmation fails. The chance of choosing an unsafe setting is then at most α whatever the true risks, and the figure in the record comes from a draw that did not choose it. Scanning the grid for the least costly setting that passes gives up that control in proportion to the number of unsafe settings near the line, which a committee cannot know in advance. The fixed sequence costs utility only when the bounds are out of order past the first failure, and on the credit register it cost nothing. It does not change what the menu can achieve: when no setting on it meets the tolerance, as for the insider, the sequence says so at its first step, and the answer is a different product.

IN PRACTICE

Before running any attack, write into the decision record the menu of settings, its order from most to least protective, the tolerance, the error level, the declared adversaries and subgroups, and the number of targets in each draw. Run the selection draw down the order and stop at the first failure. Run the confirmation on a fresh draw at the chosen setting, and if it fails, move only to more protective settings. Report the confirmed figure with its bound, and state any floor that was selected but not confirmed. Do not present, in any table the committee reads, results for settings below the first failure.

In the software. The selection used for the credit register is the function `fixed_sequence` in `scripts/programme_robustness.py`, which also computes the scan's choice (`smallest_passing`) for comparison and runs the confirmation draw; `scripts/robustness_table.py` applies the fallback of Definition 5.6 in `confirmed_floor`. The release chain of Corollary 5.8 is `privacy_risk.adaptive_audit.ReleaseChain`, which refuses to open a setting before the previous one has passed. The household grid and its frontier are produced by `privacy_risk.experiments.mechanism_grid`. The selection rule is short:

```
def fixed_sequence(levels: list[int], passed: dict) -> int | None:
    """Test from the strictest level down and stop at the first failure."""
    chosen = None
    for k in sorted(levels, reverse=True):
        if not passed[k]:
            break
        chosen = k
    return chosen
```

For a minimum cell count a larger value is more protective, so sorting in reverse puts the strictest floor first.

5.11 Proofs

Proof of Proposition 5.1. Let $g(\theta) = \sup_{t \in T} \overline{R}_{1-\alpha}(t, \theta)$ and $\Theta_\tau = \{\theta \in \Theta : g(\theta) \leq \tau\}$. Lower semicontinuity of g means that every sublevel set of g is closed, so Θ_τ is closed. A closed subset of a compact space is compact, and Θ_τ is non-empty by assumption. Let $u^* = \sup_{\Theta_\tau} U$ and take $\theta_n \in \Theta_\tau$ with $U(\theta_n) \to u^*$. By compactness a subsequence converges to some $\theta^* \in \Theta_\tau$, and upper semicontinuity gives $U(\theta^*) \geq \limsup U(\theta_n) = u^*$. Hence u^* is finite and attained at θ^*, which solves (5.1). □

Proof of Proposition 5.2. The scan chooses a setting only if that setting passes, so it chooses an unsafe setting only if some $\theta \in H$ passes. By Boole's inequality, $\mathbb{P}\{\text{some } \theta \in H \text{ passes}\} \leq \sum_{\theta \in H} \mathbb{P}\{\theta \text{ passes}\} \leq |H|\alpha$, and every probability is at most one.

Now suppose every unsafe setting is less protective than every safe one. If some unsafe setting passes, the least protective passing setting is either that setting or another setting still less protective, and every setting less protective than an unsafe one is unsafe, since all safe settings are more protective than all unsafe ones. So the scan chooses an unsafe setting. Conversely, if no unsafe setting passes, the scan chooses a safe setting or none. The scan therefore chooses an unsafe setting exactly when some unsafe setting passes. With independent draws the pass events are independent, so the probability that none of the unsafe settings passes is $\prod_{\theta \in H}(1 - p_\theta)$, and its complement is (5.2). □

Proof of Proposition 5.4. Condition on the order, which is independent of the selection draw, so that each bound keeps its level-α property given the order. If no setting is unsafe there

is nothing to prove. Otherwise let i^* be the smallest index with θ_{i^*} unsafe. The procedure chooses θ_j only if $\theta_1, \ldots, \theta_j$ all pass, so the set of settings that the procedure passes is an initial segment of the order. The procedure chooses an unsafe setting θ_j only if $j \geq i^*$, and then θ_{i^*} lies in the initial segment and passed. Hence

$$\mathbb{P}\{\text{an unsafe setting is chosen}\} \leq \mathbb{P}\{\theta_{i^*} \text{ passes}\} \leq \alpha,$$

the last step because θ_{i^*} is a fixed unsafe setting given the order and its bound is valid at level α. Integrating over the order preserves the bound. The argument uses only the identity of the first unsafe setting, not the order of the true risks (Maurer et al., 1995). □

Proof of Proposition 5.5. Fix a realisation of the selection draw. If θ_1 fails, fixed-sequence selection chooses none; the scan chooses the least protective passing setting if there is one, and 'none' counts as the most protective outcome, so the claim holds. Otherwise let θ_m be the fixed-sequence choice, so that $\theta_1, \ldots, \theta_m$ pass and θ_{m+1}, if it exists, fails. The scan's choice is the passing setting with the largest index, which is at least m because θ_m passes. So the scan's choice is at least as far down the order, that is, no more protective. The two coincide if and only if no θ_j with $j > m + 1$ passes, which is the condition stated. If utility increases strictly along the order, the solution of (5.1) over the grid with the realised bounds is the passing setting with the largest index, which is the scan's choice, and under the stated condition it equals the fixed-sequence choice. □

Proof of Proposition 5.7. The reported setting is $\hat{\theta}$ or a setting earlier in the order than $\hat{\theta}$. By Definition 5.3, every setting from θ_1 to $\hat{\theta}$ passed the selection draw. Hence the reported setting passed the selection draw. If it is unsafe, some unsafe setting passed the selection draw; as in the proof of Proposition 5.4, the settings that pass form an initial segment, so the first unsafe setting θ_{i^*} passed. That event has probability at most α. The confirmation draw enters only through which of the passed settings is reported, and every passed setting is covered by the same event, so its outcome does not affect the bound. □

Proof of Corollary 5.8. Call θ_ℓ unsafe if some coordinate has true risk above τ. Let θ_{i^*} be the first unsafe setting in the order. By Proposition 4.5, applied to the coordinates of θ_{i^*}, the probability that θ_{i^*} passes is at most α. The rest of the proof of Proposition 5.4 applies without change, since it used only the level-α property of the first unsafe setting's pass event. □

Chapter 6

Monitoring an attack while it runs

An analyst who simulates an attack on a release can look at the results whenever she likes. If the bound she computes at each look is exact, the argument runs, then whatever she concludes at that look is exact as well. Looking more often uses more of the evidence and never less of it. If an audit fails narrowly, a second audit on freshly drawn targets is new and independent evidence, and it deserves its own hearing. Each step of this argument holds up when it is examined alone. The one-sided Clopper–Pearson bound of Chapter 4 has exactly the coverage it claims at the sample size where it is computed, and a fresh draw of targets owes nothing to the draw before it. An analyst who worked this way would be doing what careful people do with any measurement that comes in slowly, and a committee that read her record would find a valid bound beside the decision to release.

The argument fails, and the size of the failure is easy to measure. Take a release whose true risk is 0.201 against a tolerance of 0.2, so that any certificate it receives is false. Audit it with 500 simulated attacks and a one-sided bound at the 5 per cent level, and the bound passes in about 3.5 per cent of audits. Now let the analyst try again on fresh targets after each failure, up to 20 attempts, each judged as a new test at the 5 per cent level. Over 20,000 simulated programmes the release was certified in 51.0 per cent of them. Had the same 5 per cent been divided among the 20 attempts before the first one ran, the rate would have been 3.9 per cent. Every single bound in the first procedure was exact. The decision built from them was a coin toss.

At the Bank, the problem arrived in the ordinary way. Naledi had been asked to audit the full pooled attack on the credit register at a minimum cell count of 150, the attack in which the adversary holds all six monthly vintages, a reading of the loan's exposure in each, and the loan's history. The chair wanted the result for the committee on Thursday. Naledi started the simulation on Monday evening, and on Tuesday morning she opened the output to see how it was going. Nothing in her training told her that this was a decision.

This chapter answers one question. What must an audit fix before it starts, and what may it leave open, so that the analyst can look at the evidence as it arrives, stop when it is clear and try again when it is not, and still hand the committee a certificate whose error probability is the one written on it?

6.1 Why valid looks add up to an invalid decision

The retry experiment has a short explanation. Write A_j for the event that attempt j certifies a release whose true risk is at or above the tolerance. Each attempt, taken alone, has $\mathbb{P}(A_j) \le \alpha$.

The committee, however, does not act on one attempt. It acts on the event that some attempt certified, which is the union $A_1 \cup \cdots \cup A_n$, and a union of events can be far more probable than any one of them.

Proposition 6.1 (Retrying an audit)**.** *Let a release have true risk $r \geq \tau$, and let an analyst run up to n audits, stopping at the first that certifies.*

1. *If the audits use independent draws of targets and each certifies with probability $f > 0$, the probability that the release is certified is $1 - (1 - f)^n$, which tends to one as n grows.*

2. *If audit j is run at a level α_j fixed before the first audit begins, with $\sum_{j=1}^{n} \alpha_j \leq \alpha$, the probability that the release is certified is at most α, whatever the dependence between the audits and whatever rule the analyst uses to decide whether to run another.*

The first part reproduces the repository's figures. With 500 attacks and a tolerance of 0.2, the 95 per cent bound passes only when the count of successes is small enough, and at a true risk of 0.201 that happens with probability $f \approx 0.035$. The single-audit rate sits below 5 per cent because the binomial count is discrete and the exact bound is conservative. Twenty independent attempts give $1 - (1 - 0.035)^{20} \approx 0.51$, which is what the simulation found. The second part is the remedy, and it rests on nothing more than the union bound. Its condition is that the levels are written down before any outcome is seen. A level chosen after the third failure, once the analyst has a feeling about the fourth, is a level chosen with the data, and the union bound no longer applies to it.

Figure 6.1 shows the two rules side by side at one, five and twenty attempts. With one attempt they coincide. With five, resetting the level each time already certifies an unsafe release in 16.5 per cent of programmes, against 4.1 per cent when the level is divided in advance. With twenty the gap is the one reported above. The pre-allocated rule stays close to its single-attempt rate however many attempts it allows, because the extra attempts are paid for out of a budget that does not grow.

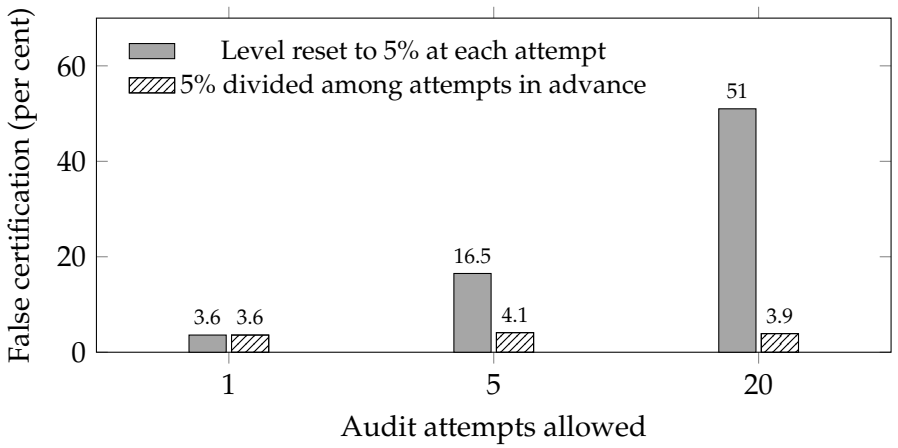


FIGURE 6.1: Retrying a failed audit. A release with true risk 0.201 is audited against a tolerance of 0.2 with 500 attacks per attempt, so every certificate is false. Solid dark bars reset the level to 5 per cent at each attempt; hatched light bars divide 5 per cent among the attempts before the first one runs. Each bar is a share of 20,000 simulated programmes.

Two features of the experiment deserve a plain statement. First, the analyst in it is honest. She does not fabricate an outcome or hide a bound; she runs a valid audit, and when it fails

she runs another valid audit. The error is in the record, which shows the last audit and not the attempts before it. Second, the experiment assumes independent attempts, which is the most favourable case for the analyst. If the attempts share anything, such as a pool of targets from which the failures are quietly removed, the inflation can be worse, and Section 6.6 returns to that.

The same mechanism works inside a single run. An analyst who computes the fixed-sample bound after every batch of attacks and stops at the first batch where it passes has also turned one test into many, though the tests are now dependent, because each look contains the looks before it. The repository measures this on the credit register. The reference attack is the pooled attack against a release that keeps at least five loans in every cell, whose success rate is 0.269. An institution simulates 4,000 attacks and computes the 95 per cent bound 20 times along the way. In 500 such runs, the bound fell below the true risk at some look in 21 per cent of them. Any of those runs could have ended in a release on a bound that was wrong. The stated level of a fixed-sample bound is a statement about one sample size chosen in advance. With unlimited looks the inflation does not stop at 21 per cent: the law of the iterated logarithm implies that a fixed-sample bound, recomputed forever, eventually falls below the truth with probability one. Online experiments in commerce met the same problem when analysts watched their tests daily and stopped on a good day (Johari et al., 2022).

On Tuesday morning Naledi’s bound for the floor of 150 stood just above the tolerance. On Wednesday it stood just below. The chair asked whether Wednesday’s figure was the answer. Naledi could not say yes, because the 95 per cent printed beside it referred to a number of attacks that nobody had fixed, and she had already looked once.

6.2 Betting against the hypothesis that the release is unsafe

The way out begins by asking the question the other way round. The committee wants to release only when there is strong evidence that the risk lies below the tolerance. The hypothesis to be rejected is therefore that the release is unsafe,

$$H : r \geq \tau,$$

and a certificate is a rejection of H. The tool that allows looking at will is a measure of evidence against H that cannot grow large, on average, while H is true. A betting game gives one.

Let $X_b \in \{0, 1\}$ record whether the bth simulated attack succeeds, each on a freshly drawn target. Before attack b the auditor chooses a number $q_b \in [0, \tau)$, using only what she has seen so far. She starts with wealth $E_0 = 1$ and updates it after each attack by

$$E_b = E_{b-1} \Big(\frac{q_b}{\tau} \Big)^{X_b} \Big(\frac{1 - q_b}{1 - \tau} \Big)^{1 - X_b}. \tag{6.1}$$

The factor is the ratio of the probability of the observed outcome under a success rate of q_b to its probability under a success rate of τ, as in the sequential test of Wald (1945), with an alternative that may change from draw to draw. A failed attack multiplies the wealth by $(1 - q_b)/(1 - \tau) > 1$; a successful one multiplies it by $q_b/\tau < 1$. The auditor is betting that attacks fail more often than H allows, and she gains when they do.

If H is true, the bet is unfavourable to her. For a success probability $p \geq \tau$ and any $q < \tau$, the expected factor is

$$p\,\frac{q}{\tau} + (1-p)\,\frac{1-q}{1-\tau} = 1 + \frac{(p-\tau)(q-\tau)}{\tau(1-\tau)} \leq 1, \tag{6.2}$$

because $p - \tau \geq 0$ and $q - \tau < 0$. Her wealth then has the structure of a gambler's fortune in a game that does not favour her.

Definition 6.2 (Test supermartingale)**.** A sequence $(E_b)_{b\geq 0}$ of nonnegative random variables, adapted to the information $(\mathcal{F}_b)$ available after each attack, is a test supermartingale for H if $E_0 = 1$ and $\mathbb{E}[E_b \mid \mathcal{F}_{b-1}] \leq E_{b-1}$ for every b whenever H is true.

The process (6.1) is a test supermartingale for H provided each attack, given everything recorded before it, succeeds with probability at least τ when H is true, and provided each q_b is fixed before X_b is seen. The first condition concerns how targets are drawn, and it is not automatic; Section 6.6 shows how it fails. The second is the rule that a bet is placed before the card is turned. Under both, a classical inequality limits how often a test supermartingale can ever become large.

Proposition 6.3 (Ville's inequality)**.** *If (E_b) is a test supermartingale for H and H is true, then for every $c > 0$,*

$$\mathbb{P}\Big\{ \sup_{b\geq 0} E_b \geq c \Big\} \leq \frac{1}{c}. \tag{6.3}$$

The inequality is due to Ville (1939), and its force lies in the supremum. The bound holds for the largest value the wealth ever reaches, so it holds at the moment the auditor chooses to stop, whatever rule she used to choose it. The audit rule follows at once: certify the release the first time $E_b \geq 1/\alpha$, and never otherwise. The chance of certifying an unsafe release is then at most α, and the auditor may look after every attack, stop when the evidence is clear, or continue when a reviewer asks for more, without changing that figure. Looking costs nothing in validity, because the guarantee was never tied to a sample size.

The simplest bet shows what the guarantee costs. With $q_b = 0$ throughout, each failure multiplies the wealth by $1/(1-\tau)$ and the first success sets it to zero for good. After B failures $E_B = (1-\tau)^{-B}$, and the audit certifies once $B \geq \log(1/\alpha)/\{-\log(1-\tau)\}$. That is the zero-success count of the fixed-sample bound in Chapter 4. At a tolerance of 0.12 and $\alpha = 0.05$ it is 24 attacks. For an audit that expects no successes, monitoring is free. The catch is in the last clause of the rule: a single success ends the audit, and an attack that succeeds even rarely will produce one. A bet that leaves room for successes has to be placed at $q_b > 0$, and then each failure earns less.

6.3 Confidence sequences

A committee often wants a number as well as a verdict: an upper bound on the risk that it can report, which remains valid however long the simulation ran and however often someone looked at it. A confidence sequence supplies one. Instead of one betting process for one tolerance, take one for every candidate risk r and mix the alternative over all values with a beta prior. With S_B successes in B attacks, the mixed process is

$$E_B(r) = \frac{\mathrm{B}(a + S_B,\; b + B - S_B)}{\mathrm{B}(a,b)\, r^{S_B}(1-r)^{B-S_B}}, \tag{6.4}$$

where B is the beta function and (a, b) are the prior's parameters (Howard et al., 2021; Ramdas et al., 2023).

Proposition 6.4 (Anytime-valid confidence sequence)**.** *If the simulated attacks are conditionally Bernoulli with risk r_0, then $E_B(r_0)$ is a nonnegative martingale with $E_0(r_0) = 1$, and the sets*

$$\mathcal{C}_B = \{r \in (0,1) : E_B(r) < 1/\alpha\} \tag{6.5}$$

contain r_0 at every B simultaneously with probability at least $1 - \alpha$.

The upper end of $\mathcal{C}_B$ is a bound the analyst may compute after every attack. In the monitoring study above, the confidence sequence fell below the true risk at some look in none of the 500 runs, against 21 per cent for the fixed-sample bound. The price is a wider interval. On the same counts the fixed-sample upper bound averaged 0.281 and the anytime bound 0.296, so the anytime bound was larger by a factor of about 1.05. For a release whose risk sits well away from its tolerance, that difference changes nothing.

Near zero risk the price is larger, and Table 6.1 gives it. To certify a tolerance of one in a hundred at the 5 per cent level with no observed success, a sample size fixed in advance needs 299 attacks, and the confidence sequence with a uniform prior needs 984. For one in a thousand the figures are 2,995 and 12,417. The ratio, between three and four, is not the price of monitoring. The zero bet of the previous section is also valid under monitoring, and it certifies after the fixed-sample count. What the larger count buys is the ability to keep going after a success. The mixture spends part of its evidence on alternatives at which successes are plausible, so a single success does not end the audit. An institution that expects some successes, as it will whenever the risk is not negligible, needs that ability and pays for it.

TABLE 6.1: Successless attacks needed to certify a tolerance at the 5 per cent level. The fixed-sample count holds at one sample size chosen in advance; the anytime count holds while the run is watched, using the confidence sequence (6.5) with a uniform prior.

Tolerance τ	Fixed sample	Anytime	Ratio
0.01	299	984	3.3
0.001	2,995	12,417	4.1

The confidence sequence and the betting rule of the previous section answer different questions, and a record should not mix them. The upper end of $\mathcal{C}_B$ is a bound on the risk. The wealth E_b of (6.1) is evidence against one hypothesis at one tolerance; neither it nor its reciprocal is an estimate of risk, and reporting it as one is an error. A decision record can carry both, in separate fields, each with its level.

Naledi switched the Thursday audit to the confidence sequence and started it again from zero on fresh targets, recording that she had done so and why. She could now look each morning without spending anything. What she could not do was erase Tuesday, and the next section is about how a programme keeps account of Tuesdays.

6.4 The registered sequential audit

A single release against a single attack is the simplest case. A programme meets a harder one. The list of attacks worth testing is not fixed when the work begins: analysts read one audit and propose another, reviewers ask for a pooled version of an attack already run, and

Chapter 9 shows why pooled attacks need audits of their own. A procedure that controls error for a family fixed in advance does not cover a family that grows in response to its own results. The journal article that accompanies this book sets out an audit for that situation, and this section gives its structure. The error control is the online form of the Bonferroni rule (Foster and Stine, 2008; Tian and Ramdas, 2021), applied to betting processes of the kind in (6.1); what is specific to disclosure is the object being audited and the conditions on how its targets are drawn.

Each pair of a candidate release and a declared attack is a *coordinate*. A coordinate is registered before its first outcome, and registration fixes four things: the release and attack, the tolerance τ_i, the attack code and target law, and a share $w_i \geq 0$ of the programme's error budget. The shares must satisfy $\sum_i w_i \leq 1$ on every path, counting coordinates that are later withdrawn. Coordinate i runs its own betting process $E_{i,b}$ and is certified the first time $E_{i,b} \geq (\alpha w_i)^{-1}$, after which it accepts no more outcomes. A release is eligible only when every coordinate declared for it has been registered and certified; an attack that has not been audited is not treated as having zero risk.

Assumption 6.1 (Conditional attack validity)**.** For each registered coordinate i whose true risk satisfies $r_i \geq \tau_i$, and for each of its draws b, the conditional probability that the attack succeeds, given everything recorded before the draw on every coordinate, is at least τ_i.

The assumption holds when each draw attacks a fresh target from the registered target law, with fresh noise in the adversary's information, independently of everything recorded so far. It is stronger than a statement about the average success rate, and Section 6.6 shows designs that break it.

Proposition 6.5 (Familywise control under adaptive registration)**.** *Suppose each coordinate's release, attack, tolerance and share are fixed at its registration, with $w_i = 0$ for a coordinate never registered; the coordinate to be audited at each step is chosen from what has already been recorded; each bet $q_{i,b} \in [0, \tau_i)$ is fixed before $X_{i,b}$ is seen; $\sum_i w_i \leq 1$ on every path; and Assumption 6.1 holds. Then, for any sequence of registrations and any rule for stopping or continuing audits,*

$$\mathbb{P}\{\textit{some coordinate with } r_i \geq \tau_i \textit{ is certified}\} \leq \alpha. \tag{6.6}$$

The retry problem of Section 6.1 now has a home. A second attempt at an audit is a new coordinate, and it needs its own share of the budget. The analyst may retry as often as she likes, provided each retry is paid for, and the sum of what she pays cannot exceed one. An open-ended programme can fix the share of the ith coordinate in advance as $w_i = 6/(\pi^2 i^2)$, which sums to one over all i, so that the threshold of the ith coordinate grows only as $2 \log i$. A programme that knows part of its family can give most of the budget to that part and hold the rest in reserve for coordinates proposed later. In log-wealth terms the threshold for coordinate i is $A_i = \ln\bigl(\pi^2/(6\alpha)\bigr) + 2 \ln i$. At $\alpha = 0.05$ the first term is 3.49, so $A_1 = 3.49$, $A_2 = 4.88$, $A_5 = 6.71$, $A_{10} = 8.10$ and $A_{20} = 9.48$. The shares should be revisited whenever the programme changes.

How the known part should be divided is a question of cost, and the article answers it. If attack i costs c_i per draw and is planned at a risk with per-draw drift μ_i, defined in Section 6.5, the allocation that minimises a bound on expected total cost is $w_i \propto c_i/\mu_i$. Costly attacks, and attacks whose risk is planned close to the tolerance, receive more of the budget. The planning values must be fixed before any outcome is seen; a plan that proves optimistic makes the audit longer and leaves it valid.

The article first ran the audit on Bernoulli streams of known risk, where Assumption 6.1 holds by construction. One coordinate had share $w = 1/8$ of $\alpha = 0.05$, a tolerance of 0.12 and a fixed bet of 0.03, so its threshold was 160 and its own error bound 0.00625. At the boundary risk of 0.12, over 2,000 runs of at most 300 attacks, it certified falsely in 0.50 per cent of runs, below its bound. A safe stream at a risk of 0.04 was certified in 95.6 per cent of runs after a median of 98 attacks, against 99.9 per cent for a Clopper–Pearson test applied once after attack 300. The monitored audit gave up a little power and finished in a third of the attacks.

The more telling run was on the credit register. The committee in that experiment declared three attacks in advance: one release with one exposure reading; six vintages with one reading and the loan's history; and six vintages with a reading in every live vintage and the history. It wanted the least protective floor, from an ordered grid running from 300 down to 1, at which all three met a tolerance of 0.12. Choosing from an ordered grid by testing in a fixed order is the subject of Chapter 5, and requiring every declared attack to pass at a setting is the intersection–union logic of Chapter 7; the article joins the two into a *release chain*, which tests each setting only after the one before it has passed, and spends one share of the budget on the whole chain with no split over settings or attacks.

The true floor, estimated from a separate draw of 100,000 targets per coordinate, was 100. There the full pooled attack succeeds at 0.116, just under the tolerance; at 75 it succeeds at 0.133, and at 150 at 0.091. Over 20 independent attack streams, the release chain chose 150 in 19 and 100 in one, and never chose an unsafe floor. Run 400 times on Bernoulli streams at the estimated risks, it never chose an unsafe floor; with every risk raised by 0.01 it did so in 6 runs, a rate of 0.015, within the budget of 0.05. Figure 6.2 sets the median number of attacks each design spent on the same 20 streams. The release chain used 6,422. Registering all 39 coordinates at the start and stopping at the first failing floor used 9,002. One chain per attack at $\alpha/3$, which is the price of choosing the adversary after the audit, used 13,764. The fixed-sample designs used 10,000 to 36,000.

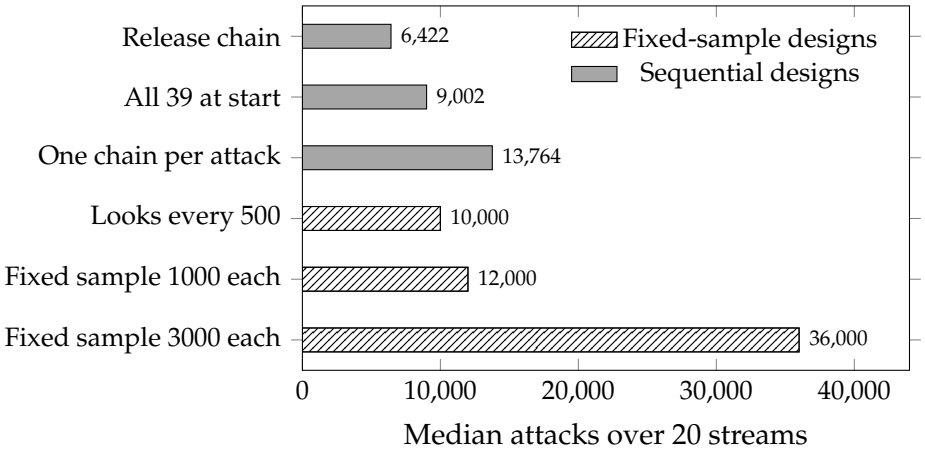


FIGURE 6.2: Attacks spent choosing a floor on the synthetic credit register, tolerance 0.12, $\alpha = 0.05$. Hatched light bars are designs that test after a planned number of attacks; solid dark bars are sequential designs that may stop at any attack. "One chain per attack" runs a separate chain for each declared attack at $\alpha/3$; "All 39 at start" registers every floor and attack before the audit. All six see the same draws, and none chose an unsafe floor.

The chain chose 150 and not the true floor of 100 for a reason the next section makes exact. At 100 the full pooled attack succeeds at 0.116, so close to 0.12 that no feasible audit

could certify it reliably. The cost of the sequential audit is not paid in error. It is paid in the floor: the audit settles on a setting it can certify in a few thousand attacks, and that setting keeps 5,553 of the 27,167 loans in the first release.

6.5 What monitoring costs

The length of an audit depends on how far the true risk lies below the tolerance. For a fixed bet q and a true risk p, each attack adds to $\log E_b$ an increment whose mean is

$$\mu(p, q) = p \log \frac{q}{\tau} + (1 - p) \log \frac{1 - q}{1 - \tau}. \tag{6.7}$$

When $\mu > 0$ the log-wealth drifts upwards and the audit will end; when $\mu \leq 0$ it may never end.

Proposition 6.6 (Expected audit length)**.** *Let the outcomes be independent Bernoulli(p) with $p < \tau$, let the bet be fixed at $q \in (0, \tau)$ with $\mu(p, q) > 0$, write $A = \log\{1/(\alpha w)\}$ for the log-threshold and $u = \log\{(1 - q)/(1 - \tau)\}$ for the gain from one failure, and let N be the number of attacks to certification. Then*

$$\frac{A}{\mu(p, q)} \leq \mathbb{E}N \leq \frac{A + u}{\mu(p, q)}.$$

Over q, $\mu(p, q)$ is largest at $q = p$, where it equals the Kullback–Leibler divergence between Bernoulli laws with parameters p and τ.

Near the tolerance that divergence is about $(\tau - p)^2 / \{2\tau(1 - \tau)\}$, so halving the margin between the risk and the tolerance roughly quadruples the audit. At a tolerance of 0.12 and an error share $\alpha w = 1/60$, an attack whose true risk is 0.10 needs about 2,100 draws with the best fixed bet, and one at 0.06 needs about 200. The bound is close to the best possible for any valid test, so no cleverer procedure escapes it. An auditor does not know p, and the credit-register run used a bet that learns it, the shrunk running rate $(S_b + 1/2)/(b + 1)$ capped at 0.11. A cap set below the true risk turns the drift negative. That is what happened at the floor of 100, where the risk of 0.116 lies above the cap: the chain could certify that floor only by chance, as it did in one stream, and even the best fixed bet would need tens of thousands of attacks on average.

So the cost of monitoring is real, and it has three parts. The first is the wider interval of the confidence sequence, small when the risk is far from the tolerance. The second is the larger zero-success count of Table 6.1, which buys the ability to continue after a success. The third is the one Proposition 6.6 describes, and it is shared with every valid procedure: a release whose risk sits just under its tolerance is expensive to certify by any method. An institution facing that cost can spend more attacks or choose a stricter setting. It cannot buy the certificate by looking more often.

6.6 When the audit process is the target

Every guarantee in this chapter rests on two conditions that the mathematics cannot check: that each attack runs on a target drawn afresh, and that each bet is placed before its outcome is seen. Both are properties of how the audit is conducted, and an operator under pressure can break either, through carelessness, a deadline or intent.

The first fails in quiet ways. Suppose the same target is attacked again after a failure. Let a hidden fact about the target decide the attack, so that every repeat gives the same result: the success probability across targets may be one half, but after one failure on this target the chance of a later success is zero. An auditor who treats the repeats as fresh draws can drive her evidence as high as she pleases. The same happens if two interleaved audits share a hidden feature of their targets, or if the adversary's information is changed after outcomes are seen. Assumption 6.1 fails in each case, and no bound follows.

The second fails when the operator has a choice the record does not show. The article measured three such choices on Bernoulli streams at a risk equal to the tolerance of 0.12, with one coordinate at the full level. An operator who can pick among seeds, or who runs shadow audits outside the record and files only the one that passed, turns a false-certification rate of 0.038 into 0.322 with ten tries and 0.979 with a hundred. An operator who chooses each bet after seeing the outcome certified an attack with true risk 0.20 in every one of 2,000 runs. These are the retry experiment of Section 6.1 in another form: the error lives in attempts the record does not hold.

The repository implements three controls that close these routes when the registry that holds them is trusted. Targets are derived from a secret the operator commits to at registration, combined with a later round of a public randomness beacon, so neither the operator nor the beacon alone chooses them. Each bet is written to the ledger before the registry runs the attack and records its outcome. The ledger is hash-chained, so an entry cannot be changed, dropped or moved without breaking the chain. With these in place the measured rates return to the honest ones. The controls are a library, and a library is not a security boundary: code in the same process can still change private state, and the hash chain detects a rewrite only if its head is published or time-stamped outside the operator's control. An institution that needs protection against its own operator must run the registry as a separate service, with the attack code signed by someone other than the analyst and every attack on the confidential register passing through it (Saltzer and Schroeder, 1975).

Some weeks later Grace, from internal audit, read the record of the Thursday decision. She did not recompute the bound. She checked that the coordinate had been registered before its first outcome, that every bet in the ledger preceded the outcome it was placed on, that the chain of hashes was unbroken from the head the Bank had published, and that the abandoned Tuesday run appeared in the ledger with the share of the budget it had spent. Each check was a question about the process, and each could be answered from the record without trusting the analyst.

6.7 The answer

The chapter asked what an audit must fix in advance so that its analyst may look at will, stop when she likes and retry when she must. The answer has four parts. The audit must fix, before the first outcome, the hypothesis it will reject, which is that the release is unsafe; the tolerance; the attack and the law from which its targets are drawn; and the share of the error budget it spends. It must use a measure of evidence that is a test supermartingale under that hypothesis, either a betting process with bets placed before outcomes or a confidence sequence. It must treat every retry and every new attack as a new coordinate that pays for itself out of a budget whose total is fixed. And it must keep a record, outside the analyst's control, of every attempt, every bet and every outcome. With these four in place the analyst may look after every attack, and the certificate means what it says. Without them, each look

can be exact and the decision can still be wrong half the time.

IN PRACTICE: WATCHING AN AUDIT

Before a simulated audit starts, write down the tolerance, the attack and its target law, the error level and, if the programme will run more than one audit, the share of the level this audit spends. Use an anytime-valid method if anyone will look at results before the planned end. Record every audit attempt, including those abandoned, and treat a retry as a new audit with its own share of the budget. Draw targets from randomness the analyst cannot choose, and log each bet before its outcome. Report the confidence-sequence bound as the risk figure and the crossing of the betting process as the verdict, in separate fields. When a release sits just under its tolerance, expect a long audit and consider a stricter setting.

In the software. The confidence sequence of Proposition 6.4 is in `privacy_risk.sequential` (`log_e_process`, `confidence_sequence`, `anytime_upper`, and `certification_replicates` for Table 6.1); `monitoring_study` runs the 20-look experiment. The registered audit is in `privacy_risk.adaptive_audit`: `AdaptiveAudit.register` spends a share of the budget and refuses one that would overspend it, `register_release_chain` opens a release chain, and `AuditCoordinate.observe` advances the betting process, refusing a bet outside $[0, \tau)$ and any outcome after certification. `zero_success_budget`, `expected_length_bound` and `drift_weights` give the cost results. `commit`, `target_seed`, `CommittedCoordinate` and `AuditLedger` are the controls of Section 6.6; the coordinate itself accepts outcomes from any caller, so the committed path binds only when a registry enforces it. The retry experiment is `audit_restarts` in `scripts/injector_experiment.py`, and the credit-register runs are `scripts/adaptive_audit_comparison.py` and `scripts/adaptive_audit_chain.py`. The update at the centre of the audit is short:

```
success = int(success)
self._log_e += (math.log(alternative / self.tolerance) if success and alternative
                else -math.inf if success else
                math.log1p(-alternative) - math.log1p(-self.tolerance))
self._trials += 1
self._successes += success
self._log.append((success, alternative))
self._crossed = self._log_e >= self._threshold
```

6.8 Proofs

Proof of Proposition 6.1. For the first part, the release escapes certification only if every one of the n audits fails to certify. The audits are independent and each fails with probability $1 - f$, so the probability that none certifies is $(1 - f)^n$ and the probability that some audit certifies is $1 - (1 - f)^n$. Since $0 < 1 - f < 1$, this tends to one as $n \to \infty$.

For the second part, imagine that the targets for all n audits are drawn in advance, whether or not the analyst will run them, and let A_j be the event that audit j, applied to its own draw at its prearranged level α_j, certifies. Because each audit is a valid level-α_j test of a hypothesis that is true, $\mathbb{P}(A_j) \le \alpha_j$. The release is certified only if some audit that was actually run certifies, and that event is contained in $A_1 \cup \cdots \cup A_n$ whatever rule decided which audits to run. By the union bound, $\mathbb{P}(\text{certified}) \le \sum_j \mathbb{P}(A_j) \le \sum_j \alpha_j \le \alpha$. No step uses independence between the audits. □

Proof of Proposition 6.3. Fix $c > 0$ and let $T = \inf\{b : E_b \ge c\}$, with $T = \infty$ if the set is empty. For each n, the stopped process $E_{b \wedge T}$ is a nonnegative supermartingale, so $\mathbb{E}[E_{n \wedge T}] \le E_0 = 1$. On the event $\{T \le n\}$ we have $E_{n \wedge T} = E_T \ge c$, and $E_{n \wedge T} \ge 0$ elsewhere, hence $c\,\mathbb{P}(T \le n) \le$

$\mathbb{E}[E_{n\wedge T}] \le 1$. Letting $n \to \infty$ and using continuity of probability gives $\mathbb{P}(T < \infty) \le 1/c$, and $\{T < \infty\}$ is the event $\{\sup_b E_b \ge c\}$. □

Proof of Proposition 6.5. Fix a coordinate i and work on the event $H_i = \{r_i \ge \tau_i\}$. By the measurability conditions, the coordinate's scenario, tolerance and share are known at its registration time σ_i, so H_i and w_i are determined by the information at σ_i. Between its own draws the process $E_{i,b}$ is constant. At draw b, the bet $q_{i,b}$ is known beforehand and, by Assumption 6.1, the conditional success probability p satisfies $p \ge \tau_i$, so by (6.2) the conditional expectation of the update factor is at most one. Hence, on H_i, the process indexed by the global sequence of recorded events and started at σ_i is a nonnegative supermartingale starting at one. Applying Proposition 6.3 conditionally on the information at σ_i, with $c = (\alpha w_i)^{-1}$, gives $\mathbb{P}(i \text{ certified} \mid \mathcal{F}_{\sigma_i})\,\mathbf{1}\{H_i\} \le \alpha w_i \mathbf{1}\{H_i\}$; when $w_i = 0$ the threshold is infinite and the left side is zero. Taking expectations, $\mathbb{P}(i \text{ certified},\ H_i) \le \alpha\,\mathbb{E}[w_i]$. Summing over i, using the union bound and monotone convergence for countably many coordinates, and then $\sum_i w_i \le 1$ on every path, gives $\mathbb{P}(\exists i : i \text{ certified},\ H_i) \le \alpha\,\mathbb{E}[\sum_i w_i] \le \alpha$, which is (6.6). □

Proof of Proposition 6.4. For fixed $u \in (0,1)$, the likelihood ratio after B attacks of a success rate u against the rate r_0 is $L_B(u) = (u/r_0)^{S_B}\{(1-u)/(1-r_0)\}^{B-S_B}$. Given the past, the next factor has expectation $r_0(u/r_0) + (1-r_0)(1-u)/(1-r_0) = 1$ under conditionally Bernoulli(r_0) outcomes, so $L_B(u)$ is a nonnegative martingale with $L_0(u) = 1$. Let π be the beta(a,b) density. By Tonelli's theorem, $\int L_B(u)\pi(u)\,du$ is again a nonnegative martingale starting at one, and evaluating the integral,

$$\int_0^1 \frac{u^{S_B}(1-u)^{B-S_B}}{r_0^{S_B}(1-r_0)^{B-S_B}} \frac{u^{a-1}(1-u)^{b-1}}{\mathrm{B}(a,b)}\,du = \frac{\mathrm{B}(a+S_B,\, b+B-S_B)}{\mathrm{B}(a,b)\, r_0^{S_B}(1-r_0)^{B-S_B}} = E_B(r_0).$$

By Proposition 6.3 with $c = 1/\alpha$, the probability that $E_B(r_0) \ge 1/\alpha$ for some B is at most α. On the complementary event $r_0 \in \mathcal{C}_B$ for every B, which proves the claim. □

Proof of Proposition 6.6. Write Y_b for the increment of $\log E_b$ at attack b. The Y_b are independent with mean $\mu = \mu(p,q) > 0$, and each is at most u, the value taken on a failure. Fix a cap B and let $T = \min(N, B)$, a bounded stopping time. Wald's identity gives $\mathbb{E}[\log E_T] = \mu\,\mathbb{E}T$. If $T = N$, then $\log E_{N-1} < A$ and $\log E_N < A + u$; if $T = B < N$, then $\log E_B < A$. In both cases $\log E_T < A + u$, so $\mu\,\mathbb{E}T < A + u$, that is $\mathbb{E}\min(N,B) \le (A+u)/\mu$. Letting $B \to \infty$ and using monotone convergence gives $\mathbb{E}N \le (A+u)/\mu$; in particular $\mathbb{E}N < \infty$, so Wald's identity applies to N itself. Since $\log E_N \ge A$ at certification, $\mu\,\mathbb{E}N = \mathbb{E}[\log E_N] \ge A$, which is the lower bound. Finally, $\partial\mu/\partial q = p/q - (1-p)/(1-q)$, which is positive for $q < p$ and negative for $q > p$, so μ is maximised at $q = p$, where $\mu(p,p) = p\log(p/\tau) + (1-p)\log\{(1-p)/(1-\tau)\}$, the Kullback–Leibler divergence of Bernoulli(τ) from Bernoulli(p). □

Part III

Decisions

Chapter 7

Governance thresholds

Many people who work on release decisions believe that the threshold is already set, either by the law or by a rule the institution has applied for years. The belief has a strong form. A central bank in the European System of Central Banks works under a precise legal duty of statistical confidentiality, written into Council Regulation (EC) No 2533/98 (Council of the European Union, 1998, 2009). Where natural persons are concerned, data protection law adds its own test (European Parliament and Council, 2016). Statistical offices have published threshold rules for decades, such as withholding a cell unless it holds a minimum number of units or contributors, and these rules are described in the standard handbook (Hundepool et al., 2012, ch. 4). On this view a committee has no tolerance to choose. It applies the law through the rule, and the rule is the threshold.

Two facts break the belief. The first is that the law gives no number. Recital 26 of the General Data Protection Regulation asks whether identification is "reasonably likely" by the means an adversary could use (European Parliament and Council, 2016), and the confidentiality duty of the European System of Central Banks forbids disclosure without saying what probability of identification counts as disclosure. The second is that a minimum cell count does not fix a risk. On the synthetic credit register of this book, at one tolerance of 0.12, the floor that meets the tolerance is 1 for an adversary who holds one release, 75 for one who pools six monthly vintages and knows when the target's loan was live, and 150 for one who also reads the loan's exposure in each vintage. Against an insider who knows exposure closely, no floor up to 300 meets it. A floor of five, applied as a rule, meets the tolerance against the first adversary and fails against the other three. The rule is the same in each case. What changed is the attack, and the rule does not say which attack it was written for.

The pattern is older than this book. Sweeney (2000) found that date of birth, sex and postcode identify most of the population of the United States. Files that had been stripped of names, as the conventions of the time required, could still be linked to people through those three fields. The convention described a procedure. It did not state a risk, so nobody could check whether the procedure delivered one.

At the Bank, the question arrived as a request. Tomás, a visiting researcher, asked for three products drawn from the credit register: the published quarterly aggregates, a loan-level research extract for six months, and the same extract again after a revision. The chair asked Naledi what the Bank's rule said. The Bank's rule was a minimum of five loans in every published cell. Naledi's answer was that the rule said how the file would be protected, and that nothing in it said how much risk the Bank would accept, against whom, or on whose authority.

This chapter answers one question. What must a release threshold contain before it can decide anything, and who must write each part of it?

7.1 What the law asks and what it leaves open

The legal basis of a release is the source of the duty and the list of those it protects. For the national central banks of the European System of Central Banks, the duty to keep statistical information confidential comes from Article 8 of Council Regulation (EC) No 2533/98, as amended by Council Regulation (EC) No 951/2009 (Council of the European Union, 1998, 2009). For European statistics produced by national statistical authorities, it comes from Regulation (EC) No 223/2009 (European Parliament and Council, 2009). Where natural persons are concerned, the General Data Protection Regulation applies to data that relate to an identifiable person, and Recital 26 says that account should be taken of all the means "reasonably likely" to be used to identify the person, directly or indirectly (European Parliament and Council, 2016).

None of these texts gives a probability. They could not sensibly do so, because the probability of identification depends on who holds the release, what else they hold, and what they stand to gain, and these differ from one product to the next. The law sets the duty and leaves the reading of it to the institution. A tolerance is the institution's written reading of "reasonably likely" for a declared adversary and a declared harm, and the record should say that this is what it is. A reading can be disputed, revised and defended; a reading that was never written down can only be inferred after the event, usually by someone hostile to the institution.

The law also fixes whom the duty protects, and for a credit register the answer has three parts. The borrowers are protected: firms whose loans, exposures and arrears the register holds. The reporting agents are protected: the lenders whose positions and customer relationships the register records, and whose contributions another lender might reconstruct from published totals. And the natural persons behind some borrowers are protected, such as sole traders, for whom a firm's credit record is a personal record and data protection law applies. The three need not share one tolerance. A lender's position reconstructed by a competitor is a commercial harm; a sole trader's arrears made public is a personal one. A tolerance that names its protected parties can be set at the level each warrants, and one that does not will be set, without anyone deciding so, at the level of whichever party the analyst had in mind.

7.2 Assembling the legal element

The legal element is the one the code cannot check, so it is written out. Record, in order: the harm class the law recognises; the protected parties; the access route by which the data reach an adversary; the argument that the attack is reasonably likely, with the evidence for each term; the horizon over which the statement holds; and the event that triggers a revisit. These are the fields the strict check lists when they are empty. The book gives no legal advice, and counsel fills these fields.

7.3 The nine elements of a threshold

A governance threshold, in the sense of this book, is a tolerance together with the things that turn a number into a decision.

Definition 7.1 (Governance threshold)**.** A governance threshold is the tuple

$$G = (\tau,\ o,\ d,\ a,\ h,\ \alpha,\ \ell,\ s,\ m), \tag{7.1}$$

where $\tau \in (0, 1)$ is the risk tolerance; o is the named owner who accepts the residual risk; d is the expiry date; a is the adversary specification; h is the harm class to which the tolerance applies; α is the accepted probability of a wrong release; ℓ is the legal basis together with the parties it protects; s is the access route; and m is the horizon, the number of releases or vintages over which a programme's pooled risk is assessed. For a threshold that governs one release only, $m = 1$.

Figure 7.1 arranges the nine elements by the question each answers. Five fix what the number refers to: the adversary, the harm class, the protected parties and legal basis, the access route and the horizon. Two fix how much risk is accepted and how sure the institution must be: the tolerance and the error probability. Two fix who answers for the decision and until when: the owner and the expiry. A risk table, however careful, supplies none of the nine. It supplies the evidence against which they are applied.

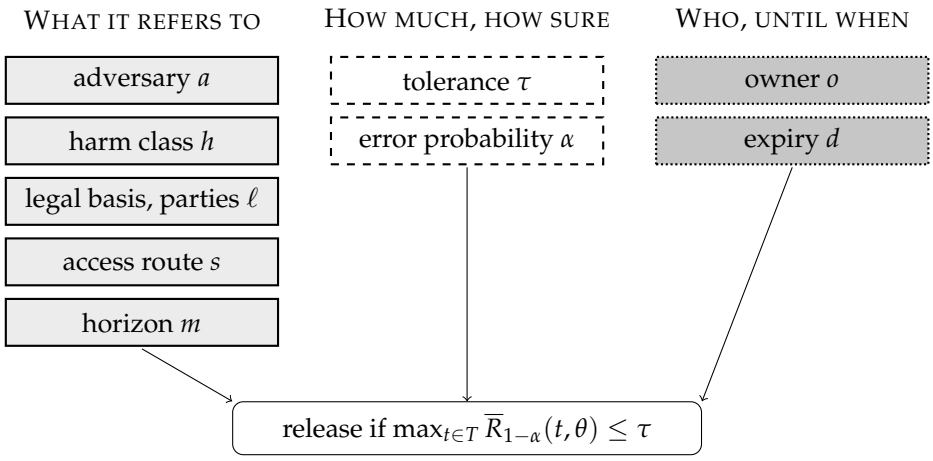


FIGURE 7.1: The nine elements of a governance threshold, grouped by the question each answers. Light boxes with solid borders fix what the risk figure refers to; white boxes with dashed borders fix how much risk is accepted and with what error; dark boxes with dotted borders fix who answers for the decision and until when. All nine feed the admissibility rule (7.2).

Each element prevents a failure that can be named. Without an owner the tolerance belongs to nobody, and nobody has the authority to revise it. Without an expiry date an assessment made on a register as it stood years ago governs a release today, after the population, the products and the adversaries have all moved. Without an adversary the number has no referent, since Chapter 2 showed that risk is defined only given an attack. Without a harm class a tolerance calibrated for commercial nuisance is applied where the consequence falls on a person. Without an error probability the institution has not said how much evidence it requires, and Chapter 4 showed that a rule based on a point estimate releases about half the time at the boundary. Without the legal basis and protected parties the tolerance

has no source of duty and no list of whom it serves. Without an access route the same tolerance is applied to products that face different adversaries. Without a horizon a programme tolerance has no fixed meaning, because Chapter 9 showed that pooled risk does not fall as releases are added, and a monthly series has no natural end.

Given the nine elements, the decision rule is short. Let $\overline{R}_{1-\alpha}(t, \theta)$ be an upper confidence bound at level $1 - \alpha$ on the risk to target t under protection setting θ, with its interpretation recorded. A release is admissible only if

$$\max_{t \in T} \overline{R}_{1-\alpha}(t, \theta) \leq \tau, \tag{7.2}$$

the assessment date precedes d, and the release, its users and its route fall within the scope of a, h, ℓ, s and m. The rule needs one more declaration, the target set T, and the next section shows why it cannot be left implicit.

When Tomás's request reached the committee, Naledi brought a single page with the nine elements as headings and blank spaces beneath them. She filled in the adversary, the harm class and the access route herself, because the analysis depended on them, and left the tolerance, the owner and the expiry blank. Those were not hers to fill.

7.4 Target sets and joint decisions

The maximum in (7.2) runs over a set T, and the choice of T is part of the threshold. A maximum over every record in the file, a maximum over the records of a protected subgroup, and an average over a declared population are three different controls. A release can pass one while failing another, and a committee that is told only that the release "passed" does not know which control it met.

The credit register shows the difference. Large firms are few, and in a release keyed on sector, region and size class they sit in thin cells. A uniform draw of targets holds only a few hundred of them, so an average over all loans says little about large borrowers. When the institution requires every firm size class to meet the tolerance of 0.12, the floor for an adversary who holds one release rises from 1 to 20, and the floor for one who pools the vintages and knows the loan's history rises from 75 to 200. For the adversary who also reads exposure in each vintage, no floor up to 200 protects every size class. The protected parties of Section 7.1 are the natural way to choose T: each party, or each subgroup the institution has a reason to protect, becomes a member of a declared family.

Requiring every subgroup to pass looks as if it should cost something in error control, since the institution now makes several statistical claims at once. It does not, and the reason is worth stating precisely, because the opposite intuition leads institutions to divide α among subgroups and so to demand far more evidence than the decision needs.

Proposition 7.2 (Joint release over subgroups)**.** *Let $T = \{1, \dots, G\}$ be a set of subgroups fixed before any outcome is seen, with true risks $r_1, \dots, r_G$, and let $\overline{R}_g$ be a one-sided upper bound for subgroup g satisfying $\mathbb{P}(\overline{R}_g < r_g) \leq \alpha$. Release if and only if $\overline{R}_g \leq \tau$ for every g. Then the probability of releasing when some subgroup has $r_g > \tau$ is at most α, whatever the dependence between the bounds.*

This is the intersection–union test of Berger (1982). The hypothesis to be rejected is a union, that some subgroup is unsafe, and the release requires every subgroup's bound to reject its own part of it. A wrong release needs at least one unsafe subgroup to pass, and the

probability of that is at most α for that subgroup alone. So each subgroup's bound may be taken at level α, with no correction for the number of subgroups. A Bonferroni correction is needed for a different claim: to report every subgroup's bound as one simultaneous statement, or to say that the worst subgroup lies above some value. Those are claims about the evidence. The release decision is a claim about the release, and it needs no correction.

The same logic covers several declared adversaries at one setting, which is how the release chain of Chapter 6 tests three attacks at each floor at the full level. It does not cover the choice of adversary after the results are read. A committee that sees the floors each adversary requires and then declares the adversary whose floor it prefers has chosen a hypothesis with the data. The journal article measured the price of making that choice honest: one audit chain per adversary at $\alpha/3$ spent a median of 13,764 attacks on the credit register, against 6,422 for one chain that tests all three adversaries declared in advance. The cheaper route requires the adversary to be part of the threshold, written down before the audit.

7.5 Declaring the adversary

The adversary specification says who attacks, what it knows, which targets it attacks, and when it acts. The last part is usually left to a convention: an attack is counted as a success when the adversary's best guess is right, or when its posterior probability on the guess exceeds one half. An adversary need not follow either convention. It acts when acting pays.

Suppose a correct allegation brings the adversary a benefit $h > 0$, a false allegation costs it $c_f \geq 0$, and making any allegation costs $c_a \geq 0$. At posterior probability p that its candidate is the target, acting has expected payoff $ph - (1-p)c_f - c_a$.

Proposition 7.3 (Economically rational attack threshold)**.** *An adversary that maximises expected payoff, and acts when acting has nonnegative expected payoff, never acts if $c_a \geq h$. If $c_a < h$ it acts if and only if*

$$p \geq p^\star = \frac{c_f + c_a}{h + c_f}, \tag{7.3}$$

and then $p^\star < 1$. The threshold equals one half exactly when $h = c_f + 2c_a$.

The convention of one half is therefore one payoff structure among many: equal stakes for right and wrong allegations, with free allegations. The repository reads one attack on the credit register at three thresholds. An adversary that gains and loses equally and pays nothing to allege acts at one half, acts on 0.179 of targets, is right 0.814 of the time and collects 0.146 correct allegations per target. An adversary that gains ten times what a false allegation costs it acts at 0.136: it acts on 0.395 of targets, is right 0.556 of the time, and collects 0.220, half as much again. An adversary for whom a false allegation is costly, through reputation or law, acts at 0.917, on 0.083 of targets, and is right 0.994 of the time.

The three rows are the same attack on the same release. A journalist who would be sued for naming the wrong firm is the third adversary. A data broker who loses little from a wrong match and sells every right one is the second. An evaluation that reports one threshold reports one point on a curve the adversary chooses from, and the record should carry the curve of precision against coverage, or state the payoffs that select the point it reports.

Pieter is the adversary the Bank's file is least able to stop. As a credit analyst at a reporting lender he knows, closely, the exposure of every loan his own bank has made, and he can see which months each was live. On the synthetic register an insider who knows exposure

to within one per cent and knows the history still names 42 per cent of targets at a floor of 300. No tolerance a committee could defend is met against him by suppression. Declaring him in the threshold does not protect his bank's customers. It makes visible that the loan-level file is the wrong product for any route Pieter can reach, which is the subject of the next section.

7.6 The access route

The same file faces different adversaries depending on how it reaches its users. A public file can be downloaded by anyone, including those who hold the richest auxiliary information and the strongest motive. A scientific-use file released under contract to accredited researchers faces users who are known, bound by agreement and limited in number. Access in a secure environment, where the researcher works on the data inside the institution's systems and takes away only checked outputs, removes the file from the adversary's hands altogether. These are three routes with three threat models. The Five Safes framework treats the route as a control in its own right, alongside safe projects, people, settings and outputs (Desai et al., 2016).

The route belongs in the threshold because it changes which adversary the tolerance must be met against. A loan-level extract that fails as a public file, because a pooling adversary with exposure readings needs a floor that destroys most of it, may pass in a secure environment, where the declared adversary no longer holds the file at all and the residual risk lies in the outputs that leave. The assessment is then a different one: the outputs must be checked, and the users must be accredited, and each of these has its own failure modes. What the route does not allow is to keep the public file's assessment and change the route afterwards, or to assess the secure route and release the public file. The tolerance holds for the route it names.

For Tomás, this settled more than the analysis did. The quarterly aggregates could be published. The loan-level extract could not be released as a file to a visiting researcher on any floor that left it useful, because a copy holding all six vintages would be dangerous in the hands of anyone who knew a firm's borrowing history, as its lenders and counterparties do; it could be made available to him in the Bank's secure environment, with outputs checked before they left. The revised extract was a third product and would need its own record when it existed.

7.7 Separating the analyst, the owner, the reviewer and audit

The elements of a threshold are written by different people, and the separation between them is part of what makes the threshold work. Central banks already use a structure that fits: the three lines of defence of operational risk.

The analyst who declares the attacks and runs the simulations, and the data owner who holds the register and knows which products are drawn from it, sit in the first line. The analyst owns the correctness of the calculation, not the decision. The reviewer, in a risk or methodology function outside the production line, checks that the declared adversaries are plausible, that none was omitted for convenience, and that the tests of the model behind the risk figure were run; this is the second line. Internal audit is the third. It checks, after the fact, that the record was filed, that the tolerance was signed before the first outcome was seen, and that the controls on the audit process described in Chapter 6 were in place. The owner

o is the senior officer who accepts the residual risk on the institution's behalf, in a central bank typically the chair of the release committee or the executive member responsible for statistics.

The reason for the separation is that each role has an incentive the others check. An analyst who has seen the results can choose, in good faith, the adversary whose floor keeps the file useful, and Section 7.4 showed what that choice costs. A reviewer who did not run the analysis has no stake in the floor it produced. An owner who signs the tolerance before the analysis has no way to fit the tolerance to the result. And an auditor who reads the record later can check the order of events, which is the one thing the people inside the decision cannot credibly certify about themselves. Where the institution is small and one person must hold two roles, the analyst and the reviewer are the pair to keep apart.

Grace read the record of the decision on Tomás's request some months later. She checked that the tolerance, its owner and its expiry had been signed by the chair on a date before the first audit was registered; that the adversaries in the record were those the reviewer had approved; that the secure-environment route was written into the threshold and not added after the public file failed; and that the expiry fell before the next scheduled revision of the register. She found one gap. The record named the borrowers as protected parties and said nothing of the lenders whose positions the extract held. The next record named them.

7.8 Expiry and horizon

The last two elements give a threshold a life. A risk assessment describes a register as it stood, a set of products as they were, and adversaries with the information available to them at the time. All three change. The register gains and loses units, new products are drawn from it, and outside sources that an adversary can link to grow. An expiry date forces the assessment to be renewed before its premises have drifted out of sight. It also makes the owner's acceptance of risk bounded in time, which is easier to give and easier to defend.

The horizon does the same work across releases. Chapter 9 proved that no function of per-release risks bounds the risk of an adversary who pools the releases, and that pooled risk does not decrease as releases are added. A programme tolerance therefore needs to say over how many releases the pooled risk was assessed. A threshold with a horizon of six monthly vintages says that the seventh vintage needs a new assessment before it is published. One without a horizon says nothing about the seventh, and the twelfth and the twentieth will be published under an assessment that never considered them.

7.9 The answer

The chapter asked what a release threshold must contain before it can decide anything, and who must write it. The law sets the duty and names the protected parties, and it leaves the probability to the institution. A minimum cell count is a protection setting, and it meets a tolerance only for a declared adversary. A threshold that can decide therefore has nine elements: a tolerance and an error probability; an adversary, a harm class, a legal basis with its protected parties, an access route and a horizon, which fix what the tolerance refers to; and an owner and an expiry date, which fix who answers for it and until when. The target set must be declared with it, and a joint decision over subgroups or adversaries declared in advance needs no division of the error probability. The analyst writes the adversary and runs the evidence; an independent reviewer checks the adversary; a senior owner signs the

tolerance before any outcome is seen; and internal audit checks afterwards that these things happened in that order.

IN PRACTICE: WRITING A THRESHOLD

For each product and each harm class, write the nine elements on one page before any risk is measured, and have the owner sign the tolerance, the error probability and the expiry. Name the legal basis and every protected party, and set separate tolerances where their harms differ. Declare the target set: all records, named subgroups, or a population average. Test every declared subgroup or adversary at the full error level, and divide the level only when reporting all the bounds as one statement. State the payoffs the adversary is assumed to face, or report the curve of precision against coverage. Name the access route, and reassess if the route changes. Give a programme tolerance a horizon and set a reassessment before the first release beyond it.

In the software. The threshold is `privacy_risk.governance.GovernanceThreshold`, and the decision is `decide`, which returns `INVALID` when the prespecified validation failed, `FAIL` when the assessment date falls on or after the expiry date or when the upper bound exceeds the tolerance, and `PASS` otherwise. The class records six of the nine elements: tolerance, owner, expiry, adversary, harm class and error probability. It has no field for the legal basis and protected parties, the access route or the horizon, so a record built from it must carry those three separately, and `decide` cannot check them. `decide` also takes one upper bound, so the maximum over the target set must be formed before it is called. The attack threshold of Proposition 7.3 is `privacy_risk.sequential.attack_threshold`, and `precision_coverage` and `at_threshold` read an attack along the curve.

```
@dataclass(frozen=True)
class GovernanceThreshold:
    tolerance: float
    owner: str
    expiry_date: str
    adversary: str
    harm_class: str
    alpha: float
```

7.10 Proofs

Proof of Proposition 7.2. If no subgroup has $r_g > \tau$ there is nothing to prove. Otherwise fix one subgroup g^* with $r_{g^*} > \tau$; since the risks are fixed parameters and T was fixed in advance, g^* does not depend on the data. A release requires $\overline{R}_g \le \tau$ for every g, and in particular $\overline{R}_{g^*} \le \tau < r_{g^*}$. Hence

$$\mathbb{P}(\text{release}) \le \mathbb{P}(\overline{R}_{g^*} \le \tau) \le \mathbb{P}(\overline{R}_{g^*} < r_{g^*}) \le \alpha.$$

The argument uses only the marginal coverage of the one bound $\overline{R}_{g^*}$, so it holds whatever the dependence between the bounds (Berger, 1982). □

Proof of Proposition 7.3. At posterior probability p the expected payoff of acting is $V(p) = ph - (1-p)c_f - c_a = p(h + c_f) - (c_f + c_a)$, which is increasing in p because $h + c_f > 0$. Its largest value, at $p = 1$, is $h - c_a$. If $c_a > h$, then $V(p) < 0$ for every $p \le 1$; if $c_a = h$, then $V(p) \le 0$ with equality only at $p = 1$, where acting gains nothing. In both cases the adversary does not act, which we record as $p^\star = +\infty$. If $c_a < h$, then $V(p) \ge 0$ if and only if $p \ge (c_f + c_a)/(h + c_f)$, and this ratio is below one because $c_f + c_a < h + c_f$. Finally, $(c_f + c_a)/(h + c_f) = 1/2$ if and only if $2c_f + 2c_a = h + c_f$, that is $h = c_f + 2c_a$. □

Chapter 8

Utility, harm and the tolerance they imply

When an institution first writes down a tolerance, it usually writes one. The same figure is applied to the quarterly series, the research extract and the public file, and the case for doing so is a serious one. A single tolerance treats every data subject alike: nobody's confidentiality is worth less because the release that contains them happens to be more useful. It is simple to explain to a board and to the public. It leaves no room for an analyst to argue a release through by finding a reason why this one is special. And it avoids the appearance, which a confidentiality office has every reason to fear, of trading people's privacy for the convenience of users. On this view, varying the tolerance from one release to the next is a step towards negotiating it.

The first fact against the view comes from the shape of the trade-off itself. On the simulated population of this book, 45 protection settings were assessed, and 11 of them are not dominated by another setting that is both safer and more useful. Along those 11, the utility gained per unit of added risk runs from 6.0 at the safe end down to 0.006 at the other. A thousandfold difference in the rate of exchange between harm and use means that one tolerance, applied to releases whose trade-offs differ, buys protection at very different prices in different places. Where protection is cheap the institution buys too little of it, and where it is dear it buys too much. The proposition and corollary below make this exact: a common tolerance is the best choice only in the special case where every release faces the same trade-off at that tolerance, and otherwise some other set of tolerances protects as well and serves users better.

The question was faced at national scale before it reached any central bank's release committee. Abowd and Schmutte (2019) argued that the level of protection a statistical agency gives is a social choice, to be made by weighing the accuracy of published statistics against the privacy of respondents, at the point where the rate at which the technology trades one for the other equals the rate at which society values them. The United States Census Bureau then had to make such a choice for its 2020 products. Its disclosure avoidance system spent a stated total privacy-loss budget and divided it among geographic levels and tables (Abowd et al., 2022), and economists who use federal statistics argued in reply that the accuracy lost had been given too little weight (Hotz et al., 2022). Whatever one thinks of the outcome, the dispute was conducted in the right terms: how much harm was being prevented, how much use was being given up, and who had the authority to weigh the two.

At the Bank, the chair was asked to sign a tolerance of 0.12 for every product drawn

from the credit register, and asked Naledi whether that was the right number. Naledi had no way to answer from the risk tables alone. What she could do was show the chair what each candidate tolerance would cost each product.

This chapter answers one question. Once an institution states how much weight it puts on a harm against the use of a release, what tolerance does that weight imply for each release, and how should a tolerance for a whole programme be shared among its releases?

8.1 Utility and the frontier

Utility needs the same discipline as risk. Let Q be a quantity users want to estimate from the confidential data, and $\widehat{Q}(Y_\theta)$ its estimate from a release produced under protection setting θ. A loss-based utility is

$$U(\theta) = -\mathbb{E}\big[\mathcal{L}\{\widehat{Q}(Y_\theta),\, Q\}\big], \tag{8.1}$$

where the expectation runs over sampling and the randomness of the mechanism, and the loss $\mathcal{L}$ is declared before any setting is evaluated. Squared error, the distortion of a confidence interval, or a weighted loss over the analyses users are known to run are all admissible (Karr et al., 2006; Snoke et al., 2018). The loss must be declared in advance for the same reason the adversary must: a setting chosen after seeing which loss it happens to do well on carries no guarantee.

For each setting, the institution now has two numbers: a risk coordinate and a utility. Following the decision rule of Chapter 7, the risk coordinate should be the upper bound $\overline{R}_{1-\alpha}(\theta)$ that the release decision will use, not the point estimate. The attainable set is

$$\mathcal{A} = \{(\overline{R}_{1-\alpha}(\theta),\, U(\theta)) : \theta \in \Theta\}.$$

A point is dominated if another attainable point has no more risk and no less utility, with one of the two strict. The *risk–utility frontier* $\mathcal{F}$ is the set of points that are not dominated, the confidentiality map of Duncan et al. (2001) drawn with a bound on the risk axis. No sensible institution chooses a dominated setting, so every choice worth discussing lies on the frontier, and the discussion is about where on it to stand.

Figure 8.1 shows the frontier for the simulated population. The 45 settings vary the width of age bands, the number of occupation groups and the minimum cell count. The frontier spans risk bounds from 0.081 to 0.564, and it is steep at the safe end: halving the bound from 0.1715 to 0.0806 costs 0.17 of utility, half the range the frontier spans, and cuts the records kept from all of them to 58 per cent. A committee looking at the figure can see at once what it is being asked to trade.

Two cautions come with the picture. The frontier is estimated, and using the upper bound as the risk coordinate moves every point to the right by an amount that shrinks as the number of simulated attacks grows. A frontier drawn with point estimates and a decision rule applied with bounds are different objects, and an institution that mixes them will choose settings that its own rule then rejects. Second, the frontier of a real mechanism family is a set of points, not a curve. Its slopes on the simulated population fall from 6.0 to 0.006 but not steadily, so the frontier is not concave, which is what a family of discrete settings produces. The propositions below are stated for a smooth concave frontier, where they are cleanest, and Section 8.4 gives the discrete version a committee actually uses.

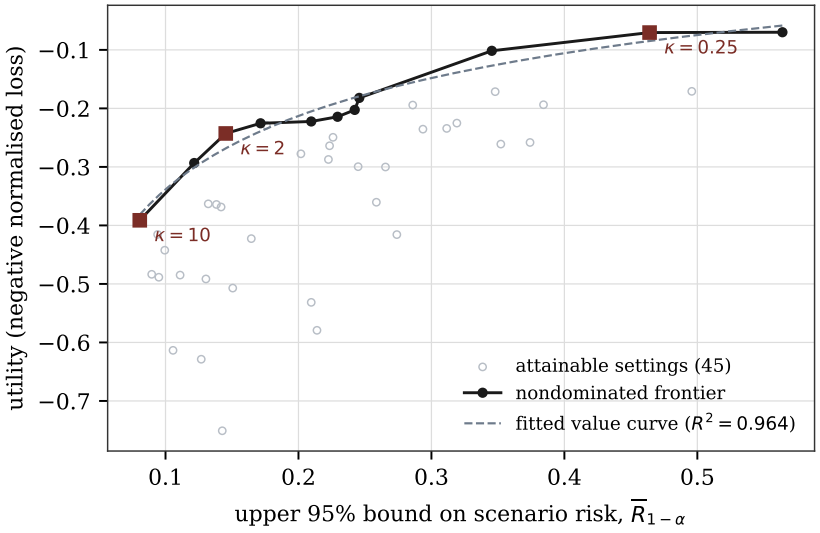


FIGURE 8.1: Risk and utility for 45 protection settings on the simulated population. The risk coordinate is the one-sided 95 per cent upper bound for a uniformly drawn target, not the point estimate. Squares mark the setting that maximises value for three values of the relative harm weight κ; the risk of each is the implied tolerance of Proposition 8.1.

8.2 The tolerance a harm weight implies

Suppose the institution values one unit of utility at $\lambda > 0$ and one unit of risk, for the declared harm class, at $c_h > 0$. No monetary valuation is needed; only the ratio enters,

$$\kappa = \frac{c_h}{\lambda},$$

which we call the *relative harm weight*. It is measured in units of utility per unit of risk: it is the utility gain the institution would require before accepting one more unit of risk. The value of a setting is

$$V(\theta) = \lambda\, U(\theta) - c_h\, \overline{R}_{1-\alpha}(\theta). \tag{8.2}$$

Proposition 8.1 (Implied tolerance)**.** *Let the frontier be the graph of a differentiable, increasing, strictly concave function $u = \varphi(r)$ on $[\underline{r}, \bar{r}]$. The maximiser of* (8.2) *over the frontier exists and is unique. If $\varphi'(\underline{r}) > \kappa > \varphi'(\bar{r})$, it is interior and its risk $\tau^\star$ satisfies*

$$\varphi'(\tau^\star) = \kappa. \tag{8.3}$$

If $\kappa \geq \varphi'(\underline{r})$ then $\tau^\star = \underline{r}$, and if $\kappa \leq \varphi'(\bar{r})$ then $\tau^\star = \bar{r}$. In every case $\tau^\star$ is nonincreasing in κ, and strictly decreasing over the interior range.

The proposition is the disclosure-control counterpart of the social-choice argument of Abowd and Schmutte (2019), and it has the form of a Bayes decision under a linear loss (Berger, 1985). Its practical use is not to compute the tolerance. A harm weight is no easier to state than a tolerance. Its use is to show that the two are the same statement made in different units. An institution that adopts a tolerance has, by (8.3), adopted the harm weight equal to the frontier's slope at that tolerance, whether or not anyone said so. An institution that adopts 0.05 on one release and 0.20 on another has, in general, put different weights on the same harm, and it should be able to say why.

Figure 8.2 reads the implied tolerance off the simulated frontier for six harm weights. At $\kappa = 0.25$ the institution would release the file without suppression, at a bound of 0.464. At $\kappa = 0.5$ it would coarsen the age bands and still keep every record, at 0.345. Doubling the weight to 1 cuts the tolerance to 0.145 and moves the release to three occupation groups with a minimum cell count of five, keeping 94 per cent of records. At $\kappa = 5$ and above the tolerance is 0.081 and the minimum cell count is 20, which keeps 58 per cent.

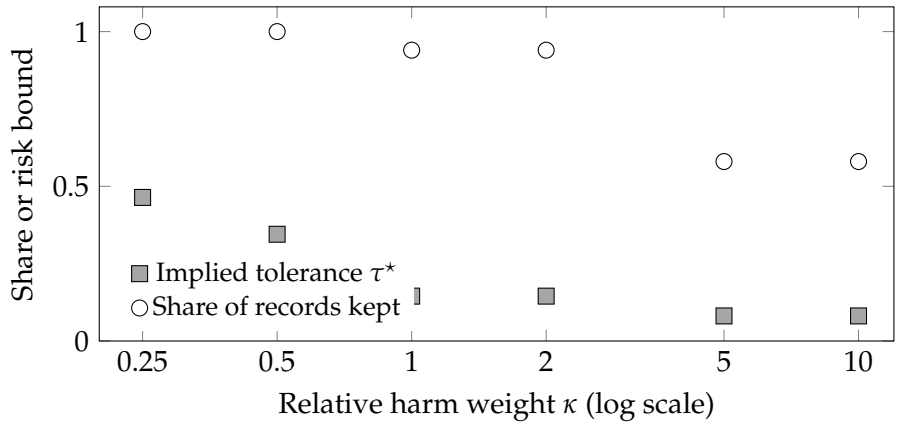


FIGURE 8.2: The tolerance implied by six relative harm weights on the simulated population (filled grey squares), and the share of records the selected setting keeps (open white circles). Only these six weights were evaluated; between two neighbouring weights with different values the setting changes at some point not shown. Each value maximises utility minus κ times the upper risk bound over the 11 settings on the frontier.

Two features of the figure carry over to practice. The implied tolerance is flat over wide ranges of the weight: one setting is best for every weight from 5 to 10. And it jumps between ranges: between weights of 0.5 and 1, a modest change in how much the institution cares about the harm moves the release from an unsuppressed file with twelve occupation groups to a suppressed file with three. The jumps come from the discrete family of settings, not from the argument. They mean that over most of the range an institution cannot fine-tune its tolerance, and that near a breakpoint a small revision of the harm weight changes the product.

The credit register shows how much rides on the choice. At a tolerance of 0.05, an adversary who holds a single release already requires a floor of 40, and against the full pooled adversary no floor up to 300 suffices. At 0.20, the three outside adversaries require floors of 1, 5 and 30. The floors move by an order of magnitude across a range of tolerances that a committee might regard as close together, which is why the tolerance deserves the same scrutiny as the adversary.

When Naledi brought the chair a figure of this kind for the credit register, the chair did not ask for the tolerance, but asked what each step down the curve would cost the users of each product, and which harm the step prevented. That was the right question, and it is the one the rest of this chapter turns into a procedure.

8.3 Why one tolerance for every release is usually inefficient

The tangency condition applies release by release. A programme whose releases share a harm class, and so a harm weight, but face different trade-offs, has a different implied toler-

ance for each.

Corollary 8.2 (Uniform tolerances are usually inefficient)**.** *Let releases $i = 1, \dots, m$ have frontiers φ_i satisfying the conditions of Proposition 8.1, and a common relative harm weight κ in the interior range of each. A common tolerance τ maximises total value $\sum_i \{\lambda \varphi_i(\tau_i) - c_h \tau_i\}$ only if $\varphi_i'(\tau) = \kappa$ for every i. Otherwise total value strictly increases when the tolerance is tightened on releases whose frontier is flat at τ and loosened on releases whose frontier is steep there.*

A frontier is flat at a tolerance when protection is cheap there: a large cut in risk costs little utility. It is steep when protection is dear. A uniform tolerance asks every release to stop at the same risk, and so it stops the cheap releases short of the protection they could give at little cost, and pushes the dear ones beyond the point where the protection is worth what it destroys. Moving tolerance from where protection is dear to where it is cheap leaves the institution better off on both counts it cares about.

The fairness argument for a uniform tolerance deserves a direct reply. Equal treatment of data subjects is the right principle, but a uniform tolerance across releases is not what delivers it. The quantity that should be the same for everyone exposed to the same harm is the weight the institution puts on that harm, κ, because that is the statement of how much a person's exposure counts against the public's use of the data. Applying one weight consistently gives different tolerances on releases whose trade-offs differ, and that is the consistent treatment. A uniform tolerance, by contrast, puts different implied weights on the same harm in different releases, so it is the uniform figure that treats people unequally, through the back door of the frontier's slope. The principle also needs the protected parties of Chapter 7: a weight is set per harm class and per party, and a borrower's identification and a lender's reconstructed position need not carry the same weight. And it needs the programme view of Chapter 9, because one person appears in many releases, and equal per-release tolerances say nothing about the risk to that person once the releases are pooled.

Two cautions belong with the corollary. The harm weight is not found in the data. Choosing it is a governance and ethical decision, and no estimation procedure will supply it. What the corollary does is force the choice into the open, because a tolerance already implies a weight. The second caution follows from the first. The owner of the threshold has to be senior enough to state the weight on the institution's behalf and answer for it in public, which in a central bank means the release committee or the executive member who chairs it, and not the analyst who runs the simulation. An owner who cannot defend the weight in public is not the right owner.

8.4 A discrete frontier and a practical way to state a weight

Real settings form a finite grid, and Proposition 8.1 needs a smooth, concave curve. The general existence result covers both.

Proposition 8.3 (Existence)**.** *Let T be the declared target set. If Θ is compact, U is upper semicontinuous, $\theta \mapsto \max_{t \in T} \overline{R}_{1-\alpha}(t, \theta)$ is lower semicontinuous, and some setting meets the tolerance τ, then the problem of maximising $U(\theta)$ subject to $\max_{t \in T} \overline{R}_{1-\alpha}(t, \theta) \leq \tau$ has a solution.*

On a finite grid every function is continuous in the trivial sense and the maximum exists, so the result adds nothing there. What a finite grid needs is a statement of which settings any harm weight could select, and what a chosen setting says about the weight. Order the

frontier points by risk, $(r_1, u_1), \dots, (r_K, u_K)$ with $r_1 < \cdots < r_K$ and therefore $u_1 < \cdots < u_K$, and define the trade between two points $i < j$ as

$$s_{ij} = \frac{u_j - u_i}{r_j - r_i},$$

the utility gained per unit of risk added in moving from i to j.

Proposition 8.4 (What a chosen setting says about the weight)**.** *Frontier point j maximises $u - \kappa r$ over the frontier if and only if*

$$\max_{i>j} s_{ji} \;\le\; \kappa \;\le\; \min_{i<j} s_{ij}, \tag{8.4}$$

with the maximum over an empty set read as 0 and the minimum over an empty set as $+\infty$. The set of weights that select j is therefore an interval, and it is empty exactly when (r_j, u_j) lies below the upper concave hull of the frontier.

The proposition gives the committee a procedure that needs no number it cannot supply. The analyst lists the settings on the upper concave hull of the frontier, in order of risk. For each neighbouring pair, she writes one line: moving from setting A to the safer setting B lowers the upper risk bound from r_A to r_B, for the declared adversary and harm, and costs this much utility, described in terms users recognise, such as the share of records kept, the occupation detail lost or the error added to a published ratio. The owner answers each line yes or no: is the reduction in risk worth the loss? Each answer places κ above or below one trade, and because the trades decline along the hull, the answers locate the weight in one interval and select one setting. On the simulated population the hull has at most 11 points, so a few questions suffice, and the flat stretches of Figure 8.2 mean that most of the range of possible weights leads to the same handful of settings.

The procedure produces three things the record should keep: the questions as put and the owner's answers, the interval of weights they imply, and the setting selected. It also catches two errors. A setting that lies below the hull cannot be selected by any weight, so a proposal to release it is a proposal to accept a setting that some other setting beats for every view of the harm; the proposition says so, and the analyst can point to the setting that beats it. And when two releases of the same harm class and protected party carry weight intervals that do not overlap, the institution has put different weights on the same harm, and the record must say why.

When Grace read the records of the Bank's first year under this procedure, she did exactly that comparison. Two research extracts drawn from the credit register, for the same harm and the same protected party, had been signed with weight intervals that did not overlap. The record of the second named a reason: the second extract went only to a secure environment, under a different access route, with a different declared adversary. That was a reason the procedure allows. Without it the difference would have been a finding.

8.5 A risk budget across a programme

A programme raises the question of tolerance in a second form. Suppose an institution has decided how much risk the whole programme may carry and must divide it among its releases. Chapter 9 showed that the probability that at least one release, used alone, exposes a target is at most the sum of the releases' risks. An institution that governs that union event

can therefore impose $\sum_i r_i \le \tau_{\text{prog}}$ and allocate the budget. Release i has value weight $w_i > 0$, and for simplicity the releases share one frontier shape φ.

Proposition 8.5 (Risk budget)**.** *Let φ be increasing, differentiable and strictly concave on $(0, \infty)$, with $\varphi'(0^+) = \infty$. The problem of maximising $\sum_i w_i \varphi(r_i)$ subject to $\sum_i r_i \le \tau_{\text{prog}}$ and $r_i \ge 0$ has a unique solution, characterised by*

$$w_i \, \varphi'(r_i^\star) = \mu \quad \text{for every } i, \qquad \sum_i r_i^\star = \tau_{\text{prog}}, \tag{8.5}$$

for some $\mu > 0$. If $\varphi(r) = \log r$, then $r_i^\star = \tau_{\text{prog}} w_i / \sum_j w_j$. If $\varphi(r) = a + br^\gamma$ with $b < 0$ and $\gamma < 0$, then $r_i^\star \propto w_i^{1/(1-\gamma)}$.

Every release ends with the same weighted marginal value of risk. The equal split, $r_i = \tau_{\text{prog}}/m$, which is what an institution does by default when it applies one tolerance to every release, satisfies (8.5) only when all the value weights are equal. The proportional rule is just as particular: it is right for the logarithmic value curve and for no other. For the power family, the exponent $1/(1-\gamma)$ is below one when $\gamma < 0$, so the allocation is flatter than value, and it tends to the equal split as $\gamma \to -\infty$.

The repository fits the power curve to the simulated frontier, with $R^2 = 0.964$ and $\gamma = -0.23$, which gives an exponent of 0.81. For four releases with value weights 1.0, 0.7, 0.5 and 0.3 and a programme budget of 0.5, the allocation is 0.186, 0.139, 0.106 and 0.070, against 0.125 each under the equal split, and on the fitted curve it cuts the total loss by 4.5 per cent relative to the equal split. The curve is fitted by nonlinear least squares to the frontier points, and the code returns the point estimates and R^2 only, with no standard errors. With few frontier points the exponent γ is weakly identified, so the implied tolerance is conditional on the fitted curve, and its sensitivity to γ should be shown before it is used in a signed threshold. Figure 8.3 sets the three rules side by side. The proportional shares, 0.200, 0.140, 0.100 and 0.060, follow from the same weights and budget. The figure is a stylised illustration: the frontier it rests on is not concave, the fitted curve smooths it, and the 4.5 per cent is a gain on the curve, not on the discrete settings. Its lesson is qualitative. The default rule is optimal only by coincidence, and the gain from doing better depends on how unequal the releases' values are and how sharply the frontier bends.

The scope of the budget must be stated with it. Proposition 8.5 governs the union event, the chance that some single release, used alone, exposes a target. It does not govern an adversary who pools the releases, and Chapter 9 proved that no statement in terms of the releases' separate risks can. A risk budget allocates attention and protection across releases. It does not replace the joint simulation of the pooled attack, and a record that cites a budget as evidence about pooled risk misstates what the budget shows. The Census Bureau's division of its privacy-loss budget among geographic levels differs on exactly this point: a formal privacy accountant composes across releases without any assumption about how they depend on one another (Dwork and Roth, 2014), which is the subject of Chapter 10, whereas a budget on the union of attack events composes only for the event it names.

8.6 The answer

The chapter asked what tolerance a stated harm weight implies for each release, and how a programme's tolerance should be shared. For each release, the implied tolerance is the point on its risk–utility frontier where the utility gained per unit of added risk equals the

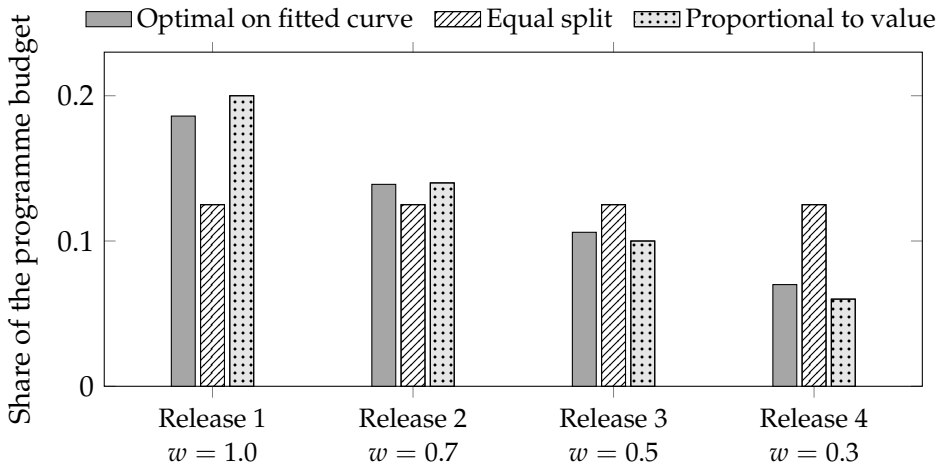


FIGURE 8.3: Three ways to divide a programme budget of 0.5 among four releases with value weights 1.0, 0.7, 0.5 and 0.3. Solid dark bars: the allocation that maximises value on the power curve fitted to the simulated frontier. Hatched white bars: the equal split. Dotted light bars: shares proportional to value, which would be optimal for a logarithmic value curve.

institution's harm weight; on a discrete frontier it is the setting whose interval of weights contains the institution's weight. Since frontiers differ, so do the implied tolerances, and one tolerance for every release is the right choice only when every frontier has the same slope at it. Fairness is served by applying one harm weight, per harm class and protected party, to every release, which gives tolerances that differ. Across a programme governed on the union event, the budget should be divided so that every release ends with the same weighted marginal value of risk; the equal split does this only when the releases are of equal value. In practice a committee need not state a weight as a number. It answers a short sequence of questions about whether one step down the frontier is worth its cost, and its answers fix both the weight and the release.

IN PRACTICE: SETTING TOLERANCES ACROSS RELEASES

Declare the utility loss for each release before any setting is evaluated, and plot the frontier with the upper bound the decision will use. Present the owner with the settings on the upper concave hull, in order of risk, and with one line per neighbouring pair stating the risk removed and the use lost. Record the owner's answers, the interval of harm weights they imply and the setting selected. Use one harm weight per harm class and protected party across releases, and write down the reason whenever two releases with the same harm and party carry weights that disagree. When dividing a programme budget, give more to the releases of greater value, but do not cite the budget as evidence about an adversary who pools the releases.

In the software. `privacy_risk.frontier` holds `nondominated` for the frontier, `choose_max_utility` for the constrained choice, `fit_value_curve` for the power curve and `concavity_diagnostic`, which reports the successive slopes and whether they decline. `privacy_risk.governance.implied_tolerance` returns the setting that maximises utility minus κ times the risk bound, and `risk_budget` and `budget_value` implement Proposition 8.5 for the power curve. `privacy_risk.experiments.programme_budget` checks the closed form against a numerical optimum. The interval of Proposition 8.4 is not yet implemented; it follows from the slopes that `concavity_diagnostic` returns. The allocation itself is short:

```
exponent = 1.0 / (1.0 - gamma)
shares = weights ** exponent
```

```
return programme_tolerance * shares / shares.sum()
```

8.7 Proofs

Proof of Proposition 8.1. On the frontier the problem is to maximise $J(r) = \lambda\varphi(r) - c_h r$ over $[\underline{r}, \bar{r}]$. J is continuous on a compact interval, so a maximiser exists, and it is strictly concave because φ is, so the maximiser is unique. Its derivative $J'(r) = \lambda\{\varphi'(r) - \kappa\}$ is strictly decreasing. If $\varphi'(\underline{r}) > \kappa > \varphi'(\bar{r})$, then J' is positive at $\underline{r}$ and negative at $\bar{r}$, so it has a unique zero in the interior, which is the maximiser and satisfies (8.3). If $\kappa \geq \varphi'(\underline{r})$, then $J'(r) < 0$ for every $r > \underline{r}$, so J is decreasing and the maximiser is $\underline{r}$. If $\kappa \leq \varphi'(\bar{r})$, then $J'(r) > 0$ for every $r < \bar{r}$, and the maximiser is $\bar{r}$. Over the interior range $\tau^\star = (\varphi')^{-1}(\kappa)$, and the inverse of a strictly decreasing function is strictly decreasing. Combining the three ranges, $\tau^\star$ equals $\bar{r}$ for small κ, decreases strictly through the interior range, and equals $\underline{r}$ for large κ, so it is nonincreasing throughout. □

Proof of Corollary 8.2. Total value is a sum of terms each depending on one τ_i, so it is maximised by maximising each term separately. By Proposition 8.1, with κ in the interior range of every release, the unique maximiser of term i is the $\tau_i^\star$ solving $\varphi_i'(\tau_i^\star) = \kappa$. A common tolerance τ is optimal only if $\tau = \tau_i^\star$ for every i, that is $\varphi_i'(\tau) = \kappa$ for every i. If this fails for some i, then $\tau \neq \tau_i^\star$, and by strict concavity term i increases strictly when τ_i moves from τ towards $\tau_i^\star$. When $\varphi_i'(\tau) < \kappa$ the frontier is flat at τ and $\tau_i^\star < \tau$, so the move tightens the tolerance; when $\varphi_i'(\tau) > \kappa$ it is steep and $\tau_i^\star > \tau$, so the move loosens it. The other terms are unchanged, so total value increases strictly. □

Proof of Proposition 8.3. The feasible set $\Theta_\tau = \{\theta : \max_{t \in T} \overline{R}_{1-\alpha}(t, \theta) \leq \tau\}$ is a sublevel set of a lower semicontinuous function, hence closed, and as a closed subset of the compact Θ it is compact. It is nonempty by assumption. An upper semicontinuous function attains its maximum on a nonempty compact set, so U has a maximiser on Θ_τ. □

Proof of Proposition 8.4. Point j maximises $u - \kappa r$ if and only if $u_j - \kappa r_j \geq u_i - \kappa r_i$ for every i. For $i > j$ we have $r_i > r_j$, and the inequality rearranges to $\kappa \geq (u_i - u_j)/(r_i - r_j) = s_{ji}$. For $i < j$ we have $r_i < r_j$, and it rearranges to $\kappa \leq (u_j - u_i)/(r_j - r_i) = s_{ij}$. Taking all i together gives (8.4); with no $i > j$ the lower condition is only $\kappa > 0$, and with no $i < j$ there is no upper condition. The set of κ satisfying finitely many inequalities of these two forms is an interval. It is empty exactly when some $i < j$ and some $k > j$ have $s_{ij} < s_{jk}$, which says that the point (r_j, u_j) lies strictly below the segment joining (r_i, u_i) and (r_k, u_k), that is, below the upper concave hull. Conversely, if (r_j, u_j) lies on the hull, a supporting line of the hull through it has a slope $\kappa > 0$ satisfying every inequality, so the interval is nonempty. □

Proof of Proposition 8.5. The objective is strictly concave, as a positive combination of strictly concave functions of separate coordinates, and the constraint set is compact and convex, so a unique maximiser exists. Because φ is increasing, the budget constraint binds at the optimum: otherwise some r_i could be raised and the objective would increase. Because $\varphi'(0^+) = \infty$, no coordinate is zero at the optimum, since moving a small amount of budget to a coordinate at zero raises its term at an unbounded rate while lowering another at a finite rate. The nonnegativity constraints are therefore slack, and the Lagrangian $\sum_i w_i \varphi(r_i) - \mu(\sum_i r_i - \tau_{\text{prog}})$ gives the first-order conditions $w_i \varphi'(r_i) = \mu$, which is (8.5); by concavity they are sufficient, and $\mu > 0$ because $\varphi' > 0$. For $\varphi = \log$ the conditions read $w_i / r_i =$

μ, so $r_i \propto w_i$, and the budget fixes the constant at $\tau_{\text{prog}} / \sum_j w_j$. For $\varphi(r) = a + br^\gamma$ with $b < 0$ and $\gamma < 0$, we have $\varphi'(r) = b\gamma r^{\gamma-1} > 0$ and $\varphi''(r) = b\gamma(\gamma-1)r^{\gamma-2} < 0$, so φ is increasing and strictly concave with $\varphi'(0^+) = \infty$. The conditions read $w_i r_i^{\gamma-1} = \mu/(b\gamma)$, so $r_i \propto w_i^{1/(1-\gamma)}$. Without the condition $\varphi'(0^+) = \infty$, the Karush–Kuhn–Tucker conditions (Boyd and Vandenberghe, 2004) add a multiplier for each nonnegativity constraint, and a release then receives no budget exactly when its weighted marginal value at zero, $w_i\varphi'(0^+)$, is at most μ. □

Part IV

Programmes

Chapter 9

Several releases to the same user

At the Lisbon workshop where this work was first presented, a participant asked what happens to the risk when an institution gives several data sets to the same researcher. Most institutions have an answer ready, and it is a reasonable one: if each file has passed its own disclosure review, the set passes too. A careful institution may go further and add up the risks of the files, on the grounds that the chance that at least one of them exposes a person can never exceed the sum of the separate chances. That sum is a true bound for the event it describes, and on this view a set of files each reviewed to a small risk carries at most a modest one.

The answer fails on a simple simulation. Four releases are drawn from one simulated register, each keyed on ordinary demographic variables, each keeping at least three records in every key class, and none containing a unique record. Their separate risks lie between 0.017 and 0.088, and their sum is 0.253. An adversary who holds all four and can match records across them identifies the target with probability 0.458, and a quarter of all targets are identified uniquely, although none was unique in any single file. The sum bounded the wrong event.

At the Bank the question arrives as a request. Tomás, a visiting researcher, wants three products from the credit register for one study: a loan-level research file covering six months, a table of lending by sector and region, and the public monthly series. Naledi has reviewed each product before, for other users, and each passed. This chapter asks one question on Tomás's behalf: *what is the disclosure risk of giving several releases to the same user, and what must an institution assess to know it?* It shows that the risk of the combination cannot be read off the risks of the parts, proves that no rule based only on the parts can bound it, measures the gap on simulated and public data, and sets out what to assess instead.

9.1 Two different questions about a programme

Let an institution publish releases $Y_1, \dots, Y_m$. For each release let C_i be the event that the release, used on its own, allows a correct identification under its declared attack, and let $R_i = \mathbb{P}(C_i)$ be its marginal risk. A natural first question is the probability that at least one release, used alone, exposes the target, which is the probability of the union $C_\cup = C_1 \cup \cdots \cup C_m$. The marginal risks say something about this event, though less than one might hope.

Proposition 9.1 (Sharp bounds from marginal risks)**.** *Without assumptions on how the events*

C_i depend on one another,

$$\max_i R_i \;\le\; \mathbb{P}(C_\cup) \;\le\; \min\Big(1,\; \sum_{i=1}^{m} R_i\Big), \tag{9.1}$$

and both bounds are attained by suitable joint distributions, so the marginal risks alone cannot identify a narrower interval.

The interval is wide. If the releases were independent, the union would have probability $1 - \prod_i (1 - R_i)$, but releases cut from one register are ordinarily dependent, since they describe the same people, and that expression needs evidence before it can be used. In the simulation reported below the independence value is 0.231 inside an interval of $[0.088, 0.253]$. It happens to sit near the upper bound, and that agreement is a feature of the design and says nothing about independence.

The second question is the one the participant asked, and it concerns a different event. An adversary who holds all m releases is not obliged to use them one at a time. It can apply a joint linkage rule that looks at all the releases together, and the programme risk is the probability R_{pool} that this joint rule names the target. Success of the joint rule is not contained in the union of single-release successes, so the bounds of Proposition 9.1 do not apply to it.

9.2 Pooling intersects candidate sets

Figure 9.1 shows the mechanism in its smallest form. Each of two releases groups the target with two other records, so an adversary using either release alone has a one in three chance. If both releases carry the same pseudonym for each record, the adversary can keep only the records that appear in both candidate sets, and here only the target does. Each release is as safe as a key class of three makes it, and the pair identifies the target with certainty. The sum of the marginal risks, two thirds, is no bound on the pooled risk.

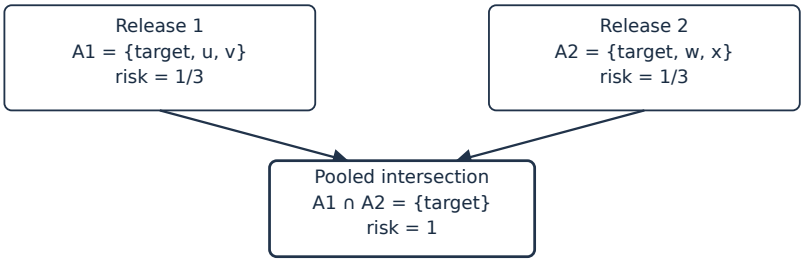


FIGURE 9.1: Pooling two releases intersects their candidate sets. Each release alone leaves three candidates for the target; with a pseudonym common to both releases the adversary keeps only records in both sets, which leaves one.

The small example generalises without limit, which is the central result of the chapter.

Proposition 9.2 (Marginal risks do not bound pooled risk)**.** *Let $m \ge 2$. Suppose each released record carries a pseudonym that is the same in every release, and the adversary knows the target's full set of quasi-identifiers, from which it forms the target's key in every release. For every $\nu > 0$*

there exist a population, a sample and m releases such that $R_i \le \nu$ *for every i while* $R_{\text{pool}} = 1$. *Consequently no upper bound on* R_{pool} *can be written as a function of* $(R_1, \dots, R_m)$ *alone. If each release publishes the candidate set implied by its key and the adversary chooses uniformly within the intersection of those sets, then* R_{pool} *does not decrease as releases are added, and* $R_{\text{pool}} \ge \max_i R_i$.

Neither the phenomenon nor its direction is new. Sweeney (2002) observed that complementary releases, each *k*-anonymous, can be joined to re-identify; differencing between linked tables is an old problem in disclosure control (Hundepool et al., 2012, ch. 4); and Ganta et al. (2008) showed that independent releases of overlapping populations, each protected by a syntactic rule, can be combined into a successful attack. Work on sequential releases of one table (Wang and Fung, 2006; Xiao and Tao, 2007) designs protection that survives republication. What the proposition adds is a statement in the terms an institution uses when it certifies a release: the quantity a programme is certified against, the per-release risk, is not the quantity that governs it.

A persistent pseudonym also has a simpler reading. If every release carries the same pseudonym, the adversary can join the releases into one table keyed on all the variables any of them publishes. The pooled attack is then an attack on one release with a finer key, and the proposition says that a finer key can carry far more risk than any of its parts. That reading points to a remedy as well as a warning, and Section 9.5 returns to it.

Naledi read the proposition with Tomás's request open beside it. The loan-level file carries a pseudonym for each loan, and it is the same pseudonym in every monthly vintage, because the Bank's users had asked for a panel. Her past reviews had each looked at one vintage. None of them had looked at what a user holding all six could do with a key that persists.

9.3 How large the gap is

Proposition 9.2 is an existence result and says nothing about the size of the gap in a real programme. To measure it, the pipeline draws four releases from a simulated register of 200,000 units with a 5 per cent sample. Each release is keyed on ordinary demographic variables, each keeps at least three records in every key class, and none contains a single unique record. The four share a record identifier.

Figure 9.2 reports the result. The marginal risks lie between 0.017 and 0.088, so the marginals bound the chance that any one release succeeds between 0.088 and 0.253. An adversary who intersects only the first two releases already succeeds with probability 0.411, which is 2.8 times the upper bound implied by those two releases. With all four the pooled risk is 0.458, 1.8 times the bound implied by all four. A quarter of targets, 26 per cent, are identified uniquely by the intersection, although none was unique in any single release.

The gap is not an artefact of synthetic data. On a public file, an extract of the US Census 2000 microdata, the same construction gives a pooled risk of 0.709 against a bound of 0.381 from the marginals. Part V reports both studies in full.

9.4 The credit register

The running example shows the same effect with an operational lever attached. The synthetic register releases loans over six monthly vintages, each loan carrying a persistent

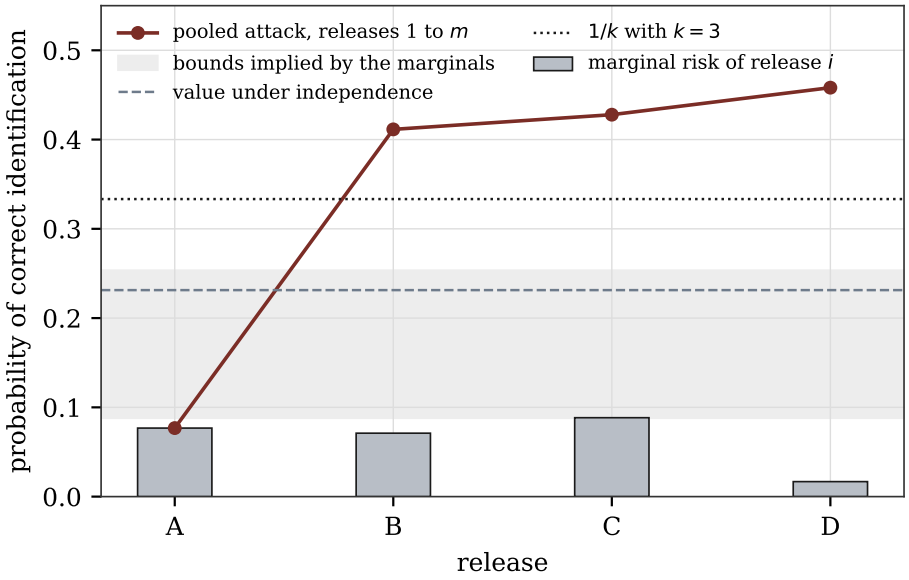


FIGURE 9.2: Four 3-anonymous releases from one simulated sample: marginal risks, the interval the marginals imply, and the risk of an adversary that pools releases 1 to m, from 20,000 target replicates. Marginal risks are unconditional, so a target suppressed from a release contributes zero to it.

pseudonym, its borrower's sector, region and size class, and its exposure. Judged one release at a time at a tolerance of 0.12, the file passes with no suppression. The outcome changes once the adversary pools the vintages, but only under a condition that is worth stating precisely.

Proposition 9.3 (Without the history, the other vintages are uninformative)**.** *Suppose each loan's published key and exposure are the same in every vintage in which it appears, that whether a loan is live in a vintage is drawn independently of its key, its exposure and the adversary's auxiliary information, and that the adversary knows only that the target is live in the first release. Then, for any fixed auxiliary information, the adversary's posterior over candidates computed from all vintages equals its posterior computed from the first vintage alone.*

The result explains why pooling needs something extra to bite. Repeating the same published fields adds nothing, and the pattern of months in which each loan is live carries no information about which loan is the target unless the adversary knows the target's own pattern. An adversary who does know in which vintages the target's loan was live, as a counterparty or an informed observer might, can strike out every candidate whose activity does not match. Against that adversary the decision rule requires a minimum cell count of 75. An adversary who also holds a reading of the loan's exposure in each vintage requires 150, and an insider who reads exposure with small error is not stopped by any floor up to 300.

The proposition changed what Naledi needed to know about Tomás. Whether he could pool the six vintages was settled by the pseudonym; what she had to establish was whether he, or anyone who later saw his file, could know in which months a given firm's loan was live. A lender knows this about its own customers, and so would a researcher who had already studied the firm. Naledi declared the history-aware adversary, and the floor rose from 1 to 75.

9.5 What to assess instead

Three consequences follow for an institution that publishes programmes. First, marginal risk figures may still be reported, because they describe what an adversary achieves with one release, but they license no claim about the programme. Second, a programme must be assessed by simulating the joint attack directly, with targets and auxiliary information shared across the releases. Separate simulations of each release cannot recover the dependence between them, because they never form the intersection. Third, the programme needs a horizon: the number of releases over which the joint risk is assessed, written into the governance threshold, with a reassessment due before the next release beyond it. Pooled risk does not decrease as releases are added, so a programme tolerance without a horizon has no fixed meaning.

The design of the identifier is the second lever. A pseudonym that persists across releases is what allows the intersection, so rotating it between releases forces any linkage to run through the published fields instead. The pseudonym itself should be keyed, for example a keyed hash of the record identifier under a key held apart from the release team, so that nobody can recompute it from the identifier. Rotation is a control on linkability and not a guarantee. On the credit register it does not help while exposure is published exactly, because the published key and exposure relink every loan from one vintage to the next. With exposure rounded to one significant figure they relink about one loan in ten. Whether rotation is worth its cost therefore depends on what else the releases publish, and the joint simulation is what tells the institution which case it is in.

How to act on the gap between pooled and single-release risk depends on its size, and the owner fixes the cut-offs in the signed tolerance. If the pooled risk is inside the tolerance, the programme is released as designed and the gap is recorded in the file. If the gap is moderate, the institution coarsens the keys, rotates pseudonyms (which only helps if the linking attributes are rounded), reduces the release frequency or moves the data to a secure environment. If the gap is large, the product is the problem: change it and assess the changed product from the start. An adversary who holds only part of the key still gains, so partial keys should be tested. The pooled attack is computed from the sets of consistent candidates and never needs an enumeration of 2^m combinations.

A formal privacy accountant is the one device that composes across releases without any assumption about how they depend on one another (Dwork and Roth, 2014). Chapter 10 sets out what that guarantee says and what it leaves unsaid. A programme tolerance can also be divided among releases as a budget on the union event, but by Proposition 9.2 such a budget governs the union only. It allocates attention across releases and does not replace the joint simulation.

9.6 The answer

The question was what the disclosure risk of giving several releases to one user is, and what an institution must assess to know it. The risk is the probability that an adversary using all the releases together names the target, and it is a different event from any one release succeeding. When records keep a pseudonym across releases, no function of the separate risks bounds it, and on a simulated register four files that each look safe identify the target far more often than their separate risks suggest. To know the risk, the institution must simulate the joint attack directly, over a declared horizon, with the targets and the

adversary's information shared across the releases; it may then rotate keyed pseudonyms, coarsen the linking fields or move the user into a secure environment, and simulate again. For Tomás the committee did the last: the loan-level file at the floor his adversary required kept too little exposure for his study, and Chapter 13 follows him into the secure room.

IN PRACTICE: GIVING SEVERAL FILES TO ONE USER

List every release from the same register that the user can access, now and within the assessment horizon, and every identifier or published field that could link records across them. If records keep a pseudonym across releases, assess the pooled attack directly by simulation rather than combining per-release figures. Record the horizon in the decision record and set a reassessment before the first release beyond it. If the pooled attack fails the tolerance at every acceptable floor, consider rotating keyed pseudonyms, coarsening the linking fields, or moving the user to a secure environment, and simulate again for each option.

In the software. The bounds and the pooled attack are in `privacy_risk.composition`: `marginal_bounds` and `independent_union` for Proposition 9.1, and `pooled_attack` (or `pooled_attack_from_sets` for any release format) for the intersection attack behind Figure 9.2.

9.7 Proofs

Proof of Proposition 9.1. Each C_i is contained in $C_\cup$, so $R_i \le \mathbb{P}(C_\cup)$ for every i, which gives the lower bound. Subadditivity of probability gives $\mathbb{P}(C_\cup) \le \sum_i R_i$, and $\mathbb{P}(C_\cup) \le 1$ always. For attainment of the lower bound, let the events be nested, with every C_i contained in the one of largest probability; then $\mathbb{P}(C_\cup) = \max_i R_i$. For the upper bound, if $\sum_i R_i \le 1$ let the events be pairwise disjoint, so that $\mathbb{P}(C_\cup) = \sum_i R_i$; if $\sum_i R_i > 1$, place the events on a probability space $[0,1)$ with Lebesgue measure as consecutive arcs of lengths R_i wrapped around the circle, so that together they cover the whole space and $\mathbb{P}(C_\cup) = 1$. □

Proof of Proposition 9.2. Fix $\nu > 0$ and choose an integer $k \ge 1/\nu$ with $k \ge 2$. Let the population and the sample both consist of k^m units, each labelled by a distinct vector $(v_1, \dots, v_m) \in \{1, \dots, k\}^m$, and let these labels be the units' quasi-identifiers. Release i publishes every unit with its persistent pseudonym and a key made of all coordinates except the ith. Units share a key in release i exactly when they agree on every coordinate other than i, so every key class in release i has k members, and under Proposition 2.2 of Chapter 2 the marginal risk of each release is $R_i = 1/k \le \nu$.

The adversary knows the target's label $(v_1, \dots, v_m)$, so in each release it can find the target's key class and, through the pseudonyms, the corresponding set of units. A unit in the class of release 1 agrees with the target on coordinates $2, \dots, m$; a unit in the class of release 2 agrees with it on coordinate 1 and on coordinates $3, \dots, m$. A unit in both classes therefore agrees with the target on every coordinate, and since labels are distinct it is the target. The intersection of the candidate sets is the target alone, so $R_{\text{pool}} = 1$. Since ν was arbitrary, any upper bound on R_{pool} written through the marginals alone must equal one at marginals as small as we please, so no bound below one exists.

For the last claim, let A_i be the candidate set of release i, which contains the target. The adversary's set after j releases is $A_1 \cap \cdots \cap A_j$, which contains the target and shrinks as j grows, so the uniform success probability $1/|A_1 \cap \cdots \cap A_j|$ does not decrease. Since the intersection of all the sets is contained in each A_i, $1/|\bigcap_i A_i| \ge 1/|A_i| = R_i$ for every i, and taking expectations over targets gives $R_{\text{pool}} \ge \max_i R_i$. □

Proof of Proposition 9.3. Let $\mathcal{C}$ be the set of candidates the adversary forms from the first vintage. For a candidate j, write w_j for its published key and exposure and a_j for its pattern of activity across vintages. Under the hypothesis that j is the target, the joint probability of everything the adversary observes factorises into a term for its auxiliary information given w_j, which is the term the first vintage alone supplies, and a term for the activity patterns of all loans. The later vintages contribute only copies of the w_ℓ, which are already known from the first, and the patterns a_ℓ. Since activity is drawn independently of keys, exposures and auxiliary information, and the adversary knows nothing of the target's pattern beyond its presence in the first vintage, which every candidate shares, the activity term is the same for every hypothesis $j \in \mathcal{C}$. It cancels from the posterior, which therefore equals the posterior computed from the first vintage alone. □

Chapter 10

Formal privacy is a different regime

Many institutions now meet disclosure control through the vocabulary of differential privacy. A statistical office announces a privacy-loss budget, a central bank asks whether its noisy tables could carry a parameter ε, and a release committee hears that $\varepsilon = 1$ is a cautious setting used in published work. The belief that follows is natural and, in its strongest form, well argued. Differential privacy is the one framework with a theorem behind it. It holds against every adversary, whatever that adversary knows, and a small ε makes the output almost the same whether or not any one person's record was used. An institution that adopts $\varepsilon = 1$, on this view, has accepted a low re-identification risk and can say so to its board.

The first fact that breaks the belief is a calculation that takes one line. Take an adversary who has narrowed a target down to three candidate records from sources outside the release, so that its prior on any one of them is one third. Against a release that is ε-differentially private with $\varepsilon = 1$, the guarantee permits that adversary to end with a posterior of 0.58 on the right record. A posterior above one half is a probable identification by any standard a release committee would use. The guarantee was kept in full, and nothing about the result is a failure of the mechanism. The adversary arrived with most of its knowledge already in hand, and the guarantee bounds only how far the release moved that knowledge.

This chapter asks one question. What does a privacy-loss parameter commit an institution to, when the commitment is stated in the units a release decision uses, the probability that a declared attack names the right record? The answer has three parts: a bound on posterior odds that depends on the adversary's prior, a ceiling on the success of every attack at once that depends on a baseline the institution must still argue for, and an accounting rule that works across a programme where scenario assessment has none.

10.1 A committee meets a parameter

The Bank's statistics release committee had a proposal on its table. The supervision department wanted to publish, each month, the number of live loans in every cell of sector, region and borrower size class drawn from the credit register, with noise added so that the counts carry a formal guarantee. The proposal set $\varepsilon = 1$ per month and described the setting as conservative. The chair asked Naledi a direct question before the meeting: if the Bank adopts this, what re-identification risk has it accepted? Naledi said she would answer in writing, and that the answer would not be a single number.

Her difficulty was that the proposal and the Bank's own decision rule speak about different objects. The rule set out in Chapter 1 compares an upper confidence bound on the

success probability of a declared attack with a tolerance, for the Bank's credit products 0.12. The proposal states a bound on how much a mechanism's output distribution can change when one record changes. Both are probabilities in some sense, and both can be small, but one is about an attack and the other is about a pair of databases. The rest of the chapter connects them.

10.2 What the definition says

A randomised mechanism M maps a database x to a random output. Two databases are neighbours, written $x \sim x'$, when they differ in the record of one individual. The mechanism is (ε, δ)-differentially private (Dwork, 2006; Dwork et al., 2006; Dwork and Roth, 2014) if, for every pair of neighbours and every measurable set E of outputs,

$$\mathbb{P}\{M(x) \in E\} \leq e^{\varepsilon}\, \mathbb{P}\{M(x') \in E\} + \delta. \tag{10.1}$$

The case $\delta = 0$ is called pure differential privacy. The probability in (10.1) is over the mechanism's own randomness only. The databases are fixed, and nothing in the definition refers to an adversary, to what it knows or to how it links records.

Two choices hide inside the word "neighbour", and each changes what a given ε protects. The first is the relation itself. Under the replace-one relation, x' is x with one record altered; under the add-or-remove relation, x' is x with one record deleted. A histogram of counts changes in one cell when a record is removed and in two cells when a record is replaced, so the same noise gives a guarantee under one relation that is twice as strong as under the other. The second choice is the unit. In a credit register a record can be a loan, a lending relationship or a borrower. A guarantee for one loan does not carry over to a firm with several loans, as Section 10.7 shows.

The standard example is the Laplace mechanism (Dwork et al., 2006). If a query q changes by at most Δ in absolute value between neighbours, publishing $q(x)$ plus Laplace noise of scale Δ/ε satisfies (10.1) with $\delta = 0$. The supervision department's proposal is of this kind: one count per cell, noise of scale $1/\varepsilon$, a fresh draw each month.

The reason for the definition's shape lies in an older result. Dinur and Nissim (2003) showed that a database answering too many queries too accurately can be reconstructed from its answers. Garfinkel et al. (2019) explain how that line of work bears on published statistical tables, and describe how the US Census Bureau came to regard its own tabulations as open to reconstruction. The Bureau's response for the 2020 census was a disclosure avoidance system built on a formal privacy guarantee, the TopDown algorithm described by Abowd et al. (2022), and the debate among data users that followed is set out by Hotz et al. (2022). The episode is the best documented case of an institution choosing this regime, and it shows both why the regime exists and why its parameter is hard to explain. A reconstruction attack needs no auxiliary file to link against. It recovers records from the published numbers themselves, and the scenario assessments of earlier chapters, which start from a declared linkage, do not see it coming.

10.3 A posterior, not a probability of success

Return to the chair's question and read the definition from the adversary's side. Fix one individual and a hypothesis H about that individual's record: that the record is in the file, or

that it carries a given value, or that a given released row is theirs. Under H the database is x, and under its negation it is a neighbour x'. Bayes' rule in odds form says that the posterior odds on H are the prior odds times the likelihood ratio of what the adversary saw, and (10.1) with $\delta = 0$ caps that likelihood ratio at e^{ε}.

Proposition 10.1 (What ε permits)**.** *Let M satisfy* (10.1) *with $\delta = 0$ under the replace-one relation. Let H be a hypothesis about one individual such that the databases under H and under its negation differ only in that individual's record, the other records having the same distribution under both. For an adversary with prior $\mathbb{P}(H) = p$, and any output set E of positive probability,*

$$\mathbb{P}(H \mid M \in E) \leq \frac{e^{\varepsilon} p}{e^{\varepsilon} p + 1 - p}. \tag{10.2}$$

When $\delta > 0$ no bound of this form holds uniformly over output sets.

The bound is the Bayesian reading of differential privacy given by Kasiviswanathan and Smith (2014). Its content is easiest to see in a picture. Figure 10.1 plots the right side of (10.2) against the prior for three values of ε. Each curve lies above the diagonal, which is where the posterior would sit if the release told the adversary nothing, and the vertical gap between a curve and the diagonal is the most the release can add. The gap is largest for priors in the middle of the range. At small priors the curves hug the diagonal in absolute terms but not in relative terms, since the bound is close to $e^{\varepsilon} p$, and e^{ε} is 2.7 at $\varepsilon = 1$.

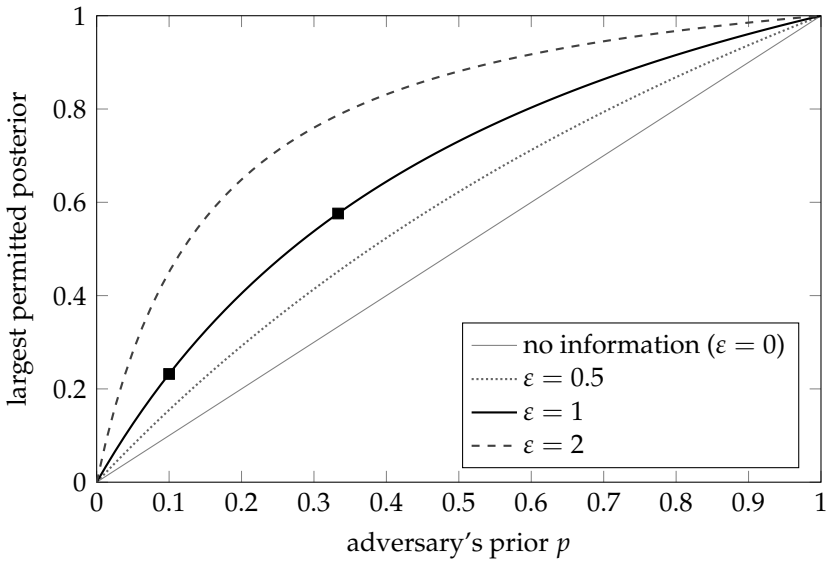


FIGURE 10.1: The largest posterior an ε-differentially private release permits, against the adversary's prior, from (10.2). The curves are told apart by line style as well as shade. The two squares mark the cases in the text: a prior of 0.1 rises to at most 0.232, and a prior of one third to at most 0.576, at $\varepsilon = 1$.

The pipeline tabulates the bound for five values of ε and three priors. Take an adversary whose side information has narrowed the target to ten candidates, so that its prior on each is 0.1. The guarantee permits a posterior of 0.155 at $\varepsilon = 0.5$, 0.232 at $\varepsilon = 1$, 0.451 at $\varepsilon = 2$ and 0.943 at $\varepsilon = 5$. With a prior of one third, $\varepsilon = 1$ permits 0.576. With a prior of one in a hundred, which is what an adversary holds if it knows only that the target lives in a district of a hundred sampled people, $\varepsilon = 1$ permits 0.027 and $\varepsilon = 5$ permits 0.600.

The misconception therefore confuses a multiplier with a level. An institution that adopts $\varepsilon = 1$ has not chosen a tolerance. It has chosen a factor of e^{ε} on the odds of

whatever prior an adversary brings, and the re-identification risk it has implicitly accepted depends on what the adversary knew before the release. The parameter cannot carry that information, because the definition is built to hold for every prior at once. That is its strength as a guarantee and the reason it cannot be read as a risk figure.

The last sentence of Proposition 10.1 deserves attention, because most deployed mechanisms have $\delta > 0$. With $\delta > 0$ the likelihood ratio on a rare output set can be arbitrarily large, so a posterior bound of the form (10.2) holds only for most outputs, with ε inflated, and fails on a set of outputs of small probability (Kasiviswanathan and Smith, 2014). The usual reading of δ as "the probability that the guarantee fails" is loose, but it points in the right direction: a mechanism with $\delta > 0$ may, with small probability, publish something that identifies a person outright. A committee that signs off on δ should know what it is signing for, and δ should be small against the reciprocal of the number of people the release covers (Dwork and Roth, 2014).

Naledi's first page to the chair said this in plain words. The parameter in the proposal was a ceiling on how much any reader of the monthly counts could learn about one loan from the counts themselves. It was not the chance that someone would identify a loan. That chance depended on what the reader already knew, and the Bank's own scenarios, written for its loan-level files, said who those readers were likely to be.

10.4 From beliefs to attacks: a ceiling for every attack at once

The posterior bound speaks about beliefs. A release committee needs a statement about attacks, because its tolerance is a bound on the probability that a declared attack succeeds. The bridge is a second reading of (10.1), as a limit on hypothesis tests (Wasserman and Zhou, 2010; Kairouz et al., 2017; Dong et al., 2022; Balle et al., 2020). Any attack is a decision made from the output, and its success is an event; the definition bounds the probability of every event under one database by its probability under the neighbour.

Fix a target t whose record sits in a known position of the database, and let an attack be any rule g that takes the release and the adversary's auxiliary information Z_t and names a position. Write x for the database with the target's record and $x^{(t)}$ for the database in which that record is replaced by another drawn from a reference distribution. The attack's success probability is

$$R = \mathbb{P}\{g(M(x), Z_t) = t\}, \tag{10.3}$$

and its baseline is

$$\kappa = \mathbb{P}\{g(M(x^{(t)}), Z_t) = t\}, \tag{10.4}$$

the probability that the same rule names the same position when the record there is not the target's. The baseline is what the adversary achieves from everything except the target's own contribution to the release: its auxiliary information, the other records, and luck.

Proposition 10.2 (A ceiling for every attack)**.** *Let M satisfy* (10.1) *under the replace-one relation, and let the adversary's auxiliary information Z_t be independent of the mechanism's randomness. Then for every attack rule g,*

$$R \le \min\bigl(1,\ e^{\varepsilon}\kappa + \delta\bigr). \tag{10.5}$$

Consequently a tolerance τ is met by all attacks together whenever $e^{\varepsilon}\kappa^{*} + \delta \le \tau$, where κ^{*} is the largest baseline success the institution admits over the attacks it is willing to consider.

This is the inequality at which the two regimes meet, and it is worth reading slowly. The privacy parameter supplies the multiplier e^{ε} and the additive slack δ, and it does so for every rule g at once, including rules nobody at the Bank has thought of. What it does not supply is κ. The baseline is a property of the adversary's knowledge and the rest of the data, and an argument about it is exactly the kind of argument that scenario assessment makes. A differentially private release therefore does not remove the need for a declared adversary. It changes the question the scenario must answer, from "how often does this attack succeed?" to "how well could anyone do without the target's own record?", and it turns the answer into a ceiling that holds for every attack.

The inequality also yields a test that a committee can run on the back of an envelope. Hold the Bank's tolerance of 0.12 and set δ negligible. At $\varepsilon = 1$ the ceiling meets the tolerance whenever the baseline is below $0.12/e$, about 0.044; at $\varepsilon = 2$ the baseline must be below about 0.016. A baseline of 0.044 is the success of a uniform guess among about 23 candidates. If the Bank can argue that no plausible reader of its counts can narrow a loan down below that many candidates without the loan's own contribution, the monthly counts meet the tolerance against every attack. If it cannot, the guarantee is still true, and it still says nothing the committee can use.

Kifer and Machanavajjhala (2014) make the adversary's prior an explicit input to a privacy definition in the Pufferfish framework, which is the formal counterpart of a declared scenario, and Kifer and Machanavajjhala (2011) show that no privacy definition can be free of assumptions about the data-generating process if it is to bound what an adversary learns about a person. These results give the ceiling its place. The guarantee in (10.1) compares two databases and needs no such assumption. The moment one asks what an adversary believes about a person, which is what (10.2) and (10.5) do, an assumption enters: in Proposition 10.1 that the other records have the same distribution under both hypotheses, and in Proposition 10.2 a reference distribution for the replacement record that fixes κ.

10.5 Noisy counts on the credit register

The pipeline runs the supervision department's proposal on the synthetic credit register of Chapter 13, and the results show both readings of the guarantee at work. The release publishes, for the first vintage, a noisy count of live loans in each of 814 occupied cells of sector, region and size class, of which 10 hold a single loan. The declared adversary attacks the cells, not the loans: it reads each noisy count, computes the posterior that the true count is one against a prior built from the distribution of cell sizes, and declares a cell to hold a single, identifiable loan when that posterior reaches one half. Its success is a correct declaration.

At $\varepsilon = 1$ for one month the adversary never acts: no cell reaches the threshold, and the exact one-sided 95 per cent bound on its success over the 814 cells is 0.0037. At $\varepsilon = 5$ it acts on 11 cells and is right on 10 of them, so it finds every singleton the release contains, with a precision of 0.909. The contrast follows the curves of Figure 10.1. The adversary's prior that a cell is a singleton is small, about 10 in 814, and at $\varepsilon = 1$ a reading of the count cannot move it to one half. At $\varepsilon = 5$ the noise has scale 0.2, a reading near one leaves little doubt, and the prior no longer protects.

Over six monthly vintages the programme spends six times the monthly parameter under basic composition, 6 at $\varepsilon = 1$ and 30 at $\varepsilon = 5$. The pooled adversary, which multiplies the likelihoods of all six readings, acts on 7 cells at $\varepsilon = 1$ and is right on 4. The accounting

predicted that the programme would leak more than one month, and it did.

One result in the run cuts the other way and is reported because it is instructive. At $\varepsilon = 5$ the pooled adversary does worse than the single-month one: it acts on 6 cells and is right on 4, against 10 of 11 from one month. The reason is in the attack, not the mechanism. The pooled adversary assumes that a cell's count is the same in every vintage, while in the register loans leave and arrive, so its likelihood is wrong and averaging over six readings blurs a count that one reading had nearly pinned. A scenario assessment measures the attack it was given. The ceiling of Proposition 10.2 holds for the better attack nobody ran, which is why the two figures must not be read against each other on one scale.

The cost side is just as plain. On the utility scale used for the credit products, where the unprotected loan-level file scores 0 and the 5-anonymous loan-level file scores -0.050, the noisy counts score -0.40 at $\varepsilon = 1$ and -0.36 at $\varepsilon = 5$. A product with no microdata cannot serve the lender-level and concentration analyses the register is collected for, whatever its noise. The comparison is not like for like, since the targets of the counts attack are cells and those of the loan-level attacks are loans, and Chapter 13 keeps them out of its comparison of policies for that reason.

The chair read the results with a question of her own. If the monthly counts at $\varepsilon = 1$ defeat the declared attack, why could the Bank not simply say so and stop? Naledi's answer was that the declared attack was one reader. The guarantee covered all readers, but only through the baseline, and the Bank had not yet written down what a lender that knows its own borrowers could achieve without the counts. That was the next piece of work.

10.6 Accounting across a programme

Chapter 9 proved that no function of per-release risks can bound the risk of an adversary who pools the releases (Proposition 9.2). Formal privacy is the one framework in which the opposite holds: guarantees add up across releases with no assumption about how the releases depend on one another.

Proposition 10.3 (Basic composition)**.** *Let mechanisms $M_1, \dots, M_m$ act on one database, where M_i may be chosen in the light of the outputs of $M_1, \dots, M_{i-1}$, and suppose each M_i is $(\varepsilon_i, \delta_i)$-differentially private under the same neighbouring relation whatever those earlier outputs were. Then the joint output is $(\sum_i \varepsilon_i, \sum_i \delta_i)$-differentially private. Combined with Proposition 10.2, any attack that uses all m outputs succeeds with probability at most*

$$R_{\text{pool}} \le \min\Big(1,\; e^{\sum_i \varepsilon_i}\, \kappa_{\text{pool}} + \sum_i \delta_i\Big), \tag{10.6}$$

where κ_{pool} is the baseline of the pooled attack.

Inequality (10.6) is the composition theorem that scenario assessment lacks, and its price is visible at once. The exponent adds, so the multiplier grows geometrically. Six months at $\varepsilon = 1$ give a multiplier of e^6, about 403, and the Bank's tolerance of 0.12 is then met only if the pooled baseline is below about 0.0003. The ceiling is valid, and for most practical baselines it is vacuous. That is not a defect of the theorem. The basic rule is loose, and much of the later work on formal privacy has gone into tightening it.

Advanced composition (Dwork et al., 2010) shows that m mechanisms each with parameter ε compose, at the cost of a small extra δ, to a total that grows like $\sqrt{m}\,\varepsilon$ for small ε, not like $m\varepsilon$. Kairouz et al. (2017) give the exact composition of (ε, δ) guarantees, read as limits on

tests. Rényi differential privacy (Mironov, 2017) and concentrated differential privacy (Bun and Steinke, 2016) track a whole curve of divergences between the output distributions on neighbours, which adds exactly across mechanisms and converts to an (ε, δ) statement only at the end; Gaussian differential privacy (Dong et al., 2022) does the same with a single parameter for mechanisms that add Gaussian noise. These accountants are what a programme should run. They are not different definitions of privacy, and each converts back into (10.1), so Proposition 10.2 applies to whatever they report.

Two further devices address a problem that is particular to programmes. The parameters of later releases are often chosen after earlier ones are out: a user asks for one more table, a department wants a finer breakdown. A privacy filter is a rule that, given a total budget fixed in advance, allows each new release only if the running account stays inside the budget, and halts the programme otherwise; the composition guarantee then holds for the total even though the parameters were chosen as the programme ran. A privacy odometer goes one step further and reports, at every point, a running bound on the loss so far that is valid whenever it is read. The two play the role for privacy accounting that the anytime-valid bounds of Chapter 6 play for attack simulation: they keep a guarantee true when the analyst looks at the account and decides what to do next. A committee that lets parameters be set release by release needs one of them in its record, because basic or advanced composition stated for fixed parameters does not cover choices made in the light of earlier outputs.

The contrast with Chapter 9 can now be stated without overstatement. For scenario assessment the per-release figures say nothing about the pooled attack, and the institution must simulate the joint attack directly. For formal privacy the per-release parameters bound the joint guarantee, and the institution must still argue for a pooled baseline before the guarantee becomes a risk figure. In both regimes the programme needs a horizon. Without one, the pooled risk in the first regime and the accumulated budget in the second have no fixed meaning.

Tomás's request made this concrete for the Bank. He asked for the monthly counts for the full six months, and also for a finer table by lender. Under the Bank's current practice the second table would have been assessed on its own. Under a formal accountant it had to come out of the same budget as the first, and Naledi's note to the chair said the Bank could release the second table only by spending budget that the first had already used.

10.7 What the guarantee does not name

A differential privacy guarantee is precise about one thing, the effect of one record, and silent about several others that a release committee must still govern.

The first is the unit. The guarantee protects whatever the neighbouring relation changes. If the mechanism treats a loan as the record, a firm with several loans is protected only as a group, and the group guarantee degrades with its size.

Proposition 10.4 (Groups)**.** *If M is ε-differentially private, then for databases x and x'' that differ in the records of g individuals, $\mathbb{P}\{M(x) \in E\} \leq e^{g\varepsilon}\, \mathbb{P}\{M(x'') \in E\}$ for every measurable E.*

In the synthetic credit register a borrower holds between one and four lending relationships. A loan-level guarantee at $\varepsilon = 1$ is therefore, for the largest borrowers, a borrower-level guarantee at $\varepsilon = 4$, and by Figure 10.1 a prior of 0.1 can then rise much further. The largest borrowers are also the ones most easily singled out by other means, so the weakest guarantee falls on the most exposed firms. A committee should state the unit of protection in its

record and choose it to match the party the law protects, which for a credit register is usually the borrower.

The second is inference from the population. Differential privacy bounds what the release reveals about a person beyond what it would reveal if their record were absent. If the release shows that firms in a sector are mostly in arrears, a reader can infer that a given firm in that sector is probably in arrears, whether or not its loan was in the data (Dwork and Roth, 2014). That inference is the purpose of statistics, and no mechanism that keeps the release useful can prevent it. It is also a form of attribute disclosure that a committee may care about for small groups, and the guarantee does not address it.

The third is dependence between records. Proposition 10.1 assumes that the other records have the same distribution whether or not H holds. In a register of linked firms, or of a borrower's several loans, removing one record does not remove the information the others carry about it, and the posterior reading can fail even though (10.1) holds (Kifer and Machanavajjhala, 2011).

The fourth is the legal notion of identification. Cohen and Nissim (2020) study how the data protection notion of singling out can be stated formally and how differential privacy relates to it. Their analysis is a reminder that a guarantee about output distributions has to be connected, by an argument, to the words a regulator uses, and that the connection is not automatic.

10.8 Hazards in implementation

A guarantee is a property of a mechanism as specified. A deployed mechanism can fail to have it for reasons that have nothing to do with the mathematics, and a committee that relies on the guarantee should ask about each of them.

The sensitivity may be stated for the wrong relation or the wrong unit. The noise scale $1/\varepsilon$ used for counts gives ε under add-or-remove for a single loan. Under replace-one, a loan that moves from one cell to another changes two counts, and the same noise gives 2ε; for a borrower with four loans the factor is four. Each of these is a correct guarantee for some statement, and a record that names the parameter without the relation and the unit has not said which.

The noise may not be the noise in the proof. A Laplace or Gaussian draw computed in floating-point arithmetic is not exactly the continuous distribution the proof assumes, and naive sampling code can leave patterns in the low-order digits of the output that reveal the value underneath. Implementations designed for formal privacy sample from discrete distributions with exact arithmetic for this reason. Rounding or clamping after noise is added is safe, since anything computed from the output alone keeps the guarantee; clamping the data before the noise, to bound sensitivity, must be part of the stated mechanism.

The budget may be spent where nobody counts it. Choosing ε, the query, the cells or the release date after looking at the confidential data is itself a use of the data, and so is a data-dependent decision to suppress or republish. Releasing a noisy figure, discarding it and drawing again because the first looked odd spends the budget twice. Reusing one noise draw across months, in the other direction, breaks the independence that the per-month accounting assumes; the pipeline's monthly counts draw fresh noise each vintage for this reason.

The code may have errors that no review catches. Empirical auditing turns the ceiling of Proposition 10.2 around: an attack whose success exceeds $e^{\varepsilon}\kappa + \delta$ for a measurable base-

line is evidence that the mechanism does not have the claimed parameter. Jagielski et al. (2020) use attacks of this kind to put lower bounds on the privacy loss of private training of machine learning models, and Steinke et al. (2023) show how to do so with a single run of the training procedure. The same logic applies to a noisy-count release, and the attack simulation machinery of Chapter 4 is what it needs. A falsification test of this kind belongs among those of Part VI.

Finally, some products carry the vocabulary without the guarantee. Synthetic data generated by a model fitted without a privacy mechanism are not protected by any parameter, and Stadler et al. (2022) show that such data can leak the records they were trained on. A release described as private should state a mechanism, a relation, a unit and an accountant, and if it cannot state all four it should be assessed as a scenario release.

10.9 Two regimes in one record

The chapter asked what a privacy-loss parameter commits an institution to, in the units a release decision uses. The answer is now available.

Adopting $\varepsilon = 1$ commits the institution to a release whose output changes by at most a factor of e when one record changes. In the units of a release decision this is a commitment of two kinds. Against beliefs, it multiplies the odds an adversary already holds by at most e: a prior of 0.1 can become 0.232, and a prior of one third can become 0.576. Against attacks, it caps the success of every attack at $e^{\varepsilon}\kappa + \delta$, where κ is what the attack could achieve without the target's own record. Neither statement is a low re-identification risk until someone argues that the adversary's prior, or the baseline, is small. That argument is a declared scenario, and the institution must write it down in exactly the form earlier chapters require.

So the two regimes answer different questions and both belong in the record. Scenario assessment is adversary-specific: it says what a named adversary achieves against this release, the question a governance committee, a data protection authority and an affected person all ask. Formal privacy is adversary-agnostic: it says how far any observer's beliefs can move because one record was included, and it is the only device that composes across a programme without an assumption about how the releases relate. An institution that wants both should hold both, running a formal accountant for the programme budget and a scenario assessment for each release decision, reporting them separately with their assumptions. They meet at (10.5), and for a differentially private release the scenario assessment is then needed only to argue for κ.

Grace read the decision record for the monthly counts some months later. It named ε, the add-or-remove relation, the loan as the unit, the accountant and the six-month horizon. Beside them it named the adversary the Bank had in mind, the baseline it had argued for that adversary, and the ceiling that followed. She had no question to ask about the parameter. Her one note concerned the unit, and asked why the record protected loans when the law protected borrowers.

IN PRACTICE: A RELEASE THAT CARRIES A PRIVACY PARAMETER

Record four things beside ε and δ: the neighbouring relation, the unit of protection, the accountant used across the programme, and the horizon over which the budget is spent. Then translate the parameter into the Bank's own units. For each declared adversary, state its prior or its baseline success κ without the target's record, and report the posterior bound (10.2) or the ceiling $\min(1, e^{\varepsilon}\kappa + \delta)$ beside the tolerance. If the ceiling exceeds the tolerance, the guarantee does

not support the release, however small ε looks. If parameters may be chosen as the programme runs, use a filter or an odometer, and count every look at the confidential data, including decisions to redraw or withdraw, against the budget.

In the software. The posterior bound (10.2) and basic composition are in `privacy_risk.dp` (`posterior_bound`, `posterior_bound_table` and `basic_composition`); the table quoted in this chapter is produced by `experiments.formal_privacy`. The noisy-count release and its single-month and pooled attacks are in `privacy_risk.evaluation` (`counts_attack`). The package does not yet compute the ceiling (10.5) or any accountant tighter than the basic one; both are short additions. The bound itself is a few lines:

```
def posterior_bound(epsilon: float, prior: float) -> float:
    """Largest posterior an (epsilon, 0)-differentially private release permits.

    With delta > 0 no uniform posterior bound exists, so this function covers the
    pure case only.
    """
    if epsilon < 0:
        raise ValueError("epsilon must be nonnegative")
    if not 0 <= prior <= 1:
        raise ValueError("the prior must be a probability")
    factor = math.exp(epsilon)
    return float(factor * prior / (factor * prior + (1 - prior)))
```

10.10 Proofs

Proof of Proposition 10.1. Write W for the records other than the individual's, with the same law under H and under its negation, and let r and r' be the individual's record under H and under its negation (each may itself be random, independent of W). For each value w of the other records, the databases (w, r) and (w, r') are neighbours under the replace-one relation, so by (10.1) with $\delta = 0$, $\mathbb{P}\{M(w,r) \in E\} \le e^{\varepsilon}\mathbb{P}\{M(w,r') \in E\}$. Integrating both sides over the common law of W and over the laws of r and r' gives $\mathbb{P}(M \in E \mid H) \le e^{\varepsilon}\mathbb{P}(M \in E \mid \neg H)$. Bayes' rule in odds form then gives

$$\frac{\mathbb{P}(H \mid M \in E)}{1 - \mathbb{P}(H \mid M \in E)} = \frac{p}{1-p} \cdot \frac{\mathbb{P}(M \in E \mid H)}{\mathbb{P}(M \in E \mid \neg H)} \le \frac{p}{1-p}\, e^{\varepsilon},$$

and since $u \mapsto u/(1-u)$ is increasing on $[0,1)$, solving for the posterior gives (10.2). If $\mathbb{P}(M \in E \mid \neg H) = 0$ then $\mathbb{P}(M \in E \mid H) = 0$ as well and E has probability zero, which is excluded.

For $\delta > 0$ the same steps give

$$\frac{\mathbb{P}(H \mid M \in E)}{1 - \mathbb{P}(H \mid M \in E)} \le \frac{p}{1-p}\Big(e^{\varepsilon} + \frac{\delta}{\mathbb{P}(M \in E \mid \neg H)}\Big),$$

and the last term is unbounded as $\mathbb{P}(M \in E \mid \neg H) \to 0$. To see that no uniform bound survives, take a mechanism that with probability δ publishes the individual's record and otherwise publishes nothing. It satisfies (10.1) with $\varepsilon = 0$ and this δ, and on the output set in which the record is published under H the posterior on H is one whatever the prior. □

Proof of Proposition 10.2. Fix a value z of the auxiliary information and a value of the database apart from the target's position. The event $E_z = \{y : g(y,z) = t\}$ is a measurable

set of outputs, and the databases x and $x^{(t)}$ differ only in the record at the target's position, so they are neighbours. By (10.1),

$$\mathbb{P}\{M(x) \in E_z\} \leq e^{\varepsilon}\mathbb{P}\{M(x^{(t)}) \in E_z\} + \delta.$$

Because Z_t is independent of the mechanism's randomness, integrating both sides over the law of Z_t, over the other records and over the reference law of the replacement record leaves the inequality intact and gives $R \leq e^{\varepsilon}\kappa + \delta$ by (10.3) and (10.4). Since R is a probability, $R \leq 1$, which gives (10.5). If $e^{\varepsilon}\kappa^* + \delta \leq \tau$ and every admitted attack has baseline at most κ^*, every admitted attack has $R \leq \tau$. □

Proof of Proposition 10.3. We prove the case $\delta_i = 0$ for all i with outputs in countable sets, which carries the idea; the general case, including $\delta_i > 0$ and adaptive choice, is in Dwork and Roth (2014, ch. 3). Let $x \sim x'$ and let $y = (y_1, \ldots, y_m)$ be a joint output. By the chain rule,

$$\mathbb{P}\{(M_1, \ldots, M_m)(x) = y\} = \prod_{i=1}^{m} \mathbb{P}\{M_i(x) = y_i \mid y_1, \ldots, y_{i-1}\},$$

where the ith factor is the probability that the mechanism chosen after $y_1, \ldots, y_{i-1}$ outputs y_i on x. Each factor is at most e^{ε_i} times the corresponding factor for x', by the hypothesis on M_i applied to the singleton $\{y_i\}$. Multiplying, the joint probability under x is at most $e^{\sum_i \varepsilon_i}$ times that under x', and summing over the outputs in any set E gives (10.1) for the joint mechanism with $\varepsilon = \sum_i \varepsilon_i$ and $\delta = 0$. The bound (10.6) follows by applying Proposition 10.2 to the joint mechanism, with the pooled attack as the rule g. □

Proof of Proposition 10.4. Let $x = x_0, x_1, \ldots, x_g = x''$ be a chain of databases in which each consecutive pair differs in the record of one of the g individuals, so that consecutive members are neighbours. Applying (10.1) with $\delta = 0$ along the chain gives $\mathbb{P}\{M(x_0) \in E\} \leq e^{\varepsilon}\mathbb{P}\{M(x_1) \in E\} \leq \cdots \leq e^{g\varepsilon}\mathbb{P}\{M(x_g) \in E\}$. □

Part V

Evidence from data

Chapter 11

A simulation study

A careful reader may arrive at this chapter with a reasonable doubt. The earlier chapters derived their results as propositions, with assumptions and proofs. A simulation, on this view, can only confirm them. If the reciprocal of the key frequency is the right risk under four assumptions, a simulation built on those assumptions returns the reciprocal of the key frequency, and a simulation built on other assumptions returns whatever those assumptions imply. Either way the formula already knew the answer, and the computer time was spent on arithmetic. In its strongest form the doubt adds that simulated populations are chosen by the analyst, so a simulation that disagrees with a formula has shown only that the analyst chose a population on which it disagrees.

The first fact that breaks the doubt comes from a release that satisfies the most widely used rule in the field. On a synthetic population of 200,000 units with a 5 per cent sample, a release that is 10-anonymous on its declared key, so that no released key class has fewer than ten members and the formula puts every record's risk at or below 0.1, contains a record that the declared adversary identifies with probability 0.744. Its one-sided 95 per cent lower bound, corrected for having picked the worst of 230 records, is 0.638. No formula in the earlier chapters produced that number. Proposition 2.3 said that the implication from k-anonymity to a risk of $1/k$ can fail once the release publishes a continuous variable the adversary also reads. It did not say whether the failure is a curiosity at the edge of the parameter space or a property of an ordinary release. Only an attack run against a population whose truth is known can say which.

This chapter asks one question. What does an attack simulated on a known population show that the formulas cannot? The answer comes in five pieces: the sizes of gaps the formulas only sign, the direction of effects the formulas leave open, the behaviour of the worst record where formulas speak of averages, the calibration of the adversary's own model, and the size of the programme effect of Chapter 9 at registers of different sizes.

11.1 A file for a visiting researcher

Tomás had asked the Bank for a research file of firm-level records: district, size, age of the firm in years, sector, and a continuous measure of the firm's borrowing. Naledi proposed to release it with a minimum class size of ten on the four categorical fields. The chair asked why the assessment plan included a simulated attack at all. A class size of ten meant a risk of at most one in ten, she said, and that was the Bank's working tolerance for research files. Naledi agreed that it meant one in ten under the formula. Her question was whether the

formula's assumptions held for a file that also published borrowing, which a lender such as Pieter's would know, roughly, for its own customers.

The study in this chapter is the one Naledi would need to answer her own question, built on a population where the answer is known. It is synthetic, its parameters are chosen, and every result in it describes that population. Its value is that every estimate can be compared with a truth, which is impossible on a real file (Chapter 12).

11.2 Design

The population has $N = 200{,}000$ units. Each carries a district (40 categories with a skewed size distribution), sex, age in single years from 18 to 87, and an occupation group (12 categories, also skewed). District and occupation are drawn with probabilities proportional to $j^{-0.9}$ and $j^{-0.8}$, and ages are thinned towards younger values so that the oldest bands are sparse. The adversary's key is the cross-classification of district, sex, five-year age band and occupation group. It has 13,440 possible cells, of which 12,596 are occupied, and 1,305 cells are population unique, so 0.65 per cent of the population would be identifiable from the key alone in a census.

Each unit also carries a confidential continuous variable,

$$y_i = 8 + \psi_{d(i)} + \omega_{o(i)} + 0.018\,(\mathrm{age}_i - 50) + 0.10\,\mathrm{sex}_i + \epsilon_i, \qquad \epsilon_i \sim N(0, 0.6^2), \tag{11.1}$$

with district effects ψ and occupation effects ω drawn once from $N(0, 0.45^2)$ and $N(0, 0.35^2)$. The variable plays the part of Tomás's borrowing measure, or of an income in a household survey. A Bernoulli sample with inclusion probability $\pi = 0.05$ gives $n = 10{,}035$ records with 4,679 distinct keys, of which 2,729 are unique, so 27.2 per cent of released records are sample unique.

The declared adversary observes district and sex without error. It reads the released age band with probability 0.15 of a one-band error and the occupation group with probability 0.10 of a wrong group, and it holds a noisy reading $z_t = y_t + N(0, 0.75^2)$ of the confidential variable. It links by maximum posterior, in the sense of (2.6) in Chapter 2, under the same error model that generates its errors, with a flat prior over the records that share the target's district and sex. A second adversary, identical except that its reading of y is twice as precise, forms the uncertainty set for the worst-case figures. All results come from one program with one seed, so that every number in the chapter can be regenerated.

One lever was built and then removed. Swapping districts between records reduced the simulated risk, but only against an adversary that takes the released district at face value. The declared scenario gives no reason to assume that, so the apparent protection was a property of the assumed adversary and not of the release, and the study dropped the lever instead of reporting a figure that depended on it.

11.3 The measures side by side

Chapter 3 showed that measures reported under one label answer different questions. The simulation puts sizes on the differences. Figure 11.1 shows the conventional summaries for the unprotected release on a logarithmic scale.

The mean prosecutor risk, the average of $1/f_k$ over released records, is 0.4663. It equals, to every digit, the marketer measure D_I/n, the number of distinct released keys over the

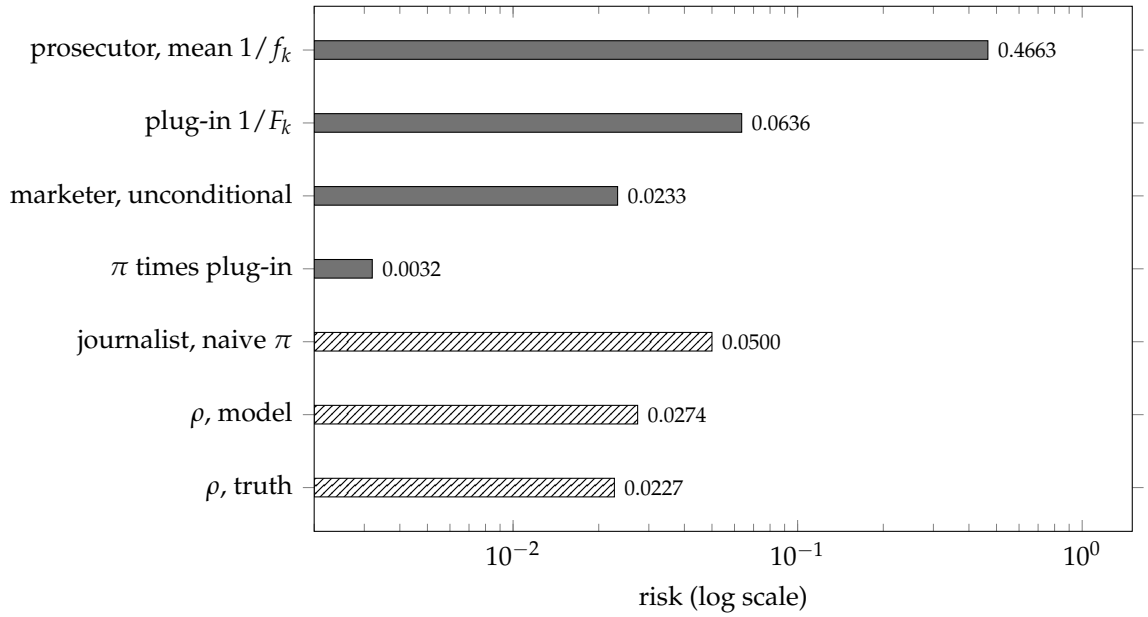


FIGURE 11.1: Conventional risk summaries for the unprotected synthetic release ($N = 200{,}000$, $\pi = 0.05$, $n = 10{,}035$). Solid grey bars are record-linkage measures; hatched bars are journalist measures of population uniqueness. The bars span more than two orders of magnitude, and several have been published under the same label.

number of records, which is Proposition 11.1 below and the marketer identity of Chapter 3. Replacing sample frequencies with population frequencies gives 0.0636, and applying the inclusion probability, as the factorisation (2.3) of Chapter 2 requires when membership is unknown, gives 0.0233. The first is 7.3 times the second and 20 times the third. Multiplying the population-frequency measure by π as well gives 0.0032, which answers no experiment at all and has been reported anyway.

The formulas of Chapter 3 order these measures. They do not say by how much they differ on a file of a given shape, and on this file the spread is large enough that the label on a risk figure decides whether a committee sees a coin toss or a one in forty chance. That is the first thing the simulation adds: the formulas sign the gaps, and the simulation sizes them.

11.4 The uniqueness probability

The journalist's question is whether a record unique in the sample is unique in the population, $\rho = \mathbb{P}(F_k = 1 \mid f_k = 1)$. It is the one measure that needs a model for something no release shows, the population frequencies. The pipeline fits a negative binomial law to the population cell sizes from the sample counts, using the property that binomial thinning leaves a negative binomial law negative binomial with the same dispersion (Bethlehem et al., 1990; Rinott and Shlomo, 2006). On the synthetic population the fit recovers a mean population cell size of 14.83 against a truth of 14.88.

Figure 11.2 compares the model-based ρ with the truth across five sampling fractions. The estimate is conservative at every fraction, with a relative error between 8 and 25 per cent. At $\pi = 0.05$ the truth is 0.0227 and the model gives 0.0274. The sampling fraction itself, which is sometimes used in place of ρ, overstates the truth at every fraction, by a factor of 2.20 at $\pi = 0.05$ and 1.64 at $\pi = 0.5$.

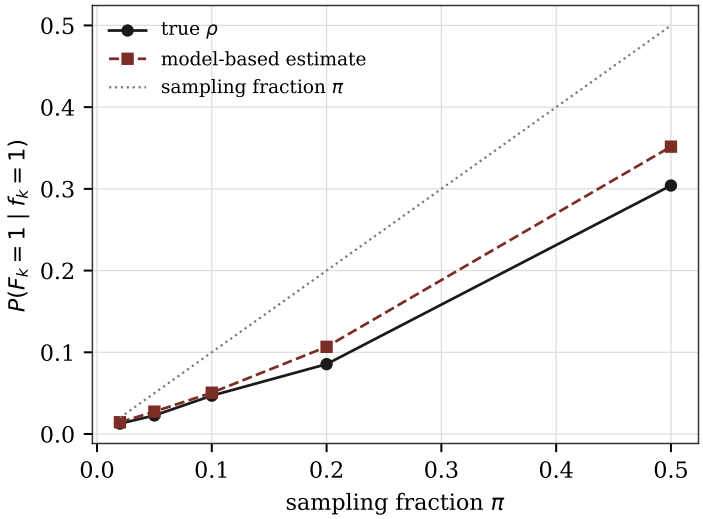


FIGURE 11.2: The uniqueness probability $\rho = \mathbb{P}(F_k = 1 \mid f_k = 1)$ against the sampling fraction on the synthetic population: the truth (solid line, circles), the model-based estimate from a fitted negative binomial law (dashed line, squares), and the sampling fraction (dotted diagonal), which is not an approximation to ρ.

Here the simulation shows something a formula could not, because the model's own check fails. If the negative binomial law were right, the fitted dispersion would be the same at every sampling fraction. It rises from 0.33 at $\pi = 0.02$ to 0.51 at $\pi = 0.2$ and 0.59 at $\pi = 0.5$. The drift is evidence of misfit, and the estimate of ρ was conservative here by good fortune, not by design. On a real file, where ρ has no truth to be compared with, the drift is the only warning available, which is why Part VI includes a test that fits the law at several subsample fractions and rejects it when the dispersion moves.

Naledi kept this result in mind for the research file. Tomás's file was a sample, the Bank knew its sampling fraction, and she could compute ρ. What she could not do on the Bank's own data was check it. The simulation told her how far off a fitted law could be on a population of this shape and which symptom to look for.

11.5 Measurement error does not simply reduce risk

Practitioners usually expect errors in the adversary's information to reduce its success, and the expectation is half right. Under exact exchangeable matching, the predicted success for a uniformly drawn released target is the mean prosecutor risk, 0.4663. Simulating the declared adversary with its reporting errors on age band and occupation, but without the covariate, gives 0.3812 over 5,000 replicates. That is the reduction practitioners expect.

Adding the single noisy reading of y returns the success rate to 0.4642 over 20,000 replicates, with a one-sided 95 per cent upper bound of 0.4701. The reciprocal-frequency figure is therefore close to the truth for this release, but by cancellation between an error that spreads the posterior across candidates and a covariate that concentrates it on one. An assessment that modelled only the reporting error would have understated the risk by 18 per cent.

The formulas of Chapter 2 said that both effects exist and that their net direction is open. They could not say that on an ordinary file the two nearly cancel, so that a practitioner who uses the reciprocal frequency is right by accident, and one who corrects it for reporting error

alone is wrong by a fifth. That is the second thing the simulation adds: the direction and size of effects that the formulas leave open.

The same run tests the adversary's posterior itself. Figure 11.3 groups the targets by the maximum posterior the adversary computed and plots, for each group, the share of its guesses that were right. When the adversary's model matches the process that generates its information, the mean gap between the binned posterior and the empirical success rate is 0.007. When the adversary assumes a covariate 2.5 times noisier than the truth, the gap rises to 0.049, and the error is one-sided: the posterior understates the success rate in the lower bins. The overall success rate barely moves, from 0.4642 to 0.4571.

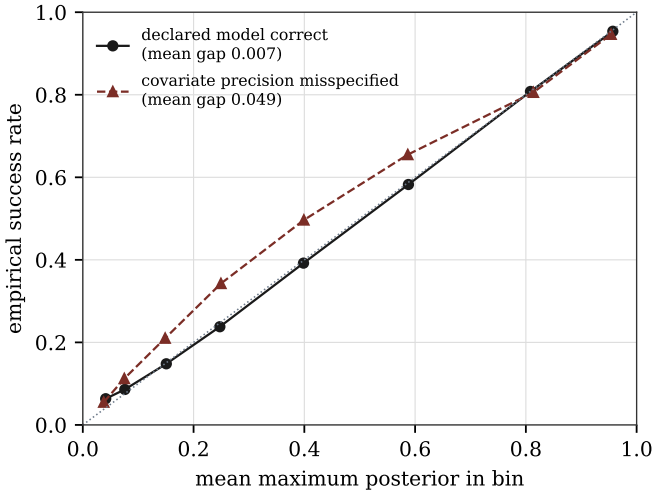


FIGURE 11.3: Calibration of the adversary's maximum posterior against the empirical success rate, 20,000 replicates per curve: the declared model (solid line, circles) and a model that misstates the covariate's precision (dashed line, triangles). The two models give overall success rates that differ by 0.007, while their calibration gaps differ by a factor of seven.

The practical point is that a single risk number need not reveal that the linkage model is wrong. Two models that disagree about the adversary's precision give the same headline rate to two decimal places. A calibration check on matches whose truth is known does reveal it, and the check needs exactly what a simulation on a known population supplies. An institution cannot run it on its real file, but it can run it on held-out records from its own data when it plants targets whose links it knows, which is the first falsification test of Part VI.

11.6 k-anonymity does not deliver $1/k$

The chapter's opening fact comes from applying minimum class sizes of 3, 5, 10 and 20 to the declared key, suppressing records in smaller classes, and attacking what remains. Two controls are measured, because they answer different questions. The first is the average risk of a target drawn uniformly from the released records, from 12,000 replicates. The second is the risk of the worst target among those in the smallest released classes, each estimated from 250 replicates of its own and reported with a lower bound that carries a correction for the selection. The decision rule of Chapter 1 is stated for every target in a declared set, so the second control is the one the rule needs.

Figure 11.4 shows both against the nominal bound $1/k$. The average stays below $1/k$ at $k = 3$ and $k = 5$, at 0.2573 and 0.1852, and exceeds it at $k = 10$ and $k = 20$, at 0.1129 and 0.0787. The worst target exceeds $1/k$ at every level, and so does its corrected lower bound, which is 0.82, 0.78, 0.64 and 0.68 as k runs from 3 to 20. Between 27 and 75 per cent of the targets in minimum-size classes have an individual lower bound above $1/k$.

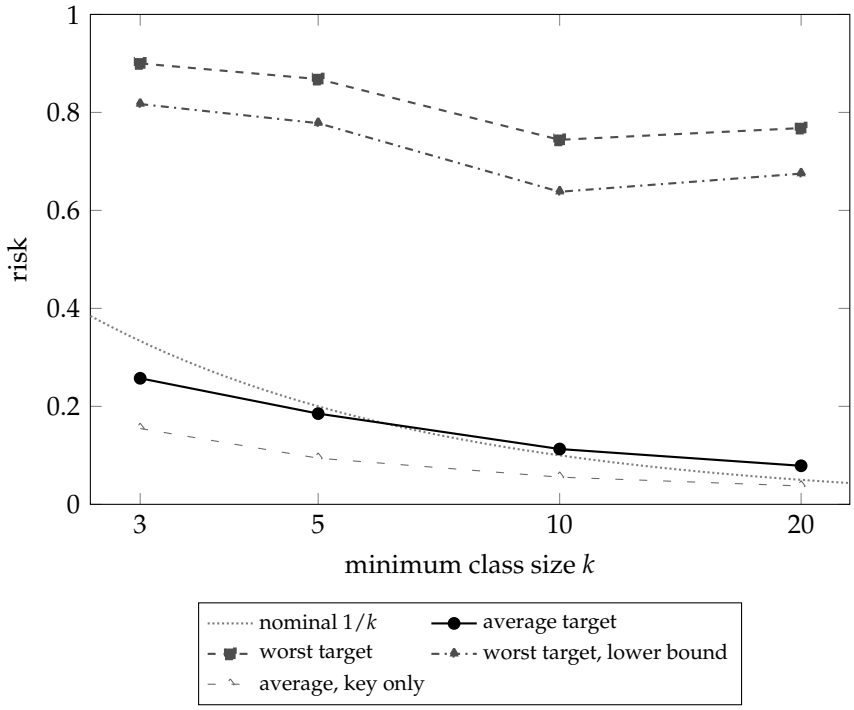


FIGURE 11.4: Minimum class size against the simulated risk of the declared adversary on the synthetic population. The worst target and its lower bound, corrected for selection, stay far above the nominal bound $1/k$ at every level; the average crosses it between $k = 5$ and $k = 10$. Without the reading of the continuous variable (key only) the average stays below $1/k$. Series differ by line style and marker as well as shade.

The lowest curve in the figure identifies the cause. The same adversary without its reading of y succeeds on average with probability 0.0555 at $k = 10$, below the nominal 0.1, and stays below $1/k$ at every level. The failure is a failure of key selection, not of the criterion. The covariate separates members of a key class, and raising k on a key that leaves it out shrinks the share of the adversary's information that the criterion controls while leaving the covariate untouched. One reading of why the average overtakes $1/k$ at the larger levels is that the nominal bound falls in proportion to $1/k$ while the covariate's power to separate the members of a class falls more slowly; the study did not test this.

Proposition 2.3 predicted that this could happen. The simulation shows that it happens at every level of k on a population with nothing contrived about it, and that the size of the failure is not small: the worst record in a 20-anonymous release is identified about three times in four. That is the third thing the simulation adds. Formulas about averages are silent about the worst record, and the worst record is what a rule stated as a supremum must control.

The cost is as plain as the failure. Figure 11.5 shows the distribution of the formula's own risk $1/f_k$ across released records under each minimum class size. Reaching $k = 10$ keeps 15.5 per cent of the released records, and reaching $k = 20$ keeps 4.8 per cent. A release

governed by its maximum prosecutor risk alone is pushed towards discarding most of its content, and against this adversary it still fails.

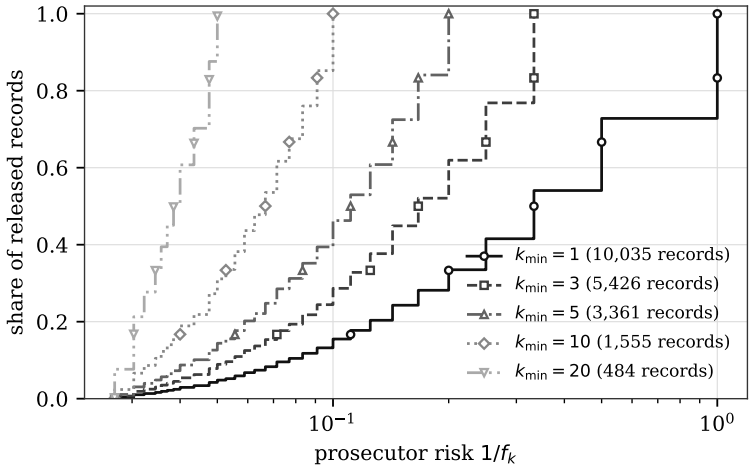


FIGURE 11.5: Cumulative distribution of the prosecutor risk $1/f_k$ across released records under minimum class sizes from 1 to 20, with the number of records each keeps, as labelled in the legend. Governing the maximum of this measure alone forces most of the file out of the release.

The chair saw the figure at the next committee meeting and asked the question Naledi had expected. If ten was not enough, what number was? Naledi's answer was that no number on that key would do, because the variable doing the damage was not in the key. The choices were to coarsen or perturb the borrowing measure, to leave it out of the file, or to give Tomás access in the Bank's secure room, where the question of linkage by an outsider did not arise in the same form.

11.7 A programme of four releases

Chapter 9 used this population for its measurement of the pooling gap, and the result needs only a short restatement here. Four releases from the same sample, each keyed on ordinary demographic fields and each 3-anonymous, have marginal risks between 0.0168 and 0.0884, so the marginals bound the chance that any one release succeeds between 0.0884 and 0.2530. An adversary that intersects the candidate sets of all four, through a record identifier the releases share, succeeds with probability 0.4581, and 26.0 per cent of targets are identified uniquely by the intersection although none was unique in any single release (Figure 9.2).

What the simulation adds to the proof is a test of whether the gap is an artefact of the size of register chosen. Figure 11.6 rebuilds the same four releases, with the same keys and the same minimum class size, on populations of 50,000, 100,000, 200,000 and 400,000 units with the sampling fraction held at 5 per cent.

Two things move in opposite directions. Release by release, larger registers are safer: the largest marginal risk falls from 0.113 to 0.062, because key classes grow with the population. The pooled attack also weakens, but more slowly, so the ratio of the pooled risk to the bound from the marginals rises from 1.57 at 50,000 units to 2.05 at 400,000. Size does not dissolve the problem. It makes the marginals a worse guide to the programme, the opposite of the reassurance a large register is usually taken to give. One reading, which the study has not tested, is that growth adds records to every key class, which protects each release, while the

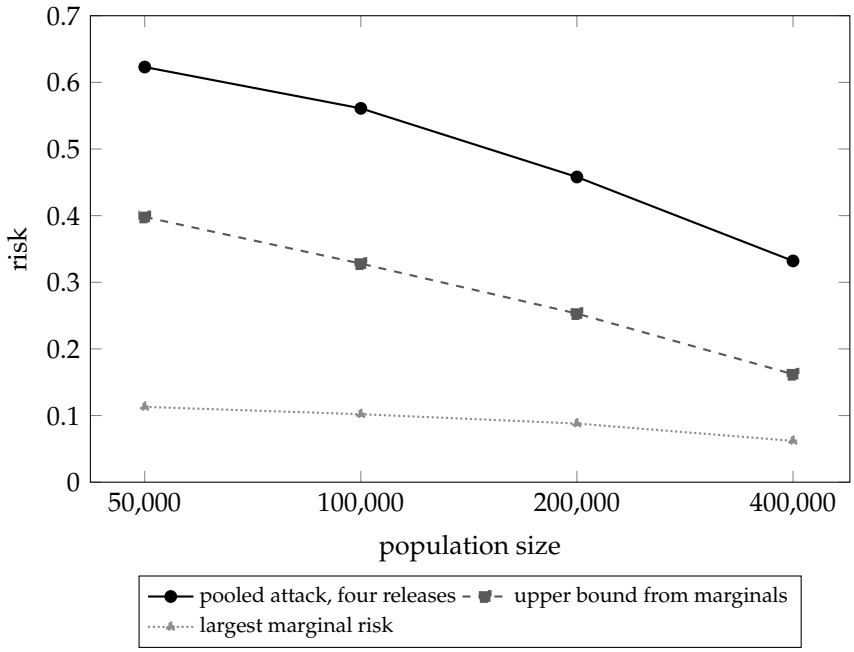


FIGURE 11.6: The pooling gap at four register sizes, synthetic population, sampling fraction 0.05, one draw per size. Each release becomes safer as the register grows, and the pooled attack weakens more slowly than the bound the marginals imply, so the ratio of the two rises from 1.57 to 2.05. Series differ by line style and marker.

intersection of four candidate sets can still close on one record.

Each size rests on one draw. At 50,000 units the Bernoulli sample gave 2,362 records against an expected 2,500, about 2.8 standard deviations low, and that row should be read with this in mind. A register of several million would need its own run; the pipeline supports it by changing one number in the configuration. That is the fourth and fifth thing the simulation adds: it sizes the programme effect, and it shows how the size moves with a design choice the proof does not contain.

11.8 What the simulation says about its own uncertainty

A simulation is itself an estimate, and Chapter 4 separates two sources of its error: the Monte Carlo error from a finite number of simulated attacks, which the binomial interval describes, and the variance carried by the sample and the fitted inputs, which it does not. The synthetic population allows the separation to be measured. In a nested design, 120 outer replicates redraw the 5 per cent sample from the same population, apply the same mechanism, and run 500 attacks each. The within-replicate standard deviation is 0.0112. The between-replicate standard deviation comes out at 0.0000 after correction, and the standard error of the mean is 0.0010 whether the design term is included or not. The interval for the share of variance the design contributes runs from 0 to 0.19.

The binomial interval is adequate on this design, then, and the data could not have told a design contribution of up to a fifth from none. Three features keep the term small: the population is fixed and large, the sampling fraction is modest, and the adversary's model is not refitted between replicates. None of the three is guaranteed in practice. The simulation also measured how much worse the more capable adversary does. Across the 45 settings

of the protection grid of Chapter 8, the adversary with the twice-as-precise reading always attains the larger risk, from 1 to 76 per cent above the declared one, with a median gap of 40 per cent. A risk figure reported for one adversary therefore carries an uncertainty about the adversary that is far larger than its Monte Carlo error, and the record must say which adversary it assumed.

The repository holds one further synthetic benchmark, for the multiscale extension of the model that proposes protection settings in Chapter 14. It uses its own seed and a deliberately nonlinear synthetic law with 400 training, 150 validation and 150 test rows, and reports held-out counted log losses of 0.2452 for a linear model, 0.2252 for a deep model on raw inputs and 0.2247 for a deep model on multiscale features. These are one-seed results on a law built to reward nonlinearity. They do not test a release, they do not establish an advantage on real data, and they belong with the proposal models of Chapter 14, which reports them in that context.

11.9 What the simulation shows

The chapter asked what an attack simulated on a known population shows that the formulas cannot. The answer is that the formulas are statements of sign and possibility, and the simulation turns them into statements of size on a population of a stated shape. The formulas said that conventional measures are ordered; the simulation found them spread across two orders of magnitude on one file. The formulas said that measurement error could move risk either way; the simulation found that on an ordinary file a reporting error and a single covariate nearly cancel, so that the familiar formula is right by accident and its obvious correction is wrong by a fifth. The formulas said that k-anonymity can fail once the adversary reads a published continuous variable; the simulation found that it fails at every level tried, with the worst record's lower bound more than six times $1/k$ at $k = 10$ and $k = 20$, and that the failure disappears when the covariate is removed. probability a model; the simulation found the model's own check failing. And the proof of Chapter 9 said that marginal risks cannot bound a pooled attack; the simulation found that the gap widens as the register grows.

None of this replaces an assessment of a real release. Every number in the chapter is conditional on a synthetic population whose parameters were chosen. What the simulation supplies is the knowledge of which formulas to distrust, by how much, and which symptoms to look for when no truth is available. Naledi's plan for Tomás's file took that form. She would not simulate the Bank's population, which she could not observe. She would plant known links among the Bank's own records, test the adversary's calibration on them, fit the frequency law at several fractions, and report the worst-record bound with its selection correction. When Grace read the file a year later, the record named the adversary, the borrowing measure as the variable that defeated the class-size rule, and the decision to release that measure only in bands.

IN PRACTICE: WHAT TO SIMULATE BEFORE A RELEASE

Simulate the declared attack whenever the release publishes a variable outside the key that an adversary may also know, and report the worst-record bound with a correction for selection, beside the average. Run the attack with and without each auxiliary field to see which field drives the risk; if the risk is driven by a field outside the key, raising the class size will not fix it. Check the adversary's calibration on records whose links are known. If a frequency model is used for

the uniqueness probability, fit it at several subsample fractions and treat a moving dispersion as a failed check. State the population, the seed and the parameters of any synthetic study, and do not carry its numbers over to a real register.

In the software. The population and its sample are built by `privacy_risk.synthetic` (`make_register`, `apply_mechanism`, `suppress_small_cells`). The attacks are in `privacy_risk.linkage` (`attack`, `per_target_risk`, `minimum_cell_targets`), the measures in `privacy_risk.risk`, and the selection-corrected bound in `uncertainty.selected_maximum_bound`. The studies of this chapter are the functions `conventional_measures`, `attack_sequence`, `k_anonymity`, `programme`, `programme_scaling` and `design_variance` in `privacy_risk.experiments`. The heart of the k-anonymity study is short:

```
for minimum in config.get("kanonymity_levels", [3, 5, 10, 20]):
    mechanism = Mechanism(register.fine_band_width, register.occupations, int(minimum))
    release = apply_mechanism(register, mechanism)
    average = attack(register, release, model, replicates(config, "kanon", 12000),
                     np.random.default_rng(seed + 300 + minimum))
    targets = minimum_cell_targets(release, int(minimum), cap,
                                   np.random.default_rng(seed + 7))
    risks = per_target_risk(register, release, model, targets, reps,
                            np.random.default_rng(seed + 400 + minimum))
    worst = selected_maximum_bound(risks, reps, alpha)
```

11.10 Proofs

The chapter states two results that its figures rely on.

Proposition 11.1 (The uniform target)**.** *Under Assumptions 2.1 to 2.4 of Chapter 2, the success probability of an adversary attacking a target drawn uniformly from the n released records is*

$$\frac{1}{n}\sum_{i\in I}\frac{1}{f_{K(X_i)}} = \frac{D_I}{n}, \tag{11.2}$$

where D_I is the number of distinct keys in the release.

Proof. By Proposition 2.2, a target with key k is identified with probability $1/f_k$, and a uniformly drawn target is record i with probability $1/n$, which gives the left side of (11.2). Grouping the sum by key value, each occupied key k contributes f_k terms each equal to $1/f_k$, so it contributes exactly one, and the sum equals the number of occupied keys, D_I. □

Proposition 11.2 (A lower bound for the worst selected target)**.** *Let T targets be attacked, target j with B independent replicates and success count $K_j \sim \mathrm{Bin}(B, p_j)$, conditional on the release. Let L_j be the one-sided Clopper–Pearson lower bound at level α/T, that is $L_j = 0$ if $K_j = 0$ and otherwise the α/T quantile of the $\mathrm{Beta}(K_j, B - K_j + 1)$ law. Let $\hat{\jmath}$ be any target chosen after seeing all the counts, for instance the one with the largest K_j. Then*

$$\mathbb{P}\Big\{\, p_{\hat{\jmath}} \geq L_{\hat{\jmath}} \text{ and } \max_j p_j \geq L_{\hat{\jmath}} \Big\} \geq 1 - \alpha.$$

No independence between targets is needed.

Proof. First, each L_j is a valid one-sided bound at level α/T: for every p_j, $\mathbb{P}\{L_j > p_j\} \leq \alpha/T$. Write $\gamma = \alpha/T$ and $G_p(k) = \mathbb{P}_p\{K \geq k\} = \sum_{s\geq k}\binom{B}{s}p^s(1-p)^{B-s}$. For $k \geq 1$, $G_p(k)$

is continuous and strictly increasing in p, and the binomial–beta identity gives $G_p(k) = I_p(k, B-k+1)$, so $L(k)$ is the unique p with $G_p(k) = \gamma$. Hence $L(k) > p$ if and only if $G_p(k) < \gamma$, and this set of counts is an upper tail $\{k \geq k^*\}$ because $G_p(k)$ decreases in k. Therefore $\mathbb{P}_p\{L(K) > p\} = G_p(k^*) < \gamma$, or zero if the set is empty.

By the union bound, $\mathbb{P}\{\exists j : L_j > p_j\} \leq \sum_j \alpha/T = \alpha$, whatever the joint law of the counts. On the complementary event every $L_j \leq p_j$ holds at once, and in particular $L_{\hat{\jmath}} \leq p_{\hat{\jmath}} \leq \max_j p_j$ for whichever target was selected, since the selection only picks an index among inequalities that all hold. □

Proposition 11.2 is what licenses the figure of 0.638 at $k = 10$. The correction divides α by the 230 targets examined, which costs width but is needed: an uncorrected bound for the target with the largest observed count would be biased upwards, because that target was chosen for having done well.

Chapter 12

Public microdata

Every statistical office has files that have been public for a long time. They sit in teaching packages, on data portals and in the replication archives of journals, and they are downloaded every week. When an institution designs a new public file, one of these is often the template, and the argument for it runs as follows. The older file has been public for years. Thousands of people have used it. If its records could be linked to named people, someone would have done it and someone would have said so, and no incident has been reported. Long exposure without harm is evidence, perhaps the best evidence an institution can get, that a file of that shape is safe.

The first fact that breaks this argument is about the file, not about its history. The extract of the US Census 2000 public use sample used in this chapter has been in teaching use for years. On a key made of four fields that an outsider could plausibly know, state, area, years of schooling and years of potential experience, 86.3 per cent of its 29,501 records are unique, and the mean of the reciprocal key frequency is 0.928. Inside the file the key almost determines the record. Whatever has protected the people behind it, it was not the shape of the file. The second fact is about the history. An adversary who links a record for gain, curiosity or harm has no reason to announce it, and the incidents that the field knows about were reported by researchers who chose to publish them.

This chapter asks one question. What can an attack on a public file tell an institution about its own confidential register? It runs the pipeline of the earlier chapters on two real public files, reports what it finds, and then sets out what such results license and what they do not.

12.1 A template for a public extract

The Bank planned a public extract from its survey of household finances, and the chair had a suggestion for the release committee. A census teaching file of the same general shape had been public for years without complaint, she said, and the Bank could follow its design and save itself a long assessment. Naledi asked for a week. She would run the Bank's own pipeline on the teaching file and report what a declared adversary achieved against it. If the file was as safe as its history suggested, the results would show it.

Her request rested on a distinction the chair accepted at once. A file's history records what was reported. An assessment records what an attack achieves. The two coincide only if every successful attack is reported, and the documented cases suggest the opposite.

12.2 What the documented cases show

The best known re-identification in the literature was an exercise, not a crime. Sweeney (2002) describes how hospital discharge records for state employees in Massachusetts, released with names removed, were linked to the voter list of Cambridge through date of birth, sex and postal code, and how the linkage found the record of the Governor of Massachusetts. The records had already been released; the linkage became known because the researcher who performed it published it. Her earlier study (Sweeney, 2000) showed that a few simple demographic fields identify most people in the United States uniquely, and Golle (2006) revisited that estimate with later census data and found the fields still highly identifying.

Narayanan and Shmatikov (2008) reached the same kind of result with a different kind of file. Netflix had released the film ratings of its subscribers, with identities replaced by numbers, as the data for a public prize. The authors showed that a few ratings with approximate dates, of the kind people post publicly elsewhere, are enough to pick out a subscriber's record, and they demonstrated the match against public ratings on another site. Again the attack became known because its authors wrote it up. de Montjoye et al. (2013) showed that four points in space and time identify most people in a large mobility data set, and Rocher et al. (2019) showed that releasing only a sample of a population does not prevent a generative model from estimating, with good accuracy, whether a matched record is the right one.

None of these is evidence that an attack on any particular file has taken place. They are evidence of something more useful for the present argument: that the files were open to linkage from the day they were released, and that the world learned of it only when someone with an interest in publishing chose to publish. Section 12.11 turns that observation into a statement about how much a quiet history can show.

12.3 Two public files

The pipeline runs on two files that ship with the `wooldridge` package for Python, version 0.5.0, whose contents are identified in the replication record by their SHA-256 hashes. The first is an extract of the US Census 2000 public use microdata sample: 29,501 records with state, public use microdata area, years of schooling, years of potential experience and log weekly income. The second is the Boston Fed study of mortgage applications: after dropping incomplete rows, 1,986 applications with area, applicant income, marital status, dependants, occupancy, the lending decision and a record of late payments. Both files are already public and already protected. Nothing confidential is read, and the exercise asks the question an agency asks before publishing a file of that shape: what would a declared adversary achieve against it?

Three things change when the data are real, and each was a claim of the earlier chapters. The population behind the file is not observed, so the population frequency measure cannot be computed and the uniqueness probability has no truth to be checked against. The analyses that define utility are the ones the file is used for, not ones the analyst chose. And the risks are far higher than on the synthetic population of Chapter 11, so the choice of tolerance bites harder.

The figures in the next sections are prosecutor risks computed inside the file, for an adversary who knows that its target is in the extract. They are not estimates of the risk the

file posed to the US population, which an office would assess with the sampling fraction, the population frequencies and the access conditions that this study does not have.

12.4 What the key alone separates

On the full key, the 29,501 census records occupy 27,390 classes. Of those classes 25,454, or 92.9 per cent, hold a single record, and they hold 86.3 per cent of the records. The key space, counting the areas actually observed within each state, has 665,896 cells, so the file fills about 4 per cent of it. The finding is not that someone blundered. It is that the key an outsider could reconstruct almost determines the record inside the file, and that any claim that the file is protected must rest on something other than the key: on the sampling, on what outsiders actually know, or on the access conditions.

The first of these is the journalist's question of Chapter 3: is a record unique in the file also unique in the population? On a real file the answer needs the sampling fraction, which for this extract is not known. The pipeline assumes $\pi = 0.05$. Under that assumption a negative binomial law fitted to the observed counts puts the mean population cell at 0.886 people, empty cells included, and a record unique in the file is unique in the population with probability 0.225. With the dispersion held at its fitted value, the same calculation gives 0.035 at $\pi = 0.01$, and lower values at smaller fractions. The file is a small subsample of the 2000 public use sample, so its fraction of the US population is far below 0.05. The figure of 0.225 is therefore a consequence of an assumed input and not a finding about the file. Chapter 11 showed that a fitted law of this kind can be off by a quarter even when its input is right; here the input is not known, and nothing can be checked. That is the position an agency is in when it does not know its own sampling fraction, and it is the first limit on what a public file can show.

12.5 A declared attack on the census file

The pipeline then treats the census file as a candidate release and searches over protection settings. It plans $3 \times 2 \times 2 \times 3 = 36$ settings, varying the width of the area, schooling and experience bands and the minimum class size, and keeps the 25 that retain at least 500 records. The declared adversary reads the key exactly but allows in its likelihood for a misreading probability of 0.10, and holds a noisy reading of log weekly income with standard deviation 0.5. A second adversary reads income twice as precisely. Each is attacked on 1,500 freshly drawn targets per setting, and the risk reported for a setting is the larger of the two.

Figure 12.1 sets out the evaluation in the order it runs. The order is the point. The frontier of settings is chosen after all the attack results are in, and a bound computed for each setting separately would not survive that choice. The pipeline therefore computes one exact upper bound for each of the 72 planned setting and adversary pairs at level $0.05/72$, so that all 72 hold together with probability at least 0.95 (Proposition 12.1). Any setting chosen after looking at the results then carries a valid bound.

Proposition 12.1 (A familywise bound survives selection)**.** *Let J setting and adversary pairs be planned before any attack is run. For pair j, let K_j be the number of successes in n_j independent attacks with success probability p_j, conditional on the released file and the attack rule, and let*

$$U_j = \begin{cases} B^{-1}_{1-\alpha/J}(K_j + 1,\ n_j - K_j), & K_j < n_j, \\ 1, & K_j = n_j, \end{cases}$$

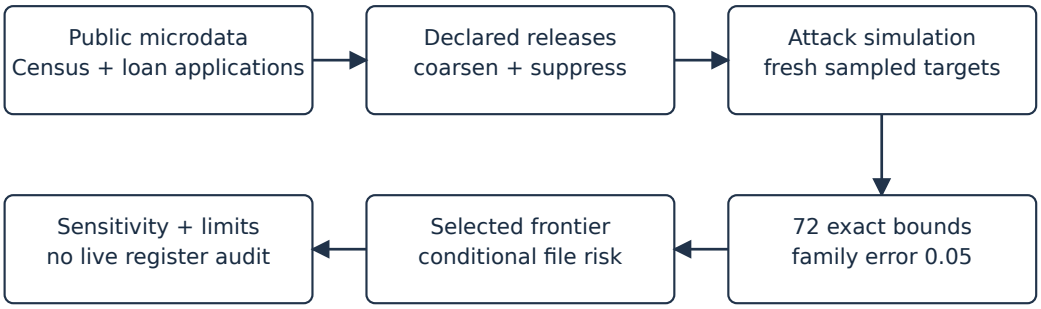


FIGURE 12.1: The public-file evaluation in the order it runs: public files, declared releases, attack simulation on fresh targets, 72 exact bounds sharing one familywise error of 0.05, the frontier chosen from them, and sensitivity checks, which lie outside the family.

where $B_q^{-1}(a,b)$ *is the* q *quantile of the beta law. Then* $\mathbb{P}\{p_j \le U_j \text{ for all } j\} \ge 1-\alpha$*, and for any setting* $\hat{\theta}$ *chosen as a function of all the counts, the reported bound* $\max\{U_j : j \text{ belongs to } \hat{\theta}\}$ *is at least the larger risk of the two adversaries at* $\hat{\theta}$ *on that event.*

Eight of the 25 settings are nondominated in the plane of the familywise upper bound and utility, and their bounds run from 0.276 to 0.942. Utility charges for coarsening as well as for suppression. Each of the three analyses, mean income by state, by schooling and by experience, is estimated at the fine level of its variable, and a release that publishes bands supplies the band mean for every fine level inside it.

Figure 12.2 compares, for the eight frontier settings, the risk the formula of Chapter 2 gives under exact matching with the simulated risk of the declared attack and its familywise bound. The simulated attack beats exact matching at every setting, by a factor between 1.05 and 4.19. At the widest bands with a minimum class size of ten, which keeps 28 per cent of the records, exact matching gives 0.062 and the simulated attack 0.239.

These are conditional attack success rates against the tested public file, not population re-identification rates and not certificates against attacks that were not tested. The pipeline also reads implied tolerances off this frontier, in the manner of Chapter 8, and the book does not report them: a tolerance for a file of this kind is a policy choice for the office that publishes it, and in-file prosecutor risks are not the quantity it would govern.

Naledi's note to the chair after the first run was short. The template file, judged by the Bank's own method, had a declared-attack risk above one fifth at every setting the pipeline could find, and above one half at every setting that suppressed nothing. Its history said nothing about that, because nothing in its history measured it.

12.6 How much the attacker's knowledge moves the figure

The figures above assume one reading precision. The attacker's knowledge is an input, and an exploratory run makes its weight visible. On one setting of the census grid, with 3,000 targets per level, the declared attack succeeds with probability 0.371, 0.271 and 0.186 as the standard deviation of its log-income reading rises from 0.25 to 0.5 to 1.0. On the mortgage file the corresponding rates are 0.179, 0.137 and 0.093 for loan-amount readings with standard deviations of 20, 40 and 80. These rates come from an archived run and carry individual 95 per cent intervals only; they lie outside the 72-comparison family. Cutting

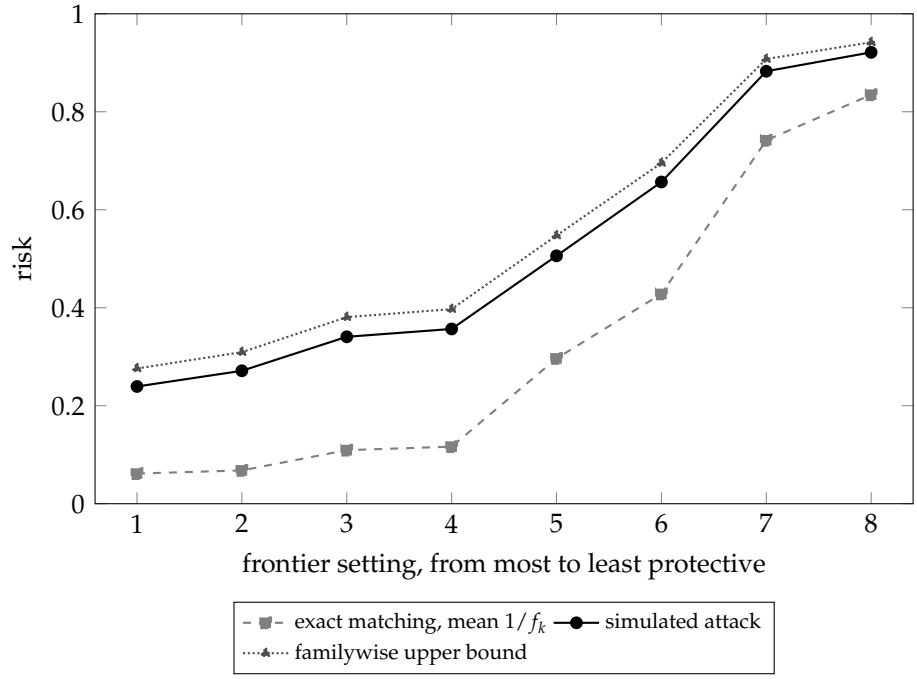


FIGURE 12.2: The eight nondominated settings for the census extract. Settings 1 to 4 apply a minimum class size of 10 or 5 with wide bands; settings 5 to 8 suppress nothing and narrow the bands in turn. The attack that uses a noisy reading of income beats exact matching on the key at every setting. Series differ by line style and marker as well as shade.

the error of the attacker's reading by a factor of four doubles its success on the census file, and the figure an institution reports is therefore a statement about the attacker as much as about the file. That is why the decision record of Chapter 7 names the adversary beside the number.

12.7 The worst records

The decision rule of Chapter 1 is stated for every target in a declared set, and on a real file the worst records are easy to find. The pipeline takes the middle setting of the census frontier, with areas grouped in hundreds, four-year schooling bands, ten-year experience bands and no minimum class size, and examines 80 records that sit in classes of one. The attack names them with mean probability 1.000; the worst is identified with probability 1.00 and a lower bound, corrected for selection as in Proposition 11.2, of 0.964. On this setting the key alone separates them, so the check shows how exposed an unsuppressed release of the file is. It does not test a k-anonymity promise, since the setting makes none, and it should not be set beside the synthetic result of Section 11.6.

12.8 Pooled releases from the census file

Chapter 9 reported, in one sentence, that the gap between pooled and marginal risk appears on real data as well as on synthetic data. The census file is where it was measured. Four releases are built from it, each with a minimum class size of three: geography and schooling; geography and experience; state, schooling and experience; and the full key with experience

in ten-year bands. As in the synthetic study, the releases are taken to share a record identifier, so that a record can be matched across them. Four public-use files cut from one census sample would not ordinarily carry a common identifier, and an adversary would then have to link them through the published fields. That case was not measured.

Figure 12.3 shows the result for 5,000 targets. The marginal risks are 0.138, 0.030, 0.118 and 0.096. The second release keeps only 2,812 of the 29,501 records, because geography with single years of experience leaves most classes below three, which is why its marginal risk is small. The marginals bound the chance that any one release succeeds between 0.138 and 0.381, and independence would give 0.332.

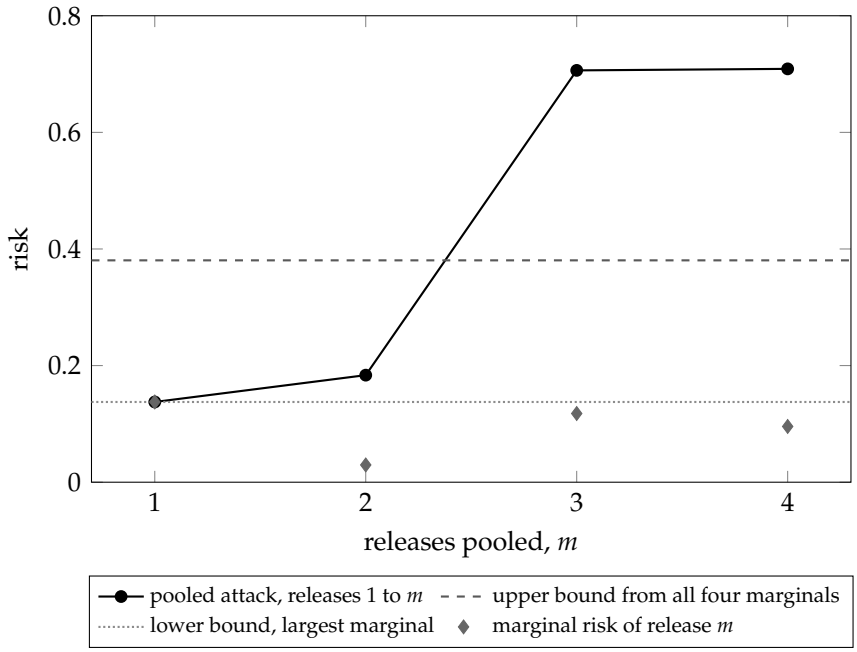


FIGURE 12.3: Four 3-anonymous releases from the census extract, assumed to share a record identifier, 5,000 targets. The pooled attack leaves the interval implied by the marginals once the third release is added. Series differ by line style and marker as well as shade.

The pooled attack reaches 0.709 with all four releases, 1.86 times the upper bound, and 60.7 per cent of targets are identified uniquely by the intersection although no release contains a unique record. The shape of the curve explains the mechanism. The first two releases together add little, 0.184, because the second covers so few records. The jump comes with the third: the first release carries the area and the third carries experience, and between them they rebuild almost the whole key, which Section 12.4 showed to be nearly unique. This is the remark of Chapter 9 in a concrete form. Releases that share an identifier are one release keyed on the union of their keys, and here the union is the full key.

On the synthetic population the same comparison gave 1.8 times the bound. Each figure comes from one draw, so the two do not show that the effect is larger on either kind of data. They show that it is not an artefact of synthetic data.

12.9 The mortgage applications

The second file is small and makes concrete three risks that the census file does not. On the key of area, income band, marital status, dependants and occupancy, the 1,986 applications

occupy 150 classes, of which 63, or 42 per cent, are unique; those classes hold 3.2 per cent of the applications. An adversary holding a noisy reading of the loan amount, with standard deviation 40 in the file's units, identifies 0.134 of applications over 1,500 attacks, with a one-sided upper bound of 0.149.

Attribute disclosure is the risk this file shows most clearly. 12.2 per cent of applications sit in a class where every application received the same decision, and 0.65 per cent sit in a class where every application was denied, so for those applicants the denial is disclosed by the class without any record being identified. For late payments the shares are 29.6 per cent and 0.35 per cent. Both harmful directions occur here, where the synthetic credit register of Chapter 13 had none, which is the difference between a base rate chosen for a simulation and one that occurred.

Published totals leak by subtraction. Aggregating loan amounts by area, occupancy and income band gives 48 cells, of which 16, a third, contain a single application, so that application's amount is the published total. Those cells hold 2.2 per cent of the lending in the file, and a rule requiring three records per cell would suppress 20 of the 48 cells, 42 per cent of them. The file was designed for the study of lending discrimination and serves it well. The point of these figures is that the same table, published by an agency without an assessment, would have disclosed the amounts of sixteen individual applications.

12.10 Models fitted on the public file

The repository also holds two exploratory runs that fit models to attack outcomes on the census file, holding out whole states for testing, as a check on the proposal models of Chapter 14. On three overlapping state-held-out splits, the held-out binomial log loss of a linear logistic model is between 0.607 and 0.621, and a logistic model with quadratic terms does slightly better than one- and three-layer networks on every split, at between 0.602 and 0.614. The differences are in the third decimal place. These runs predict one declared attack on public data, the models were never trained on final audit outcomes, and their predictions are not risk bounds. Chapter 14 discusses what such models are for and reports the negative results beside the positive ones; nothing in this chapter rests on them.

12.11 What a quiet history can show

The chair's argument can now be put in a form that can be tested. Suppose attempts to link the file's records to named people arrive over time at some rate, each succeeds with the probability the file's risk gives, and each success becomes publicly known with some probability. The institution observes the number of reported incidents. How far should a count of zero over many years move its belief that the file is safe?

Proposition 12.2 (A quiet history is weak evidence)**.** *Suppose attempts arrive as a Poisson process of rate λ per year, each succeeds independently with probability p, and each success is reported independently with probability r. Compare the hypotheses that the file's risk is p_0 and that it is $p_1 > p_0$. After T years with no reported incident, the posterior odds of the second hypothesis are at least*

$$\frac{\mathbb{P}(p_1)}{\mathbb{P}(p_0)}\,\exp\{-\lambda r T (p_1 - p_0)\} \;\geq\; \frac{\mathbb{P}(p_1)}{\mathbb{P}(p_0)}\,e^{-\lambda r T}. \tag{12.1}$$

In particular, as the reporting probability r tends to zero the quiet history leaves the prior odds unchanged, whatever the number of years and the rate of attempts.

The proposition says that a quiet history is evidence of safety only in proportion to $\lambda r T$, the expected number of reported attempts. Every factor in that product is uncertain, and one of them is likely to be small. The adversaries a release committee most needs to worry about, a lender reading a competitor's customers, a fraudster, a journalist with a story not yet written, gain nothing from announcing a successful link and may lose a great deal. The documented cases were reported by researchers who wanted them known. A file can be open to linkage for years, as the Netflix ratings and the Massachusetts hospital records were, and become known as such only on the day someone chooses to say so.

The proposition also says what would make the history informative. If an institution monitored the uses of its file, required registration for access, or ran periodic attacks of its own and published the results, the reporting probability would rise and a quiet history would start to carry weight. That is a design choice for the access route, and it is one reason the decision record of Chapter 7 names the route.

12.12 What public files can and cannot show

The chapter asked what an attack on a public file can tell an institution about its own confidential register. The answer has a positive part and a limiting part.

The positive part is that a public file is a real file with a real distribution, and the pipeline's findings on synthetic data reappear on it. A covariate the adversary reads beats exact matching at every setting. When releases share an identifier, the pooled attack leaves the interval the marginals imply. Attribute disclosure and leakage by subtraction occur at rates that nobody chose. These are findings about files of that shape, and an institution designing a file of the same shape learns from them which attacks to run and roughly how strong they are.

The limiting part has four elements. The population behind a public file is not observed, so the uniqueness probability rests on an assumed sampling fraction and a fitted law that cannot be checked, and utility is measured against the file and not against the population. The adversary is declared and not observed: the misreading rate and the precision of the income reading are assumptions, and the attacker's knowledge moves the risk by a factor of two. The files are public, so the exercise measures what a release of that shape discloses and says nothing about whether a wrong was done in publishing them. And only the demographic path of the pipeline has run on them; the credit evaluation of Chapter 13 has run only on the synthetic register.

Together these limits say that a public file can teach an institution which attacks to run, but cannot run them for it. The confidential register has its own key, its own population and its own adversaries, and the assessment must be done on it. A long history without reported incident adds little, because the people most able to exploit a file are the least likely to report having done so.

Naledi's report to the committee said this in four sentences. The template file was highly identifying on a key an outsider could know. Its history was silent because nobody had an interest in speaking. The Bank's extract would be assessed on the Bank's survey, with the Bank's own adversaries declared. And the template was useful for one thing: it showed which attacks to run first. The chair accepted the report. When Grace read the decision record, the line that cited the template file had been replaced by the name of the adversary the Bank had assessed, and the template was cited only as the source of the attack design.

IN PRACTICE: USING A PUBLIC FILE AS A TEMPLATE

Do not cite a file's public history as evidence that a release of the same shape is safe; a quiet history is evidence only in proportion to the expected number of reported attempts, which is rarely known and often small. Use the public file to choose and test attacks: run the declared attack on it, with and without each auxiliary field, and with several precisions for the adversary's readings. Bound every setting in the grid together before choosing among them. Check attribute homogeneity and published totals as well as identification. Then repeat the assessment on the confidential register, and record which results came from the template and which from the register.

In the software. The two files are loaded by `privacy_risk.realdata` (`load_census2000`, `load_loan_applications`); releases are built by `build_release` and attacked by `attack` and `per_target`. Attribute and residual disclosure are `attribute_disclosure` and `residual_disclosure`, and the pooled attack is `composition`, which calls `composition.pooled_attack_from_sets`. The whole study is `experiments.real_data`, and `scripts/realdata_replication.py` reruns it with source hashes and the sensitivity checks. The familywise level is set in one place:

```
grid_size = (len(settings.get("puma_widths", [1, 10, 100]))
             * len(settings.get("education_widths", [1, 4]))
             * len(settings.get("experience_widths", [1, 10]))
             * len(settings.get("min_cell_counts", [1, 5, 10])))
family_alpha = alpha / (2 * grid_size)
```

12.13 Proofs

Proof of Proposition 12.1. Fix j and write $b = n_j$, $k = K_j$, $\gamma = \alpha / J$, and $F_p(k) = \sum_{s=0}^{k} \binom{b}{s} p^s (1-p)^{b-s}$. For $k < b$, $F_p(k)$ is continuous and strictly decreasing in p, and the binomial and beta laws are linked by $F_p(k) = 1 - I_p(k+1, b-k)$, so U_j is the unique solution of $F_{U_j}(k) = \gamma$. For a fixed true p_j, the event $\{U_j < p_j\}$ therefore equals $\{F_{p_j}(K_j) < \gamma\}$, which is a lower tail $\{K_j \le k_*\}$ of the count because $F_{p_j}(k)$ increases in k. Its probability is $F_{p_j}(k_*) < \gamma$, or zero if the tail is empty; when $K_j = b$ the bound is 1 and cannot fail. Hence $\mathbb{P}\{p_j > U_j\} \le \gamma$ for every j.

By the union bound, $\mathbb{P}\{\exists j : p_j > U_j\} \le J\gamma = \alpha$, with no assumption about how the counts for different pairs depend on one another. On the complementary event, every inequality $p_j \le U_j$ holds at once. A setting $\hat{\theta}$ chosen from the counts is one of the planned settings, so both of its pairs are among the J, and the larger of its two risks is at most the larger of its two bounds. Selection cannot move the realised counts off the event on which all bounds hold, so the guarantee is unaffected by it. □

Proof of Proposition 12.2. Attempts form a Poisson process of rate λ, and each is independently a reported success with probability pr. By the thinning property of the Poisson process, reported incidents form a Poisson process of rate λpr, so the number N reported in T years is Poisson with mean λprT, and $\mathbb{P}(N = 0 \mid p) = \exp(-\lambda prT)$. Bayes' rule in odds form gives

$$\frac{\mathbb{P}(p_1 \mid N = 0)}{\mathbb{P}(p_0 \mid N = 0)} = \frac{\mathbb{P}(p_1)}{\mathbb{P}(p_0)} \cdot \frac{e^{-\lambda p_1 rT}}{e^{-\lambda p_0 rT}} = \frac{\mathbb{P}(p_1)}{\mathbb{P}(p_0)} \exp\{-\lambda rT(p_1 - p_0)\}.$$

Since $0 \le p_0 < p_1 \le 1$, the difference $p_1 - p_0$ is at most one, which gives the second inequality in (12.1). As $r \to 0$ the exponential factor tends to one for fixed λ and T, so the

posterior odds tend to the prior odds. □

Chapter 13

A granular credit register

Ask a disclosure control officer what a minimum cell count costs, and the answer usually comes as a count of records. A floor of ten removes 4 per cent of the loans; a floor of twenty removes 15 per cent. The answer is honest and it is the one release documents print, because records are what the rule acts on and what the software counts. Behind it sits a belief that most practitioners hold without stating it: suppression costs data in proportion to the records it removes. On that belief a file that keeps four loans in five has kept four fifths of what its users wanted, and a committee can weigh protection against loss by reading one percentage.

The belief is reasonable for a file whose records are roughly alike in the analyses they feed. A credit register is not such a file. On the synthetic register of this chapter, the floor that a history-aware adversary requires keeps 20 per cent of the loans and 4 per cent of the exposure, and it keeps no loan to a large firm. The records that a floor removes first are the ones in thin cells, and in a credit register the thin cells hold the largest borrowers, whose loans carry most of the money. Counting records measures the wrong loss.

This chapter asks one question: *for each adversary an institution might declare against a loan-level release of its credit register, what minimum cell count does a tolerance require, and what is left of the file once that floor is applied?* The answer runs through the register itself, the five harms it can cause, an adversarial evaluation of five protection policies, the floors that fixed-sequence testing selects for six adversaries, what those floors keep, and how far the floors move when the tolerance, the register draw and the adversary's precision change.

13.1 Why the register is synthetic

> The numbers here belong to the example Every floor, tolerance and reading precision in this chapter was chosen or generated for the synthetic register. None is a standard, and none transfers to a real programme. Take the procedure from this chapter and not the numbers.

Chapter 1 introduced the register briefly. It is built to share the structure of a supervisory loan-level collection. In the euro area, AnaCredit collects, mostly monthly, data on loans by credit institutions to legal entities whenever the debtor's commitment with the reporting institution is at least EUR 25,000, and each record ties a borrower to a lender (ECB, 2016; Israël et al., 2017). Registers of this kind support research on credit supply that could not be done any other way: Khwaja and Mian (2008) use a credit register to separate supply from

demand, and Jiménez et al. (2012) use loan applications in the Spanish register to identify the bank balance-sheet channel. Loan-level data of this kind are not released for general use, and research access is restricted. That is why the register here is synthetic, and it is also why its parameters are chosen and not estimated: there is no released file from which to estimate them.

The generator draws 20,000 borrowers, each with a sector from 18, a region from 12 and a size class from four, so the borrower key has $18 \times 12 \times 4 = 864$ cells. Size classes are drawn with probabilities 0.62, 0.24, 0.11 and 0.03, so large firms are three borrowers in a hundred. Each borrower has between one and four lending relationships, drawn as one plus a binomial count on three trials whose success probability rises from 0.10 for the smallest firms to 0.70 for the largest, so most small firms have one lender and most large firms have several. Lenders are chosen from 60 with skewed shares; the largest holds 23 per cent of relationships. Exposure is lognormal within size class, with log means 11, 12.5, 14 and 16 and a log standard deviation of 0.8, so a typical large-firm loan is about 150 times a typical micro-firm loan. Arrears are drawn at a rate between 3 and 10 per cent that varies with sector and size, about 7 per cent overall.

Six monthly vintages are generated as a panel with a stable stock. Most loans are outstanding when the panel opens; the others are originated in a month drawn uniformly from the second to the sixth. Every loan then leaves with a monthly hazard of 0.04 and does not return. Over the six months 32,581 loans are observed, the first release holds 27,167 of them, the stock stays near 27,000, and 81 per cent of the loans in the first release are live in all six vintages. The difference, 5,414 loans, is the sixth of the observed loans that were originated during the panel.

Each choice was made to reproduce a structural feature that bears on disclosure: a skewed lender market, several relationships per large borrower, a thin top size class, and monthly entry and exit. None was fitted to data. Two simplifications bear directly on the results and must be kept in view. A loan's exposure is the same in every vintage in which it is live, where a real register reports outstanding amounts that change as loans are drawn and repaid. And whether a loan is live in a month is drawn independently of its key and exposure, where in a real register large loans may run longer. Every floor in this chapter is a floor for this design. It describes a class of release systems with these features, and the way to refute it is to run the same evaluation inside an institution that holds the real file.

Tomás arrives at the Bank with a research proposal on credit to large firms. He asks for the loan-level file for six months: each loan with its sector, region, size class and exposure, under a pseudonym that lets him follow a loan from month to month. Naledi reads the request twice. The fields are the ones the register was built around, and his research question is about the borrowers in its thinnest cells.

13.2 Five harms from one register

A single risk figure covers one harm, and a credit register can cause five. A borrower may be identified. A borrower–lender relationship may be alleged, whether or not the borrower is identified. Arrears status may be inferred for a class of borrowers without identifying anyone. One lender's contribution may be reconstructed from published totals by another participant. And two consecutive vintages may be differenced. Table 13.1 reports all five against one lever, a minimum cell count $k_{\min}$ on the borrower key, applied to the first release.

TABLE 13.1: Five harms and one utility on the synthetic credit register under a minimum cell count on the borrower key. Identification is the share of released relationships the attack recovers, with its exact one-sided 95 per cent bound, and again for the largest size class. The predictive value is that of an alleged borrower–lender relationship. Status disclosed is the share of large borrowers whose key class is homogeneous in arrears status. Utility is the negative weighted loss over the four analyses the register is collected for.

$k_{\min}$	Kept	Ident.	Bound	Large firms	Relationship PPV	Status	Utility
1	1.000	0.0855	0.0931	0.247	0.328	0.444	−0.000
5	0.990	0.0770	0.0843	0.273	0.324	0.381	−0.050
10	0.958	0.0705	0.0775	0.119	0.320	0.262	−0.139
20	0.851	0.0563	0.0626	0.155	0.313	0.119	−0.254

Identification is a subgroup problem. The attack recovers the exact relationship for 0.0855 of released rows, with a bound of 0.0931. Recovering the right borrower succeeds for 0.1095, because a borrower with several lenders can be found through any of them. Both look comfortable as averages. Split by size class, the relationship figure is 0.050, 0.091, 0.128 and 0.247, so a large borrower faces about five times the risk of a micro borrower. The large-firm column of Table 13.1 is not monotone in the floor: it is 0.273 at $k_{\min} = 5$ and 0.119 at $k_{\min} = 10$. The class rests on few targets in a uniform draw, and its interval at $k_{\min} = 1$, $[0.192, 0.309]$, overlaps the interval at $k_{\min} = 20$, $[0.085, 0.250]$. The large-firm figure at no suppression agrees across independent draws: 0.248 in the policy comparison below, and about 0.24 on a stratified draw of 4,000 large-firm targets. This is why the subgroup floors later in the chapter use a stratified draw.

An alleged relationship is a base-rate statement. Suppose the adversary names, for every released borrower, the lender holding the largest share of that borrower's key class. What counts for harm is the probability that an allegation is true, which Bayes' theorem gives from the prevalence p of a borrower–lender pair, the sensitivity s of the rule and its specificity q:

$$\mathbb{P}(\text{true} \mid \text{alleged}) = \frac{ps}{ps + (1-p)(1-q)}. \tag{13.1}$$

On the register $p = 0.0243$, $s = 0.225$ and $q = 0.989$, and (13.1) returns 0.328, the realised share of correct allegations. A specificity of 0.989 sounds protective and is not, because the prevalence is low. An allegation right a third of the time is 1.4 times the success rate of always naming the largest lender in the market, whose share is 0.23, and it is wrong about two borrowers in three. The floor barely moves it: 0.328 at $k_{\min} = 1$ and 0.313 at $k_{\min} = 20$. The lever that governs identification does not govern relationship disclosure.

Status can be disclosed without identification. A key class whose members all share an arrears status discloses that status for every member, whatever the size of the class. For large borrowers 44.4 per cent of released relationships sit in such a class at $k_{\min} = 1$, falling to 11.9 per cent at $k_{\min} = 20$. In this register every homogeneous class is homogeneous in the benign direction, because arrears are rare, so what is disclosed is the absence of arrears. That is a property of the chosen base rate. On a real register a small sector–region cell in distress is the case to look for, and the attribute criteria of Machanavajjhala et al. (2007) and Li et al. (2007) address it by changing the risk model, which a cell count does not.

Published totals give contributions away by subtraction. Totals by region, sector and size class form a linear system. In the first vintage 2.0 per cent of the 814 published cells have a single contributor, so that lender's position is the published total, and 3.9 per cent have two, so either participant recovers the other's amount by subtraction. Together they hold 2.8 per cent of total exposure. Requiring five contributors in place of three suppresses 14.4 per cent of cells in place of 5.9 per cent. With lender shares up to 23 per cent, one contributor can estimate another's amount closely even when three or more contribute, which is what the dominance and p% rules for magnitude tables are designed to catch (Hundepool et al., 2012, ch. 4). They are not evaluated here. The loan-level floor leaves this risk untouched.

Monthly vintages can be differenced. When a published count moves by one between vintages, the change in the published total is the arriving or departing amount, provided one loan moved. The adversary does not see how many loans moved, and Proposition 13.1 gives the posterior that a unit change was a single move.

Proposition 13.1 (Differencing posterior)**.** *Let the arrivals A and departures D of a published cell between two vintages be independent Poisson variables with common mean $\lambda > 0$. Then*

$$\mathbb{P}\big(A + D = 1 \,\big|\, |A - D| = 1\big) = \Big(\sum_{d=0}^{\infty} \frac{\lambda^{2d}}{d!\,(d+1)!}\Big)^{-1} = \frac{\lambda}{I_1(2\lambda)}, \tag{13.2}$$

where I_1 is the modified Bessel function of the first kind. The posterior is strictly decreasing in λ and tends to one as $\lambda \to 0$.

The proposition turns an intuition into a number: a unit change in a quiet cell almost certainly reveals one loan, and in a busy cell it may be offsetting moves. Across consecutive months 36.1 per cent of comparable cell pairs show a unit change at a count floor $c_{\min} = 1$, and 39.0 per cent at $c_{\min} = 10$, because a higher count floor keeps only the busier cells, where counts move more. Which rate enters (13.2) decides the answer. Pooled over all cells, arrivals and departures together average 2.7 per cell per transition, so $\lambda = 1.35$ and the posterior is 0.45. Evaluated at the rate of each cell that shows a unit change and averaged, it is 0.65, and the realised share of unit changes that were in fact a single move is also 0.65. The pooled rate understates the exposure, and the institution, which holds the register, can count realised single moves directly. At $c_{\min} = 10$ a unit change still reveals one loan's exposure in 56 per cent of cases.

Naledi takes Table 13.1 to the chair before any floor is chosen. The table has one row per floor and five columns of harm, and only the first column moves much. "Which of these is Tomás's file?" the chair asks. The honest answer is the identification column, because his file is the loan-level one; the other four belong to the aggregates and the monthly series the Bank already publishes.

13.3 An adversarial evaluation

The rest of the chapter fixes an adversary and a set of attacks and runs them against five protection policies on the same register, with one body of auxiliary information. The reporting follows the practice for attacks on trained models: how often an attack succeeds, how often it is right when it acts, and what it must know before it can act at all (Shokri et al., 2017; Carlini et al., 2022).

13.3.1 The adversary and the attack

The adversary targets one borrower. It holds the sector, region and size class exactly, which is the natural case for a credit register, where these are public facts. It holds a reading of the loan's log exposure with Gaussian error of standard deviation 0.6, either once or once for every vintage in which the loan is live. In some attacks it also knows the loan's history, the set of vintages in which the loan is live.

The history needs an argument, because it is what lets pooling help. The borrower knows its own loan, and so do the lender and any counterparty to it, such as a guarantor or a buyer of the loan. Public records can reveal it too, where a register of charges over company assets is public or a firm reports its borrowing in a filing. A journalist or researcher who knows a firm's sector, region and size will not in general know in which months a particular loan was drawn or repaid. The evaluation therefore reports every pooled attack with and without the history, and the institution declares which adversary it guards against.

The attack blocks on region, keeps the loans in the first release whose key agrees with the target's, scores each candidate by the log density of the exposure readings under the adversary's error model, and names the highest-scoring candidate. The pooled attack uses every vintage. With the history, it excludes every candidate whose pattern of appearances differs from the target's in any vintage, through a penalty of 30 on the log score for each mismatch; since the target always matches, a mismatched candidate is in effect removed. With one reading per vintage, the extra readings amount to one more precise reading of the same exposure. Figure 13.1 shows the history test on four loans in one cell.

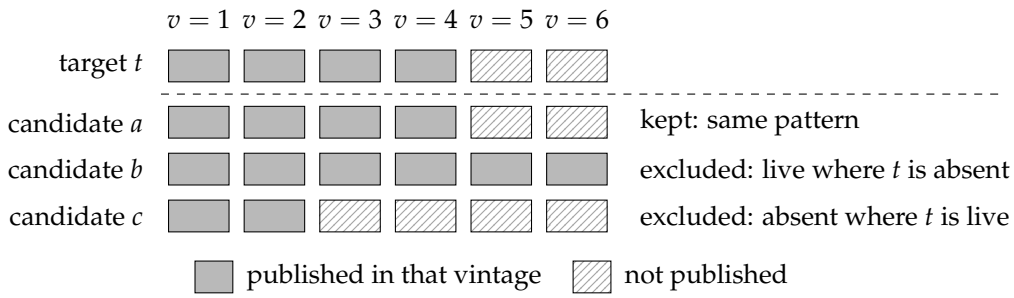


FIGURE 13.1: How the pooled adversary uses a loan's history. All four loans share the target's key cell. An adversary who knows the vintages in which the target is live keeps only candidates whose pattern matches the target's in every vintage, and then ranks the survivors by its exposure readings. Shade and hatching both mark the two states.

Without the history the other vintages add nothing. Proposition 9.3 of Chapter 9 proves this under the two simplifications of the generator: published values repeat across vintages, and whether a loan is live is independent of them. The gain from pooling on this register therefore lives in the history, and on a register whose published fields change from month to month the proposition would not apply.

13.3.2 Five policies

Five loan-level policies enter the comparison. They are stylised versions of rules described by Hundepool et al. (2012), not copies of any institution's practice: no suppression; a threshold rule that withholds a cell unless at least three lenders contribute; the same rule with a heuristic secondary suppression that also withholds the smallest surviving cell in each region; and k-anonymity on the borrower key at $k = 5$ and $k = 10$. The secondary suppression

is not audited against the sector and size margins, so it is not secondary suppression in the sense of Cox (1980) and Fischetti and Salazar-González (2001). Table 13.2 gives the outcomes, each from 4,000 attacks on independently drawn targets.

TABLE 13.2: Attack outcomes by policy on the synthetic register, tolerance 0.12. Release and pooled give the share of targets named correctly for one release and over six vintages (pooled attack with a reading per vintage and the history). Precision is over the attempts whose posterior passes one half, and acts is how often that happens. The last two columns give the size class with the highest rate and its exact 95 per cent upper bound.

Policy	Kept	Utility	Release	Pooled	Precision	Acts	Worst class	Bound
No suppression	27,167	−0.000	0.089	0.260	0.716	0.020	0.248	0.297
Three contributors	27,068	−0.019	0.084	0.255	0.822	0.018	0.180	0.228
Secondary suppression	27,027	−0.028	0.080	0.259	0.746	0.015	0.156	0.203
k-anonymity, $k = 5$	26,885	−0.050	0.074	0.269	0.771	0.012	0.183	0.234
k-anonymity, $k = 10$	26,019	−0.139	0.070	0.244	0.800	0.009	0.141	0.193

Four results come out of Table 13.2. Every suppression policy reduces the per-release risk and none removes it: from 0.089 with no suppression to 0.070 under $k = 10$. The reduction is modest because the exposure reading separates candidates inside a cell however large the cell is, and every policy meets the tolerance of 0.12 on one release.

The same policies do far less against the pooled attack, which lies between 0.244 and 0.269 for all five. Under $k = 5$ the per-release 0.074 becomes 0.269, the largest gap in the table at 0.195. Table 13.3 separates the two sources of the gap on the same targets. Under no suppression, six readings against one release raise the rate from 0.089 to 0.158; pooling six vintages with one reading raises it to 0.191, with a bound of 0.201, already above the tolerance; with both the rate is 0.260. On this register the two effects add with almost no interaction, and for the adversary with the history the programme finding does not rest on the extra readings.

TABLE 13.3: Where the pooled excess comes from: share of targets named correctly, with the exact 95 per cent upper bound in brackets, changing one factor at a time on the same targets. All pooled attacks know the history.

	One release		Six vintages	
Policy	one reading	six readings	one reading	six readings
No suppression	0.089 (0.097)	0.158 (0.168)	0.191 (0.201)	0.260 (0.272)
k-anonymity, $k = 5$	0.074 (0.081)	0.141 (0.150)	0.183 (0.194)	0.269 (0.281)
k-anonymity, $k = 10$	0.070 (0.076)	0.130 (0.139)	0.179 (0.190)	0.244 (0.256)

The adversary is often right when it acts, and it seldom acts on one release. Under $k = 5$ it passes the decision threshold on 48 of 4,000 attempts and is right on 37 of those, a precision of 0.771 resting on 48 decisions, or about 108 attempts per correct identification. An evaluation that reports only the average hides the precision, and one that reports only the precision hides how few attempts it rests on.

The average also conceals a subgroup. With no suppression the largest-firm class reaches 0.248, with a bound of 0.297, and no row of Table 13.2 passes a subgroup control at 0.12 even on one release. Large borrowers are few, so their cells are small and their exposures far apart, and both effects push the same way. The three other attacks are indifferent to the

lever: under $k = 5$ an alleged relationship is true 0.324 of the time, a unit change is a single move 0.623 of the time, and a confident reading of arrears status is right 0.946 of the time against 0.929 for guessing no arrears for everyone, a gain of 0.017.

Naledi's assessment of Tomás's file, one release at a time, passes with no suppression. She writes the result into a draft record and then adds a line she has learned to add: the file goes out monthly, and the pseudonym persists. The release she assessed is not the release Tomás will hold.

13.4 The floor each adversary requires

Fix the tolerance at $\tau = 0.12$ and choose the minimum cell count by the exact upper bound. Because the level is chosen from a grid by the same bounds that certify it, the choice uses fixed-sequence testing (Chapter 5): levels are tested from $k_{\min} = 300$ down on one draw of 4,000 targets, the procedure stops at the first level whose bound exceeds the tolerance, and the floor is the last level that passed. The chosen level is then audited on a second, independent draw. Where confirmation fails, the next stricter level is confirmed on the same fresh draw.

The adversary who pools six vintages with one reading and knows the history shows how the procedure runs. On the selection draw the upper bounds, from the strictest level down, are 0.042 at 300, 0.059 at 200, 0.073 at 150, 0.081 at 100, 0.101 at 75, 0.111 at 50 and 0.139 at 40. The sequence stops at 40 and selects 50. On the fresh draw, 50 gives 458 successes in 4,000, a rate of 0.1145 with a bound of 0.1231, and fails. The next stricter level, 75, gives 384 in 4,000, a rate of 0.096 with a bound of 0.104, and is confirmed. The selection step carries the error bound; the confirmation guards against reporting a floor that passed by chance near the tolerance, which is what happened at 50. Table 13.4 gives the result for six adversaries.

TABLE 13.4: The floor each adversary requires on the synthetic register, tolerance 0.12. Floors are chosen by fixed-sequence testing from $k_{\min} = 300$ down on 4,000 targets and confirmed on a second draw (rate, with exact 95 per cent upper bound). Loans and exposure are the percentages of the first release kept. The subgroup floor requires every size class to meet the tolerance, with the largest firms on a stratified draw, from $k_{\min} = 200$ down; it was not confirmed on a second draw. "None" means no level up to the top of the grid passes.

Adversary	Floor	Confirmation	Loans	Exposure	Subgroup
One release, one reading	1	0.084 (0.092)	100	100	20
Six vintages, one reading, no history	1	0.083 (0.091)	100	100	–
Six vintages, one reading, history	75	0.096 (0.104)	45	11	200
Six vintages, reading per vintage, no history	20	0.105 (0.114)	85	52	–
Six vintages, reading per vintage, history	150	0.101 (0.109)	20	4	none
Insider: exact exposure, history	none	–	–	–	–
Exposure published to one significant figure					
One release, one reading	1	0.079 (0.086)	100	100	–
Six vintages, one reading, history	50	0.111 (0.119)	58	20	–
Six vintages, reading per vintage, history	150	0.085 (0.092)	20	4	–
Insider: exact exposure, history	none	–	–	–	–

Read down the first block and the floor climbs with what the adversary knows. Against one release the floor is 1, no suppression, with a confirmation rate of 0.084 and a bound of 0.092. Pooling six vintages without the history adds nothing, as Proposition 9.3 predicts: the

floor stays at 1 and the confirmation is 0.083 (0.091). With the history the floor is 75. With a reading per vintage and no history it is 20, and with both it is 150, confirmed at 0.101 (0.109). The insider, who knows the history and reads exposure with a standard deviation of 0.01 on the log scale, is not stopped by any floor: its success is 0.772 at $k_{\min} = 1$ and still 0.416 at $k_{\min} = 300$.

The insider's result is not an accident of the grid. A precise reading separates candidates in a cell of any size, and Proposition 13.2 says by how much.

Proposition 13.2 (A precise reading defeats any floor)**.** *Let the candidates for a target t have published log exposures $x_1, \dots, x_n$, let the adversary read $y = x_t + e$ with $e \sim N(0, \sigma^2)$, and let it name a candidate that maximises the Gaussian likelihood, breaking ties uniformly at random.*

1. *If the gap $g = \min_{j \neq t} |x_j - x_t|$ is positive, the attack names the target with probability at least $2\Phi\big(g/(2\sigma)\big) - 1$, whatever the number n of candidates; for fixed published values this tends to one as $\sigma \to 0$.*

2. *If $r - 1 \geq 1$ other candidates publish the same value as the target, the attack names the target with probability at most $1/r$, whatever σ.*

The first part is why a floor cannot stop the insider: a floor bounds n from below, and the bound in part 1 does not depend on n. A larger cell does shrink the typical gap g, which is why the insider's success falls from 0.772 to 0.416 across the grid without approaching the tolerance. The second part is why rounding works on the insider where floors do not. Published to one significant figure, exposures tie, and the insider's success at $k_{\min} = 1$ halves, to 0.388. The outside adversaries, whose readings are already coarse, barely notice the rounding: their floors are 1, 50 and 150.

Pieter is a credit analyst at one of the Bank's reporting lenders. He knows, to the euro, the exposure of every loan his bank has reported, and the months in which each was live. In the language of Table 13.4 he is the last row. Naledi does not need to assume that Pieter would misuse a file to see what his row implies: whatever a loan-level file attaches to a record, a reporting lender can find that record, and no floor changes this.

13.5 What each floor keeps

The floors have a cost, and it falls on credit volume more than on loan counts. Figure 13.2 plots the share of first-release loans and of exposure kept at each floor of the grid. The two curves start together and separate at once. At a floor of 20 the file keeps 85 per cent of loans and 52 per cent of exposure. At 75 it keeps 45 per cent of loans and 11 per cent of exposure, and from 75 up it keeps no loan to a large firm. At 150 it keeps 20 per cent of loans and 4 per cent of exposure. Even the floor of 50 that selection first picked for the history adversary keeps 58 per cent of loans and 20 per cent of exposure.

The gap between the curves has a simple source, which Proposition 13.3 states. A floor removes whole cells, the cells with the fewest loans; if those cells hold the larger loans, the exposure share kept must fall below the loan share kept.

Proposition 13.3 (Suppression costs exposure faster than loans)**.** *Let cell c of a release hold n_c loans with mean exposure m_c, and let a floor keep the set S of cells with $n_c \geq k$. Write $L_S = \sum_{c \in S} n_c / \sum_c n_c$ for the share of loans kept, E_S for the share of exposure kept, and μ_S and μ for the*

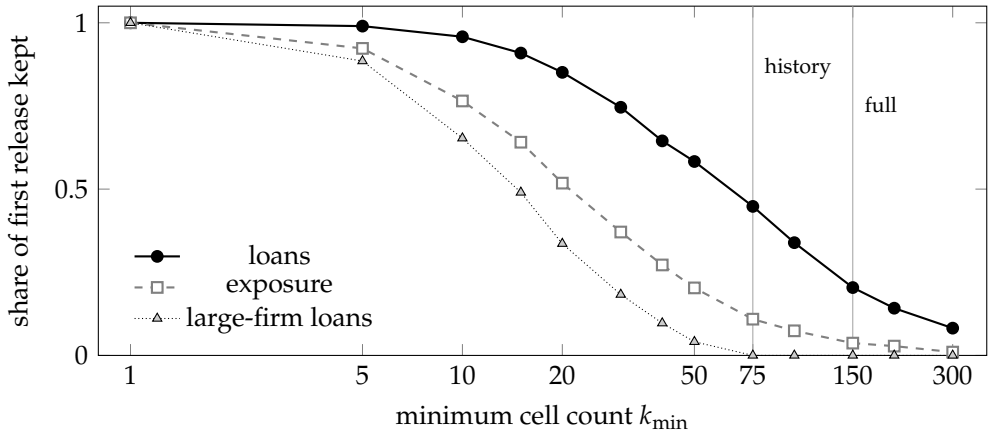


FIGURE 13.2: What each floor keeps of the first release on the synthetic register: loans (solid, filled circles), exposure (dashed, open squares) and loans to large firms (dotted, triangles). Vertical lines mark the floors for the pooled adversary with the history (75) and with a reading per vintage as well (150).

mean exposure per loan in S and in the whole release. Then

$$E_S = L_S \frac{\mu_S}{\mu}. \tag{13.3}$$

If every removed cell has a mean exposure at least as large as every kept cell, then $E_S \leq L_S$, with equality only if $\mu_S = \mu$.

The identity holds for any floor on any file. On the register, at the floor of 150, $E_S/L_S = 0.037/0.204 = 0.18$, so the loans that survive the floor carry on average less than a fifth of the mean exposure of the first release. The condition of the second part is sufficient and not necessary, and on this register it holds approximately: large firms are 3 per cent of borrowers, their log mean exposure is two log units above the next class, and their few loans sit in the thinnest cells. The misconception the chapter opened with assumes $\mu_S = \mu$, which is the one case in which counting records measures the loss.

A file that meets the tolerance against a history-aware adversary is therefore not the file its users asked for. It has lost the borrowers that concentration and supervisory analysis are about, and most of the borrowers with several lenders, who fall from 6,942 at no suppression to 2,747 at 75 and 1,194 at 150. The utility measure of Table 13.1 makes the same point from the side of the analyses: at $k_{\min} = 20$ the weighted loss is 0.254, almost all of it from sectoral exposure and lender-level supervision, at 0.531 and 0.435, while concentration and the arrears ratio lose 0.015 and 0.016. Against such an adversary a loan-level release is the wrong product. Aggregates with a contributor rule, or access to the loan-level file in a secure setting under the Five Safes (Desai et al., 2016), where the declared adversary no longer holds the file, are the realistic options.

Size-class floors. The subgroup control asks more. When every size class must meet the tolerance, with the largest firms audited on their own stratified draw of 4,000 targets, the per-release floor is 20, pooling with the history needs 200, and for the full pooled attack no level up to 200 passes (last column of Table 13.4). From $k_{\min} = 75$ no large-firm loan is released, so that class has no target to protect and the subgroup floors above 75 are set by the other classes. These floors were selected by the same fixed-sequence rule and not confirmed

on a second draw. Under the subgroup control the per-release answer is therefore 20 and not 1.

Tomás reads the figure when Naledi sends it to him. His study is about large firms, and the curve for large-firm loans reaches zero at the floor his file would need. He asks whether he could work on the full file in the Bank's secure room instead. That is the answer Naledi had expected to give him, and it is now in the record with its reason.

13.6 How far the floors move

Every floor so far rests on one tolerance, one register draw and one value of the adversary's reading precision. Each of the three can move it.

The tolerance. Table 13.5 repeats the selection at four tolerances from the same draw, before confirmation, which is why pooling with the history shows 50 at a tolerance of 0.12 where Table 13.4 shows the confirmed 75. At 0.05 the per-release floor is 40, pooling with the history needs 300, and the full attack has no floor up to 300; at 0.20 the three floors are 1, 5 and 30. The insider has no floor at any of them. A committee unsure of its tolerance should read the whole table, since the floor moves by an order of magnitude across this range, and Chapter 8 sets out how a stated harm weight implies a tolerance.

TABLE 13.5: Floor selected at four tolerances on the selection draw, before confirmation. "None" means no level up to 300 passes.

Tolerance	One release	Pooling, history	Full pooled	Insider
0.05	40	300	none	none
0.09	5	100	200	none
0.12	1	50	150	none
0.20	1	5	30	none

The register draw and the reading precision. Table 13.6 repeats the choice over 20 independent register draws and three values of the standard deviation of the adversary's log-exposure reading, with 2,000 targets per level. At 0.6, the value used so far, the per-release floor is 1 in every draw, pooling with the history gives a median of 75 with a range of 40 to 75, and the full attack a median of 150 with a range of 100 to 150. With a reading twice as precise, at 0.3, the medians are 20, 150 and 300, and for the full attack no level up to 300 passes in 3 of the 20 draws. At 1.0 they are 1, 40 and 75.

TABLE 13.6: Floors over 20 independent register draws, 2,000 targets per level, by the standard deviation of the adversary's log-exposure reading. Median floor with the smallest and largest in brackets; the superscript counts draws in which no level up to 300 passed.

Reading SD	One release	Six vintages, history	Six vintages, history, reading each
0.3	20 [10, 20]	150 [150, 200]	300 [300, none][3]
0.6	1 [1, 1]	75 [40, 75]	150 [100, 150]
1.0	1 [1, 1]	40 [30, 40]	75 [50, 100]

The reading precision moves the floor more than anything else varied, and it is the input an institution most needs to argue for. At fixed precision the floor stays in a narrow

band across register draws. For the full attack at $k_{\min} = 150$ the rate has a between-register standard deviation of 0.0073 over the 20 draws, with a chi-square interval on 19 degrees of freedom from 0.006 to 0.011. The binomial standard error of one draw of 2,000 targets at the mean rate is 0.0064, inside that interval, so the register draw may add nothing or may add a design standard deviation of up to about 0.009. Twenty draws do not estimate the design term well. Chapter 4 treats this split of variance between the simulation and the design.

A chain of audits. The same choice can be run as one release chain, a fixed sequence of tests from the strictest floor down with all three outside attacks declared at once (Chapter 6). Over 20 independent target streams it chose 150 in 19 and 100 once, against an estimated true floor of 100, and no stream chose an unsafe floor. The gap to 100 comes from the full attack's risk at that floor, 0.116, which lies so close to the tolerance that the audit's capped bet certifies it only by chance.

Rotating the pseudonym. The pooled attacks match loans across vintages through a pseudonym that persists. Chapter 9 discussed rotation as a remedy; the register measures it. With exposure published exactly, the pair of key and exposure is unique for every first-release loan at every floor tested, a share of 1.000, so the published fields alone relink every loan and rotation gives no protection. With exposure rounded to one significant figure the pair relinks 0.104 of loans at $k_{\min} = 1$, falling to 0.057 at 20 and 0.010 at 150. Rotation together with rounding would then remove most of the linkage that the persistent pseudonym gives. These shares count loans that could be joined; the attacks were not rerun with rotation, so they bound linkability and are not risk figures.

What the attack needs. Table 13.7 withdraws the adversary's auxiliary information one field at a time under k-anonymity at $k = 5$, then keeps all four fields and misstates the dispersion of the exposure reading in the adversary's own likelihood.

TABLE 13.7: What the attack needs, and an adversary with a wrong model, under k-anonymity at $k = 5$. Calibration error is the gap between the average posterior the adversary reports and the share of its guesses that are right.

Adversary holds	Fields	Risk	Precision	Calibration error
All four fields	4	0.081	0.672	0.000
No exposure reading	3	0.025	–	0.002
No size class	3	0.042	0.500	0.002
Region and exposure only	2	0.004	–	0.001
Assumed $\sigma \times 0.5$	4	0.079	0.384	0.073
Assumed $\sigma \times 1$	4	0.079	0.667	0.003
Assumed $\sigma \times 2$	4	0.079	1.000	0.039

Without the exposure reading the adversary reaches 0.025 and never passes the decision threshold, because a released cell has at least five members and nothing separates them. Holding only region and the reading it reaches 0.004. The key narrows the candidates to a cell and the reading separates the members of that cell, and neither does much alone. When the adversary misstates its own dispersion, its success stays at 0.079 whether it halves or doubles the true value, because the highest-scoring candidate does not change when the same density is rescaled over the same candidates. What moves is calibration, from 0.003 to

as much as 0.073. An overconfident attacker still identifies the same records. An institution that estimates risk under a misspecified adversary model reports a number whose stated confidence is wrong, and the falsification tests of Chapter 15 are the check that catches it.

Grace reads the decision record some months later. She finds the selection at 50, the failed confirmation at 50, the confirmed 75, and the note that 150 is required if the adversary also tracks the loan's exposure month by month. What she checks is whether the record says which adversary the Bank declared and why. It does: the outside researcher without the history, for the secure-room file, and the full pooled adversary for anything published.

13.7 Arguing the adversary's reading precision

The floors depend on how well the adversary reads the attribute it uses, and that precision cannot be estimated from the synthetic register. It has to be argued, and the argument has five steps. First, list the sources an outsider could hold for the key attributes: public records, a commercial file, an earlier release, a colleague's knowledge. Second, for each source, state its error rate or noise and the evidence for it. Third, state how an adversary would combine sources and whether combining lowers the error. Fourth, run the floors across the plausible range of precision and not at one value, and report the range, in the form of Table 13.6. Fifth, sign the scenario with the range and the reason for its upper end. A floor that is stable across the range can be carried into a signed threshold. A floor that moves a lot is reported with its range.

13.8 What the evaluation does not settle

The register is synthetic, and every result in this chapter holds on the synthetic register only. Its parameters are chosen, and two of the generator's simplifications bear on the floors. Exposure is constant across vintages, so without the history the other vintages carry no information; with amortising balances they would. And whether a loan is live is independent of its key and exposure. The credit module would take a real register in place of the simulated one, but it has not been run on one.

The adversary is one adversary, strong in some respects and weak in others. It holds the key exactly, uses a persistent pseudonym, knows the true error model of its reading and, in the pooled attacks with history, the loan's live vintages. It does not use arrears status, lender identity or a longer history. The evaluation is therefore neither a lower nor an upper bound on what an adversary can achieve. Its value is comparative, since every policy faces the same attacker, and conditional, since every floor holds for the adversary declared.

Finally, the policies are compared as they are usually configured. A contributor rule calibrated on the pooled programme would do better than the row for it in Table 13.2: the same lever, calibrated on the programme, changes the outcome.

13.9 The answer

The floor depends on what the declared adversary knows. At a tolerance of 0.12 on this register it is 1 against an adversary with one release or without the loan's history, 75 against one who knows the history, 150 against one who also tracks the exposure in every vintage, and no floor at all against an insider who reads exposure precisely. Protecting every size

class moves the per-release floor from 1 to 20 and the pooled floors to 200 or beyond. What is left of the file falls much faster than the count of loans: at 75 the file keeps 45 per cent of loans and 11 per cent of exposure, at 150 it keeps 20 per cent and 4 per cent, and from 75 up it keeps no loan to a large firm. Suppression removes thin cells, and in a credit register the thin cells carry the money, so the cost of a floor has to be stated in exposure and in the analyses lost. Against a history-aware adversary the loan-level file is the wrong product, and the institution should offer aggregates with a contributor rule or access in a secure setting.

IN PRACTICE: A LOAN-LEVEL CREDIT FILE

Declare the adversaries before choosing a floor, and for each say whether it knows the loan's history and how precisely it reads exposure, since these two inputs move the floor more than anything else. Choose the floor by fixed-sequence testing on the exact upper bound and confirm it on a fresh draw. Report the cost of the floor in exposure, in loans to each size class and in the analyses the register serves, not in records alone. Assess the other four harms with their own levers: an attribute rule for status, a predictive-value tolerance for alleged relationships, a contributor or dominance rule for aggregates, and a count rule for differencing. If the floor a declared adversary requires removes the borrowers the users need, offer a secure setting or aggregates in place of the file. Rotating a keyed pseudonym helps only when the published fields cannot relink loans on their own, so test the relink share first.

In the software. The register and the five harms are in `privacy_risk.credit` (`make_credit_register`, `borrower_identification`, `relationship_disclosure`, `status_inference`, `reconstruction_from_totals`, `differencing_risk`). The policies and attacks are in `privacy_risk.evaluation` (`identification_attack`, `programme_attack`, `relink_share`), and `scripts/programme_robustness.py` produces the floors, the tolerance table, the replication and the relink shares. The relink share is short enough to show whole:

```
def relink_share(register, policy, exposure_digits=None):
    key = register.key()
    first = released_rows(register, policy, 0)
    published = round_significant(register.exposure[first], exposure_digits)
    pairs = np.stack([key[first], np.log(published)], axis=1)
    _, inverse, counts = np.unique(pairs, axis=0, return_inverse=True,
                                   return_counts=True)
    return float(np.mean(counts[inverse.ravel()] == 1))
```

13.10 Proofs

Proof of Proposition 13.1. Since A and D are independent Poisson variables with mean λ, $\mathbb{P}(A = a, D = d) = e^{-2\lambda}\lambda^{a+d}/(a!\,d!)$. The event $|A - D| = 1$ is the disjoint union of $\{A = d+1, D = d\}$ and $\{A = d, D = d+1\}$ over $d \geq 0$, and the two families have equal probability by symmetry, so

$$\mathbb{P}(|A - D| = 1) = 2e^{-2\lambda} \sum_{d=0}^{\infty} \frac{\lambda^{2d+1}}{d!\,(d+1)!}.$$

The event $A + D = 1$ within it is the case $d = 0$ of both families, with probability $2e^{-2\lambda}\lambda$. Dividing gives

$$\mathbb{P}\big(A + D = 1 \,\big|\, |A - D| = 1\big) = \frac{\lambda}{\sum_{d\geq 0} \lambda^{2d+1}/(d!\,(d+1)!)} = \Big(\sum_{d=0}^{\infty} \frac{\lambda^{2d}}{d!\,(d+1)!}\Big)^{-1}.$$

The series $\sum_{d\geq 0}(\lambda)^{2d+1}/(d!\,(d+1)!)$ is the power series of $I_1(2\lambda)$, which gives the second form. Each term $\lambda^{2d}/(d!\,(d+1)!)$ with $d \geq 1$ is strictly increasing in $\lambda > 0$ and the $d = 0$ term equals one, so the sum is strictly increasing and its reciprocal strictly decreasing, and the sum tends to one as $\lambda \to 0$. □

Proof of Proposition 13.2. The Gaussian log likelihood of candidate j is $-(y - x_j)^2/(2\sigma^2)$ up to a constant, so the attack names a candidate nearest to y. For part 1, suppose $|e| < g/2$. Then $|y - x_t| = |e| < g/2$, and for every $j \neq t$, $|y - x_j| \geq |x_j - x_t| - |e| > g - g/2 = g/2$. The target is then the unique nearest candidate and is named. Hence the probability of success is at least $\mathbb{P}(|e| < g/2) = 2\Phi(g/(2\sigma)) - 1$, which does not involve n and tends to one as $\sigma \to 0$ for fixed $g > 0$. For part 2, the r candidates that publish the target's value have the same likelihood for every y. Whenever the target is among the maximisers, so are the other $r - 1$, and a uniform choice among at least r maximisers names the target with probability at most $1/r$. Whenever the target is not among the maximisers, the attack fails. Averaging over e gives the bound. □

Proof of Proposition 13.3. Let $N = \sum_c n_c$ and $N_S = \sum_{c\in S} n_c$. The total exposure is $\sum_c n_c m_c = N\mu$ and the exposure kept is $\sum_{c\in S} n_c m_c = N_S \mu_S$, so $E_S = N_S\mu_S/(N\mu) = L_S\,\mu_S/\mu$, which is (13.3). For the inequality, let $\bar{S}$ be the removed cells and $\mu_{\bar{S}}$ their mean exposure per loan, defined when $\bar{S}$ holds a loan; if it holds none, $L_S = E_S = 1$. Under the condition, every m_c with $c \in S$ is at most every $m_{c'}$ with $c' \in \bar{S}$, so the loan-weighted mean over S is at most that over $\bar{S}$: $\mu_S \leq \mu_{\bar{S}}$. Since $\mu = L_S\mu_S + (1 - L_S)\mu_{\bar{S}}$ is a convex combination of the two, $\mu_S \leq \mu$, and (13.3) gives $E_S \leq L_S$. Equality requires $\mu_S = \mu$. □

Chapter 14

Proposing attacks with a model

Every assessment in this book rests on a declared list of attacks, and the list is written by people. People miss things. A natural response is to let a model write part of the list: a network that learns from past attacks where the risk is high, or a large language model that has read the literature on re-identification and can propose scenarios an analyst would not think of. The belief behind that response has two parts, and in its strongest form both sound right. A model that proposes attacks makes the audit stronger, because more and better-aimed attacks can only find more of what is there. And a large language model will find attacks that random search misses, because it brings knowledge that a random draw does not have.

The repository tested both parts, and neither survived as stated. On 30 state samples of a public Census file, a neural search that learned from measured attacks confirmed almost every setting it chose as high risk, 31.57 of 32, and covered fewer distinct kinds of high-risk scenario than a scrambled Sobol sequence that learned nothing: 10.07 scenario cells against 13.83. A small open-weight language model given the same budget covered 6.05 cells against 13.65 for Sobol on the states where it produced a valid answer, because it copied the front of the list it was shown. And on the credit register of Chapter 13, a logistic model saved as many audit draws as the network it was meant to be a baseline for.

This chapter asks one question: *what can a model that proposes attacks add to a release decision, and what has to be kept out of its reach for the decision to stay valid?* The answer is built in four steps. A network can plan an audit and make it cheaper without touching its error bound. A language model can enter only as an injector of scenarios into a closed network, through review and measurement. The experiments that test whether either finds more risk are reported in full, including the ones that failed. And the provenance ledger and a penetration test show what the record must carry for an auditor to trust any of it.

14.1 Two jobs a model might do

A model can enter an audit in two places. It can help *plan* an audit whose attacks are already declared: predict how each declared attack will fare at each candidate setting, and so choose where to start and how much to bet on each draw. Or it can help *search*: propose new scenarios, new adversaries, new settings, and so change what is declared. Planning changes the cost of the audit. Search changes its scope. Neither job, done well or badly, is the job of deciding.

The book's decision rule fixes where the decision lies. For a proposed setting θ and a

finite declared set $\mathcal{J}$ of scenarios, each fixing an adversary, a harm, a target group and a vintage, let $r_j(\theta)$ be the probability that one fresh attack under scenario j succeeds, and let $\bar{r}_j(\theta)$ be an upper confidence bound from an audit. The release is admissible when every audited margin

$$z_j(\theta) = 1 - \frac{\bar{r}_j(\theta)}{\tau_j}, \qquad j \in \mathcal{J}, \tag{14.1}$$

is nonnegative, where τ_j is the tolerance for scenario j. A coordinate that was never audited receives $\bar{r}_j = 1$ and fails. The vector of margins is what the decision record carries, because its minimum alone hides which group or which pooled attack constrains the release. A model's output is a predicted attack frequency. It is not an audited margin, and nothing a model writes can stand in for $\bar{r}_j$.

The Bank's data innovation group sends the release committee a proposal: a language model that drafts attack scenarios for each release, so that Naledi's lists would no longer depend on what she happens to think of. The chair forwards it to her with one line: does this make the audit stronger? Naledi answers that she does not know yet, and that the repository has the experiments that would tell her.

14.2 A network that plans the audit

The release chain of Chapter 6 audits floors from the strictest down, and at every draw it needs a bet, and it needs a floor at which to start. The neural release pipeline supplies both from a network fitted to earlier attacks. It runs in four stages, shown in Figure 14.1, and keeps their evidence apart.

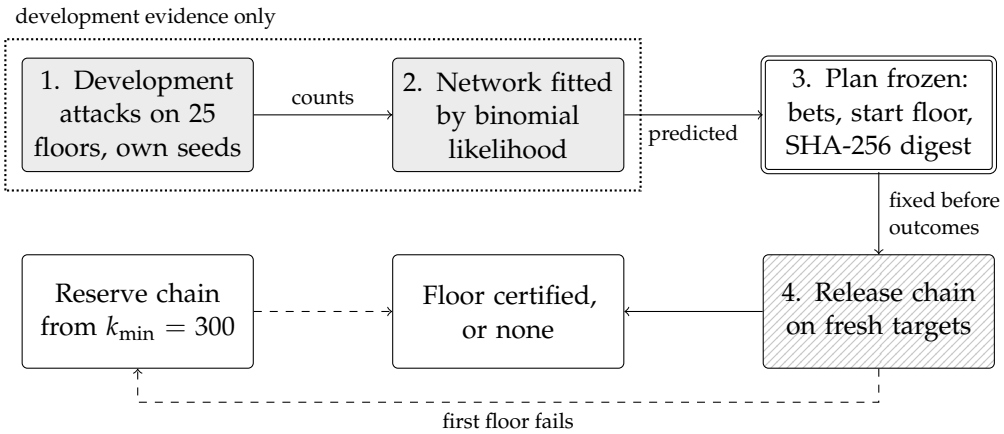


FIGURE 14.1: The neural release pipeline. Shaded boxes use development attacks only; the double box is the plan, frozen with its digest before any final outcome exists; the hatched box is the only stage whose outcomes can certify a floor. Dashed arrows are the reserve path, used only if the chain fails at its first floor.

In the first stage the three declared attacks of Chapter 13 (one release with one reading; six vintages with one reading and the history; six vintages with a reading per vintage and the history) are run on a dense grid of 25 floors with 400 development targets each, on seeds the final audit never uses. In the second, a network with two hidden layers of 16 units is fitted to these counts by the binomial likelihood, one output per declared attack, with a third of the floors held out for early stopping; a logistic model is fitted to the same counts

as a baseline. The features are ones an institution can compute from the register without running an attack: the floor, the share of loans kept, the mean size of the released cells, and the share of released loans not live in every vintage.

In the third stage, before any final outcome exists, the fitted model predicts every attack's success at every floor of the 13-floor release grid. The plan records these predictions, a bet for each draw derived from them, the predicted floor (the last floor, from 300 down, at which every predicted risk lies below the tolerance) and a start floor one step stricter. The plan is serialised and hashed with SHA-256; the network's plan in the reported run has the digest beginning `982f0d60` and the logistic plan the digest beginning `ac74ebfb`. In the fourth stage the release chain audits the plan on fresh targets. If it fails at its first floor, a reserve of a fifth of the error budget runs a second chain from $k_{\min} = 300$.

Because every bet is fixed before its outcome and the order of the chain is fixed at registration, the network's accuracy does not enter the error bound. A poor network costs attacks. It cannot raise the chance of certifying an unsafe floor. The argument for the betting chain is the test-martingale argument of Chapter 6; the version for a fixed family of exact bounds, which SENTRY-R below uses, is Proposition 14.1.

Proposition 14.1 (Selection-safe certification)**.** *Let $\mathcal{D}$ be all development information: every development attack, every proposal, every model fitted and every tuning choice. Conditional on $\mathcal{D}$, let a finite family of M pairs (θ, j) of candidate setting and scenario, their tolerances τ_j and their trial counts $B_{\theta j}$ be fixed, and suppose each final count satisfies $K_{\theta j} \mid \mathcal{D} \sim \mathrm{Binom}(B_{\theta j}, r_j(\theta))$. Let $\bar{r}_{\theta j}$ be the exact one-sided Clopper–Pearson upper bound at level α / M. Then, with probability at least $1 - \alpha$, $r_j(\theta) \le \bar{r}_{\theta j}$ for every pair at once, and any rule that certifies only candidates whose margins* (14.1) *are all nonnegative certifies a candidate with $r_j(\theta) > \tau_j$ for some j with probability at most α. This holds however $\mathcal{D}$ was produced, including by a model or a language model used adaptively.*

Table 14.1 compares five designs over 20 attack streams on the same final draws, at a tolerance of 0.12 and $\alpha = 0.05$. Every design chose 150 in almost every stream and none chose an unsafe floor. The difference is cost. With bets from the network the chain used a median of 5,620 attacks against 6,422 with bets from the running rate; with the network's start floor as well it used 5,044, about a fifth fewer, and never needed the reserve.

TABLE 14.1: The neural release pipeline on the synthetic credit register: 20 attack streams, tolerance 0.12, $\alpha = 0.05$. All designs are release chains over the three declared attacks on the same final draws, and differ only in where the bets and the start floor come from. Floors chosen, with the number of streams in brackets, and the median number of attacks.

Source of bets and start	Floor chosen	Attacks
Running-rate bets, chain from 300	100 (1), 150 (19)	6,422
Logistic bets, chain from 300	100 (2), 150 (18)	5,572
Network bets, chain from 300	100 (2), 150 (18)	5,620
Logistic bets and start floor	100 (1), 150 (19)	5,019
Network bets and start floor	100 (1), 150 (19)	5,044

Depth does not explain the saving. The logistic model's plan saved as many attacks as the network's, 5,019 against 5,044, and its predictions were closer to the true risks. Figure 14.2 shows what the two models learned. In each panel the solid line is the network's prediction at the floors of the release grid, fixed before the final audit, and the dotted line is the true risk estimated from 100,000 separate targets per floor. The crosses are the development rates, each from only 400 attacks, which is all the models saw. Both recover the shape

of the three curves: flat and low for a single release, falling steeply with the floor for the two pooled attacks, and crossing the tolerance near $k_{\min} = 100$ for the full pooled attack, which is where the true floor lies.

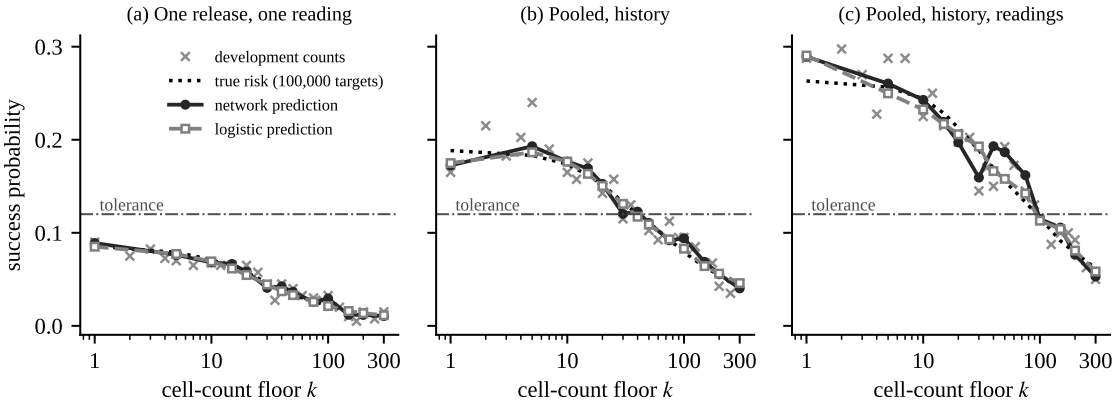


FIGURE 14.2: Frozen predictions against the truth on the synthetic credit register, for (a) one release with one reading, (b) six vintages with the history, and (c) six vintages with the history and a reading per vintage. Crosses are development rates from 400 attacks each; the dotted line is the true risk from 100,000 targets per floor; filled and open markers are the network's and the logistic model's predictions; the dash-dotted line is the tolerance.

Table 14.2 puts numbers on the figure. The two models reach almost the same held-out log loss, 0.3482 for the network and 0.3480 for the logistic model, and both predict a floor of 100 and start the chain at 150. Against the true risks over the 39 coordinates of the release grid, the network's mean error is 0.008 and the logistic model's 0.004. The network fits the development data as well as the simpler model and no better, and its largest errors sit in the middle of panel (c), where it follows the scatter of the development rates. On this register its extra capacity goes into following noise.

TABLE 14.2: What the two planning models learned from the development attacks. Log loss is per development attack; the epoch is where held-out loss was lowest. Errors compare each frozen prediction with the true risk from 100,000 targets over the 39 coordinates of the release grid.

	Log loss			Error		Floor	
Model	Training	Held out	Epoch	Mean	Largest	Predicted	Start
Network, two hidden layers	0.3240	0.3482	1146	0.008	0.031	100	150
Logistic model	0.3245	0.3480	2067	0.004	0.027	100	150

The pipeline keeps the network as its default because its architecture extends to many more features and declared attacks than this register has, and reports the logistic model beside it so that a reader can see when depth has not paid. The gain on this register comes from basing bets and the start floor on development evidence. Any model that turns development counts into reasonable predictions would have given it.

Grace's question about the pipeline is the one an auditor should ask: how does she know the plan was fixed before the final draws? The record gives her the plan's digest and the time it was written, and the final audit refuses to run against a plan whose digest has changed. Whether that is enough depends on who holds the record, a point the chapter returns to.

14.3 A language model as an injector

Planning leaves the declared attacks alone. Search does not, and a language model is the obvious candidate for search, because it can describe a new adversary, a pooled release or a new auxiliary source in words. SENTRY-R, short for scenario-enriched neural triage for release review, is the repository's workflow for letting it do so without giving it any power over the decision. The phrase the repository uses is exact: the language model is an injector to a closed neural network. *Closed* names a restricted data path. It does not mean an air gap, frozen weights or a proof of privacy.

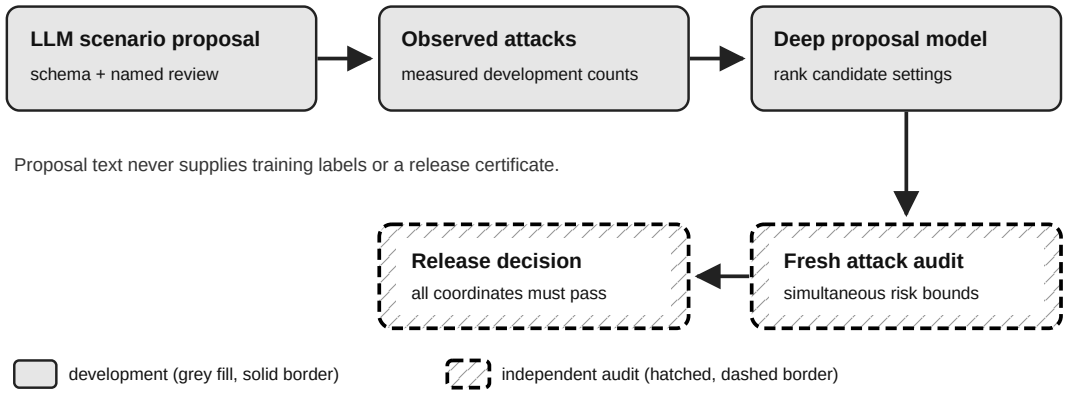


FIGURE 14.3: The SENTRY-R boundary. A scenario proposed by a language model passes a fixed schema and a named review, and can add a scenario but never an outcome. Only counted development attacks train the proposal model, whose ranking fixes a shortlist. Only the fresh final audit, with simultaneous bounds over every coordinate, reaches the release decision. The upper row is development; the lower row, drawn with a different fill and border, is the final audit.

Figure 14.3 shows the path, and the sequence is strict. The model returns a JSON array of objects with exactly three fields: an identifier, a scenario description and a vector of numeric features. A deterministic validator checks the feature names and their predeclared bounds. The calling code, not the model, writes the provenance: the model's name as the caller states it and the SHA-256 hash of the exact prompt sent. A named human reviewer approves or rejects each proposal with a reason. A separate attack evaluator runs each approved scenario and records the count of successes and the number of trials. Only those measured counts enter the training data of the proposal network, which ranks candidate settings and fixes a shortlist. Only the final audit, on draws no earlier stage has seen, decides. The software refuses success counts written by the model, refuses responses larger than 256,000 bytes before parsing them, and refuses more than 32 proposals in one batch. The proposal text is stored as data and never executed.

The reason for refusing a model's counts is statistical, and Proposition 14.2 states it. A proposal network trained on measured counts learns the attack's success frequency; trained on anything else, it learns that thing.

Proposition 14.2 (Only measured counts teach the risk)**.** *For a feature vector x and a trial count B fixed before outcomes, let the training label be a count K out of B with $\mathbb{E}[K \mid x, B] = B\pi(x)$, and let the per-pair loss of predicting $p \in (0,1)$ be $\ell(p) = -[K \log p + (B-K)\log(1-p)]/B_\bullet$ for a*

normalising constant $B_\bullet > 0$. Then

$$\mathbb{E}[\ell(p) \mid x, B] - \mathbb{E}[\ell(\pi(x)) \mid x, B] = \frac{B}{B_\bullet} \mathrm{KL}\big(\mathrm{Bern}(\pi(x)) \,\|\, \mathrm{Bern}(p)\big) \;\geq 0, \tag{14.2}$$

with equality only at $p = \pi(x)$. If K is replaced by a generated count $\widetilde{K}$ with $\mathbb{E}[\widetilde{K} \mid x, B] = B\widetilde{\pi}(x)$, the minimiser becomes $\widetilde{\pi}(x)$, whatever the attack's true success probability.

A prompt that asserts a small risk is therefore no evidence of a small risk, however plausible its prose. The same identity explains the order of the stages: the network can only be as right about risk as the measured attacks it is trained on, and the final audit, which the network never sees, is what checks it.

14.3.1 The provenance ledger

The workflow is only as good as the record that it was followed. The repository's provenance ledger appends every stage to one hash-chained file: budget, proposal, review, measurement, training, freeze, audit. Each entry carries the hash of the one before and, with a key held by the operator, an HMAC-SHA256 signature of its own hash. Entries hold digests and identifiers only, never a proposal's text and never a count. The ledger enforces three rules. Order: a review is refused unless its proposal is on the chain, a measurement unless an approving review is, a training run unless every measured proposal it uses is, a freeze unless its model is, and an audit unless its freeze is. Budget: a programme's error budget is declared once, and every audit debits its α, so a new shortlist after a failed audit costs budget and does not restart at the nominal level. Closure: once a freeze has been audited, a training run may not mix its development evidence with evidence measured after the audit, so an audit outcome cannot steer the next model.

The budget rule answers a measured danger. In a synthetic experiment with a true risk of 0.201, a tolerance of 0.2 and 500 trials per attempt, so that every certificate is false, an operator who retries a failed audit 20 times at the nominal 5 per cent level certifies in 50.975 per cent of 20,000 simulated programmes, with an exact interval of 50.280 to 51.670 per cent. Preallocating a total of 0.05 across the 20 attempts gives 3.875 per cent (3.612 to 4.152). Chapter 6 treats retries in general; the point here is that a language model in the development loop is one more source of reasons to try again.

What the ledger cannot do is stated in the repository as plainly as what it can. An operator who holds the key and can rewrite the file can rewrite the chain. The budget debit is recorded after outcomes are supplied, so it does not stop a caller from drawing outcomes outside the ledger and discarding them. Provenance is optional, base evidence is supplied by the caller, and reviewer and evaluator identities are labels the caller writes. A deployment needs a registry service with its own storage, keys and time source, a reservation of budget before any draw, and chain heads anchored outside the operator's control. That service is specified and not built. The separation applies the principle of separation of privilege: no single party should hold every key (Saltzer and Schroeder, 1975).

Four integrity probes, written against the supplied prototype, all failed before repair: altered measured counts were accepted; altered measured features were accepted; an injected copy of a validation row could pass once its identifiers were changed; and file-intake metadata could bypass the response-size limit. The repairs commit the digest of the full measured row to the keyed chain and check it when a training artifact is recorded, refuse an injected training vector identical to a validation vector, and bound file input before parsing.

The copy rule can reject legitimate collisions and does not detect near copies or semantic contamination.

Naledi reads the ledger section twice and writes a note for the chair. The tool the innovation group proposes can be fitted into the Bank's process, but only behind a review, an evaluator and a record that the release team does not control. Without those, a language model in the loop is one more way to retry an audit until it passes.

14.4 Does a model find more?

Keeping a model away from the decision makes it safe. Whether it makes the audit stronger is an empirical question, and the repository ran four experiments on it. The first two are synthetic pilots with no language model at all; they test whether adding measured scenarios to the training data helps the network. The third tests whether depth helps the proposal model. The fourth is the public-file search, where a real language model finally entered.

14.4.1 The probability-law pilot

The first pilot drew labels from a hand-written probability law. Ten paired seeds, two laws (one additive, one with an interaction), a logistic model and a network with two hidden layers, 120 base development rows, 80 early-stopping rows and 32 extra measured rows per augmentation arm, each row with 100 simulated trials, gave 120 model fits. Three arms were compared: no augmentation, random augmentation, and a hand-written stress heuristic that deliberately targets a risk corner present in the interaction law.

On the shifted test set of the interaction law, augmentation helped. The deep model's log loss fell from 0.8856 to 0.6252 with random augmentation and to 0.6597 with the heuristic, paired changes of -0.2604 (95 per cent interval -0.4582 to -0.0625) and -0.2259 (-0.3889 to -0.0629). The logistic model with heuristic augmentation had the lowest mean of all, 0.5872. The result looks favourable, and the repository warns against reading it that way. The heuristic's advantage is informed by construction, since it knows where the law puts the risk, and the intervals are exploratory t intervals over ten seeds with no adjustment for multiplicity.

14.4.2 The executable linkage pilot

The second pilot replaced the probability law with attacks that were actually run. Each trial built a synthetic population of 64 people with three quantised quasi-identifiers, chose a target, produced a noisy auxiliary observation and noisy released keys from one to four releases, and counted whether nearest-neighbour linkage named the target. Five paired seeds, 250 shifted test configurations with 1,000 trials each, and separate streams for base, stopping, test and augmentation gave 1,382,000 linkage trials and 30 more model fits.

The result is negative. Table 14.3 gives the test log losses. Augmentation did not establish an improvement over the base for the deep model: the random-minus-base change was -0.0133 with an interval from -0.0342 to $+0.0075$, and the heuristic-minus-base change -0.0127 (-0.0331 to $+0.0077$). The heuristic made the logistic model worse, by $+0.0145$ ($+0.0081$ to $+0.0209$). This is the pilot the preface refers to, in which augmenting the training data did not improve the model. Once the labels came from executed attacks in place of a law written to reward the heuristic, the advantage went away.

TABLE 14.3: Executable linkage pilot: shifted-test log loss by model and augmentation arm, mean over five seeds; 1,382,000 linkage trials. Lower is better.

Model	No augmentation	Random augmentation	Heuristic augmentation
Logistic	0.2849	0.2908	0.2994
Two hidden layers	0.3038	0.2905	0.2911

14.4.3 Depth in the proposal model

The third experiment asked whether depth helps SENTRY-R's proposal model on a toy law where nothing else is in question. For each of ten seeds it drew three disjoint programmes: 320 training rows, 100 rows for early stopping and 2,000 test rows. Both models trained until 150 epochs passed without improvement. The logistic model reached a test log loss of 0.2556 (standard deviation 0.0043 across replicates) and the three-layer network 0.2576 (0.0047). The paired deep-minus-logistic difference was 0.0020 (0.0014), and the deep model was lower in none of the ten replicates. The table shows that the pipeline runs. It does not show that depth helps, and the test programmes come from the same law as the training data, so they do not test shift.

A comparison on public Census records with whole states held out, in which one- and three-layer networks reached lower test loss than a plain logistic model and a quadratic logistic model did better still, is in the repository too. Its linear baselines were not trained to convergence and its differences lie within bootstrap intervals, so the long-form paper treats it as evidence neither for nor against the procedure, and so does this book.

Naledi sends the chair the linkage table with a two-line summary. The pilots show that a network can be trained on measured attacks and that the plumbing works. They do not show that adding proposed scenarios makes the network better at finding risk, and none of them involved a language model.

14.4.4 Searching a public file

The fourth experiment measures search directly, on real records. The source is the 29,501-row public `census2000` extract distributed with the Python `wooldridge` package, version 0.5.0. The first 30 encoded state groups with at least 100 rows were selected before any attack outcome was seen. Within each state the attacker uses a matching-key prior and an independently perturbed reading of log income to identify a record drawn uniformly from the released file. A setting is a tuple of four declared choices: geographic coarsening (PUMA width 1, 10 or 100), education coarsening (width 1, 2, 4 or 8), the minimum released cell (1, 3, 5 or 10) and the standard deviation of the auxiliary reading (0.1, 0.25, 0.5, 1 or 2), 240 settings in all before excluding those that release fewer than 30 records.

Each state shares 16 warm-start settings and eight early-stopping settings measured with 64 trials each. Each of four search arms then chooses 32 further settings, measured with 64 trials each, and every chosen setting receives 300 fresh confirmation trials. The arms are random search; a domain heuristic that ranks settings by reciprocal key frequency plus $0.2/(0.1 + \mathrm{sd})$; a scrambled Sobol sequence mapped onto the catalogue; and neural search, which fits a two-hidden-layer network of width 16 to the counts and acquires batches of eight by predicted risk plus 0.15 times the distance to the nearest measured setting. The total is $30\{24(64) + 4(32)(64 + 300)\} = 1{,}443{,}840$ attack trials, with 120 neural fits. Evaluator cost is matched across arms; total compute is not, since only the neural arm fits a model.

A chosen setting is confirmed high risk only if the exact one-sided lower limit from its 300 fresh trials exceeds 0.2, at a per-test level of 0.05/128 for the 32 settings of each of four arms within a state. Conditional on the full search history, the probability of any false high-risk confirmation within a state is then at most 0.05; the argument is Proposition 14.1 with the direction of the bound reversed. This is a lower-bound confirmation of excessive risk, not a certificate of safety, and it controls error within each state, not across all 30.

The primary endpoint is coverage: the number of distinct cells, out of a fixed taxonomy of 24, that contain at least one confirmed high-risk setting. The cells cross the three geography levels, education width above or below two, a minimum cell equal to one or larger, and a reading standard deviation above or below 0.5. The secondary endpoint is yield: how many of the 32 chosen settings were confirmed. Table 14.4 gives both.

TABLE 14.4: Public-file search over 30 Census state samples: mean confirmed high-risk cells (of 24) and confirmed settings (of 32) per state, with the paired-state bootstrap difference in cells from Sobol and its 95 per cent interval. Neural fits capped at 600 epochs.

Search	Cells	Settings	Difference from Sobol
Random	13.20	22.50	−0.63 [−1.23, −0.03]
Domain heuristic	11.97	32.00	−1.87 [−2.37, −1.33]
Scrambled Sobol	13.83	22.03	–
Neural search	10.07	31.57	−3.77 [−4.43, −3.07]

The table splits the arms into two pairs. Sobol and random search fill the space and cover the most cells, 13.83 and 13.20, while confirming about two thirds of their settings. The heuristic and neural search aim at risk: the heuristic confirms all 32 settings on average and neural search 31.57, and they cover fewer cells, 11.97 and 10.07. Neural search covers 3.77 fewer cells than Sobol, with a paired interval of −4.43 to −3.07. Four of the 120 neural fits had their best epoch at the 600-epoch cap; a rerun with a cap of 2,000 epochs kept the neural mean at 10.07 cells and the difference at −3.77 (−4.40 to −3.07). It reused the same data and non-neural draws, so it is a sensitivity check and not a replication.

A model that learns where the risk is will keep choosing settings near the risk it has already found, and so confirm many settings in few places. That is the trade-off between yield and coverage, and it is the first half of the misconception. A model that proposes attacks makes the audit more efficient at confirming what it already suspects; it does not by that fact make the audit find more kinds of failure. The coverage score favours space filling by construction, and the cells are bins of declared choices, not attack families, so the result does not show neural search to be worse for every purpose. It shows that "finds more high-risk settings" and "finds more of the high-risk space" are different claims.

14.4.5 A language model as the fifth arm

The first attempt to add a language model failed before it started. A pilot with a small hosted model saved its prompts in advance and sent its first request, which returned an HTTP 429 error with no proposals; the account showed no remaining credit, and collection stopped with no automatic retry. The record keeps the plan, the prompt hashes and the failed attempt, and states that no result exists from it.

The second attempt ran an open-weight model on the author's own machine: `qwen3:4b`, a 4.0-billion-parameter model quantised to Q4_K_M, served by Ollama 0.40.9, so that no account, key or charge was involved and nothing left the machine. The saved per-state

prompt was sent byte for byte, with the model's thinking mode off, temperature 0.3, and decoding constrained to the JSON shape the prompt asks for. Each state was allowed three attempts with distinct seeds. Generation took about 36 seconds per state. The constraint fixed the shape of the output and nothing else; every list was still checked for its count, the distinctness of its tuples and membership of the catalogue.

The model returned a valid list of 32 tuples for 20 of the 30 states, seven of them at the first attempt. Of 70 attempts, 50 were rejected: 43 for repeated or off-catalogue tuples and seven for repeated identifiers. All five arms were then rerun on exactly those 20 states, 1,195,520 attack trials, with the per-test level at 0.05/160. Figure 14.4 plots each arm's coverage against its yield.

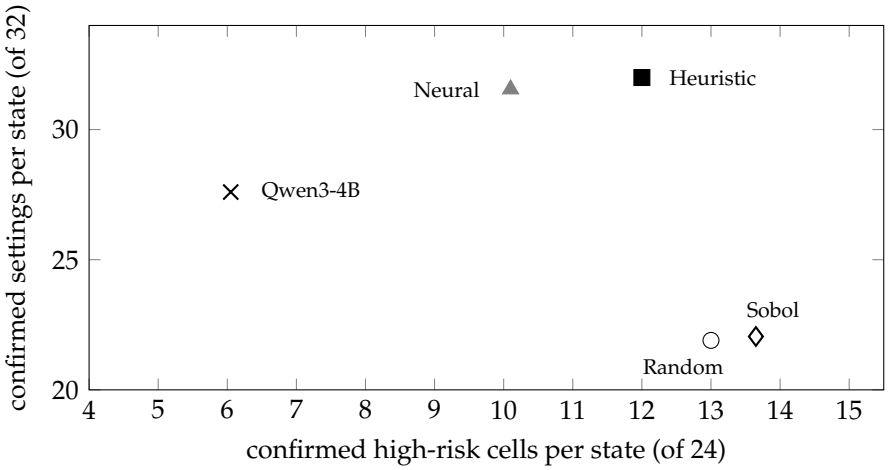


FIGURE 14.4: Coverage against yield for five search arms on the 20 Census states for which the language model returned a valid list (1,195,520 attack trials). Each arm has its own marker shape and label. Space-filling arms sit to the right, yield-maximising arms at the top; the language model sits in neither place.

The baseline ordering of the 30-state run holds on this subset: Sobol 13.65 cells, random 13.00, heuristic 12.00, neural 10.10. The language model confirmed 27.60 of its 32 settings per state, between the two yield-maximising arms and the two space-filling ones, but covered only 6.05 cells, 7.60 fewer than Sobol with a paired interval of -8.30 to -6.80. The proposals show why. In 19 of the 20 states every tuple the model chose sat at the finest geography, and the lists followed the catalogue's own order. The model read the catalogue as a list to copy and not as a space to search. Its tuples were high risk, because the finest geography is where the risk is, and they occupied a quarter of the cells.

This is the second half of the misconception, and the experiment settles less of it than the first. It settles that a four-billion-parameter open-weight model, under this fixed prompt, adds no coverage over quasi-random or neural search here and often fails to produce a valid list at all. It does not settle what a larger model, a prompt that names the coverage goal, or a worked example would do. The prompt was held fixed so that the comparison with the baseline arms stayed exact, and any change to it is a new experiment that must be declared before its confirmation outcomes are read. GPU decoding is not bit-reproducible across hardware, so a rerun may yield a different set of valid states; the recorded responses are the evidence.

14.5 Findings from a penetration test

A language model is one source of new adversaries. A security test of the dissemination system is another, and the repository treats its findings the same way. A portal, a download interface and the database behind them are software, and a flaw in the software can return what the release mechanism removed. A finding is therefore a change in what the adversary can see, and it belongs in the audit.

The repository's red-team module declares a closed list of capabilities a finding can grant: read suppressed cells (the floor becomes one), read fine geography, read fine education, sharpen the auxiliary reading (its standard deviation halves, never below the finest declared level), and read microdata (all of these at once). The adversary view of a release under a set of findings applies every granted capability, and each component of the view is the minimum of its published value and the value a capability sets.

Proposition 14.3 (Monotone composition of findings)**.** *Let $v(r, F)$ be the adversary view of release r under a set of findings F. Then $v(r, F)$ does not depend on the order in which the findings are listed; $v(r, F \cup \{f\})$ is at least as relaxed as $v(r, F)$ in every component; and $v(r, F \cup \{f\}) = v(r, F)$ whenever f grants no capability or only capabilities already granted.*

The proposition is what makes the pooled adversary of a test cycle well defined, and it stops a tester from weakening that adversary by listing findings in a convenient order. Chapter 9 explains why the pooled adversary must be audited and not inferred: two findings that each leave a release within tolerance can fail together.

The worked example in the repository runs this on the first state group of the public Census extract, 425 records, released at the coarsest setting of the catalogue: geography width 100, education width 8, minimum cell 10 and reading standard deviation 2. Four advisories from the 2026 changelog of the PHP language serve as the findings, mapped to the capability each would grant on a hypothetical portal: an access-control bypass in the process manager that serves the tables (read suppressed cells), a credential leak across origins on redirect (sharpen the reading), an injection flaw in the database driver (read microdata), and a cross-site scripting flaw in a status page (nothing relevant to disclosure). The bypass and the leak are marked confirmed, the injection theoretical, and the scripting flaw confirmed but granting nothing. The tolerance is 0.2, the cycle's budget share 0.05, split equally over the five audited rows, and each row gets 600 fresh trials.

TABLE 14.5: Penetration-test findings audited as adversaries on one public Census state group: 600 trials per row, tolerance 0.2, error budget 0.05 split equally over the rows. The view gives geography width, education width, floor and reading standard deviation. The bound is the exact one-sided binomial upper bound.

Adversary	View	Successes	Rate	Bound	Within
As published	(100, 8, 10, 2)	60	0.100	0.132	yes
Access-control bypass	(100, 8, 1, 2)	222	0.370	0.417	no
Credential leak	(100, 8, 10, 1)	87	0.145	0.182	yes
Status-page scripting flaw	(100, 8, 10, 2)	60	0.100	0.132	yes
Bypass and leak pooled	(100, 8, 1, 1)	257	0.428	0.477	no

Table 14.5 gives the result. The release passes as published, with 60 successes in 600 and a bound of 0.132. The bypass, which lets the adversary read the cells the floor suppressed, takes the bound to 0.417, and the release fails under it. The leak, which halves the reading

noise, takes the bound to 0.182, within tolerance on its own. The two together take it to 0.477, worse than either. A remediation plan that fixed the bypass and left the leak would still have to re-audit, and one that reasoned from the two single rows would have been wrong. The scripting flaw leaves the view and the bound where they were. The injection flaw, the most severe by its own rating, is absent because it was not confirmed. The ordering by measured disclosure risk is therefore not the ordering by severity score: a severity score ranks a flaw by its consequences for the system, and the audit ranks it by its consequences for the people in the register. The portal in the example is hypothetical, no advisory was exploited, and exploitability in a given deployment is the tester's finding.

The Bank's own security team runs its annual test of the statistics portal in the same quarter. Naledi asks for the findings as capabilities with a status, not as a ranked list, and declares the confirmed ones as adversaries for the next release cycle with a share of the programme's budget. Grace finds both the findings and the audited rows in the record, and the ranking that came out of the audit is the one she checks remediation against.

14.6 What remains open

The gaps are part of the result. No experiment in the repository shows a language model finding more risk than simple search; the one genuine comparison is one small model under one fixed prompt, and it covered fewer cells than every baseline. The public-file evaluator executes one attack family, so proposals cannot create new executable attacks without a separately specified evaluator, and a scenario described well in prose is not an attack that has been run. The synthetic pilots exercise the fitting of a closed network from measured outcomes, and they do not exercise the human review, the full provenance chain, a genuine language model or generalisation across attack families. No neural or language-model performance has been established on real institutional release programmes. The registry service that would hold keys, clock and budgets apart from the release team is specified and not built. And the confirmation in the public-file study controls error within each state, not across the study.

The repository also says what would close the first gap. Its proposed confirmatory protocol compares a constrained language-model proposer with random search, an informed heuristic and a strong numerical search at a fixed number of evaluator trials, on at least 30 independent programme and seed blocks, with distinct confirmed failures per 10,000 development trials as the primary endpoint and token and provider costs reported apart from evaluator trials. A benefit may be claimed only if the predeclared contrast improves with its interval and total cost disclosed; otherwise the negative finding is published. That protocol has not been run.

The chair's question gets its answer at the next committee meeting. The planning network stays, because it made audits cheaper without touching their validity and its logistic twin is reported beside it. The language-model proposal is accepted as a pilot under the injector rules, with its proposals reviewed, measured and recorded, and with no claim in any decision record that it found what a person would have missed.

14.7 The answer

A model that proposes attacks can make an audit cheaper and broader in what it considers, and it can do so without weakening the decision, provided three things are kept out of its

reach: the outcomes of the final audit, the right to supply labels or counts, and the record. Within those limits a network fitted to development attacks chose the same floor on the credit register as an audit without it and used about a fifth fewer attacks, and a logistic model did as well. Whether a model makes the audit stronger, in the sense of finding failures the declared list would have missed, is so far unsupported. Neural search confirmed many high-risk settings in few places and covered fewer kinds of risk than a Sobol sequence; a small language model covered fewer still, by copying its catalogue; and the pilot that added measured scenarios to the network's training data did not improve it once the labels came from executed attacks. The audit is as strong as its declared scenarios and its independent final draws, and a model's contribution has to be measured against simple search before anyone claims more.

IN PRACTICE: LETTING A MODEL PROPOSE

Let a model plan an audit freely: fit it to development attacks on seeds the final audit never uses, freeze its predictions, bets and start floor with a digest before any final draw, and report a logistic model beside it. Let a language model propose scenarios only through a fixed schema, a named reviewer and a separate evaluator, and train on nothing but measured counts. Never accept a count, a risk estimate or a label from the model. Charge every audit, including retries, to one programme budget held by someone outside the release team, and anchor the record outside the operator's control. Before claiming that a model finds more, compare it with random, Sobol and a domain heuristic at the same evaluator budget, and report coverage and yield separately. Treat penetration-test findings as capabilities with a status, audit each confirmed finding and their combination, and order remediation by the audited bound.

In the software. The planning pipeline is `privacy_risk.neural_pipeline` (`run`, entry point `python -m privacy_risk.neural_pipeline`). SENTRY-R is `privacy_risk.sentry_r` (`propose_from_llm`, `stage_llm`, `review_proposals`, `join_evidence`, `DeepRiskNet`, `shortlist`, `final_audit`) with the `sentry-r` command line; the ledger is `privacy_risk.provenance.ProvenanceLedger`. The public-file search is `privacy_risk.injector_search` and `scripts/public_injector_search.py`, the local model arm `privacy_risk.llm_pilot_local`, and the penetration-test protocol `privacy_risk.redteam` (`adversary_view`, `run`). The core of the injector, which sets provenance itself and refuses anything but the three permitted fields, reads:

```
prompt = llm_prompt(schema, question)
prompt_hash = hashlib.sha256(prompt.encode("utf-8")).hexdigest()
text = call(prompt)
if len(text.encode("utf-8")) > max_bytes:
    raise ValueError(f"LLM response exceeds {max_bytes} bytes; refused unparsed")
result = json.loads(text)
if not isinstance(result, list):
    raise ValueError("LLM response must be a JSON array")
rows = []
for item in result:
    if not isinstance(item, dict) or set(item) != LLM_RESPONSE_FIELDS:
        raise ValueError("LLM response objects must hold exactly proposal_id, "
                         "scenario and features")
    rows.append({**item, "llm_model": model_name, "prompt_hash": prompt_hash})
return stage_llm(rows, schema, provenance=provenance)
```

14.8 Proofs

Proof of Proposition 14.1. Fix a pair with $b = B_{\theta j}$ trials, write $\gamma = \alpha/M$, and let $F_p(k) = \sum_{t=0}^{k} \binom{b}{t} p^t (1-p)^{b-t}$. For $k < b$, $F_p(k)$ is continuous and strictly decreasing in p, and the exact upper bound $u(k)$ is the unique solution of $F_{u(k)}(k) = \gamma$; set $u(b) = 1$. For a fixed true value p, the event $\{u(K) < p\}$ consists of the counts k with $F_p(k) < \gamma$, which by monotonicity in k form a lower set $\{0, \dots, k_*\}$ (possibly empty). Its probability is $F_p(k_*) < \gamma$, or zero if the set is empty. Hence, conditional on $\mathcal{D}$, $\mathbb{P}\{r_j(\theta) > \bar{r}_{\theta j} \mid \mathcal{D}\} \le \gamma$ for each pair, and a union bound over the M pairs gives $\mathbb{P}\{\exists(\theta, j) : r_j(\theta) > \bar{r}_{\theta j} \mid \mathcal{D}\} \le \alpha$. No independence between pairs is needed, and the number and identity of the pairs may depend on $\mathcal{D}$, since the bound is conditional on it. On the complementary event every certified candidate satisfies $r_j(\theta) \le \bar{r}_{\theta j} \le \tau_j$ for all j, whichever certified candidate the rule selects. So the conditional probability of certifying a candidate that breaches a tolerance is at most α, and taking expectations over $\mathcal{D}$ gives the unconditional statement. □

Proof of Proposition 14.2. Given x and B, linearity gives

$$\mathbb{E}[\ell(p) \mid x, B] = -\frac{B}{B_\bullet}\big[\pi \log p + (1-\pi)\log(1-p)\big], \qquad \pi = \pi(x).$$

Subtracting the same expression at $p = \pi$ leaves

$$\frac{B}{B_\bullet}\Big[\pi \log \frac{\pi}{p} + (1-\pi)\log \frac{1-\pi}{1-p}\Big] = \frac{B}{B_\bullet}\,\mathrm{KL}\big(\mathrm{Bern}(\pi) \,\|\, \mathrm{Bern}(p)\big),$$

which is nonnegative by Gibbs' inequality and zero only when $p = \pi$; this is (14.2). The calculation used the label only through $\mathbb{E}[K \mid x, B] = B\pi(x)$. Repeating it with $\widetilde{K}$ in place of K replaces π by $\widetilde{\pi}$ throughout, so the expected loss is minimised at $p = \widetilde{\pi}(x)$, which bears no relation to the attack's success probability unless $\widetilde{\pi} = \pi$. □

Proof of Proposition 14.3. Each component of the view is the minimum of the published value and the values set by the capabilities that the findings in F grant. The minimum of a finite set does not depend on the order in which its elements are listed, which gives the first claim. Adding f to F can only add values to the set over which each minimum is taken, and a minimum over a larger set is no larger, which gives the second. If f grants no capability, or only capabilities already granted by members of F, the sets are unchanged, because the minimum is idempotent, and the view is unchanged, which gives the third. □

Part VI

Practice

Chapter 15

Testing the model behind the figure

Many statisticians hold a reasonable belief about risk models fitted inside an institution. The model that produces a disclosure risk figure is fitted to the institution's own register: its key frequencies, its error rates, its records. Checking that model against the same register looks circular. The fit was chosen to agree with those data, so a test on those data can only confirm the fit, and any real check must wait for an outside attack that nobody hopes to see. On this view the risk model is an assumption to be argued about, and the register has nothing more to say.

The belief fails because a linkage model does more than describe the register. It predicts how often an attack will name the right record, and it predicts this separately for each level of confidence the adversary reaches. Those predictions can be compared with attack outcomes that the fit never used. On the simulated population of Chapter 11, two linkage models were run side by side. One used the adversary's true error precision. The other assumed the adversary's covariate was noisier than it was. The two risk figures differed by 0.007. Their stated posteriors, set against the realised success rates on held-out attacks, differed by a factor of seven: a mean gap of 0.007 for the correct model and 0.049 for the wrong one. The register did contradict one of the models. The risk figure could not see it, and a calibration test could.

This chapter asks one question. What evidence can show that the model behind a risk figure is wrong, and what should the institution do with the figure when it is?

15.1 An implementation check is not a validation

The pipeline that produces this book's numbers reproduces every equation of the earlier chapters in code, and a test suite confirms that the code computes what the equations say. That is an implementation check. It shows that the reciprocal of a key frequency is computed correctly when Assumptions 2.1 to 2.4 of Chapter 2 hold. It cannot show that they hold. A model can be coded without error and still describe an adversary nobody faces, a population that does not exist, or a mechanism that behaves differently from the way it was described.

Empirical validation needs a different kind of evidence. A test of the model must name a quantity that the model predicts, a source of evidence the model was not fitted on, a band inside which agreement counts as agreement, and a consequence if the evidence falls outside it. All four are written down before the evidence is read. A test without a stated band is a description, and a test without a consequence is a report that nobody has to act on.

At the Bank, Naledi keeps a folder for each assessment, and the first sheet in it is the list

of tests she will run, with their bands, signed and dated before she starts the simulations for Tomás's request. The chair once asked why the sheet had to come first. Her answer was that a band chosen after the result is read can always be drawn to include the result.

The long form of the paper behind this book proposes six tests. Figure 15.1 sets them out by the part of the model each one examines and by what follows from a failure. Four of them, linkage calibration, population-frequency transport, adversary stress and mechanism monotonicity, test the model of a single release, and a failure of any of them makes the release-level figure invalid. The fifth, the composition test, tests whether the release is the right object of assessment at all. The sixth, temporal transport, cannot be run when the assessment is made, and the expiry date of the governance threshold enforces it instead.

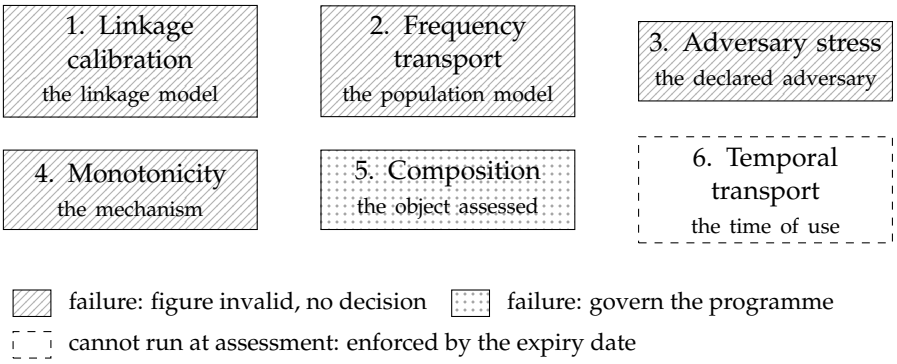


FIGURE 15.1: The six falsification tests, each with the part of the model it examines (small type) and the consequence of a failure. Grey fill with hatching marks the four tests whose failure makes a release-level figure invalid; dotted fill marks the composition test, whose failure moves the assessment to the programme; the dashed outline marks the test that waits for the passage of time.

15.2 Linkage calibration

The first test examines the linkage model itself. When the declared adversary attacks a target, the model gives a posterior probability that the record it names is the right one. If the model is correct, then among all attacks in which the adversary named a record with posterior near 0.3, about 30 per cent should be right. The test bins the attacks by stated posterior, computes the realised success rate in each bin from held-out outcomes, and reports the mean gap between the two.

Figure 15.2 shows the test on the simulated population. With the adversary's error model stated correctly, the eight bins lie close to the diagonal and the mean gap is 0.007. When the model assumes that the adversary's covariate is 2.5 times noisier than it is, the stated posteriors fall below the realised rates in the middle of the range: in the bin from 0.3 to 0.5, the model says 0.399 and the attacks succeed 0.496 of the time. The mean gap is 0.049. Meanwhile the success rate of the attack hardly moves: 0.464 with the right model and 0.457 with the wrong one.

The near equality of the two success rates is not an accident of this simulation. Proposition 15.1 shows why a wrongly scaled error model leaves the adversary's choice, and so its success rate, unchanged, while it changes the confidence the adversary attaches to that choice.

Proposition 15.1 (A misstated error scale leaves success unchanged)**.** *Let the target t lie in a candidate set C of released records, all equally likely a priori in the adversary's information, with*

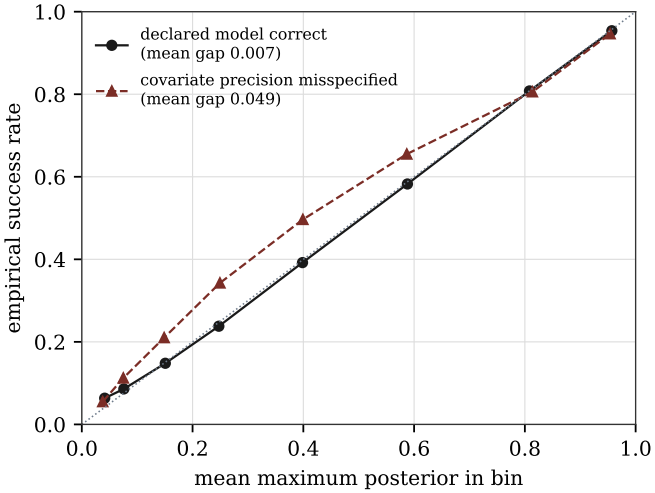


FIGURE 15.2: Linkage calibration on the simulated population: mean stated posterior against realised success rate in eight bins, 20,000 attacks per curve. Circles on a solid line show the declared model; triangles on a dashed line show a model that overstates the adversary's covariate noise by a factor of 2.5. The dotted diagonal is perfect calibration.

published values x_j for $j \in C$. The adversary observes $z = x_t + e$ with $e \sim N(0, \sigma^2)$, assumes a standard deviation $s > 0$, forms the posterior $p_s(j) \propto \exp\{-(z - x_j)^2/(2s^2)\}$ and names $\hat{\jmath}$, the candidate of largest posterior, with ties broken by a fixed rule. Then

(i) $\hat{\jmath}$ does not depend on s, so the success probability $\mathbb{P}(\hat{\jmath} = t)$ is the same for every $s > 0$;

(ii) when $s = \sigma$ the stated posterior is calibrated: $\mathbb{P}(\hat{\jmath} = t \mid z) = p_\sigma(\hat{\jmath})$;

(iii) the stated posterior $p_s(\hat{\jmath})$ is non-increasing in s, tends to $1/|C|$ as $s \to \infty$, and, when the nearest candidate is unique, tends to 1 as $s \to 0$.

Part (i) says that the risk figure, which counts how often the named record is right, cannot reveal a wrong error scale. Parts (ii) and (iii) say where the evidence lies instead: in the posteriors, which are right only under the right scale. An adversary who believes its covariate is noisier than it is reports posteriors too close to $1/|C|$, as the dashed curve in Figure 15.2 does through most of its range. The attack in the pipeline also carries error in the key fields, so the candidate sets differ slightly between the two scales, and the success rates differ by 0.007 instead of by nothing. The conclusion is the same.

This answers the belief stated at the start of the chapter. A model calibrated on the register can be refuted by the register, because the model makes a family of predictions and the fit pins down only some of them. The calibration test reads the predictions the risk figure ignores.

The test had a weakness in an earlier version of the repository, and the way it was repaired is part of the lesson. The configuration that drove the paper's run set the band for the mean calibration gap at 0.06, and the misspecified model's gap of 0.049 lay inside it, so the pipeline as then configured would have passed the wrong model. A band has to be set from the size of error that would change a decision, and a band of 0.06 on a gap whose correct value is 0.007 was too wide to catch an error of the kind this chapter studies. The band now stands at 0.03, and the pipeline counts the calibration test as valid only if the band also rejects the deliberately misspecified model: a band that passes a known wrong model

cannot show the declared one wrong either. The value of 0.03 was chosen after the wider band was found wanting, so it was not written down before the evidence was seen; its first prespecified use is the next run on fresh draws.

Naledi's sheet for Tomás's request sets the band at 0.02 and records why: on the simulated population a correctly specified model sits well below it, and the misstated scale sits well above it. The chair asked whether 0.02 was a standard figure. It is not. It is Naledi's reading of what separates the two cases, signed before she runs the attacks, and the reviewer can disagree with it in writing.

15.3 Population-frequency transport

The second test examines the model of the population. Journalist risk, the chance that a sample-unique record is also unique in the population (Chapter 3), cannot be computed from the sample alone. It rests on a fitted law for the population cell sizes, here a negative binomial, and the fitted law makes predictions about quantities that can be checked against held-out or external counts.

On the simulated population, where the truth is known, the fitted law recovers the mean population cell size closely: 14.83 against a true 14.88 at a 5 per cent sample. It recovers the uniqueness probability less well, with errors between 8 and 25 per cent across five sampling fractions, and every error lies on the conservative side, since the model's uniqueness probability exceeds the truth at each fraction. The test also exposes a misfit that a single number hides. A negative binomial law that is closed under random thinning would keep the same dispersion whatever the sampling fraction. The fitted dispersion moves instead, from 0.33 at a 2 per cent sample to 0.51 at 20 per cent and 0.59 at 50 per cent. The family does not describe this population, though it summarises it well enough at one fraction.

The paper draws the consequence that its own test demands. A model-based uniqueness probability that fails transport is replaced by a bound, or reported beside the truth and not used. No decision in the long form rests on the model-based figure, and on the public Census extract, where there is no truth to check against, the uniqueness probability is reported across a range of sampling fractions instead of as one value (Chapter 12).

Here too the code is narrower than the text. The pipeline's frequency test checks the absolute error of the fitted mean cell size against a band of 1.5 cells. The measured error is 0.055 cells, so the test passes, but a test of the mean cell size would pass a model whose uniqueness probability was wrong by a factor of two. An institution that uses the journalist measure for a decision should test the quantity it uses, the uniqueness probability or the share of small cells, against external counts, and should set the band on that quantity.

15.4 Adversary stress

The third test examines the declared adversary. Every risk figure in this book is conditional on what the adversary is declared to know, and the declaration is a guess about sources outside the institution's control. The stress test asks whether a materially stronger adversary that is still plausible would change the decision. If it would, the release fails or the scenario is revised in writing; the figure for the weaker adversary is not quietly kept.

The best-known demonstration that a declared adversary can be too weak is Sweeney (2000). Hospital discharge records released without names still carried date of birth, sex and postcode, and a public voter list carried the same fields with names attached. A release

reviewed against an adversary who held only the published file faced one who held a second file. The stress test is the discipline of asking, before release, what the second file could be.

On the simulated population the stronger adversary holds a covariate twice as precise as the declared one. Across the 45 mechanism settings of the grid, it raises the simulated risk by between 1 and 76 per cent, with a median of 40 per cent. The stronger adversary is never weaker at any setting, which is what the pipeline checks: a capability that lowered the risk would point to a misdeclared scenario or a coding error. The size of the movement is the more useful result. A setting that passes the tolerance against the declared adversary with a margin of less than 40 per cent may well fail against an adversary with a better reading, and that is why the long form takes the risk under the stronger adversary, the robust risk, as the coordinate of the risk–utility frontier (Chapter 8).

Capability and modelling are different things, and the test keeps them apart. An adversary that keeps its information but misjudges its own error, the case of Proposition 15.1, succeeds as often as before, so the stress test cannot see it; the calibration test can. An adversary with better information succeeds more often, so the calibration test may pass while the stress test fails. Neither test replaces the other.

The test also removed a protection lever from the pipeline. An early version offered record swapping between districts. It lowered the simulated risk, but only against an adversary that read the released district at face value, and nothing in the declared scenario justified that assumption. An adversary who knew swapping was used, or who held a second source for the district, gained the protection back. The lever was taken out, and the paper reports the removal. Reporting its protection would have turned an assumption about the adversary into a property of the release.

When Naledi ran the stress test for Tomás's request, the stronger adversary was not hypothetical. Pieter, a credit analyst at a reporting lender, knows the exposure of his own bank's loans to the last unit. The declared outside adversary reads exposure with error; Pieter does not. On the synthetic register an insider who reads exposure with a log error of 0.01 and knows the loan's history succeeds against 77 per cent of targets with no suppression, and against 42 per cent at a minimum cell count of 300. No floor up to 300 stops him. Naledi's record therefore names him as a separate adversary with his own row, governed by the access route and not by the floor.

15.5 Mechanism monotonicity

The fourth test examines the protection mechanism. When an institution claims that a stronger setting reduces risk, a larger minimum cell count or a coarser key, the claim is empirical and can fail. The test looks for settings where stronger protection is followed by higher measured risk, and requires that each violation be investigated and not smoothed away.

The long form reports four kinds of finding under this test. The first is a large, systematic violation of a different claim: every row of the k-anonymity table breaks the nominal bound $1/k$ on the worst target, with the worst target at $k = 10$ identified with probability 0.744 against a nominal 0.1. The investigation of that violation produced a proposition, that k-anonymity delivers $1/k$ only when the adversary holds nothing beyond the key (Chapter 3). The other three are small breaks in order. The worst target rises from 0.744 at $k = 10$ to 0.768 at $k = 20$, estimates that rest on 250 replicates each. The pooled rate on the credit register rises from 0.260 with no suppression to 0.269 under k-anonymity with $k = 5$, about

one standard error at 4,000 attacks each. The large-firm identification rate rises from 0.247 to 0.273 between the same two levels, on a few hundred targets. None of the three exceeds what sampling noise explains, and the long form treats them as noise after looking for a mechanism and finding none.

The decision depends on this test less than it seems to, for a reason worth stating. The choice of a floor in this book does not assume that risk falls as the floor rises. Fixed-sequence testing (Chapter 5) tests the floors from the strictest down and stops at the first failure, and its error guarantee holds whatever the order of the true risks. Monotonicity is tested because a claim that a lever protects should be checked, not because the selection rule needs it.

The pipeline's version of the test runs over the minimum cell counts at the finest key, with a band of 0.025 for simulation noise, and finds no violation on the paper's run.

15.6 The composition test

The fifth test asks whether the single release is the right thing to assess. It runs the joint attack on the programme, compares the pooled risk with the interval the marginal risks imply (Proposition 9.1), and reports the pooled risk against a programme tolerance of its own.

On the simulated population the test fails clearly. Two releases, each 3-anonymous, pooled by an adversary who intersects their candidate sets, identify the target with probability 0.411, which is 2.8 times the upper bound of 0.148 that their two marginal risks imply. Four releases reach 0.458, 1.8 times the bound of all four.

A failure of this test does not invalidate the release-level model. The linkage model, the population model and the mechanism may all be right, and each release may be assessed correctly on its own. What fails is the assumption that the release is the object to govern. Proposition 9.2 says that the marginal risks cannot bound the pooled risk, so the consequence is a change of object: the programme needs its own simulation, its own tolerance and its own horizon. The pipeline records the composition result beside the validation report and keeps it out of the validity flag for that reason. It applies the result in the gate instead: on the paper's run, the chosen setting passes on its own with an upper bound of 0.081, and the gate still returns a fail, because the upper bound on the pooled risk of the declared four-release programme, 0.464, exceeds the release tolerance of 0.12.

The decision record now carries one programme tolerance. In an earlier version the validation step judged the pooled bound against a programme tolerance of 0.5 while the gate judged it against 0.12, and both statements sat in the same file. The validation step and the gate now read the same declared value, 0.12, and the record names it; the figure of 0.5 survives only as the total of the risk budget on the union event of Chapter 8, recorded under that name.

Tomás's request is a composition question before it is anything else. He asked for a research extract in March, a second in June and a third with the year's vintages. Each was assessed when it came in. Naledi's first act on the new request was to assess the three together, with his auxiliary information held fixed across them, because the earlier records did not license any statement about the set.

15.7 Temporal transport

The sixth test cannot be run when the assessment is made. An assessment is calibrated on the register, the auxiliary sources and the linkage methods of its date. New public sources appear, a register changes its coverage, and attack methods improve. The test asks that the assessment fail or be re-estimated when any of these moves observed attack performance outside the range the assessment declared.

Because the evidence for this test arrives later, the governance threshold enforces it through the expiry date d and the re-assessment triggers. The pipeline refuses a decision on or after the expiry date. A bound from Chapter 6 shows how little shift a certificate absorbs. If the deployed attack distribution lies within a Kullback–Leibler divergence ρ of the one assessed, the deployed risk is bounded through the Bernoulli divergence. The programme floor of 150 on the credit register has a confirmed upper bound of 0.109, and it can absorb a divergence of about 6×10^{-4} before that bound reaches the tolerance of 0.12. Halving the standard deviation of the adversary's exposure reading, from 0.6 to 0.3, is a divergence of about 0.32 per reading, three orders of magnitude larger, and it moves the floor from 150 to a median of 300 over twenty register draws. The certificate holds for the declared reading precision and for little else. The expiry date exists because the world the certificate describes will move.

15.8 Why a failed test blocks the decision

When a test fails, the tempting response is to keep the figure and add a margin. If the calibration gap is too large, raise the risk figure by a fifth and compare that with the tolerance. The response is tempting because it is cheap and seems prudent. It is wrong, and the reason is that a failed test does not say the figure is too small by some amount. It says the model that produced the figure does not describe the attack, and nothing in the evidence says by how much it errs.

Proposition 15.2 makes both halves of the argument precise. A rule that refuses to decide when a test fails can only remove passes, so it keeps the error guarantee of the upper bound whatever the dependence between the test and the bound. A fixed margin cannot repair a model that omits what the adversary knows, because such a model can understate the risk by any factor.

Proposition 15.2 (Blocking keeps the level; no margin repairs a wrong model)**.** *Let $\overline{R}$ be an upper bound for the model risk R with $\mathbb{P}(\overline{R} < R) \le \alpha$ when the model is correct, let V be any event, possibly dependent on the same simulations, on which the validation tests pass, and let the decision pass when V occurs and $\overline{R} \le \tau$.*

(i) If the model is correct and $R > \tau$, the decision passes with probability at most α.

(ii) For every margin $c \ge 1$, tolerance $\tau \in (0,1)$ and $\varepsilon > 0$, there is a release and a model, satisfying the assumptions of Proposition 2.2 for an adversary who holds only the key, such that $c\,\overline{R} \le \tau$ with probability at least $1 - \varepsilon$, while an adversary who also holds one published attribute exactly identifies every target with probability one.

Part (ii) is the k-anonymity finding in its plainest form. The model is correct for the adversary it describes and wrong for the one who attacks, and no scaling of its output closes the gap, because the gap depends on what the adversary holds and not on the model's own

error. The only repair is a new model, which means a new declaration, new simulations and a new set of tests.

The pipeline encodes the rule. When a release-level test fails, the gate returns `INVALID` and the process exits with a non-zero status, so a failed validation cannot be presented as a release decision. The status differs from a fail. A fail says that the evidence, read by a model the institution accepts, shows the risk above the tolerance; the remedies of Chapter 16 apply. An invalid result says the institution has no accepted model and so no evidence either way. The release cannot go out on it, and the work returns to the declaration.

The chair raised the margin idea at the committee meeting on Tomás's request. A draft calibration check on a new auxiliary source had come back with a gap above Naledi's band, and the chair proposed adding a fifth to the figure and moving on. Naledi showed the committee the dashed curve of Figure 15.2 and the two success rates beside it. A larger figure would have looked prudent and would have been computed from the same wrong model. The committee recorded the result as invalid, and Naledi rebuilt the declaration.

15.9 What the tests can and cannot show

The six tests do not prove that a model is right. A model that passes all six can still be wrong about an adversary nobody thought to declare, and no test can range over adversaries that are not written down; Chapter 23 returns to this limit. What the tests do is give the model a stated chance to fail, so that a figure that survives them has survived something. The claims-and-evidence register of the repository states for each claim what would refute it, and the last row is the governance protocol itself, whose claim to improve auditability rests on an institutional argument and has no comparative evidence yet.

The answer to the chapter's question is short. Evidence can show that the model behind a risk figure is wrong whenever the model predicts something beyond the figure: posteriors that can be compared with outcomes, population counts that can be compared with external benchmarks, an ordering of risk across settings, a response to a stronger adversary, and a range of conditions over time. The institution's own data can supply most of that evidence, so the belief that a model fitted to the register cannot be refuted by it is false. When a test fails, the figure is not adjusted. The assessment is invalid, no decision is taken on it, and the model is rebuilt from a new declaration.

IN PRACTICE: TESTING THE MODEL BEFORE THE DECISION

Before running any attack, write the test sheet: the six tests, the quantity each compares, the evidence it uses and the band for each, signed and dated. Set each band from the size of error that would change the decision, not from habit. Test the quantity the decision uses: if the journalist measure enters the decision, test the uniqueness probability against external counts, not only the mean cell size. Run a stronger plausible adversary for every scenario and name any insider as a separate adversary. Treat a failed release-level test as an invalid assessment, not as a high figure. Treat a failed composition test as a reason to assess the programme. Put the expiry date and the re-assessment triggers in the record, since they carry the sixth test.

In the software. The tests are in `privacy_risk.validation`: `calibration_table` and `calibration_gap` for linkage calibration, `frequency_transport_error` for the population model, `stress_ratios`, `monotonicity_violations` and `composition_test`, combined by `full_validation_report`. `privacy_risk.pipeline.validate` runs them on a pipeline run, and `governance.decide` returns `INVALID` when the report has not passed. The bands come from the configuration file:

```
report = full_validation_report(ece, float(config.get("calibration_ece_limit", 0.03)),
                                transport,
                                float(config.get("frequency_mae_limit", 1.5)),
                                violations, stress, composition,
                                misspecified_ece=(max(misspecified)
                                                  if misspecified and config.get(
                                                      "require_calibration_power", True)
                                                  else None))
```

15.10 Proofs

Proof of Proposition 15.1. (i) For every $s > 0$ the map $u \mapsto \exp\{-u^2/(2s^2)\}$ is strictly decreasing in $u \geq 0$, and the prior is uniform over C. The posterior $p_s(j)$ is therefore largest exactly for the candidates that minimise $|z - x_j|$. This set does not depend on s, and the fixed tie-breaking rule picks the same member of it for every s, so $\hat{\jmath}$ is a function of z and the x_j alone. The event $\{\hat{\jmath} = t\}$ and its probability are then the same for every s.

(ii) When $s = \sigma$ the adversary's likelihood is the true conditional density of z given that j is the target, and with a uniform prior on C that matches the target's distribution in the adversary's information, Bayes' theorem gives $\mathbb{P}(t = j \mid z) = p_\sigma(j)$ for each $j \in C$. Since $\hat{\jmath}$ is a function of z, $\mathbb{P}(\hat{\jmath} = t \mid z) = p_\sigma(\hat{\jmath})$.

(iii) Write $d_j = (z - x_j)^2$ and $d_* = d_{\hat{\jmath}} = \min_j d_j$. Then

$$p_s(\hat{\jmath}) = \Big(\sum_{j \in C} \exp\{-(d_j - d_*)/(2s^2)\} \Big)^{-1}.$$

Each term with $d_j > d_*$ increases with s and tends to 1 as $s \to \infty$, and each term with $d_j = d_*$ equals 1, so $p_s(\hat{\jmath})$ is non-increasing in s and tends to $1/|C|$. If the nearest candidate is unique, every other term tends to 0 as $s \to 0$, so $p_s(\hat{\jmath}) \to 1$. □

Proof of Proposition 15.2. (i) The decision passes on $V \cap \{\overline{R} \leq \tau\}$, which is contained in $\{\overline{R} \leq \tau\}$. When $R > \tau$, $\{\overline{R} \leq \tau\} \subseteq \{\overline{R} < R\}$, whose probability is at most α when the model is correct. No assumption on the dependence between V and $\overline{R}$ is used.

(ii) Choose an integer $k \geq 2$ with $1/k < \tau/c$, and let $\delta = \tau/c - 1/k > 0$. Build a release with $n = kM$ records, M key classes of exactly k records each, and one further published attribute whose values are distinct across all n records. For an adversary who holds only the key, Assumptions 2.1 to 2.4 hold and Proposition 2.2 gives risk $R = 1/k$ for every target. Let $\overline{R}$ be the exact one-sided upper bound from B simulated attacks of that adversary. Its success count is binomial with mean B/k, and the exact bound converges to $1/k$ in probability as $B \to \infty$, so for B large enough $\mathbb{P}(\overline{R} \leq 1/k + \delta) \geq 1 - \varepsilon$, and on that event $c\,\overline{R} \leq c/k + c\delta = \tau$. An adversary who also holds the target's value of the distinct attribute exactly finds a single released record with that value, which is the target's, so it identifies every target with probability one. □

Chapter 16

An operational protocol

Experienced release committees often regard a written protocol with suspicion, and the suspicion has a fair basis. The safety of a release comes from the protection applied to it and from the judgement of the people who know the data and its users. Committees made sound decisions long before anyone asked them to fill in a record with nine elements. A protocol adds forms, sign-offs and waiting time, and none of that changes a single cell of the file. On this view a protocol is paperwork that slows releases without making them safer.

The view fails on the evidence of the method's own audit. Suppose a disclosure analyst runs an honest audit of an attack whose true risk sits exactly at the tolerance. The audit certifies it falsely in 3.8 per cent of runs, inside its stated error level of 5 per cent. Now let the analyst run the same audit ten times with different seeds and report the first that passes. The false certification rate becomes 32 per cent, and with a hundred tries 98 per cent. Nothing in the file changed, and no single audit was computed wrongly. What changed is that the choice of seed was made after the outcomes could be seen. The control that restores the rate of 3.8 per cent is a matter of order: the targets are fixed before any outcome exists, and every audit is recorded where it cannot be dropped. That is a protocol. The error statement that makes a risk bound worth computing holds only when certain things are written down before certain other things happen.

This chapter asks what an institution must write down, in what order and by whom, for a release decision to carry the error statement the method promises.

16.1 The order is the control

Every guarantee in the earlier chapters has the same shape. A bound computed from simulated attacks lies below the true risk with probability at most α, and so a release whose true risk exceeds the tolerance passes with probability at most α. The probability is taken over the attacks, with everything else held fixed: the tolerance, the declared adversary, the target set, the setting under test and the number of attacks. If any of these is chosen after the outcomes are seen, the probability statement is about a different experiment, and in the worst case it says nothing.

Proposition 16.1 states this for the two choices an institution is most tempted to make late: the tolerance itself, and which of several audits to report.

Proposition 16.1 (The order of declaration)**.** *Let $\overline{R}$ be an upper bound with $\mathbb{P}(\overline{R} < R) \leq \alpha$ for the true risk R of a declared attack.*

(i) If the tolerance τ is fixed before the attacks are run and $R > \tau$, the release passes with probability at most α.

(ii) If the tolerance may be set after the attacks, as any value in a range that contains $\overline{R}$, then a rule that sets $\tau = \overline{R}$ passes every release, whatever its risk.

(iii) If the operator can run N independent audits of the same attack and report one that passes, and a single audit passes with probability f, the reported audit passes with probability $1 - (1 - f)^N$, which tends to one as N grows whenever $f > 0$.

Part (iii) produces the figures of the opening paragraph: with $f = 0.038$, ten tries give $1 - 0.962^{10} = 0.32$ and a hundred give 0.98. Table 16.1 reports the full measurement from the repository, for an attack at the tolerance of 0.12 and for one slightly above it at 0.14, and adds a third way of breaking the order: an operator who chooses the bet of a sequential audit after seeing the outcome certifies an unsafe attack in every run. With the controls in place, each row returns to the rate of a single honest audit.

TABLE 16.1: Attacks on the audit process: the share of runs in which an attack at or above the tolerance of 0.12 is certified, with one coordinate at the full level $\alpha = 0.05$. "Tries" is the number of seeds or unregistered shadow audits the operator can choose among. Without the control, the rate for several tries is $1 - (1 - f)^{\text{tries}}$ for the single-try rate f; with it, the operator has one stream of targets fixed in advance.

Attack on the audit	True risk	Tries	Without control	With control
Seed grinding or shadow audits	0.12	1	0.038	0.038
Seed grinding or shadow audits	0.12	10	0.322	0.038
Seed grinding or shadow audits	0.12	100	0.979	0.038
Seed grinding or shadow audits	0.14	10	0.066	0.007
Seed grinding or shadow audits	0.14	100	0.495	0.007
Bet chosen after the outcome	0.14	1	1.000	0.007
Bet chosen after the outcome	0.20	1	1.000	0.000

The table does not describe dishonest analysts. Each row describes something an analyst under a deadline might do without intending harm: rerun an audit that failed because the seed looked unlucky, try a few settings in a scratch copy before registering the one that works, or adjust the audit's design after a bad early result. A protocol that fixes the order of work removes these choices, and with them the inflation they cause. The paperwork is the means by which the order can be checked afterwards.

Grace, from the Bank's internal audit function, reads decision records after the fact, and she reads them for order. Her first question on any record is not whether the risk figure is small. It is whether the tolerance was signed before the first attack outcome existed, and whether every audit run on the register appears in the registry. A record that cannot answer both questions, she reports, carries no error statement at all, however careful its arithmetic.

16.2 Four roles and three lines

The protocol needs four roles, and the first and third should not be held by the same person. The *release owner* is a named senior officer, at the Bank the chair of the statistics release committee, who sets the tolerance, signs the decision and owns the residual risk. The *data owner* is the business area that collects the source register and knows which products are drawn

from it, how often and under what legal basis. The *disclosure analyst*, Naledi at the Bank, declares the scenarios, runs the attacks and computes the bounds; she owns the correctness of the calculation and not the decision. The *reviewer*, outside the production line in a risk, audit or methodology function, checks that the scenarios are plausible and that the falsification tests of Chapter 15 were run.

The roles map onto the three lines of defence that central banks already use for operational risk. The analyst and the data owner sit in the first line, the reviewer in the second, and internal audit, which checks the filed record and the controls on the audit process, in the third. The separation is the point. The person who computes the bound should not be the person who decides whether the residual risk is acceptable, and neither should be the person who checks afterwards that the work was done in the stated order. Chapter 7 sets out the legal basis for this division and the elements of the threshold the owner signs.

16.3 Seven steps

Figure 16.1 shows the seven steps as a decision process, with the role that leads each step. Two loops leave the main line. The dotted loop is the re-assessment at each new vintage or at expiry, which begins again at the planning step and draws on what remains of the programme's error budget. The dashed loop is a change of product: when no setting meets the tolerance, the institution moves to a different product, such as aggregate tables or access in a secure setting, and assesses it from the start.

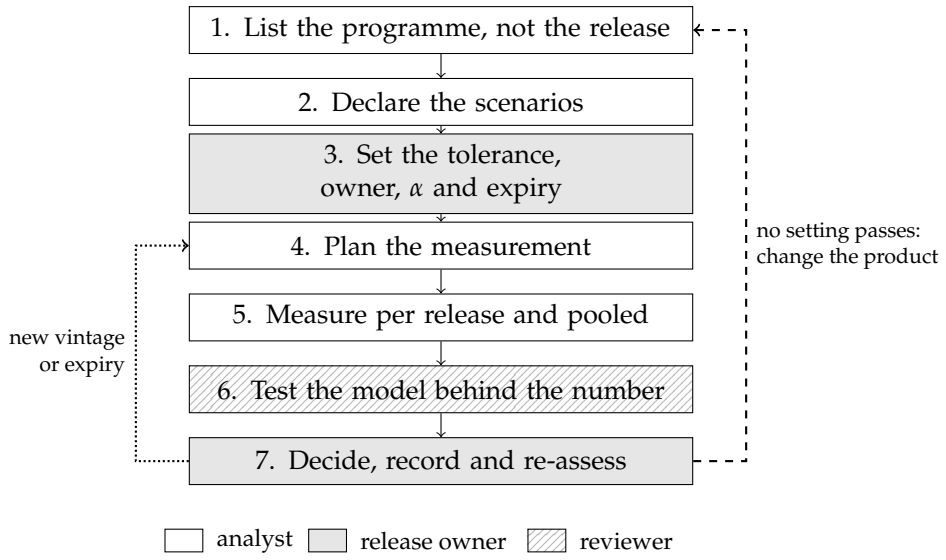


FIGURE 16.1: The seven steps of the protocol and the role that leads each one. Plain boxes are led by the disclosure analyst, grey boxes by the release owner and hatched boxes by the reviewer. The dotted arrow is the re-assessment at a new vintage or at expiry; the dashed arrow is a change of product, which needs a new assessment.

Step 1. List the programme, not the release. The analyst starts from the source register and not from the product about to be published. Every product drawn from the register over the planning horizon is listed: monthly or quarterly aggregates, supervisory publications, research extracts, data shared with other authorities and answers to ad hoc requests.

For each, the inventory records its frequency, the population it covers, the variables it publishes and whether any identifier or stable ordering persists from one vintage to the next. A persistent pseudonym, a stable row order or a rare combination of fields that seldom change lets an adversary match records across vintages, which is the condition of Proposition 9.2. The step ends with a declared horizon m, the number of vintages the assessment covers.

Step 2. Declare the scenarios. For each product, and for each combination of products an outsider could hold together, the analyst writes down who the adversary is, what it knows, which records it targets and what harm follows if it succeeds. For a pooled scenario the declaration says whether the adversary knows in which vintages the target is present, and why, because on the credit register that single piece of knowledge decides whether pooling adds anything (Proposition 9.3). The declaration should be specific enough that someone can disagree with it. Chapter 17 gives starting declarations for six release types.

Step 3. Set the tolerance. For each harm class the release owner sets a tolerance, an error level, an expiry date and the horizon of Step 1, and records the legal basis, the parties the tolerance protects and the access route. The tolerance can be stated directly, as the largest probability of successful exposure the institution accepts for that harm, or indirectly, as a relative weight on harm against lost utility from which the tolerance follows (Chapter 8). Either way it is written down before any risk is measured. Where no tolerance has been agreed for a harm, the record says so and no decision is made on that harm.

Step 4. Plan the measurement. Before running anything, the analyst fixes how many attacks each scenario needs. With no observed success, certifying a tolerance τ at $\alpha = 0.05$ needs about $3/\tau$ independent attacks (Hanley and Lippman-Hand, 1983); exactly, 299 for a tolerance of 0.01 and 2,995 for 0.001. If a floor is to be chosen from a grid, the order in which the floors are tested is fixed now (Chapter 5). If the release must pass every one of a fixed set of scenarios, each may be assessed at the full level, since that is an intersection–union test. If scenarios may be added later, for example because a reviewer proposes a stronger adversary, the plan reserves part of the error budget for them (Chapter 6).

Step 5. Measure, per release and pooled. The analyst runs the declared attacks on the release as it would be published, on fresh targets, and computes the exact upper bound for each scenario. The pooled attack is then run over the programme of Step 1 with the same auxiliary information. Pooled attacks that add knowledge, such as more readings or the target's history across vintages, are reported separately, because a committee needs to know which adversary it is protecting against. Where a level is chosen from a grid, it is audited again on fresh targets. The step reports what the chosen level keeps, in records and in volume.

Step 6. Test the model behind the number. Before the bound is used, the reviewer confirms that the six falsification tests of Chapter 15 were run against bands fixed in advance. A failed release-level test makes the assessment invalid, not failed: the institution cannot release on it and must repair the model first.

Step 7. Decide, record and re-assess. The release passes a scenario only if the upper bound, not the point estimate, is at or below the tolerance, and it passes overall only if it passes every declared scenario, pooled ones included. When the programme fails, the release fails, even if it passes on its own. The owner signs the decision record. Re-assessment is due when the programme is about to pass its horizon, when a new product is drawn from the register, when a new public source could serve as auxiliary information, when the mechanism changes and at expiry.

The long form of the paper lists six stages where this list has seven. The difference is one of presentation: the long form folds the inventory of Step 1 into its first declaration stage, and the planning of Step 4 into its computation stage. The seven-step version is the one an institution can staff, because each step has one output and one lead.

When Grace audited the record for Tomás's third request, she checked the steps against their dates. The inventory listed the three extracts and the quarterly aggregates drawn from the same register. The scenario sheet was dated a week before the tolerance was signed, and the tolerance was signed two days before the registry showed its first attack. She noted one finding: the reviewer's sign-off on the falsification tests was dated after the decision. The tests had been run on time; the sign-off was late. She recorded it as a control weakness, because a late sign-off cannot show that the decision waited for it.

16.4 A filled decision record for the credit register

Table 16.2 fills the record for the synthetic credit register of Chapter 13, for the first vintage and for the programme of six. It rebuilds the decision record of the long form of the paper and uses only figures that the long form reports from the repository's runs. The register is synthetic, so the record shows how the method fills a record, not what any central bank would find. The tolerance of 0.12 is illustrative, as are the owner, the dates and the legal basis.

The declarations common to every row sit outside the table, because they are the same for each harm. The legal basis is Council Regulation (EC) No 2533/98, Article 8, as amended by Council Regulation (EC) No 951/2009 (Council of the European Union, 1998, 2009), given here for illustration. The protected parties are the borrowers, the reporting agents and the natural persons behind borrowers, such as sole traders. The access route is a scientific-use file for accredited researchers. The error level is $\alpha = 0.05$. The owner is the chair of the statistics release committee. The record expires one year after the assessment, and it is re-assessed earlier on a vintage beyond the horizon $m = 6$, on a new auxiliary source available to the adversary, on a change of mechanism or on a change of access route. The floor of 150 was chosen by fixed-sequence testing from 300 down and confirmed on a second, independent draw of targets.

The record is worth reading row by row, because its value lies in what the rows disagree about. The first row is the assessment a committee would have made under per-release review: one release, the outside adversary with one reading of exposure, and a bound of 0.017 at the chosen floor. On its own this row would need no suppression at all. The second row is the programme. The record declares the full outside adversary, who pools six vintages, reads exposure in each, and knows in which of them the target's loan was live, and against that adversary the floor of 150 passes with a confirmed bound of 0.109. The row then states what the floor costs, a fifth of the loans and 4 per cent of the exposure, because the owner has to weigh that cost and cannot do so from a risk figure alone.

TABLE 16.2: Decision record for the synthetic credit register, on the first vintage as released and on the programme of six vintages, tolerance 0.12 where one is set. Declarations common to every row are given in the text. The bound is the exact one-sided 95 per cent upper bound from B simulated attacks; where a row reports a census of released cells instead of a simulation, the measure is named.

Harm	Target set	Adversary	τ	Evidence	Setting	Decision
Identification, one release	Released loans, first vintage	Linkage attack with one exposure reading	0.12	At $k_{\min} = 150$: confirmation 0.0135, bound 0.017, $B = 4{,}000$; this row alone would need no floor	$k_{\min} = 150$	Pass
Identification, programme	Loans over six vintages, $m = 6$	Pooled attack, one reading per live vintage, knows the loan's history	0.12	Floor 150 by fixed sequence; confirmation 0.101, bound 0.109, $B = 4{,}000$ fresh targets; keeps 20% of loans, 4% of exposure	$k_{\min} = 150$	Pass on risk; owner decides whether the file is still worth releasing; re-assess before vintage 7
Identification, every size class	Each size class, largest firms on a stratified draw	As in the second row	0.12	At $k_{\min} = 150$ the worst class bound is 0.184; no floor up to 200 passes	None found	Fail; offer aggregates or secure access
Identification by an insider	Released loans	A party to the loan, log exposure error 0.01, knows the history	Not set	Success 0.772 at $k_{\min} = 1$, 0.416 at $k_{\min} = 300$	Not governed by $k_{\min}$	Fails at every floor once a tolerance is set; restrict access
Relationship allegation	Released borrowers	Names the largest lender in the key class	Not set	Predictive value 0.328 at $k_{\min} = 1$, 0.313 at $k_{\min} = 20$	Not governed by $k_{\min}$	No decision until τ is set
Arrears status	Large borrowers	Reads status off a homogeneous key class	Not set	Homogeneity share 0.119 at $k_{\min} = 20$	Attribute rule needed	No decision until τ is set
Contribution from totals	814 published cells	A participating lender, by subtraction	–	5.9% of cells recoverable	Three-contributor rule; dominance rule not assessed	Rule-based; not a probability
Differencing	Consecutive cell pairs	Claims a single move on a unit change	Not set	A unit change is a single move in 65% of cases at $c_{\min} = 1$, 56% at $c_{\min} = 10$	$c_{\min} = 10$	No decision until τ is set

The third row fails. When every size class must meet the tolerance, with the largest firms drawn on purpose, the worst class bound at the floor of 150 is 0.184 and no floor up to 200 passes. The fourth row names the insider, Pieter's position, whose success is 0.772 with no suppression and 0.416 at a floor of 300. No floor governs him; access does. The next two rows, the relationship allegation and arrears status, have no tolerance, because the institution has not decided how much of those harms it will accept. The record says so and makes no decision on them. The seventh row is governed by a rule and not by a probability: a participating lender can recover 5.9 per cent of the 814 published cells by subtraction, and the three-contributor rule answers it. The last row shows a differencing finding with a floor applied and no tolerance yet.

Read together, the rows decide the file. One row passes, one passes at a cost the owner must accept, one fails at every floor tested, one would fail at every floor once a tolerance is set, three cannot reach a decision, and one is governed by a rule. The decision on the file as a whole follows from those rows and not from the one that passes. The long form draws the conclusion for this synthetic register: against an adversary who knows loan histories, a loan-level file is the wrong product, and aggregates or secure access are the options a committee can defend.

The record also shows what per-release review would have hidden. The per-release reading of the rule releases all 27,167 loans of the first vintage; the programme reading against the full adversary releases 5,553, with 4 per cent of the exposure. Whether the adversary knows the loan's history moves the programme floor from 1 to 75, and whether it also holds a reading in every vintage moves it from 75 to 150. These are the differences between declarations, and a record is the only place where a committee can see them side by side.

The chair's first reaction to the record was that it made the decision harder. A single pass would have been easier to sign. Naledi's answer was that the single pass existed, in the first row, and that it described an adversary the committee did not believe in. The chair signed the record with the third row's remedy: Tomás would work with the loan-level file inside the Bank's secure environment, and the extract that left the building would be aggregate.

16.5 When a release fails

A failed scenario does not end the matter. It tells the committee what to change, and the record states which remedy was chosen and why. The remedies fall into five groups, in roughly increasing order of what they cost the user.

The first is to change the key: coarsen the published breakdown, or withdraw the field the adversary uses most. On the synthetic register the exposure reading, not the key, carries the attack; withdrawing it lowers identification from 0.081 to 0.025. The second is to raise the floor, choosing the level by fixed-sequence testing on the upper bound of the pooled scenario and confirming it on fresh targets. The third is to break the link between vintages by rotating pseudonyms, changing the row order or publishing some products less often. Rotation controls linkability and does not make a record anonymous in the legal sense, and the pseudonym should be a keyed hash under a key held apart from the release team. It helps only if the published fields cannot relink the records: on the synthetic register the key and the exact exposure relink every loan from one vintage to the next, and with exposure rounded to one significant figure about one loan in ten (Chapter 9). The fourth is to add noise under a formal mechanism and account for its budget over the horizon, remembering that the guarantee bounds posterior odds and does not name a linkage probability (Chapter 10).

The fifth is to change the access route or the product: move the file from public or scientific-use release to a secure research environment, where access controls and output checking govern the risk (Desai et al., 2016), or publish aggregates in place of loan-level records.

Shortening the horizon is also a remedy, but only if the institution is prepared to stop the series or re-assess it at the new horizon. A horizon shortened on paper while the series continues is no remedy at all.

16.6 Starting small

An institution does not need to apply the protocol to every product at once. A workable start is one register and one product family over one quarter. Choose the register whose products are published most often. Complete the programme inventory for it and agree a horizon. Agree one tolerance, for identification, with a named owner. Run one per-release attack and one pooled attack, with the attack counts of Step 4. File one decision record and present it to the release committee beside the current method, so that the committee can see where the two disagree.

The pilot's purpose is that comparison. A committee that sees its current method and the protocol reach the same answer has learnt that its method was sound for that product and can say why. A committee that sees them disagree has found the case that the protocol was built for, and has found it on one product, before it affects many. The pipeline that produces this book runs the attacks and writes the record; it needs, for each product, the published file or its specification, the key fields, the auxiliary variables the adversary is declared to hold and the tolerance. Its demographic path has run on two public files (Chapter 12); its credit path has so far run only on the synthetic register.

16.7 What the method adds to the security of a release system

A statistical release system can be assessed in the terms used for any other information system, and readers in information security, operational risk and internal audit will find the method easier to place in those terms. This section restates it so, following the long form. It does not claim that the method meets, or is certified under, any security standard.

The asset is the confidential register: its units and their attributes. Disclosure review faces two adversaries. The first, $\mathcal{A}_{\text{data}}$, holds the releases and the declared auxiliary information and tries to identify a unit; on the credit register the record declares three outside classes of it and reports the insider beside them. The second, $\mathcal{A}_{\text{op}}$, is whoever runs the audit and wants a release to pass, through carelessness, deadline pressure or intent. Per-release review has a model of neither. The method gives a claim about the first that holds only if the second is contained, so both belong in the record.

The claim is this. For each declared adversary and harm class, with probability at least $1 - \alpha$, no release is certified whose declared attack succeeds at a rate at or above the tolerance. The probability is over the simulated attacks, given the register and the declared attack. It holds against $\mathcal{A}_{\text{op}}$ only if five things are trusted: the registry that stores the audits and the error budget, one per programme; the randomness from which targets are drawn, which the operator cannot predict when it registers an audit; the attack code, fixed by a digest when it is registered; the list of attacks declared for each release, fixed before its first outcome; and a time source the operator does not control. An operator who chooses the seed knows every outcome in advance, and then no error statement holds, which is the content of Table 16.1.

The repository implements matching controls in its adaptive audit module. Targets are derived from a secret committed at registration and a later round of a public randomness beacon, so that neither the operator nor the beacon alone chooses them. Each bet is recorded in a hash-chained ledger before the attack runs, and its outcome after. The ledger's entries cannot be changed, dropped or reordered without breaking the chain or changing its head. With the controls in place, and the registry trusted, the rates in the table return to those of an honest single audit.

Against the outside adversary the method adds four properties to per-release review. It assesses the pooled attack over the declared horizon, which per-release figures cannot bound (Proposition 9.2). It compares an upper bound with the tolerance, so the chance of false assurance is at most α, where a point estimate passes a release at the boundary about half the time; and when the evidence is watched as it arrives, it uses a sequence that stays valid, where twenty looks at a fixed-sample bound put it below the true risk in 21 per cent of runs. It fails closed: an attack that has not been audited is given a bound of one, and the pipeline returns a fail when the pooled bound exceeds the tolerance even if each release passes on its own. And a new attack proposed by a red team enters with its own share of the error budget, without voiding certificates already issued, if budget was reserved for it. Against the operator, the method makes audit shopping, the repetition of an audit or the scanning of settings until one passes, either impossible or paid for out of the error budget. That holds only under complete mediation (Saltzer and Schroeder, 1975): every attack run on the confidential register passes through the registry, and development work uses a separate extract.

What the method does not give should be stated as plainly. The implementation is a library, not a security boundary: read-only fields in Python are not a security control, since code in the same process can change them. The SHA-256 hashes a run writes show that a file has changed since the run, but they carry no key or trusted time stamp, and the ledger's chain detects changes only if its head is published or time-stamped outside the operator's control. A deployment that must hold against the operator needs the registry as a separate service with its own storage, signing keys and time stamps, which the repository specifies and has not built. It needs separation of duties, in which an independent team writes and signs the attack code, a data custodian holds the register and the randomness commitment, and the owner signs the tolerance before the first outcome. And it needs verification by internal audit inside the secure environment, since the log of which targets were identified is itself confidential. The method gives no guarantee against attacks that were not registered, and no differential privacy guarantee.

In the language of security practice, the declared scenario is a threat model (Shostack, 2014): asset, adversary, capability and harm written down before testing. The attack simulation is red-team testing with a statistical stopping rule. The decision record is an assurance case: a claim, the evidence for it and the argument that links them, with a named owner and an expiry date. The mapping helps an institution place the method within its risk and audit processes. It is not a claim of conformity.

For Grace, this section is the part of the method she can test. She cannot judge whether the declared adversary is realistic; that is the reviewer's work. She can check that the registry holds every audit, that the targets were derived from a commitment made before the beacon round, that the ledger's head matches the copy published at registration, and that the tolerance's signature predates the first entry. Each of those checks either passes or fails on the documents.

16.8 What the protocol does not decide

The protocol makes a decision reproducible. It does not make the decision for the institution. It cannot say whether a declared adversary is realistic, whether a tolerance is legally or ethically right, or whether a release is worth its risk. Those remain judgements for the release owner. The repository's own register of claims puts the protocol's benefit in the same terms: that it improves auditability is an institutional argument, supported by a machine-readable record, generated tables and hashes that show it can be done, and it would be refuted by comparative evidence that it does not help or costs too much. No such comparison has been run.

The answer to the chapter's question can now be given plainly. For a release decision to carry the error statement the method promises, the institution must write down, before any attack outcome exists, the programme and its horizon, the declared adversaries, the tolerance with its owner, error level and expiry, and the plan of measurement, including the order in which settings are tested. The analyst then measures and the reviewer tests the model, each recorded in a registry that holds every audit. The owner decides from the full record, and internal audit checks the order afterwards. The writing is not overhead on a safe release. Without it the probability statement attached to the bound does not hold, and with ten unrecorded tries an audit at the tolerance passes nearly a third of the time.

IN PRACTICE: RUNNING THE PROTOCOL

Assign the four roles to named people, and keep the analyst and the owner apart. Date every output: the inventory, each scenario sheet, the signed tolerance, the measurement plan, every registered audit, the falsification results and the signed decision. The tolerance's signature must predate the first audit outcome, and every attack run on the confidential register must appear in the registry, including failed and abandoned runs. File a record with one row per harm, including the harms for which no tolerance has been set. When a release fails, record the remedy and assess the changed product from Step 1. Start with one register and one product family, and compare the protocol's record with the current method before extending it.

In the software. `privacy_risk.pipeline.gate` writes the decision record (`decision.json`) with the release and programme outcomes, and `governance.decide` applies the tolerance and the expiry date. The audit controls are in `privacy_risk.adaptive_audit`, and `scripts/audit_process_attacks.py` produces Table 16.1. The governance record the code writes holds six of the nine elements of the threshold. The other three, the legal basis with the protected parties, the access route and the horizon, have to be added by hand (Appendix A, Chapter 7).

```
@dataclass(frozen=True)
class GovernanceThreshold:
    tolerance: float
    owner: str
    expiry_date: str
    adversary: str
    harm_class: str
    alpha: float
```

16.9 Proofs

Proof of Proposition 16.1. (i) With τ fixed before the attacks, the pass event $\{\overline{R} \leq \tau\}$ is contained in $\{\overline{R} < R\}$ whenever $R > \tau$, and the latter has probability at most α.

(ii) If τ may take any value in a range containing $\overline{R}$, the rule $\tau = \overline{R}$ is admissible, and under it $\overline{R} \leq \tau$ holds on every outcome, so the release passes with probability one whatever R is.

(iii) Let $P_1, \ldots, P_N$ be the pass indicators of N independent audits, each with $\mathbb{P}(P_i = 1) = f$. The operator can report a passing audit exactly when at least one passes, which has probability $1 - \prod_i \mathbb{P}(P_i = 0) = 1 - (1 - f)^N$. For $0 < f \leq 1$, $(1 - f)^N \to 0$ as $N \to \infty$. $\square$

Chapter 17

Worksheets for six release types

A common view among central bank statisticians is that their release types are too different for one method. Aggregate tables from a credit register are protected by contributor and dominance rules, survey microdata by recoding and the journalist measure, ad hoc requests by a reviewer's judgement, and each tradition has its own literature (Hundepool et al., 2012). On this view a single framework can only be a lowest common denominator. The view has a fair basis in the tools, which do differ. It does not hold for the decision. Every one of the six release types in this chapter fills the same four slots: whom the adversary targets, what it holds, which harm follows and which measure answers the question. What changes from type to type is the content of the slots, not their number.

The chapter asks one practical question: what should an institution write in the worksheets of Chapter 16 for the release types it actually publishes? It gives starting declarations for six types, then the blank worksheets and a checklist. The declarations are illustrations to be adapted to each institution's products, legal basis and users. They do not describe the practice of any institution.

17.1 Starting declarations

Table 17.1 maps six release types common in central banks to the objects of Steps 2 to 5 of the protocol. The first column names the product. The others say whom the adversary targets, what it plausibly holds, which harm is at stake, which measure in this book applies and where a pooled scenario arises. The last column is the one most often left empty in practice, and it is the one Proposition 9.2 says cannot be left empty when records or cells persist across vintages.

The six rows are discussed in turn below. Each discussion says what the declaration must settle before any attack is run, what evidence the earlier chapters offer, and where the assessment usually goes wrong.

Aggregates from a credit register

Tables of credit by sector, region and size class are the most frequent product of a credit register, and they face two adversaries at once. An outsider who knows a firm's sector, region and size from its public filings looks for the cell where the firm sits alone or nearly so. A participating lender knows its own contribution to every cell and can subtract it from a published total to read what remains. The first is measured by the prosecutor risk of

TABLE 17.1: Starting declarations for six release types common in central banks. The entries are illustrations to be adapted to each institution's products, legal basis and users; they do not describe the practice of any institution.

Release type	Targets	Adversary holds	Harm	Measure	Pooled scenario
Aggregates from a credit register	Borrowers in small cells; lenders in concentrated cells	A firm's sector, region and size from public filings; a lender's own book	Identification of a borrower; a lender's position revealed	Prosecutor risk by cell; residual disclosure from totals	Monthly vintages; differencing of consecutive months
Loan-level research extract	Every borrower in the extract	Key fields and a noisy reading of exposure from other sources	Identification; disclosure of arrears	The linkage attack of Chapter 13; an attribute rule	Successive extracts for one project; extracts for different projects
Supervisory statistics by bank group	Individual banks	Each bank's own figures; published annual reports	A bank's confidential figure recovered	Contribution from totals; dominance and p% rules	Quarterly series; tables that share a margin
Payment and transaction statistics	Merchants or payers in thin cells	Merchant category and location; own transactions	Identification; commercial information revealed	Prosecutor risk by cell; minimum contributor count	Daily or weekly series of the same cells
Household or firm survey microdata	Respondents	Demographic or firm characteristics from registers or social media	Identification; disclosure of wealth or debt	Journalist and marketer measures; the frequency model	Survey waves; a public file and a scientific-use file
Answers to ad hoc requests	Units in the requested breakdown	Everything already published from the same register	Identification by differencing	Differencing posterior	Each answer pooled with every earlier publication

each cell (Chapter 3). The second is a matter of arithmetic, not probability: on the synthetic register a participating lender can recover 5.9 per cent of 814 published cells by subtraction, which is what the three-contributor rule is written to prevent. A dominance rule usually sits beside the contributor rule, and the record should say whether it was assessed. The pooled scenario is the monthly series: a cell that changes by one unit between two months may reveal a single loan's movement, and the record should carry the differencing posterior for consecutive months (Chapter 13).

Loan-level research extracts

A research extract puts every borrower in the file, so the target set is the whole extract and the adversary holds the key fields plus whatever it can read of the exposure from other sources. The declaration that decides the result is the precision of that reading. On the synthetic register, halving its standard deviation from 0.6 to 0.3 on the log scale moves the programme floor from a median of 150 to a median of 300 over twenty register draws. The worksheet should therefore record the reading precision and the argument for it. The pooled scenario arises when one researcher receives several extracts, for one project over time or for different projects, and the declaration must say whether the researcher can know in which

vintages a target's loan was live. Arrears status needs an attribute rule as well, since a key class in which every large borrower is in arrears discloses that status without naming anyone.

Supervisory statistics by bank group

When a table reports figures by bank group, the targets are individual banks and the adversary is often another bank, which knows its own figures and can read the others' published annual reports. The harm is the recovery of a confidential figure, and the relevant tools are the contributor and dominance rules of the sensitivity tradition (Hundepool et al., 2012). These are rules, not probabilities, and the record should report them as such, in the way the seventh row of the decision record in Chapter 16 does. The pooled scenario is the quarterly series and any pair of tables that share a margin, since a margin shared by two tables can be subtracted from both.

Payment and transaction statistics

Payment statistics are published at high frequency, and their thin cells, a merchant category in a small town, can contain one or two merchants. The adversary knows the category and the location and, if it is a participant, its own transactions. The measures are the prosecutor risk by cell and a minimum contributor count. The pooled scenario is the defining feature of this release type: a daily or weekly series of the same cells gives an adversary many readings of each, and the horizon of Step 1 should be set in days or weeks, not in years.

Household or firm survey microdata

Survey microdata are the case for which most of the classical measures were built. The targets are respondents, and the adversary holds demographic or firm characteristics from registers or social media. Because the file is a sample, the journalist measure, which asks whether a sample-unique record is unique in the population, needs a model for the population cell sizes, and the transport test of Chapter 15 applies to that model. The marketer measure, the expected share of records an adversary would link correctly, answers a different question and protects a different party, and the worksheet should say which of the two the tolerance is for. The pooled scenario is a sequence of waves, or a public file and a scientific-use file cut from the same survey.

Answers to ad hoc data requests

An ad hoc request is the release type most often assessed alone and least suited to it. The adversary is anyone who holds everything already published from the same register, so every answer is pooled with every earlier publication. The harm is identification by differencing, and the measure is the differencing posterior. The practical consequence is that the inventory of Step 1 has to include past answers, which many institutions do not keep in one place.

When Naledi filled the first worksheet for the credit register, the row for ad hoc answers took the longest. Three departments had answered requests from the register in the previous year, and two kept no list. She recorded what could be found and noted the gap in the record, since an inventory that is known to be incomplete is more useful to the owner than one that looks complete.

17.2 Blank worksheets

The three worksheets below follow the steps of Chapter 16. They are meant to be copied into an institution's own templates. The decision record of Table 16.2 shows the third worksheet filled for the synthetic credit register, with the declarations common to its rows given in the text.

Worksheet 1. Programme inventory (Step 1)

One row per product drawn from the source register over the planning horizon. The fifth column is the one that decides whether pooled scenarios are needed.

Product drawn from the register	Frequency	Population covered	Variables published	Link across vintages (pseudonym, row order, stable attributes)	Access route

Source register: _____ Horizon m (vintages): _____ Data owner: _____

Worksheet 2. Scenario declaration (Step 2), one per scenario

One sheet per declared adversary, single-release or pooled. The stress-test row names the stronger adversary that the falsification test of Chapter 15 will run.

Item	Entry
Scenario name and products it holds	
Adversary (who, and why it would attack)	
Auxiliary information held, with its error	
Target set (prosecutor, journalist or marketer)	
Success event (what counts as a correct attack)	
Harm class	
Single release or pooled over the horizon	
For a pooled scenario: does the adversary know in which vintages the target is present, and why	
Stronger adversary for the stress test	
Declared by, and date	

Worksheet 3. Decision record (Steps 3 to 7), one row per scenario

One row per scenario and harm, including those for which no tolerance has been set. The decision column takes one of four values: pass, fail, invalid (a falsification test failed) or not set (no tolerance has been agreed).

Scenario	Legal basis and protected parties	Access route	τ	α	Attacks B	Upper bound	Model tests passed	Decision	Remedy if failed

Release owner: _____ Signed on: _____ Expiry: _____
Re-assess when the horizon is reached, a new product is drawn from the register, a new public source could serve as auxiliary information, the mechanism changes, or the record expires.

17.3 Checklist before release

The checklist is for the reviewer and, afterwards, for internal audit. Each item can be answered yes or no from the documents.

1. Every product drawn from the register over the horizon is listed.
2. Every scenario names its adversary, auxiliary information, target set and harm.
3. A tolerance, error level, owner and expiry are recorded for every harm, or the record says that none has been set.
4. The legal basis, the protected parties and the access route are recorded for every product.
5. The number of attacks, the error level of each scenario and the order in which any grid of levels is tested were fixed before any result was seen.
6. Pooled scenarios were run with the same auxiliary information as the single-release ones; any pooled scenario that adds knowledge is reported separately; all are reported.
7. The decision uses upper bounds, not point estimates.
8. All six falsification tests were run against bands fixed in advance, and their results are filed.
9. The reviewer is not the analyst, and the release owner signed the record.

The answer to the chapter's question is the worksheets themselves, filled from the starting declarations of Table 17.1 and changed wherever the institution's own products, users and law differ from them. The six release types differ in their targets, adversaries, harms

and measures, and they share the four slots and the pooled column. An institution that fills the slots for its own products, and leaves none empty without saying why, has done Steps 1 to 3 of the protocol.

IN PRACTICE: USING THE WORKSHEETS

Copy the three worksheets into the institution's own templates and keep the column for links across vintages and the row on the adversary's knowledge of presence, since those decide whether a pooled scenario is needed. Start each declaration from the matching row of Table 17.1 and record every change from it with a reason. Keep ad hoc answers in the programme inventory. Record rows for harms that have no tolerance yet, with the decision "not set".

In the software. The pipeline fills the decision record for the declared scenarios of a run (`privacy_risk.pipeline.gate`, written to `decision.json`); the credit register's harms are measured by `privacy_risk.credit` and attacked by `privacy_risk.evaluation`. The inventory and scenario sheets have no counterpart in code and are kept as documents.

Part VII

Continuous adversarial learning

Chapter 18

Continuous adversarial learning

Many people who build or buy security systems bring a belief about learning that the rest of computing has taught them. A system that watches for attacks, records each one that succeeds and adapts its defences grows stronger with every attempt. Spam filters work this way, and so do the fraud models of commercial banks. Carried over to disclosure control, the belief says that a system which keeps searching a register for weaknesses, learns which protection settings resist the attacks it has found, and tries again whenever a candidate fails, will in time make a release safe against hacking. The belief has a sound core. A system that never searches finds nothing, and a system that searches without learning spends its effort where it has already looked. The extension of November 2026 on which this part of the book is built takes that core as its central research direction: a system that keeps searching for disclosure weaknesses, learns from measured attack outcomes and identifies stronger protection before a release is authorised.

The belief fails at the step that looks most like diligence, which is trying again. Take a release whose true risk is 0.201 against a tolerance of 0.2, so that any certificate it receives is false. Audit it with 500 simulated attacks and judge the audit by an exact one-sided bound at the 5 per cent level. When an attempt fails, let the system learn what it can, draw fresh targets and try again, judging each new attempt at 5 per cent, up to 20 attempts. Over 2,000 simulated programmes the release was certified in 50.15 per cent of them. Had the same 5 per cent been divided among the 20 attempts before the first one ran, the share would have been 4 per cent. Chapter 6 reported the same split from an older script over 20,000 programmes, 51.0 against 3.9 per cent; the extension's evidence run reproduces it with the engine's own code.

This chapter asks one question: what may a system that keeps learning about attacks claim about a release, and what has to stand between its learning and the decision to publish? The answer takes four sections. Section 18.1 fixes what the learning system observes. Section 18.2 separates the power to propose from the power to admit, and Section 18.3 places both in a cycle. Section 18.4 shows what repeating the cycle does to the error, and Section 18.5 says what the system can do about a weakness in a file that is already out.

18.1 What the learning system observes

The system learns from development cases. For case i, let x_i describe the permitted characteristics of the data and its release history, θ_i the protection setting proposed for it, and a_i the attack scenario. A versioned attack evaluator runs the attack and returns two integers: K_i

successes in B_i trials. A predictor with parameters ϕ then estimates

$$\widehat{r}_\phi(x_i, \theta_i, a_i) \approx \mathbb{P}\{\text{attack succeeds} \mid x_i, \theta_i, a_i\}. \tag{18.1}$$

The conditioning in (18.1) is fuller than it looks. It includes the declared law from which targets are drawn and the convention on membership, whether success is counted over all targets or only over those in the release. These are the objects of Chapter 2, and Chapter 19 returns to them for releases that change over time. When an attack uses several releases, x_i has to describe their history, or the evaluator has to receive the whole permitted history. A predictor trained on single releases cannot be assumed to predict pooled disclosure, for the reason Chapter 9 proved: no function of per-release risks bounds the pooled one.

The outcome is kept as two integers because a rate throws away the information that says how much to trust it. A success rate of 0.1 measured on ten trials and the same rate measured on ten thousand carry very different weight in a binomial likelihood, and the exact bounds of Chapter 4 need K and B separately. The engine enforces this. A `MeasurementRecord` refuses a success count or a trial count that is not an integer, and each record names the scenario, the mechanism, the target law, the conditioning event, the data split, the stream of random seeds, and SHA-256 digests of the data and of the evaluator. A record in the final-audit split must also name the hash of the frozen plan it belongs to and the share of the error budget allocated to it. Two rows that differ in any of these identifiers measure different experiments, and the learning system may not pool them without saying so.

The same discipline applies across kinds of harm. Linkage, membership inference, attribute inference and reconstruction are separate harm classes, and Chapter 13 found five distinct harms in one credit register. They need not share a denominator, a tolerance or a measure of utility. A training objective that pools them has to represent each consistently first, or the predictor will learn an average of quantities that answer different questions.

A language model can enter at this point, as Chapter 14 described. It can suggest an auxiliary source the library has missed, a combination of releases, or a strategy nobody has tried. A reviewer and a deterministic evaluator then turn the suggestion into an experiment that can be run, and the label is the count the experiment returns. What the model expected to happen is not a label. Proposition 14.2 showed why: only measured counts carry information about the risk. A predictor trained on counts that a model generated learns the generator's frequency, which is the model's guess about the risk and not the risk.

Naledi's first task was an inventory of what the Bank already knew in this form. Most of it was in tables of rates from the work of Chapter 13, and an older helper in the software had rebuilt counts by multiplying rates by the number of replicates. The engine now takes counts directly and warns when it has to rebuild them. Grace, reading the inventory, asked for one thing: that each row should say how many trials stood behind it. Naledi regenerated the rows from the stored runs, and the committee's first exhibit became a list of integer pairs with their identifiers.

18.2 Learning proposes; evidence authorises

The decision to release is a constrained choice. At review epoch t, let $\mathcal{A}_t$ be the family of attacks the institution requires, Θ_t the frozen set of candidate protection settings, and $V_t(\theta)$ the usefulness of a release under setting θ for its declared analyses. For each required attack a, let $U_{a,t}(\theta)$ be a valid upper confidence bound on the attack's success and $\tau_{a,t}$ its approved

tolerance. The release gate chooses

$$\theta_t^* \in \arg\max_{\theta \in \Theta_t} V_t(\theta) \quad \text{subject to} \quad U_{a,t}(\theta) \le \tau_{a,t} \text{ for every } a \in \mathcal{A}_t. \tag{18.2}$$

This is the selection problem of Chapter 5 with one change of emphasis. There the candidates were a grid of minimum cell counts and the order of testing was fixed in advance; here the candidates may have been proposed by a learning system, and the gate has to keep its guarantee however they were found.

Four rules turn (18.2) into something an institution can operate. A missing audit coordinate is not zero risk: a candidate with no bound for a required attack fails that attack. If the feasible set is empty, the release is withheld, redesigned or moved to a more restricted access route. The learning system never relaxes a tolerance to obtain a pass. The institution may reconsider its tolerances, but through the governance process of Chapter 8, with a new decision record and a named owner. And the predictor's estimate (18.1) never enters the constraint. The network can rank candidates, draw up a shortlist or decide where audit effort should go, but the bound $U_{a,t}$ comes from fresh audit attacks under the declared sampling design, and when several candidates are inspected the bounds are simultaneous, as in Proposition 14.1, or come from a sequential procedure with its own proof, as in Chapter 6.

Proposition 18.1 (The release gate)**.** *Let $\mathcal{D}$ be all development information, including every model fitted and every candidate proposed. Conditional on $\mathcal{D}$, fix the candidates Θ_t, the required attacks $\mathcal{A}_t$, the tolerances and the audit design, and let E be the event that $r_a(\theta) \le U_a(\theta)$ for every pair (θ, a) for which an audit bound exists. Suppose $\mathbb{P}(E \mid \mathcal{D}) \ge 1 - \alpha$. Then:*

1. *for any utility V_t, the probability that the gate releases a candidate θ with $r_a(\theta) > \tau_a$ for some required a is at most α;*

2. *adding a required attack, or lowering a tolerance, never enlarges the feasible set;*

3. *if a missing coordinate were counted as zero risk, no bound on the probability in part 1 below one would hold.*

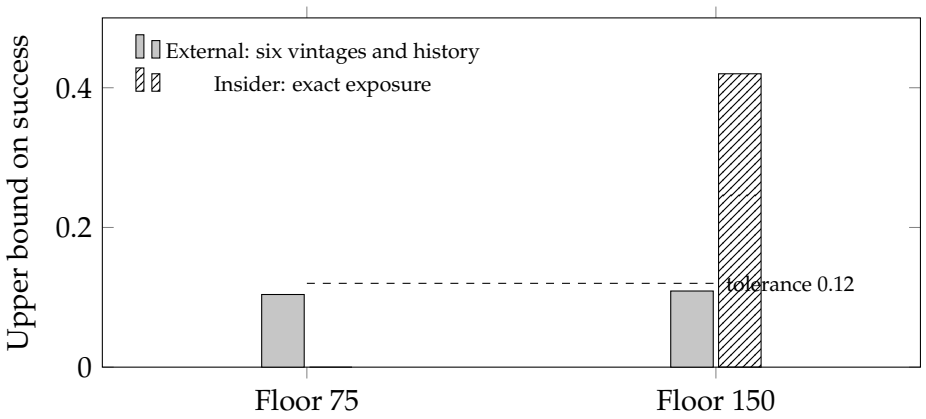


FIGURE 18.1: The bounds the release gate received in the extension's evidence run. Solid grey bars are the external adversary who pools six vintages and knows the loan's history; hatched bars are the insider. The dashed line is the tolerance. The floor of 75 has no insider audit, which the gate counts as a failure (no hatched bar is drawn there). With the insider required, no candidate is feasible and the gate withholds; with only the external adversary required, it releases the floor of 75.

The proposition puts the gate's whole guarantee in the event E. Nothing in it depends on how good the predictor was, how the candidates were proposed, or how the utility was estimated. A poor predictor costs audit effort and may leave a useful candidate off the shortlist. It cannot raise the chance that an unsafe candidate is released, because it never touches the bounds. This is the old design principle of separation of privilege, under which a protection mechanism should need two keys (Saltzer and Schroeder, 1975): learning holds one key, which is the power to propose, and the audit holds the other, which is the power to admit. Part 3 is the reason the gate refuses to treat silence as safety. A candidate for which no one has run the insider attack has not passed the insider attack.

Utility needs evidence of its own. If V_t is estimated with uncertainty, the uncertainty should be recorded and tested on the analyses the release is meant to serve. A gate can stay statistically valid while a badly estimated utility picks a release that helps nobody. That is a failure of a different kind from false certification, and it should be reported as such.

The extension's evidence run applies the gate to two candidates from the credit register of Chapter 13: a minimum cell count of 75, which keeps 45 per cent of loans, and one of 150, which keeps 20 per cent. The tolerance is 0.12 for every attack. Figure 18.1 shows the bounds the gate received. For the external adversary who pools six vintages and knows the loan's history, the bounds are 0.104 at 75 and 0.109 at 150, both below the tolerance. For the insider, the gate received a bound of 0.42 at 150, close to the 0.416 that Chapter 13 reports for the insider even at a floor of 300, and no bound at all at 75. When the insider is a required attack the gate returns WITHHOLD: the floor of 150 fails the insider and the floor of 75 is missing the insider coordinate. When only the external attack is required the gate returns RELEASE and chooses 75, the candidate that keeps more loans.

The example also shows the limit of what the gate can check. The run files both external bounds under one attack name, but they come from two rows of Table 13.4: 0.104 is the confirmed bound at 75 for the adversary who holds one reading of exposure, and 0.109 is the bound at 150 for the adversary who holds a reading in every vintage. The second adversary needs a floor of 150, so had the gate been given that adversary for both candidates, 75 would have failed it. The gate decides on the coordinates it is handed. It cannot notice that two of them answer different questions. That check belongs to the person who writes the plan, and to the identifiers on each measurement record, which is why the record names the scenario as well as the attack.

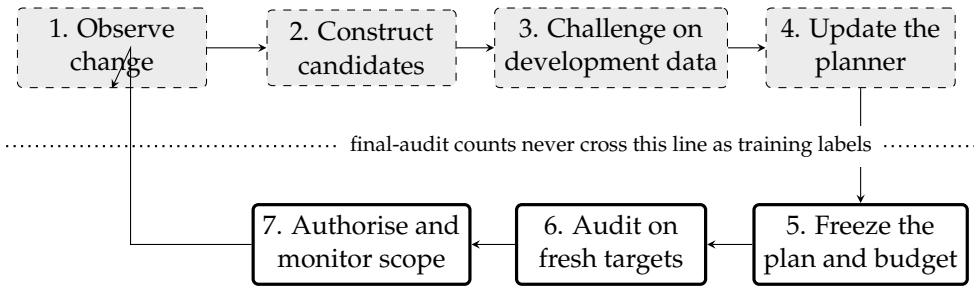


FIGURE 18.2: The seven stages of the protection cycle. Dashed boxes on a grey ground are learning: they may use any development evidence and change as often as needed. Heavy boxes on a white ground are evidence and decision: their plans are frozen before outcomes exist, and their counts are stored for governance. The dotted line marks the separation of Section 18.4.

When Naledi showed the committee the withheld decision, the chair asked the obvious question: could the system not search until it found a setting that passed the insider? Naledi

pointed to Proposition 13.2. An adversary who reads exposure precisely separates candidates in a cell of any size, so no minimum cell count can stop Pieter, and rounding exposure to one significant figure only brings his success with no floor down to 0.388. The search could go on for ever and the feasible set would stay empty. What the chair could decide, and the system could not, was whether Pieter belonged in the required family for this file. Tomás would work in the Bank's secure environment, with outputs checked before they left, and no reporting lender's staff had access to it. The chair recorded the insider as outside the scope of that access route, signed the external-only plan, and the gate chose 75 for the secure room. The decision record says who removed the insider from $\mathcal{A}_t$, and why.

18.3 The protection cycle

The gate is one stage in a cycle that the extension sets out in seven stages, drawn in Figure 18.2. The first stage observes change: new data, a shift in the distribution of the register, altered access, an additional release, a new auxiliary source outside the institution, or a newly demonstrated attack. For each change it records whether the change could invalidate the scope of an existing certificate. The second constructs candidate releases with explicit mechanism parameters and a declared assessment of their utility, while the confidential data stay protected throughout development. The third challenges the candidates. It runs approved attacks on development evidence, searches across attacker capabilities and pooled release histories, and keeps the searches that failed as well as those that succeeded. The fourth updates the planner: it fits or recalibrates the predictor, compares it with simple baselines, inspects the subgroups where it fails, and versions the data, evaluators and models so that any apparent improvement can be traced.

The last three stages are of a different kind, and the figure marks them differently. The fifth freezes the proposed decision. Before any final outcome is observed, it registers the candidate settings, the required attacks, the tolerances, the target laws, the audit sample design and the allocation of the programme's error budget. The sixth audits independently, running fresh attacks through the registered path, and approves a release only if every required condition passes and the policy authority signs. The seventh monitors the approved scope for the changes that would trigger another review, and keeps clear the difference between a certificate for a specified release and an open-ended claim about the future.

The order of stages five and six carries the statistics. Chapter 16 proved that declarations must precede the evidence they govern (Proposition 16.1), and the cycle applies that order once per pass. Learning in stages one to four may be as adaptive as the analyst likes, because Proposition 18.1 conditions on everything that happened there. What it may not do is reach forward into stage six.

Continuous, in this design, describes the search and not the signing. The predictor can be refitted weekly and the search can run every night. Release authorisation remains a controlled, versioned decision, taken at the pace at which the data and the adversary's information change. A register whose exposures move monthly and whose users receive monthly files needs a monthly review; a public table published once a year does not.

Naledi ran the first pass for Tomás's second file. Stage one found the largest change: the new file would add six months to a history Tomás already held, so the pooled adversary would see twelve vintages where the certificate had assumed six. That change took the request outside the old scope, and Chapter 19 shows why a longer history of a changing register cannot be assessed by reading the old certificate forward. Stage three searched,

stage four refitted the planner on the development counts, and on Thursday Naledi froze a plan with three candidate floors and the external attack only, as the chair had recorded.

Remark 18.2 (Check the retry study yourself). The two figures at the head of the chapter come from a short simulation, and the reader can rerun it. A release has true risk 0.201 and the tolerance is 0.2, so any certificate is false. Each attempt draws 500 attacks and certifies when the exact one-sided upper bound at level α is at most 0.2. Twenty attempts at $\alpha = 0.05$ certified the release in 0.501 of 4,000 simulated programmes; the same twenty attempts at $\alpha = 0.05/20$ did so in 0.041 of them. The Bayesian betting audit of Example 18.5 can be replayed in a few lines in the same way: with the stated forecast the wealth after the first four attacks is 1.023, 1.057, 1.101 and 0.705 and the audit certifies at attack 55, and a clean run reaches 4.0 at attack 20 and 22.2 at attack 37. A fixed audit that certifies only on zero successes needs $\lceil \ln 0.05 / \ln 0.88 \rceil = 24$ attacks. These were recomputed from the formulas, not copied from the extension's run, and they agree with it.

18.4 Leakage across successive cycles

A cycle that runs once has a clean separation between development and audit. A cycle that runs every month does not, unless the institution builds one. The danger is that the final audit of one pass becomes training data for the next, and that the next pass then certifies a candidate on the same favourable sample that taught the planner to propose it. Reusing an audit sample after retraining is not independent verification. It is the scan of Chapter 5, in which the same counts that pick the winner are asked to certify it, and Proposition 5.2 measured the cost of that. Machine learning met the same problem in the holdout set that is consulted again after every round of model changes, and found that reuse invalidates it unless access is rationed (Dwork et al., 2015).

The conservative rule, and the one the engine implements, is that final-audit outcomes are stored for governance and never used as development labels. The function that loads training rows refuses a request that includes the final-audit split, so a permissive query cannot leak final outcomes into development by accident. A failure found in an audit is not lost. It triggers a separately registered development investigation, which draws its own fresh measurements.

A more permissive design is possible, and the next proposition says what it would need. Think of the programme as a sequence of cycles, and let $\mathcal{F}_{\ell-1}$ hold everything known before the final audit of cycle ℓ begins, including the final audits of every earlier cycle.

Proposition 18.3 (Error accounting across cycles)**.** *Let a programme run cycles $\ell = 1, 2, \dots$, and let A_ℓ be the event that cycle ℓ certifies a candidate whose risk exceeds its tolerance for some required attack. Suppose that the error allocation α_ℓ of cycle ℓ is fixed by $\mathcal{F}_{\ell-1}$, that $\sum_\ell \alpha_\ell \le \alpha_{\text{prog}}$ in every outcome, and that $\mathbb{P}(A_\ell \mid \mathcal{F}_{\ell-1}) \le \alpha_\ell$. Then the probability that any cycle in the programme certifies falsely is at most α_{prog}. If instead each cycle is run at a fixed level α with no budget, the only general bound after n cycles is* $\min(1, n\alpha)$.

The proposition lets earlier audits influence later development, and it is exact about the price. Each cycle's audit must be valid conditional on everything before it, which in practice means fresh targets that the earlier cycles have not touched, and each cycle's allocation must be written down before its audit starts. The retry study at the head of this chapter is the second part of the proposition at work: twenty attempts at 5 per cent each allow a bound of one, and the simulation found 0.50. Dividing the 5 per cent among the attempts restores the

first part, and the simulation found 0.04. The conservative rule is a special case that is easier to check, because a final audit that never reaches development cannot make a later audit's targets depend on it.

Random seeds alone do not make two audits independent. The targets themselves can be shared. Two releases can overlap in ways the seed does not see. And a development path and an audit path can read the same records through different routes. Where the target is a borrower with several loans, the dependence has to be reflected in the unit of audit. On the register of Chapter 13 the attack recovered the exact lending relationship for 0.0855 of released rows and the right borrower for 0.1095, because a borrower with several lenders can be found through any of them. When the borrower is the person at risk, the audit samples borrowers.

The engine's registry makes the accounting hard to forget. A programme opens with a total error budget. A plan is registered by its hash before any outcome exists. Each attempt reserves its share of the budget in a database transaction before the audit draws a single target, and an attempt that fails or is abandoned keeps its spend. Two writers cannot both spend the last of a budget, because the reservation takes the database's write lock before it reads the amount already spent. Weaker isolation would let two transactions each read the same balance and each write a spend, an anomaly that Berenson et al. (1995) catalogue for the isolation levels of SQL. Each attempt receives its own audit random stream, from a seed drawn at random when the budget is reserved and recorded with the attempt, so that it can be replayed afterwards but not computed in advance from the plan, and it stays apart from the development streams. The registry detects a later edit to a stored plan by recomputing its hash. It does not prove who wrote a row or when; for that an external timestamp or an independently administered signed store is needed, and both remain open. Dividing a fixed error among looks taken over time is an old device of clinical trials, where an alpha-spending function fixes how much of the error each interim analysis may use (Lan and DeMets, 1983), and Foster and Stine (2008) let a testing programme earn back budget from its discoveries. The registry uses the plainest version: shares fixed in advance, each paid before its audit runs.

Example 18.4 (Four reservations against one budget)**.** Take the programme for Tomás's second file with a total budget of 0.05, and let each attempt reserve 0.0125, a quarter of it. The figures are illustrative. Naledi registers a plan for the floor of 50 and reserves the first share; the registry records the attempt and reports 0.0375 left. That audit fails and is closed as a failure. A second reservation against the same plan leaves 0.025, and it fails too. A plan for the floor of 75 is registered, its reservation leaves 0.0125, and it passes. A fourth reservation, opened by mistake, leaves nothing and is closed as abandoned. Every attempt spent its share, so the ledger shows 0.05 spent. A fifth reservation of 0.0125 is refused, and so is one of 0.001: the programme has no budget left, and only a new decision by the committee can open another programme. By Proposition 18.3, the chance that any of the four certified a release whose risk exceeds the tolerance is at most 0.05. Had each attempt been judged at 0.05 with no ledger, the only general bound would be $4 \times 0.05 = 0.2$.

Sequential audits fit the same accounting. The betting audit of Chapter 6 fixes each bet from the outcomes before it and stops when its wealth reaches $1/\alpha$, and Ville's inequality bounds the chance of a false certificate by α (Ville, 1939; Howard et al., 2021). Testing by betting has become a general method of statistics (Shafer, 2021; Ramdas et al., 2023; Grünwald et al., 2024), and betting strategies of this kind also give confidence bounds for bounded means (Waudby-Smith and Ramdas, 2024). In the extension's evidence run, at a tolerance of

0.12 and $\alpha = 0.05$, a release whose true risk sat exactly at the tolerance was certified in 46 of 1,000 runs, a rate of 0.046 with an exact upper bound of 0.058. Below the tolerance the audit stopped early: the median run used 56 attacks at a true risk of 0.03, 144 at 0.06 and 669 at 0.09, and at 0.09 it certified in 85 per cent of runs within its 1,500 attacks. A fixed audit that certifies only on zero successes would need 24 attacks at the same tolerance. The bets came from a Bayesian forecast, and the forecast set the cost. It did not set the error, which stayed within its bound.

Example 18.5 (The first attacks of a betting audit)**.** Take the evidence run's settings: tolerance 0.12, $\alpha = 0.05$, so the audit certifies when its wealth reaches 20, and bets from a $\mathrm{Beta}(1, 9)$ forecast. Before the first attack the forecast's mean is $1/10 = 0.1$, so the first bet is 0.1. If the attack fails, the wealth is multiplied by $(1 - 0.1)/(1 - 0.12) = 1.023$. The forecast moves to $1/11 = 0.091$, the second failure multiplies the wealth by $0.909/0.88$, and the wealth is 1.057; a third failure, at a bet of $1/12$, brings it to 1.101. Suppose the fourth attack succeeds. The bet was $1/13 = 0.077$, so the wealth is multiplied by $0.077/0.12 = 0.641$ and falls to 0.705. The forecast after a success rises to $2/14 = 0.143$, above the tolerance, so the engine clips the next bet to $0.12 - 0.012 = 0.108$, and each failure then adds only 1.4 per cent. The bets fall again as failures accumulate, and the audit certifies at attack 55. Had the fourth attack failed, the wealth would have reached 4.0 after 20 attacks and 22.2 after 37, and the audit would have certified there. The fixed audit that certifies only on zero successes would stop at 24 attacks with no success and would fail outright at the fourth. The betting audit is slower on a clean run and survives a success; it never certifies a release at the tolerance more often than one time in twenty.

Grace read the registry for Tomás's second file a week after the decision. She found the four attempts of Example 18.4, each with its share spent, and the verification of plan hashes returned an empty list. Her question was why a failed attempt should cost anything, since it had certified nothing. Naledi answered with the retry study: the cost of a failure is that it buys the right to try again, and the budget is what keeps the twentieth try from being a coin toss.

18.5 After a weakness is found

A system that keeps searching will sometimes find a weakness in a release that is already out. What it can do then depends on where the release sits. For a proposed release, the institution can withhold it, raise the protection, reduce the detail or change the access route before publication. For a controlled service, such as a query server or a secure environment, an authorised response can suspend new queries, restrict access or change future outputs. These are prospective controls. They act on what has not yet been disclosed.

A public file already copied by others cannot be made private again. Retraining the predictor does not recall it, deleting the source copy does not recall it, and publishing a safer replacement adds a second file to the first. The documented cases show the pattern. The film-rating data that Narayanan and Shmatikov (2008) linked to public reviews had been released for a prize contest before the linkage was published. The US Census Bureau's reconstruction experiments on the published tables of the 2010 census led it to adopt a formal privacy guarantee for 2020 (Garfinkel et al., 2019; Abowd et al., 2022); the 2010 tables remained published, and the response protected the next census. In neither case did a later decision lower the risk of the earlier file.

Proposition 18.6 (What a later decision cannot lower)**.** *Let release Y_1 be public and let $R^*(Y_1; B)$ be the success of the best attack, within a declared family, that uses Y_1 and auxiliary information B. For any later releases $Y_2, \ldots, Y_m$, chosen by any rule, the success of the best attack in the same family on the whole programme is at least $R^*(Y_1; B)$, with equality when nothing further is released. If the auxiliary information grows from B to $B' \supseteq B$, then $R^*(Y_1; B') \geq R^*(Y_1; B)$.*

The proposition is elementary, and that is its use in a committee room. It says that the risk of a published file is a floor under every future programme that includes it, and that the floor can rise when someone outside the institution publishes a new source, but never fall because of anything the institution does next. A correct response to a discovered weakness therefore documents the affected releases, reviews future publication with the earlier files counted in the pooled attack, as Chapter 9 requires, and follows the institution's incident process where one applies. It must not claim that a model update erased earlier exposure.

Three months into Tomás's study, stage one of the cycle found a new source: a commercial data vendor had begun publishing loan sizes for listed firms. The secure room was not affected, because nothing had left it except checked outputs. The Bank's public monthly series was another matter. It had been published for years, and the vendor's data could now be laid beside it. Naledi's note to the chair said what the Bank could not do, which was recall the series, and what it could do: assess the next month's release jointly with the published back series and the vendor's data, and coarsen it if the pooled attack failed.

18.6 The claim that has to be tested

The extension proposes a contribution with a precise name: continuous adversarial learning for disclosure control, with independently verified release decisions. Its test is comparative. Model-assisted search and planning should either find relevant weaknesses earlier, or keep more utility at a fixed risk criterion and a fixed compute budget, and in either case the audit's error control must hold. The measures that would show it are attack coverage, independently confirmed success rates, performance on the worst subgroup, release utility, evaluator cost, reviewer effort and the rate of false certification, all under a design declared before the experiment runs. The design has to include attacks outside the training library and later periods of data, because a system that learns can always look good on the attacks it was trained against.

The evidence so far is evidence about components. Of the extension's seventeen engineering steps, ten are done in the engine with tests, and one, the model of changing exposure, is half done. Three are open because they need a governed register or the joined end-to-end experiment, and three are not built: two wait until simpler models have been shown to fall short, and the third is a live model adapter that the Bank has chosen not to connect. Where the book has measured the learning parts directly, the results are mixed or negative. A small language model given the same budget covered 6.05 scenario cells against 13.65 for a Sobol search, on the states where it produced a valid answer (Chapter 14), and Chapter 22 examines what a deeper planner adds over a logistic one on the same counts. The claim that the whole pipeline improves institutional release decisions is therefore unestablished. It needs steps 10, 16 and 17: authorised data in place of synthetic observations, a joined synthetic release experiment, and validation on a governed register. Grace recorded those three steps as conditions on any operational claim, and the committee minuted them beside the decision for Tomás.

18.7 The answer

The chapter asked what a system that keeps learning about attacks may claim about a release, and what has to stand between its learning and the decision to publish. It may claim that it searched, what it searched, and what it found, with every finding stored as an integer count that names its experiment. It may not claim that a release is safe against hacking, because that names no event anyone can bound, and it may not let its own predictions decide. Between learning and publication stand three things. A gate admits only candidates whose fresh audit bounds meet every required tolerance, fails any candidate with a missing coordinate, and withholds when nothing passes. A frozen plan and an error budget, spent before outcomes exist, stop retries and successive cycles from buying a pass. And a person with authority decides the required attacks, the tolerances and the access route, and signs. Under those conditions the gate's chance of a false release is the α written on the plan, whatever the learning did. A file already published stays published, and the most the system can do after finding a weakness in it is protect the next one.

IN PRACTICE: RUNNING A LEARNING SYSTEM BESIDE THE RELEASE GATE

Keep every development and audit outcome as an integer count of successes and trials, with the scenario, mechanism, target law, conditioning event, split and evaluator version beside it. Before any final audit, freeze the candidates, the required attacks, the tolerances, the target law, the audit design and the error allocation, and record the plan's hash. Treat a missing audit coordinate as a failure. When no candidate passes, withhold, redesign or move to a more restricted access route, and send any change of tolerance or of required attacks to the committee as a new decision with a named owner. Spend error budget for every attempt, including those that fail or are abandoned. Never train on final-audit outcomes. When a weakness is found in a published file, document the affected releases and assess the next release jointly with what is already out.

In the software. The gate is `privacy_risk.gate.decide_release`, which takes candidate utilities, required tolerances and a list of `gate.Coordinate` bounds whose source must be `"audit"`. Measurement rows are `evidence.MeasurementRecord`, and `evidence.training_rows` refuses the final-audit split. The programme budget and frozen plans live in `registry.Registry`, a SQLite store that reserves each share inside a write transaction. The betting audit is `betting.run_audit`, with bets from `betting.predictable_bets`. The tests are in `tests/test_ext_learning.py`, and they replay the checks of Algorithm 18.1, Example 18.4 and Example 18.5.

Algorithm 18.1. The release gate.

Input. The frozen candidates $\theta \in \Theta_t$, each with a utility $V_t(\theta)$; the required attacks $a \in \mathcal{A}_t$, each with a tolerance τ_a; and a list of upper bounds $U_a(\theta)$, each naming its candidate, its attack and its source.

Output. RELEASE with the chosen candidate, or WITHHOLD with none; and, for each candidate, the attacks it failed and the attacks for which it has no bound.

1. Refuse the inputs if there is no candidate or no required attack, if a tolerance lies outside $(0, 1]$, if a bound lies outside $[0, 1]$, if a bound comes from any source other than an independent audit (a planner's prediction is refused), or if a bound names a candidate that was not frozen.

2. For each candidate θ and each required attack a: if no bound $U_a(\theta)$ was given, record a as missing for θ; if $U_a(\theta) > \tau_a$, record a as failed for θ. A bound equal to

the tolerance passes.

3. Call θ feasible when it has no failed and no missing attack, so that it meets every constraint of (18.2) with evidence (Proposition 18.1, part 3).

4. If no candidate is feasible, return WITHHOLD, name redesign and the more restricted access routes as next steps, and leave every tolerance as it was.

5. Otherwise return RELEASE with the feasible candidate of largest utility.

Checks, each run against the software by the test suite:

- Floors of 75 and 150 with utilities 0.45 and 0.20, the bounds of Figure 18.1, and only the external pooled attack required at 0.12: RELEASE.
- The same, the floor chosen: 75.
- The same with the insider also required: WITHHOLD.
- The same, the attack missing at the floor of 75: insider.
- The same, the attack failed at the floor of 150: insider.
- Only the external attack required, with the bound at 75 exactly equal to the tolerance, the floor chosen: 75.
- A bound whose source is a planner's prediction: refused.
- No required attack: refused.

18.8 Proofs

Proof of Proposition 18.1. Write C for the set of pairs (θ, a) with $\theta \in \Theta_t$, $a \in \mathcal{A}_t$ and an audit bound $U_a(\theta)$. The gate's feasible set is

$$F(\mathcal{A}_t, \tau) = \{\theta \in \Theta_t : (\theta, a) \in C \text{ and } U_a(\theta) \le \tau_a \text{ for every } a \in \mathcal{A}_t\}.$$

For part 1, let W be the event that the gate releases some θ^* with $r_a(\theta^*) > \tau_a$ for some $a \in \mathcal{A}_t$. If the gate releases θ^*, then $\theta^* \in F$, so for every required a the pair (θ^*, a) is in C and $U_a(\theta^*) \le \tau_a$. On E we then have $r_a(\theta^*) \le U_a(\theta^*) \le \tau_a$ for every required a, so W does not occur. Hence $W \subseteq E^c$ and $\mathbb{P}(W \mid \mathcal{D}) \le \mathbb{P}(E^c \mid \mathcal{D}) \le \alpha$. Taking expectations over $\mathcal{D}$ gives the unconditional bound. The argument used no property of V_t beyond the fact that θ^* is chosen from F, and E covers every pair at once, so the choice of θ^* may depend on the bounds.

For part 2, if $\mathcal{A}_t \subseteq \mathcal{A}'_t$, every condition defining $F(\mathcal{A}_t, \tau)$ appears among those defining $F(\mathcal{A}'_t, \tau)$, so $F(\mathcal{A}'_t, \tau) \subseteq F(\mathcal{A}_t, \tau)$. If $\tau'_a \le \tau_a$ for every a, then $U_a(\theta) \le \tau'_a$ implies $U_a(\theta) \le \tau_a$, so $F(\mathcal{A}_t, \tau') \subseteq F(\mathcal{A}_t, \tau)$.

For part 3, take one candidate θ, one required attack a with $r_a(\theta) = 1$ and no audit bound, and any tolerance $\tau_a < 1$. The event E holds with probability one, since it constrains no pair. If the missing coordinate is counted as $U_a(\theta) = 0 \le \tau_a$, the gate releases θ with probability one, and $r_a(\theta) > \tau_a$. So the probability in part 1 equals one while the premise holds for every $\alpha \in (0, 1)$, and no bound below one is possible. □

Proof of Proposition 18.3. Cycles that are never run have A_ℓ empty and may be given $\alpha_\ell = 0$. By countable subadditivity, $\mathbb{P}(\bigcup_\ell A_\ell) \le \sum_\ell \mathbb{P}(A_\ell)$. For each ℓ, the tower property and the hypothesis give $\mathbb{P}(A_\ell) = \mathbb{E}[\mathbb{P}(A_\ell \mid \mathcal{F}_{\ell-1})] \le \mathbb{E}[\alpha_\ell]$. Since each α_ℓ is nonnegative, monotone convergence lets the sum pass inside the expectation, and

$$\mathbb{P}\Big(\bigcup_\ell A_\ell\Big) \le \sum_\ell \mathbb{E}[\alpha_\ell] = \mathbb{E}\Big[\sum_\ell \alpha_\ell\Big] \le \alpha_{\text{prog}}.$$

For the second statement, if every cycle runs at level α, the same argument with $\alpha_\ell = \alpha$ for $\ell \le n$ gives $\mathbb{P}(\bigcup_{\ell \le n} A_\ell) \le n\alpha$, and a probability is at most one. That no better general bound exists follows from part 1 of Proposition 6.1: with independent attempts that each certify falsely with probability f, the chance of some false certificate is $1 - (1 - f)^n$, which for $f = \alpha$ and small $n\alpha$ is close to $n\alpha$ and tends to one as n grows. □

Proof of Proposition 18.6. Let δ be any attack in the declared family that uses Y_1 and B. An adversary who holds $Y_1, \dots, Y_m$ and B can run δ and ignore $Y_2, \dots, Y_m$; this attack uses the programme and has the same success as δ, whatever rule chose the later releases. The best attack on the programme therefore succeeds at least as often as δ, and taking the supremum over δ gives the first inequality. When nothing further is released the programme is Y_1 alone and the two quantities coincide. For the second statement, an attack that uses B is also an attack available to an adversary who holds $B' \supseteq B$, who can discard the extra information; the supremum over the larger set of attacks is at least the supremum over the smaller. □

Chapter 19

Releases that change over time

Every floor this book reports for the credit register was computed on a register that does not move. A loan's exposure is the same in every vintage in which it appears, and whether it appears is drawn afresh each month, independently of everything else (Chapter 23). A reader may well take that simplification to be on the safe side. Real exposures amortise, are drawn down and are restructured; loans close, reopen and are refinanced. To an adversary who has read a loan's exposure once, every later movement puts distance between what he holds and what the register publishes, much as the measurement error of Chapter 2 does. On this view a frozen register hands the adversary the easiest possible target, a value that waits for him, and an assessment made on it overstates the risk of the register that actually changes. The view has the shape of a sound rule of thumb: when in doubt, assume the adversary's task is easy.

The view fails on the extension's evidence run. On a synthetic panel of 3,000 loans in 12 key classes, an adversary who knows a loan's history of activity and holds a noisy reading of its exposure in each vintage it is live identifies 403 of 1,200 targets after six vintages when the panel is frozen, a rate of 0.336 with an exact upper bound of 0.359. On the same panel with exposures that drift and activity that persists, the same adversary identifies 703 of 1,200, a rate of 0.586 with an upper bound of 0.610. The frozen panel understates what the changing one reveals by 300 of the 1,200 targets.

At the Bank, the question came from two directions in the same week. Tomás's second file would extend his history from six vintages to twelve (Chapter 18), and the chair asked whether the certificate for the first file could simply be read forward. The supervision department, for its part, argued that the register's exposures move every month, so any figure computed on a frozen stand-in must overstate the risk, and the floor of 75 could safely be lowered for monthly files. Naledi did not know which side was right. She knew only that the old certificate had assumed the register stood still.

This chapter asks one question: what does an adversary learn from a register whose exposures and activity change over time, and can a frozen register stand in for it? It begins by fixing what is being measured, because a register that changes offers more than one denominator. It then shows that the naive way of combining several readings overstates the evidence they carry, sets out the extension's model of changing exposure and activity, measures the pooled attack on it, and describes a simpler graph model that any richer model of change must beat. It ends with what the evidence does not yet show.

19.1 One experiment, one denominator

A risk figure changes meaning whenever the target population, the auxiliary information, the attack or the conditioning changes, and a register that changes over time multiplies the ways in which these can drift apart. The objects are those of Chapter 2. Let D be the confidential register, Y_θ the release under setting θ, J the target's released identity with $J = 0$ for no match, I the set of released candidates and $S = \mathbf{1}\{J \in I\}$ the event of membership. The adversary observes $O = (Z, Y_\theta)$ with background information B_a, and the attack $\delta_a(O)$ names an element of I or abstains. For a fixed target, the unconditional and membership-conditional risks of Definition 2.1 are related by (2.3), $R^{\text{unc}} = \pi R^{\text{in}}$ with $\pi = \mathbb{P}(S = 1 \mid t, B_a)$, and no independence is needed for that. A sampling fraction may stand in for π only when conditioning on B_a leaves it unchanged; an adversary who knows that the target is in the file has $\pi = 1$ whatever the sampling fraction.

Population risk needs a declared law over targets. With weights $w_t \geq 0$ summing to one, $r_a(\theta) = \sum_t w_t R_a^{\text{unc}}(t, \theta)$. Averaging over released targets only defines a different law and must be named as such. On a credit register there are at least three natural laws, and they are not interchangeable: an average over borrowers, an average over loans, and an average weighted by lender. A borrower with five loans counts once under the first and five times under the second.

The correction of Chapter 2 belongs here too. A forced-choice attack that names the most probable released candidate succeeds, given the observations and membership, with probability $\max_{j \in I} p_j/(1 - p_0)$, where p_0 is the posterior on no match (equation (2.6) and Proposition 2.4). An earlier draft of that equation left out the denominator, and the book's figures were unaffected because its simulated attacks normalise over released candidates only. A model of change does not alter the correction. It does make the no-match term harder to ignore, because in a changing register a target can be absent from one vintage and present in the next.

That last point fixes the denominator for the experiments in this chapter. The extension's run draws its targets from loans that are live in the first vintage, takes the first 400 such loans in each of three synthetic panels, and counts success over all 1,200 of them. Membership is conditioned on at the first vintage only. A target's absence from a later vintage is not a reason to drop it from the denominator: it is a published fact the adversary can use. An assessment that conditioned on presence in every vintage would remove from the count exactly the targets whose pattern of absences makes them easiest to find.

Naledi wrote this into the plan for the twelve-vintage file before she ran anything. The target law was over loans, since Tomás's study was of lending relationships; the conditioning was on presence in the first month of the new file; and absence later was part of what the adversary saw. Grace, reading the draft plan, asked what happened to a loan that closed in the third month. Naledi's answer was that it stayed in the denominator and that its closure was part of the attack.

19.2 Many readings are not many independent pieces of evidence

An adversary who holds several readings of a target, one for each vintage or each auxiliary source, has to combine them. The simplest combination multiplies the likelihoods of the readings, as naive Bayes does, and as the record-linkage weights of Fellegi and Sunter (1969)

do when they add the evidence of each agreeing field:

$$L_j(z) = \prod_{h=1}^{m} p(z_h \mid J = j). \tag{19.1}$$

This assumes the readings are independent given the identity of the target. It does not assume they are independent in the population, which is a different and weaker statement. The assumption fails whenever readings share an error: when one stale figure is copied into several months, when two auxiliary sources took their numbers from the same filing, or when an exposure is reported twice under two loan identifiers. A Gaussian model with covariance Σ keeps the shared error in view:

$$\log L_j(z) = -\tfrac{1}{2}(z - \mu_j)^\top \Sigma^{-1}(z - \mu_j) - \tfrac{1}{2}\log|\Sigma| - \tfrac{m}{2}\log(2\pi). \tag{19.2}$$

The off-diagonal entries of Σ carry the point. Ten identical readings are one reading written ten times. A classifier built on (19.1) can still name the right candidate when the readings share errors, but the probabilities it reports are then unreliable (Domingos and Pazzani, 1997), and a risk assessment reads those probabilities.

Proposition 19.1 (Correlated readings and naive evidence)**.** *Let $z \in \mathbb{R}^m$ be readings with $z \mid J = j \sim N(\mu_j \mathbf{1}, \Sigma)$, where $\Sigma = \sigma^2[(1-\rho)I + \rho \mathbf{1}\mathbf{1}^\top]$ with $\sigma > 0$ and $0 \le \rho < 1$. For two candidates with $\mu_j - \mu_k = \Delta$, the expected log-likelihood ratio in favour of the true candidate j is*

$$\mathbb{E}\Big[\log \frac{L_j(z)}{L_k(z)} \,\Big|\, J = j\Big] = \frac{\Delta^2}{2\sigma^2} \cdot \frac{m}{1 + (m-1)\rho} \tag{19.3}$$

under the correct model (19.2)*, and $m\Delta^2/(2\sigma^2)$ under the naive model* (19.1) *with independent errors of variance σ^2. The naive figure is larger by the factor $1 + (m-1)\rho$. As m grows the correct figure tends to $\Delta^2/(2\sigma^2\rho)$ for $\rho > 0$, while the naive one grows without bound.*

The expected log-likelihood ratio is the evidence a typical reading set carries against the wrong candidate, and posterior odds move by its exponential. With a correlation of one half, twenty readings carry the evidence of fewer than two independent ones, and a naive posterior built on them is overconfident by a factor in the exponent of more than ten. The size of the overstatement depends on how the figure is computed. An assessment that reports the expected maximum posterior from a naive model overstates the adversary's confidence. A simulation that draws the adversary's reading errors independently, when the real adversary's errors are shared, credits the simulated adversary with more information than the real one has. Both errors lie on the side of overstating risk, but neither is a measurement, and the second is an assumption about the adversary that ought to be written down.

The extension's run makes that assumption. Its adversary receives a fresh, independent reading error in every vintage in which the target is live, with standard deviation 0.3 on the log scale. On a frozen panel those are several independent readings of one unchanging value, and their average has a standard deviation smaller by the square root of their number. On a changing panel each reading is of a different value, and the readings carry the path of the exposure as well as its level. That difference drives much of what follows.

Naledi asked herself which adversaries at the Bank would hold readings of each kind. Pieter reads his own bank's books, which record each month's exposure as it stands, so his readings follow the path and his errors are small and unshared. An outsider who copies a figure from a firm's annual report into each month has one reading with a shared error, repeated. She declared the two separately. For the outsider she counted the twelve months as one reading, which is what the proposition gives as ρ approaches one, since the engine's attack draws independent errors and would have credited him with twelve.

19.3 A model of changing exposure and activity

The extension proposes a first model for the two features that Chapter 23 left open: exposures that change between vintages, and activity that depends on the past. For loan j in vintage t, let $e_{j,t}$ be its exposure and $a_{j,t} \in \{0, 1\}$ its activity. Log exposure follows a first-order autoregression around a long-run level $\bar{e}$, the simplest of the dynamic panel models of econometrics (Arellano and Bond, 1991),

$$\log e_{j,t+1} - \bar{e} = \mu + \varphi(\log e_{j,t} - \bar{e}) + \epsilon_{j,t}, \qquad \epsilon_{j,t} \sim N(0, \sigma^2), \quad |\varphi| < 1, \tag{19.4}$$

and activity follows a logistic transition, a Markov regression of the kind Zeger and Qaqish (1988) set out for binary series,

$$\mathbb{P}(a_{j,t+1} = 1 \mid a_{j,t}, e_{j,t}) = \operatorname{logit}^{-1}\big(\gamma_0 + \gamma_1 a_{j,t} + \gamma_2(\log e_{j,t} - \bar{e})\big). \tag{19.5}$$

The extension also allows borrower characteristics x_j in (19.5); the engine's version omits them.

Two choices in this model need stating. First, $e_{j,t}$ is defined while a loan is inactive, as a latent potential exposure: the amount the loan would carry if it were live. Without that convention the autoregression has nothing to evolve during an inactive spell, and the model would have to specify origination and closure as separate processes. Second, the pooled likelihood of a target's readings integrates over the whole latent path. An attack that matched each date's marginal distribution separately would discard the dependence between dates, which is exactly what makes a path identifying.

Example 19.2 (Two loans that start together)**.** Take two loans in Pieter's key class, both live, both with log exposure 11.60 (about 109,000) in the first vintage. The numbers are illustrative. Let $\bar{e} = 11$, $\mu = 0$ and $\varphi = 0.95$, as in the run's changing setting, and suppose the first loan draws shocks of $+0.10$ and -0.20 in the next two months and the second draws -0.10 and $+0.15$. By (19.4) the first loan's path is $11.60 \to 11 + 0.95 \times 0.60 + 0.10 = 11.67 \to 11 + 0.95 \times 0.67 - 0.20 = 11.44$, and the second's is $11.60 \to 11.47 \to 11.60$. Pieter's target is the first loan, and suppose his three readings fall exactly on its path. The engine's attack scores each candidate by $-\frac{1}{2}\sum_t (\text{published}_t - \text{reading}_t)^2 / 0.3^2$. The first loan scores 0. The second differs by 0, 0.20 and 0.16 and scores $-\frac{1}{2}(0.20^2 + 0.16^2)/0.09 = -0.364$, so the first loan is $e^{0.364} = 1.44$ times as likely, and between the two the attack gives it probability 0.59. On a frozen panel both loans would publish 11.60 in every month, the scores would tie, and the attack could do no better than a coin toss. Each month of movement adds a term in which the loans differ. Two months moved the odds from even to 1.44; a year of movement would add ten more such terms.

With $\varphi = 1$, $\sigma = 0$ and $\gamma_1 = \gamma_2 = 0$ the model reduces to a frozen panel: exposure never changes, and each loan is live in each vintage with probability $\operatorname{logit}^{-1}(\gamma_0)$, independently of its past and its size. The engine's frozen setting takes $\gamma_0 = 2$, so a loan is live with probability about 0.88 in each vintage after the first. The changing setting of the run takes $\varphi = 0.95$, $\sigma = 0.15$, $\gamma_0 = 0$, $\gamma_1 = 1.5$ and $\gamma_2 = 0$: a live loan stays live with probability about 0.82, an inactive one returns with probability one half, and across the six vintages 79 per cent of loan-months are live. Every loan is live in the first vintage, log exposure starts from a normal law with mean 11 and standard deviation 1.2, and each loan's key class is drawn uniformly from twelve.

These are chosen parameters, not estimates, and two of them bite. With $\gamma_2 = 0$, activity in the run does not depend on exposure, so the dependence of activity on loan size that Chapter 23 named is in the model and not in the evidence. And the frozen limit is not the register of Chapter 13. That register was rebuilt after review, with originations during the panel, a monthly exit hazard of 0.04 with no return and a stock of about 27,000 loans. The extension's panel forces every loan to be live in the first vintage and lets loans vanish and return, which is the design the referees criticised. Its frozen rows are a reference for this panel only, and they should not be compared with the floors of Table 13.4.

19.4 The pooled attack, using presence and absence

The engine's pooled attack runs three adversaries on each panel. All three know the target's key class and that its loan is live in the first vintage. The first knows nothing else and chooses at random among the loans of the class. The second also knows the target's activity in every pooled vintage, and strikes out every candidate whose pattern of presence and absence differs. The third knows the history and holds, for each vintage in which the target is live, a reading of its log exposure with error of standard deviation 0.3; it scores the remaining candidates by the Gaussian likelihood of those readings and names the most likely one. A candidate that is absent in a vintage where the target is live offers no published value to compare, and the attack rules it out. Absence is information. The first published version of the history attack had used only presence, and the referees pointed out that an adversary who knows when a loan is live also knows when it is not (Chapter 23).

The history adversary can be analysed exactly, and the analysis explains its rows in the evidence.

Proposition 19.3 (Success from activity patterns)**.** *Let a key class hold n loans, all live in the first vintage, whose activity patterns over vintages $2, \dots, v$ are independent across loans with a common law P on the set Π_v of possible patterns. An adversary who knows the target's key and pattern and names a loan chosen uniformly among those in the class with the same pattern identifies a target drawn uniformly from the class with probability*

$$s_v = \frac{1}{n} \sum_{\pi \in \Pi_v} \left[1 - (1 - P(\pi))^n \right] = \frac{\mathbb{E}[\textit{number of distinct patterns among the } n \textit{ loans}]}{n}. \qquad (19.6)$$

Consequently $s_v \le \min(1, 2^{v-1}/n)$, and s_v never decreases as v grows.

The proposition turns the history attack into a count. The adversary succeeds as often as the class contains distinct patterns, divided by its size. A frozen panel whose loans are live nine months in ten produces few distinct patterns, because most loans show the same unbroken run. A panel in which loans persist when live and stay away for a while when not spreads its loans over more patterns, and the adversary does better.

Example 19.4 (Counting patterns in a class of four)**.** Naledi checked the proposition on the smallest case she could hold in her head: a key class of four loans and three vintages, so that each loan shows one of four patterns over the second and third months. On the frozen panel each later month is live with probability $\text{logit}^{-1}(2) = 0.881$, independently, and the four patterns have probabilities 0.776 (live, live), 0.105 (live, absent), 0.105 (absent, live) and 0.014 (absent, absent). On the changing panel a live loan stays live with probability 0.818 and an absent one returns with probability one half, which gives 0.668, 0.149, 0.091 and 0.091.

Equation (19.6) sums $1-(1-P(\pi))^4$ over the patterns: the expected number of distinct patterns is 1.77 on the frozen panel and 2.10 on the changing one, and the history adversary succeeds with probability $1.77/4 = 0.442$ and $2.10/4 = 0.525$. The key-only adversary, who picks among the four at random, succeeds with probability 0.25. The changing-panel pattern probabilities above are rounded, and the unrounded ones are 0.6684, 0.1491, 0.0912 and 0.0912. Simulating 200,000 such classes with the short script below gave 0.442 and 0.526. The changing panel spreads its loans more evenly over the patterns, and evenness is what the adversary needs.

```
import numpy as np
rng = np.random.default_rng(7)
def formula(P, n=4):                        # equation (s_v)
    return sum(1 - (1 - p) ** n for p in P) / n
def simulate(P, n=4, reps=200_000):         # name one of the n at random
    pats = rng.choice(len(P), size=(reps, n), p=np.array(P) / sum(P))
    same = np.sum(pats == pats[:, :1], axis=1)
    return float(np.mean(1.0 / same))
```

The first loan of each simulated class is taken as the target, and the adversary succeeds with probability one over the number of loans that share its pattern. The script and the formula agree to the third decimal, which is the check that the closed form counts what the attack does.

TABLE 19.1: Pooled attacks on the extension's synthetic panels: successes out of 1,200 targets (400 in each of three panels of 3,000 loans in 12 key classes), with the exact one-sided 95 per cent upper bound in brackets. The key-only adversary does not use the vintages and is the same for both panels. Readings have standard deviation 0.3 on the log scale. All figures are synthetic.

		Frozen panel		Changing panel	
Vintages	Key only	History	History and readings	History	History and readings
1	3 (0.006)	3 (0.006)	39 (0.042)	3 (0.006)	39 (0.042)
2	2 (0.005)	7 (0.011)	96 (0.094)	7 (0.011)	96 (0.094)
3	2 (0.005)	20 (0.024)	143 (0.136)	17 (0.021)	200 (0.185)
4	3 (0.006)	25 (0.029)	245 (0.224)	33 (0.037)	378 (0.338)
5	5 (0.009)	56 (0.058)	319 (0.288)	81 (0.081)	541 (0.475)
6	7 (0.011)	90 (0.089)	403 (0.359)	134 (0.128)	703 (0.610)

At six vintages there are at most 32 patterns, so in a class of 250 loans, the average size in a panel of 3,000 loans and 12 classes, the history adversary cannot exceed about 0.13. Evaluated at the run's parameters, (19.6) reproduces the history rows of the evidence for both panels to within their sampling error.

Table 19.1 gives the full evidence, and Figure 19.1 draws its two informative columns. Read the key-only column first, because it is a check. That adversary does not use the vintages, so its success should stay near one in 250 whatever the number pooled, and it does: between 2 and 7 successes in 1,200 for both panels. The first vintage is identical for both panels by construction, since nothing has yet changed. At two vintages the two panels give the same totals, 7 for the history adversary and 96 with readings. One transition offers only two patterns in either panel, and (19.6) gives nearly the same value for both, so the history adversary is expected to do about equally well; the equal readings totals are partly chance, since the counts differ panel by panel and happen to sum alike. From three vintages on the

panels separate, and the gap widens with every vintage. At six the history adversary succeeds for 0.075 on the frozen panel and 0.112 on the changing one, and the adversary with readings for 0.336 and 0.586.

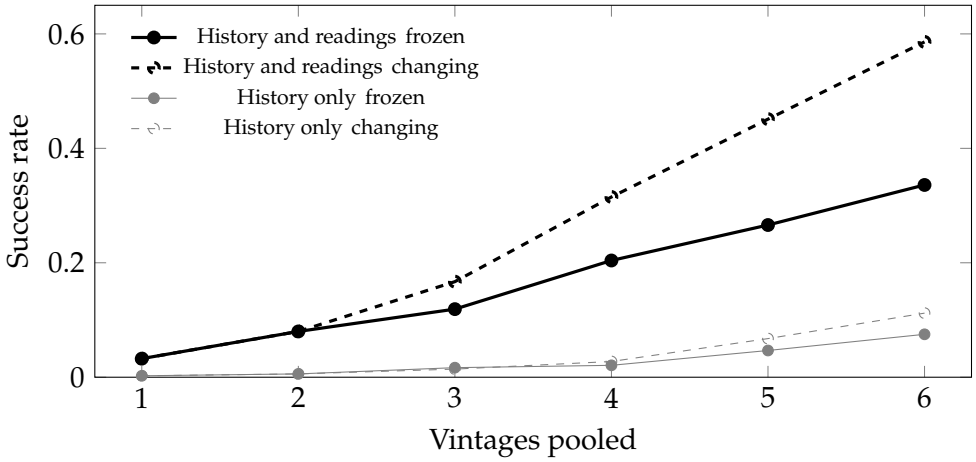


FIGURE 19.1: Success of the pooled attack against the number of vintages pooled, on the extension's synthetic panels. Solid lines with filled marks are the frozen panel; dashed lines with open marks are the changing panel. Heavy black lines are the adversary with history and a reading in each live vintage; thin grey lines are the adversary with history only. Each point is a rate over 1,200 targets; the counts and upper bounds are in Table 19.1.

Two mechanisms produce the gap, and the proposition and the readings section each account for one. The changing panel spreads loans over more activity patterns, so the history alone leaves fewer candidates. And on the changing panel each vintage's reading is of a new value along which candidates differ, so the readings accumulate evidence about a path; on the frozen panel they only refine one number. The run cannot say how much of the gap each mechanism explains, because the changing setting alters both at once: activity is less frequent and more persistent, and exposure moves. A run that varied them one at a time would split the gap. It has not been done.

The misconception treated change as noise in the adversary's information. That holds for an adversary whose information is stale, one who read a loan's exposure once and compares that reading with later published values. For him a moving exposure does blur the match, and a frozen register overstates his success. It fails for an adversary who observes the target as it changes, a lender, a counterparty or anyone with a monthly source, because for him change is signal: it gives the target a path that no other loan shares. A frozen register is conservative for some adversaries and optimistic for others, and the evidence run shows the optimism can be large. It cannot be assumed conservative without saying which adversary is meant. The documented cases point the same way. A few film ratings with rough dates were enough to pick out subscribers in the Netflix Prize data (Narayanan and Shmatikov, 2008), and four points in space and time singled out 95 per cent of the people in a mobility data set of 1.5 million (de Montjoye et al., 2013). A path is a set of such points.

Naledi took Table 19.1 to the committee with one sentence above it: the frozen figure was too low for any adversary who watched the register month by month, and those were the adversaries the Bank worried about. The supervision department's proposal to lower the floor for monthly files was withdrawn. The chair's question about reading the old certificate forward had its answer too: the certificate covered six vintages of a register assumed to stand still, and the twelve-vintage file had to be audited afresh.

19.5 A graph model that has to stay in the comparison

Changing exposure and activity are one kind of movement. Another is movement between discrete states: a borrower changes sector, a loan moves between lenders, a firm crosses a size threshold. The extension's simplest model for that movement is a continuous-time Markov chain on a graph of states, the model that credit-risk work has long used for movements between rating grades (Jarrow et al., 1997; Lando and Skødeberg, 2002). Let $c_i(t)$ be the mass of targets in state i, and let Q be a generator: $Q_{ij} \geq 0$ for $i \neq j$, and $Q_{ii} = -\sum_{j \neq i} Q_{ij}$. For masses written as a column vector,

$$\dot{c} = Q^\top c, \qquad c(t) = \exp(tQ^\top)\, c(0), \tag{19.7}$$

and with a disclosure risk g_i for a target in state i, the model risk is $r(t) = g^\top c(t) / \mathbf{1}^\top c(t)$. The engine computes the matrix exponential with SciPy's scaling-and-squaring routine (Al-Mohy and Higham, 2009); Moler and Van Loan (2003) review why the naive ways of computing it fail.

Proposition 19.5 (The graph baseline)**.** *Let Q be a generator and $c(0) \geq 0$. Then:*

1. *$c(t) \geq 0$ and $\mathbf{1}^\top c(t) = \mathbf{1}^\top c(0)$ for every $t \geq 0$;*
2. *if $\mathbf{1}^\top c(0) > 0$, the model risk $r(t)$ lies between the smallest and the largest of the g_i;*
3. *if complete paths are observed, with n_{ij} transitions from i to j and total time $T_i > 0$ spent in state i, the maximum-likelihood estimate of Q_{ij} for $j \neq i$ is n_{ij}/T_i.*

The first part is what a model of movement has to get right before anything else: no state may hold a negative number of targets, and none may be created or lost. The second says the model risk is an average of state risks, so it can never exceed the riskiest state. The third says that the rates can be read off transition counts and holding times, with no optimisation; Albert (1962) derived this estimator, and Lando and Skødeberg (2002) used it for rating transitions. Together they make the graph model cheap and hard to get wrong, and those are the reasons it has to stay in any comparison. The extension goes on to propose continuous transport models and an optional fluid closure, which Chapter 20 examines. A richer model of movement earns its place only by predicting held-out periods better than this baseline, or by supplying a structural constraint that has meaning on its own.

Example 19.6 (Mass moving across three states)**.** The supervision department, still interested in movement, asked Naledi what the graph model would say about restructured loans. She wrote down three illustrative states: performing small loans, performing large loans and restructured loans, with disclosure risks $g = (0.02, 0.05, 0.30)$, since restructured loans are few and stand out. Per year, small loans grow large at rate 0.10 and are restructured at 0.05; large loans shrink at 0.05 and are restructured at 0.05; restructured loans return to the small class at 0.20. Suppose a sample of paths had shown 6 moves from small to large and 3 to restructured in 60 loan-years spent small, 2 and 2 in 40 loan-years spent large, and 4 returns in 20 loan-years spent restructured. Part 3 gives $6/60 = 0.10$, $3/60 = 0.05$, $2/40 = 0.05$ twice and $4/20 = 0.20$, which is the generator above. Starting from masses $(0.6, 0.3, 0.1)$, the model risk is $0.6 \times 0.02 + 0.3 \times 0.05 + 0.1 \times 0.30 = 0.057$. After one year the masses are $(0.552, 0.326, 0.122)$ and the risk is 0.064; after five, $(0.449, 0.379, 0.171)$ and 0.079; in the long run, $(0.4, 0.4, 0.2)$ and 0.088. The masses sum to one at every time, and the risk stays below

the largest state risk, 0.30, as part 2 requires. The risk rises by half without any loan changing what it publishes, only because mass drifts towards the restructured state. Whether the Bank's loans move at these rates is a question for its own transition counts.

The evidence run checks part 3 on a chain of four states with known rates between 0.05 and 0.15. It simulates 400 complete paths over a horizon of 10, counts the transitions and holding times, and estimates the generator. The largest error in any estimated entry is 0.016. Started from masses of 0.4, 0.3, 0.2 and 0.1 and run forward to time 5, the true and estimated generators give the masses in Figure 19.2: 0.202, 0.360, 0.220 and 0.218 under the true rates, and 0.198, 0.369, 0.215 and 0.218 under the estimates. Both sets sum to one. The largest error in a mass, in the second state, is under 0.01. The check verifies the estimator on data generated by the model it assumes. It says nothing yet about whether lenders, sectors or size classes in a real register move as a Markov chain.

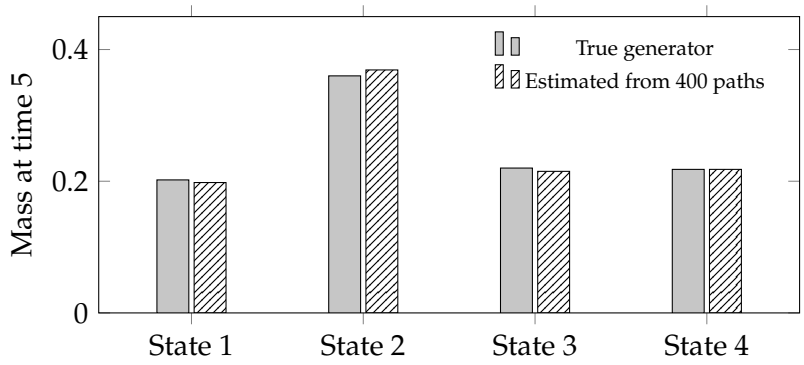


FIGURE 19.2: The graph baseline in the extension's evidence run. Masses in four states at time 5, starting from 0.4, 0.3, 0.2 and 0.1, under the true generator (solid grey) and under the generator estimated from transition counts and holding times on 400 simulated paths (hatched). Synthetic data.

19.6 What the evidence does not show

Everything in this chapter is synthetic. The panels were generated with chosen parameters, the readings with a declared error, and the graph with a generator written into the script. The evidence shows that a changing register can reveal far more than its frozen version to an adversary who watches it, and that the engine's models behave as their mathematics says. It does not show how much more a real register reveals.

For the same reason the book's credit-register floors have not been recomputed under the dynamic model, and the extension's own status table records the step as open. There are four reasons, and each would have to be removed first. The model's parameters are chosen, and a floor computed from them would encode those choices while looking like a measurement. The panel on which the model runs is not the register of Chapter 13: it lacks originations, irreversible exits, unequal lenders and the key structure of that register, and it forces every loan to be live at the start. The floors of Chapter 13 came from fixed-sequence selection with confirmation on 4,000 targets per level, and nothing of that kind has been run on the panel. And the dependence of activity on loan size, the feature that most separates a real register from the frozen one, was switched off in the run. Replacing chosen parameters with estimated ones needs a governed register, which is step 10 of the extension's plan, and validating the result needs step 17.

The direction of the evidence is still useful before its size is known. A floor that was set on a frozen register should not be lowered on the argument that real exposures move. If anything, the movement argues for a stricter floor against adversaries who watch the register, and for auditing each extension of a history afresh.

Grace's note on the twelve-vintage decision was short. The audit had run on the Bank's own twelve months inside the secure environment, so it already contained whatever movement the register had; the frozen model bore only on the old synthetic certificate, which the committee had stopped reading forward. She asked for one more thing, and the chair agreed: that the Bank estimate the autoregression and the activity transition on its own register, inside the secure room, so that the next synthetic assessment could use parameters the Bank had measured.

19.7 The answer

The chapter asked what an adversary learns from a register whose exposures and activity change over time, and whether a frozen register can stand in for it. An adversary who watches the register month by month learns the target's path: its pattern of presence and absence, and the sequence of its exposures. Change gives each loan a path that others do not share, so it adds to what this adversary learns. On the extension's synthetic panels, six vintages of history and readings identify a target 0.34 of the time when the panel is frozen and 0.59 when it changes, and the history alone identifies 0.075 and 0.112. A frozen register is conservative only for an adversary whose information is stale, and it cannot be assumed conservative without naming the adversary. Measuring the real gap needs a model fitted to a real register, a target law and conditioning event fixed before the attack runs, readings whose shared errors are modelled and not multiplied away, and a simple graph baseline that any richer model of change must beat. Until a governed register supplies the parameters, the book's floors stand as properties of a frozen synthetic register, and the dynamic model remains code and tests.

IN PRACTICE: ASSESSING A REGISTER THAT CHANGES

Fix the target law (borrowers, loans or lenders) and the vintage at which membership is conditioned before running any attack, and keep targets that later disappear in the denominator. Declare, for each adversary, whether its readings follow the target's path or repeat one stale value, and model shared errors with a covariance instead of multiplying likelihoods. Run the pooled attack with presence and absence both used. Do not read a certificate forward to a longer history; audit the new history. Do not lower a floor because exposures move. When a model of change is used, report its parameters, say whether they were chosen or estimated, and compare it with the graph baseline on held-out periods.

In the software. The panel model and attack are in `privacy_risk.dynamics`: `PanelModel` (with `PanelModel.static` for the frozen limit), `simulate_panel` and `pooled_attack`, which returns integer counts. The graph baseline is in `privacy_risk.graph`: `validate_generator`, `propagate`, `node_risk` and `estimate_generator`. The tests are in `tests/test_ext_models.py`, and `scripts/extension_evidence.py` regenerates the pooled-attack table. Run with seed 2026, it gives 434 and 740 of 1,200 for the adversary with history and readings after six vintages on the frozen and the changing panel, and 77 and 147 for history alone. Table 19.1 comes from the extension's own run, which gave 403 and 703 and 90 and 134, so the two runs agree in direction and size and differ by sampling error. Algorithm 19.1 sets out the pooled attack, including its treatment of absence.

Algorithm 19.1. The pooled attack, using presence and absence.

Input. A panel of loans over v vintages, each loan k with a key class, an activity $a_{k,t}$ and a log exposure $\log e_{k,t}$ in each vintage; the number w of vintages pooled; a list of targets; whether the adversary knows each target's activity; the standard deviation s of its readings of log exposure, if it holds any; a seeded random source.

Output. The number of targets the adversary names correctly, and the number of targets.

1. Refuse the inputs unless $1 \le w \le v$, and keep only the first w vintages.
2. For each target j, refuse the inputs if j is not live in the first vintage. Otherwise its candidates are the loans of its key class that are live in the first vintage.
3. If the adversary knows the target's activity, keep only the candidates whose pattern of presence and absence over the w vintages equals the target's in every vintage (Proposition 19.3).
4. If the adversary holds readings with $s > 0$, draw a reading $y_t = \log e_{j,t} + sZ_t$, with Z_t standard normal, for each vintage t in which the target is live. Score each candidate k by $-\frac{1}{2}\sum_t (\log e_{k,t} - y_t)^2/s^2$ over those vintages, as in Example 19.2. A candidate absent in any of those vintages offers no published value there and takes a residual of 10^6, which rules it out. Keep the candidates with the highest score.
5. Name one of the remaining candidates at random, each equally likely, and count a success if it is the target.
6. Return the number of successes and the number of targets.

Checks, each run against the software by the test suite:

- Changing panel of Table 19.1, adversary who knows the history, six vintages, successes summed over the three panels of 400 targets: 134.
- The same adversary with readings of standard deviation 0.3: 703.
- Frozen panel, adversary who knows the history, six vintages: 90.
- Frozen panel, adversary who knows only the key class, six vintages: 7.
- Two loans in one class with the same exposures, the other loan absent in the second of three vintages, an adversary with readings but no activity history, share of 200 tries that succeed: 1.000.
- The same pooling only the first vintage, where the two loans cannot be told apart: 0.480.
- A target that is not live in the first vintage: refused.
- Four vintages pooled from a panel of three: refused.

19.8 Proofs

Proof of Proposition 19.1. The vector $\mathbf{1}$ is an eigenvector of Σ: $\Sigma\mathbf{1} = \sigma^2[(1-\rho)\mathbf{1} + \rho m\mathbf{1}] = \sigma^2(1+(m-1)\rho)\mathbf{1}$. Since $0 \le \rho < 1$, every eigenvalue of Σ is positive (the others equal $\sigma^2(1-\rho)$), so Σ is invertible and $\Sigma^{-1}\mathbf{1} = \mathbf{1}/[\sigma^2(1+(m-1)\rho)]$, which gives $\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = m/[\sigma^2(1+(m-1)\rho)]$.

Under the correct model, the log-likelihood ratio between two normal laws with common covariance is

$$\log\frac{L_j(z)}{L_k(z)} = -\tfrac{1}{2}(z-\mu_j\mathbf{1})^\top\Sigma^{-1}(z-\mu_j\mathbf{1}) + \tfrac{1}{2}(z-\mu_k\mathbf{1})^\top\Sigma^{-1}(z-\mu_k\mathbf{1}),$$

the normalising terms cancelling. Write $u = z - \mu_j\mathbf{1}$, so that $z - \mu_k\mathbf{1} = u + \Delta\mathbf{1}$. Expanding,

$$\log\frac{L_j(z)}{L_k(z)} = \Delta\,\mathbf{1}^\top\Sigma^{-1}u + \tfrac{1}{2}\Delta^2\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1}.$$

Given $J = j$, $\mathbb{E}[u] = 0$, so the expectation is $\frac{1}{2}\Delta^2\mathbf{1}^\top\Sigma^{-1}\mathbf{1}$, which is (19.3). The naive model replaces Σ by $\sigma^2 I$ in the same expression, and its log-likelihood ratio is $\Delta\mathbf{1}^\top u/\sigma^2 + m\Delta^2/(2\sigma^2)$. The true law of u still has mean zero, so the naive expectation is $m\Delta^2/(2\sigma^2)$. The ratio of the two expectations is $1+(m-1)\rho$. Finally, for $\rho > 0$, $m/(1+(m-1)\rho) \to 1/\rho$ as $m \to \infty$, while $m\Delta^2/(2\sigma^2)$ grows without bound. □

Proof of Proposition 19.3. Fix the target and condition on its pattern π. The other $n-1$ loans have patterns drawn independently from P, so the number N of them sharing π is binomial with parameters $n-1$ and $p = P(\pi)$. The adversary chooses uniformly among $N+1$ loans, one of which is the target, and succeeds with probability $1/(N+1)$. Using $\binom{n-1}{k}/(k+1) = \binom{n}{k+1}/n$,

$$\mathbb{E}\Big[\frac{1}{N+1}\Big] = \sum_{k=0}^{n-1}\binom{n-1}{k}\frac{p^k(1-p)^{n-1-k}}{k+1} = \frac{1}{np}\sum_{k=0}^{n-1}\binom{n}{k+1}p^{k+1}(1-p)^{n-k-1} = \frac{1-(1-p)^n}{np}$$

for $p > 0$. The target's own pattern has law P, so averaging over π with $P(\pi) > 0$ gives $s_v = \sum_\pi P(\pi)(1-(1-P(\pi))^n)/(nP(\pi))$, which is the first expression in (19.6); patterns of probability zero contribute nothing to either side. The same answer holds for a target drawn uniformly from the class, since the loans are exchangeable. Pattern π appears among the n loans with probability $1-(1-P(\pi))^n$, and summing these indicators' expectations gives the expected number of distinct patterns, which is the second expression.

The number of distinct patterns among n loans is at most n and at most $|\Pi_v| = 2^{v-1}$, since the first vintage is fixed and each later vintage has two states; dividing by n gives the bound. For monotonicity, each pattern π on vintages up to v splits into the patterns on vintages up to $v+1$ that extend it, with probabilities summing to $P(\pi)$. The function $f(x) = 1-(1-x)^n$ is concave on $[0,1]$ with $f(0) = 0$, hence subadditive: $f(x+y) \le f(x) + f(y)$ for $x, y \ge 0$ with $x + y \le 1$. Applied to each split, the sum in (19.6) does not decrease from v to $v+1$. □

Proof of Proposition 19.5. For part 1, the rows of Q sum to zero, so $Q\mathbf{1} = 0$ and $\frac{d}{dt}\mathbf{1}^\top c = \mathbf{1}^\top Q^\top c = (Q\mathbf{1})^\top c = 0$; total mass is constant. For positivity, let $\lambda \ge \max_i |Q_{ii}|$. The matrix $M = Q^\top + \lambda I$ has nonnegative entries, and since λI commutes with $Q^\top$, $\exp(tQ^\top) = e^{-\lambda t}\exp(tM) = e^{-\lambda t}\sum_{k\ge 0} t^k M^k/k!$. Every term of the series has nonnegative entries, so $\exp(tQ^\top)$ does, and $c(t) = \exp(tQ^\top)c(0) \ge 0$.

For part 2, by part 1 the weights $c_i(t)/\mathbf{1}^\top c(t)$ are nonnegative and sum to one, so $r(t)$ is a convex combination of the g_i.

For part 3, a continuous-time Markov chain in state i waits an exponential time with rate $q_i = -Q_{ii} = \sum_{j \neq i} Q_{ij}$ and then jumps to j with probability Q_{ij}/q_i. A completed holding period ending in a jump to j therefore contributes $q_i e^{-q_i s} \cdot Q_{ij}/q_i = Q_{ij} e^{-q_i s}$ to the likelihood, and a period cut off by the end of observation contributes $e^{-q_i s}$. Multiplying over all periods of all paths, the log-likelihood is

$$\ell(Q) = \sum_i \sum_{j \neq i} \big(n_{ij} \log Q_{ij} - Q_{ij} T_i \big),$$

which separates into one term for each pair (i, j). For $n_{ij} > 0$ the term is strictly concave in $Q_{ij} > 0$ with derivative $n_{ij}/Q_{ij} - T_i$, which vanishes at $Q_{ij} = n_{ij}/T_i$. For $n_{ij} = 0$ the term $-Q_{ij} T_i$ is maximised over $Q_{ij} \geq 0$ at $0 = n_{ij}/T_i$. The diagonal is then fixed by the row sums. $\square$

Chapter 20

Dynamic models and Bayesian inference

A reader who arrives at this chapter from the physical sciences brings a belief that deserves a fair hearing. The borrowers in a credit register do not sit still. Their exposures grow and shrink, their loans start and end, and the cells of a release table fill and empty as the months pass. Physics has spent two centuries writing equations for quantities that move, and the best of them, the conservation laws and the Navier–Stokes equations of fluid flow, are among the most tested statements in science. On this view a model drawn from physics, with more equations and more structure, must forecast disclosure risk better than the counts and tables used in the rest of this book. The structure carries knowledge that the data alone cannot supply: mass is not created from nothing, densities do not go negative, change is smooth. A risk forecast that respects these rules ought to be more accurate than one that does not.

The first fact that breaks the belief is a fact about this book's own evidence. No transport model in this book has yet beaten a simpler baseline on held-out data, because that comparison has not been run. The claim ledger records the idea that Navier–Stokes describes the movement of information as a proposed analogy, unestablished, and it records the general Navier–Stokes solver as not built, conditional on a simpler model being shown to fall short. A second fact comes from the one experiment that was run. When the equations were fitted to observations that they had generated themselves, with the parameters known in advance, the observations responded about fourteen times more strongly to one parameter than to the other, in the units the model uses, and the weaker one was the drift that the model exists to forecast. Structure in a model is a set of assumptions. It helps a forecast only where the assumptions are true and the data can resolve the parameters they introduce, and both of those are things to be shown.

This chapter asks one question. What would a dynamic model of the target population have to show before its forecast could enter a release decision? The answer is built in layers: what such a model can mean for disclosure, how its equations are solved and checked, how its parameters are inferred from data and how far the data can identify them, and which simpler model it has to beat.

20.1 A proposal from a visiting researcher

Tomás, the visiting researcher whose requests for credit-register files run through earlier chapters, came to Naledi with a different kind of proposal. He had trained in applied mathematics before turning to finance, and he wanted to model how the Bank's borrowers drift across the space of exposure and lending relationships from one vintage to the next. If the drift could be forecast, he argued, the Bank could see in advance which cells of a release table were thinning out and becoming risky, and set its floors before the risk arrived. His note proposed an advection–diffusion equation for the density of borrowers, with a drift field governed by the Navier–Stokes equations.

Naledi read the note twice before replying. She did not doubt the mathematics. Her question was what each symbol stood for in the register, and what evidence would tell the committee that the forecast was better than the one it already had. She asked Tomás to answer both in writing before the proposal went any further.

20.2 What a fluid equation can and cannot mean here

The Navier–Stokes equations describe the balance of momentum in a fluid. A credit register has no mass of information, no momentum and no pressure law, and an analogy does not supply them. What the equations can offer is narrower: a declared regularisation model for how a population moves across a space of features. Such a model may help to generate coherent stress scenarios or to smooth a distribution that changes over time. Whether it forecasts disclosure risk better than a simpler model is an empirical question, and this chapter treats it as one.

The space must be built with care. Let $x \in \Omega$ be a feature vector made from continuous quantities, such as standardised log exposure and a continuous measure of how concentrated a borrower's lending relationships are. Integer codes for sectors, regions or lenders do not define a distance; sector 4 is not closer to sector 5 than to sector 12. Categorical relationships belong in a graph or in separate strata, as Section 20.11 sets out. An embedding that turns categories into coordinates must be fitted on development data, frozen, and tested for distortion, because a rotation or rescaling of the embedding would otherwise change the scientific claim without anyone deciding to change it.

On that space let $c(x,t) \geq 0$ be the density of target states at time t. It is a population distribution: it says where the borrowers are, not how exposed they are. Let $u(x,t)$ be the drift of features, $D_c > 0$ a dispersion coefficient and $s(x,t)$ the net rate at which targets enter or leave. For any fixed region $V \subset \Omega$ with outward normal n, the mass in V changes only by flux through its boundary and by entry and exit:

$$\frac{d}{dt}\int_V c\,dx = -\int_{\partial V} (c\,u - D_c \nabla c)\cdot n\,dS + \int_V s\,dx. \tag{20.1}$$

Because V is arbitrary, the divergence theorem turns this into the advection–diffusion equation

$$\partial_t c + \nabla\cdot(c\,u) = D_c \Delta c + s. \tag{20.2}$$

With periodic boundaries, or boundaries that let no mass through, and with $s = 0$, the total mass $\int_\Omega c\,dx$ is constant. With entry and exit it changes by $\int_\Omega s\,dx$ per unit time. A density that does not satisfy this balance should not be normalised and then read as a probability. Resetting the total to one after every numerical step is a common shortcut, and it hides

exactly the conservation error that the balance is there to catch; if normalisation is needed, the equation for the normalised density should be derived.

The density is not yet a risk. The link to the disclosure experiment of Chapter 2 comes through an attack response $g_a(x, \theta) \in [0, 1]$, the probability that attack a names the right record for a target with features x when the release uses protection setting θ. The model risk is the average response over the population:

$$r_a(\theta, t) = \frac{\int_\Omega g_a(x, \theta)\, c(x, t)\, dx}{\int_\Omega c(x, t)\, dx}. \tag{20.3}$$

Everything a dynamic model says about disclosure passes through g_a, and g_a must be fitted to measured attack outcomes, the counts of Chapter 4. A density that moves without a fitted response says nothing about disclosure risk. Where an attack uses history, as the pooled attacks of Chapter 13 do, either the feature vector must carry enough of the history or the response must be a function of the whole trajectory.

Tomás's written answer took a week. The density, he wrote, would be the share of borrowers in each small region of log exposure and relationship concentration. The response would come from Naledi's audits. He conceded that he had not thought about the second part: his note had shown moving densities and called them risk maps. Naledi marked the concession as the most useful line in the reply.

20.3 The optional Navier–Stokes closure

Equation (20.2) needs a drift u. The simplest choice estimates u directly from how features change between vintages. Tomás's proposal went one step further and gave the drift its own dynamics. On a two-dimensional periodic domain the incompressible Navier–Stokes equations read

$$\partial_t u + (u \cdot \nabla) u = -\nabla \pi + \nu \Delta u + f, \tag{20.4}$$

$$\nabla \cdot u = 0, \tag{20.5}$$

with $\nu > 0$ a smoothing coefficient, f a forcing term and π a Lagrange multiplier that enforces the divergence constraint, fixed by $\int_\Omega \pi \, dx = 0$. In a register these are not a measured financial viscosity and pressure. They are tuning devices of a proposed model, and the names carry no evidence.

The constraint (20.5) is strong. It says the drift preserves volume in feature space: a cluster of borrowers may move and change shape, but it may not spread into a larger region or crowd into a smaller one by drift alone. Credit cycles do the opposite. In a downturn, exposures across many borrowers fall together, and the population contracts towards small values. If observed drift expands or contracts volumes, a compressible transport model or a graph process fits the data better than an incompressible closure.

Naledi put the constraint to Tomás in the register's terms. She asked what his drift would do in a month when every lender cut its credit lines at once. Tomás agreed that the borrowers would crowd towards small exposures, which a volume-preserving drift forbids, and he wrote the constraint into his note as an assumption to be tested on the Bank's own vintages, not a fact about them.

The closure has one property that is useful whatever it means for disclosure. For smooth periodic solutions, the kinetic energy of the drift changes only through smoothing and forcing.

Proposition 20.1 (Energy identity)**.** *Let u and π be smooth, periodic on $\Omega = [0,1)^2$, and satisfy* (20.4) *and* (20.5)*. Then*

$$\tfrac{1}{2}\tfrac{d}{dt}\|u\|_2^2 + \nu\|\nabla u\|_2^2 = \langle f, u\rangle. \tag{20.6}$$

Without forcing, the energy can only fall, at a rate set by the gradients. A numerical solver that reports energy rising in an unforced run has a bug. That is the identity's whole use here: it is a diagnostic for the solver. It says nothing about whether the drift describes how borrowers behave. The identity also assumes smoothness, which holds for the two-dimensional periodic problem; whether smooth solutions persist in three dimensions is an open problem in mathematics (Fefferman, 2006), and nothing in this book depends on it.

20.4 An exact solution as the first benchmark

A solver should be tested first on a problem whose answer is known exactly. The Navier–Stokes equations have few such solutions, but a decaying shear on the periodic square is one of them.

Proposition 20.2 (Exact benchmarks)**.** *On $\Omega = [0,1)^2$ with periodic boundaries, let $k = 2\pi$ and*

$$u(x,y,t) = \big(Ae^{-\nu k^2 t}\sin(ky),\, 0\big), \qquad \pi = 0, \qquad f = 0. \tag{20.7}$$

1. *The pair (u,π) solves* (20.4) *and* (20.5) *for every amplitude A and every $\nu \geq 0$, and it satisfies* (20.6)*.*

2. *With $A = 0$ and $s = 0$, the density*

$$c(x,y,t) = 1 + e^{-D_c k^2 t}\big(0.35\cos(kx) + 0.20\sin(ky)\big) \tag{20.8}$$

solves (20.2)*, has unit mass and stays positive for all $t \geq 0$.*

The shear is a real but deliberately limited connection to Navier–Stokes. The drift moves borrowers horizontally, faster at some heights than others, and the motion dies away at a rate set by ν. It contains no turbulence, no chaos and no data: its amplitude and decay are chosen. Passing this benchmark shows that the solver handles one known drift correctly. It does not show that any drift fitted to a register would behave like this one.

The initial density in (20.8) at $t = 0$ is the engine's default: positive everywhere, with mean one, and with structure along both axes so that the shear has something to move. The engine samples it at cell centres, so every mass and error figure below refers to that discrete representation.

20.5 The finite-volume update and its bound

The engine solves (20.2) on an $n \times n$ grid of square cells of side $h = 1/n$, with time step Δt. The drift of (20.7) is horizontal and depends only on the row, so write u_j for its value in row j at the current step and $u^+ = \max(u,0)$, $u^- = \min(u,0)$. The flux through the right face of cell (i,j) takes its density from the cell upstream:

$$F_{i+1/2,j} = u_j^+ c_{i,j} + u_j^- c_{i+1,j}.$$

Taking the density from upstream is the standard first-order choice for transport in finite-volume methods (LeVeque, 2002). The update combines this upwind advection with centred diffusion and an explicit step in time:

$$c_{i,j}^{m+1} = c_{i,j}^{m} - \frac{\Delta t}{h}\left(F_{i+1/2,j} - F_{i-1/2,j}\right) + \frac{D_c \Delta t}{h^2}\left(c_{i+1,j}^{m} + c_{i-1,j}^{m} + c_{i,j+1}^{m} + c_{i,j-1}^{m} - 4c_{i,j}^{m}\right), \tag{20.9}$$

with indices taken modulo n. Explicit schemes of this kind are stable only when the time step is small enough, a condition first stated by Courant et al. (1928), and the engine enforces a bound with a margin.

Proposition 20.3 (Nonnegativity and mass)**.** *Suppose $|u_j| \le |A|$ for every row and step, and*

$$\Delta t\Big(\frac{|A|}{h} + \frac{4D_c}{h^2}\Big) \le 0.8. \tag{20.10}$$

Then each updated value $c_{i,j}^{m+1}$ is a weighted average of $c_{i,j}^{m}$ and its four neighbours, with nonnegative weights that sum to one. Consequently $\min c^{m+1} \ge \min c^{m}$, so a nonnegative density stays nonnegative, and the total mass $\sum_{i,j} c_{i,j}^{m+1}$ equals $\sum_{i,j} c_{i,j}^{m}$ exactly in exact arithmetic.

The bound (20.10) keeps the weight on the cell's own value at least 0.2, so no cell can be drained below zero by one step. The proposition gives the reason the engine never clips: a scheme that needs clipping to stay positive has already lost mass or invented it, and the clipping hides which. When a caller asks for a step that breaks the bound, `transport.transport` raises an error. A request for two steps over the horizon on a 32×32 grid, with amplitude 0.2 and dispersion 0.03, gives a stability number of 12.9, sixteen times the limit, and is refused. When the caller leaves the step count to the engine, it chooses the smallest stable one.

Example 20.4 (One step from a single cell)**.** Put all the mass in one cell of an 8×8 grid, so $h = 0.125$, with $D_c = 0.03$, amplitude 0.2 and smoothing 0.05. In that cell's row the drift at the first step is $0.2 \sin(2\pi \times 0.3125) = 0.185$. Take $\Delta t = 0.0667$. The proof of Proposition 20.3 gives the weights directly: $a|u| = (0.0667/0.125) \times 0.185 = 0.099$ and $d = 0.03 \times 0.0667/0.125^2 = 0.128$, so the cell keeps $1 - 0.099 - 4 \times 0.128 = 0.389$ of its mass, the cell downstream receives $0.099 + 0.128 = 0.227$, and each of the other three neighbours receives 0.128. The five shares sum to one, and the engine's step returns these values (0.3891, 0.2267 and 0.1281) with a stability number of 0.619. Now ask for one step of $\Delta t = 0.2$. The stability number is $0.2 \times (0.2/0.125 + 4 \times 0.03/0.125^2) = 1.856$, and the cell's own weight would be $1 - 0.2 \times (0.185/0.125 + 7.68) = -0.83$: the step would leave the cell with negative mass and pay for it by overfilling its neighbours. The engine refuses the request and names the bound it breaks.

Figure 20.1 shows the first test of the solver against an exact answer. With the drift switched off, the computed density at time 0.2 was compared with (20.8) on grids of 16, 32 and 64 cells a side. The root mean square error fell from 1.03×10^{-4} to 3.23×10^{-5} to 8.51×10^{-6}. Each halving of the cell width divided the error by 3.2 and then by 3.8, approaching the factor of four expected of a second-order method. The engine chose 8, 31 and 123 steps, with stability numbers of 0.768, 0.793 and 0.799, each just inside the bound. In all three runs the mass error was zero, and the smallest density value stayed above 0.56. The order should be read with its conditions. Centred diffusion is second order in space, and the time step shrinks with h^2 under (20.10) when dispersion dominates, so the pure-diffusion case approaches second order. Upwind advection is first order, and this study says nothing about the order of the full solver with drift.

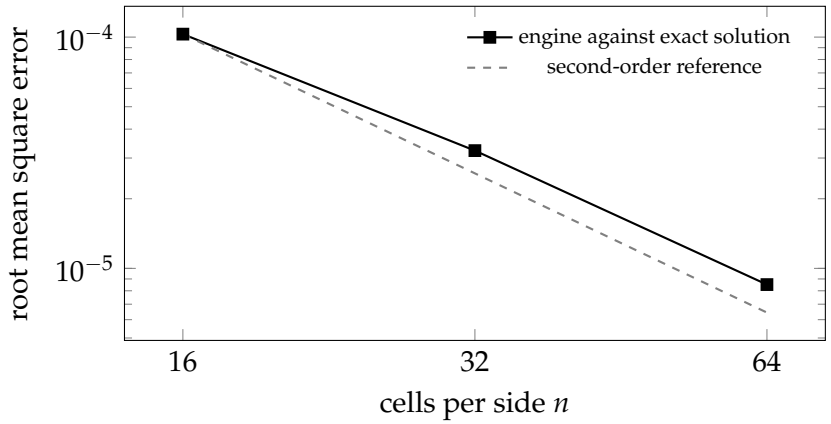


FIGURE 20.1: Error of the finite-volume solver against the exact diffusion solution (20.8) at time 0.2 with dispersion 0.03 and no drift. The measured errors (solid black line, filled squares) are set beside a reference line falling with the square of the cell width (dashed grey line, drawn through the first point). Synthetic benchmark; the measured slope steepens towards the reference as the grid is refined.

With the shear switched on at amplitude 0.2 and smoothing 0.05 on the 32×32 grid, the engine took 33 steps with a stability number of 0.784, the mass error was again zero and the smallest value was 0.573. These are the properties Proposition 20.3 guarantees. There is no exact solution for the density under shear with dispersion in the engine's tests, so the shear run checks conservation and positivity, not accuracy.

Tomás's first runs were refused. He had asked for a coarse time step to save computing time and, when the engine declined, asked Naledi whether the negative values could simply be set to zero. She showed him Proposition 20.3 and asked him to let the engine choose the step. The runs that followed conserved mass to the last digit. Tomás took this as evidence that the model was sound; Naledi took it as evidence that the solver was, and said the two claims needed separate support.

20.6 Bayesian inference over a finite prior

A solver turns parameters into a forecast. The parameters themselves, the dispersion D_c and the drift amplitude A in the simplest case, must come from data. Write G_η for the map from a parameter vector η to predicted densities at the observation points and times, and H for the observation operator that selects them. A model for continuous observations is

$$y = H G_\eta + \delta + \varepsilon, \qquad \varepsilon \sim N(0, \Sigma), \tag{20.11}$$

where δ is the discrepancy between the proposed dynamics and the world, the term that Kennedy and O'Hagan (2001) introduced for the calibration of computer models. Leaving δ out forces every systematic error into the coefficients, which then look precisely estimated and mean little. Putting it in creates its own identifiability problem, since a flexible discrepancy can absorb anything, and its prior has to be argued for; Brynjarsdóttir and O'Hagan (2014) show that physical parameters are learned correctly only when that prior carries real knowledge of the discrepancy. Stuart (2010) develops the Bayesian treatment of such inverse problems in general; the disclosure application, and the feature geometry it needs, remain proposals here.

Bayes' rule gives the posterior $p(\eta \mid y) \propto p(y \mid \eta)p(\eta)$. The engine computes it exactly for a finite prior. If the prior puts weight p_g on each of G parameter vectors η_g and $\ell_g = -\frac{1}{2}\|y - HG_{\eta_g}\|^2_{\Sigma^{-1}}$ is the log likelihood, then

$$w_g = \frac{\exp(\ell_g - \ell_{\max})\, p_g}{\sum_{h=1}^{G} \exp(\ell_h - \ell_{\max})\, p_h}. \tag{20.12}$$

Subtracting the largest log likelihood $\ell_{\max}$ changes nothing in exact arithmetic and keeps the exponentials from underflowing. The calculation is transparent: every weight can be printed and checked. It is not a sampler for a continuous posterior, and it should be read as one only after the grid has been refined and the prior range varied.

A forecast from the posterior must carry the posterior's uncertainty. If a future audit of B fresh targets yields K successes, and given η these are Binomial$(B, r_a(\eta))$, then the forecast of K is the mixture

$$\mathbb{P}(K = k \mid y) = \sum_{g=1}^{G} w_g \binom{B}{k} r_a(\eta_g)^k \big(1 - r_a(\eta_g)\big)^{B-k}. \tag{20.13}$$

It is tempting to replace this by a single binomial at the posterior mean risk $\bar{r} = \sum_g w_g r_a(\eta_g)$. That shortcut understates the spread.

Proposition 20.5 (The predictive mixture is wider)**.** *Let K have the mixture law* (20.13) *and let $\bar{r}$ be the posterior mean risk. Then*

$$\operatorname{Var}(K \mid y) = B\bar{r}(1-\bar{r}) + B(B-1)\operatorname{Var}\big(r_a(\eta) \mid y\big) \;\geq\; B\bar{r}(1-\bar{r}),$$

with equality only when $B = 1$ or the posterior puts all its weight on parameter vectors with the same risk.

The extra term grows with the square of the audit size. For the audit sizes of Chapter 4, hundreds of attempts, even modest uncertainty in the risk dominates the binomial term, and a plug-in forecast would describe the next audit as far more predictable than it is. This is the same distinction Chapter 18 draws between planning figures and certificates: the mixture is a planning figure, and no posterior, however it is computed, replaces the audit bound in the release rule.

Example 20.6 (Planning an audit of 400 attacks)**.** Suppose Tomás's posterior puts weight one half on a risk of 0.05 and one half on 0.10, and Naledi plans an audit of 400 fresh attacks. The numbers are illustrative. The posterior mean risk is 0.075, and a single binomial at that risk has mean 30, variance $400 \times 0.075 \times 0.925 = 27.75$ and standard deviation 5.3. The posterior variance of the risk is $0.025^2 = 0.000625$, so Proposition 20.5 adds $400 \times 399 \times 0.000625 = 99.75$ to the variance, which becomes 127.5, a standard deviation of 11.3. The plug-in forecast puts 95 per cent of the count between 20 and 41; the mixture puts it between 13 and 50. A count of 40 or more has probability 0.04 under the plug-in and 0.26 under the mixture. A plan sized on the plug-in would treat one audit in four as a surprise.

20.7 What the evidence run recovered

The evidence run put a finite prior on 81 pairs (D_c, A): nine dispersions equally spaced from 0.01 to 0.05 and nine amplitudes from 0.1 to 0.3. It generated noisy observations at sixteen

points of a 16×16 grid at time 0.2, from a true pair of $D_c = 0.03$ and $A = 0.2$, with noise of standard deviation 0.01, and computed the posterior (20.12).

In the first case the observations were generated by the same solver, on the same grid, that the inference used. In the extension's run the posterior mean was 0.0306 for dispersion and 0.175 for amplitude, with standard deviations of 0.0017 and 0.020, and the most probable pair was $(0.03, 0.175)$, with weight 0.44. Dispersion was recovered to within a fraction of its grid spacing. Amplitude was not: the most probable value sat one grid step below the truth, and the truth lay about one and a quarter standard deviations above the mean. With one draw of the noise this is not an error in the method. It is what a wide posterior looks like when a single realisation is reported, and it is the first sign that the observations say much less about the drift than about the dispersion.

The draw can be repeated. With the repository's solver, a lattice of 16 cell centres and three noise seeds, the posterior standard deviation of the amplitude came out at 0.019 to 0.020 and that of the dispersion at 0.0003 to 0.0009; the posterior mean of the amplitude was 0.222, 0.245 and 0.197, and the most probable amplitude was 0.225, 0.25 and 0.2. These runs used the repository's code and differ from the extension's draw, so the individual means are different and the spread is the same. One draw can land on either side of the truth, and the spread is the stable result.

In the second case the observations came from a finer, 32×32 solver run with the same true parameters, sampled at sixteen points, while the inference still used the 16×16 model. In the extension's run the posterior mean was 0.0300 for dispersion and 0.193 for amplitude, the most probable pair was the truth, $(0.03, 0.2)$, with weight 0.49, and the standard deviations were 0.00035 and 0.019. The mismatched case landed closer to the truth than the matched one. Read naively, that would say that model error helps. Two features of the run warn against the reading. The noise draws differ between the two cases, and with one draw each, which case lands closer is partly luck. And the dispersion standard deviation of 0.00035 is smaller than the prior's grid spacing of 0.005: almost all the weight sits on one dispersion value, so the reported spread measures the coarseness of the prior, not the information in the data. A narrow posterior produced by a coarse or strong prior is not identification.

Tomás brought the first posterior to Naledi as a success: the dispersion had come back almost exactly. She agreed, and asked about the drift. His model's central claim was that drift could be forecast, and the drift was the parameter the data had resolved least well. She asked him for the spectrum that the engine reports beside every posterior.

20.8 Identifiability and the singular values

Whether data can resolve a parameter is a property of the forward model and the observation design, and it can be checked before any posterior is computed. Let J be the Jacobian of the predicted observations with respect to the parameters, the matrix of sensitivities $J_{ih} = \partial (HG_\eta)_i / \partial \eta_h$. Its singular values measure how strongly the observations respond to movement along each direction in parameter space. A large singular value marks a direction the data resolve; a small one marks a direction the data barely see. Under the Gaussian model (20.11) with $\Sigma = \sigma^2 I$ and no discrepancy, the Fisher information is $J^\top J / \sigma^2$, and near the truth the posterior standard deviation along a direction with singular value s is about σ / s.

At the true pair the engine's finite-difference Jacobian had singular values of 7.13 and

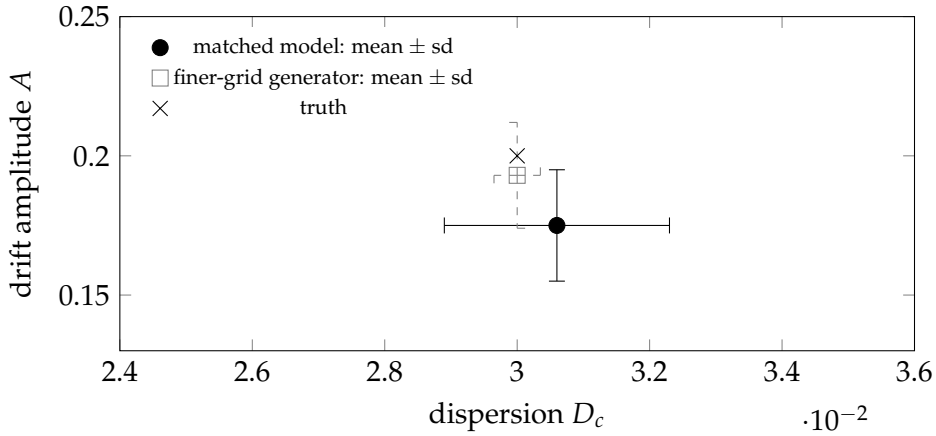


FIGURE 20.2: Posterior means and standard deviations for dispersion and drift amplitude under the 81-point finite prior, from the extension's run, shown near the truth. The matched-model posterior (filled black circle, solid bars) and the finer-grid posterior (open grey square, dashed bars) are told apart by mark, shade and bar style; the cross marks the truth. Synthetic observations.

0.515, a condition number of 13.8. The Fisher information had eigenvalues of about 508,000 and 2,650. With $\sigma = 0.01$ these give standard deviations of about 0.0014 and 0.019 along the two directions. The second figure matches the amplitude's posterior standard deviation of 0.020 in the matched case, which says that the weak direction is the amplitude. The observations at one time respond to the dispersion through the overall decay of the pattern, which is large and easy to see; they respond to the drift only through a small sideways shift of it. The repository's `bayes.sensitivity_spectrum`, on a lattice of 16 cell centres, gives 7.09 and 0.516, a condition number of 13.7, and Fisher eigenvalues of about 502,700 and 2,660. The small difference from the extension's figures comes from where the sixteen observation points sit, which the extension does not state.

A condition number of fourteen is mild. Models in systems biology commonly show sensitivity spectra spread over many orders of magnitude (Gutenkunst et al., 2007). A model with a full drift field, a forcing term, an initial condition and an attack response, fitted to one scalar risk series, would have many more directions and far smaller singular values in most of them. One risk series cannot normally identify a two-dimensional velocity field. The order of work this implies is plain: fix the attack response and the initial condition, estimate a few coefficients, report the spectrum, and add parameters only while the spectrum shows that the data can carry them.

20.9 The inverse crime

The matched case of Section 20.7's first run generated its data with the same model, the same grid and the same observation operator that the inference then assumed. In the literature on inverse problems this is called an inverse crime (Kaipio and Somersalo, 2007). It makes recovery look easier than it is, because every error the model makes when generating is made again, identically, when inferring, and the two cancel. A matched-model run is a pipeline check: it shows that the code computes the posterior it was written to compute. It is not external validation, and the engine's documentation says so beside the function. The finer-grid case is one step away from the crime. It changes the mesh, and with it the positions of the sixteen observation points, which in the finer grid sit half a coarse cell away

from where the coarse model expects them. It does not change the model family, the noise law or the drift. A real validation would generate observations independently of the model, perturb the dynamics and the observation operator, and include data the model was not designed for, which for a register means data from vintages held out of every fitting step.

Tomás had committed the crime in his first draft without knowing its name. His synthetic register had been made by his own solver, and his recovery of the drift from it was the result his note led with. Naledi asked him to move it to an appendix headed as a check of the code, and to leave the main text for data his solver had not made.

20.10 A PDE surrogate is a separate task

When a solver is slow, a common next step is to train a neural network to approximate it. A physics-informed network outputs fields $(u_\phi, \pi_\phi, c_\phi)$ and is trained to fit observations while keeping the residuals of (20.2), (20.4) and (20.5) small at chosen collocation points (Raissi et al., 2019). Such a surrogate answers a different question from the risk planner of Chapter 22. The planner predicts attack success from measured counts; the surrogate approximates a solver. Mixing the two would let a network trained on its own simulated output stand in for evidence about attacks.

Surrogates also fail in documented ways. A small residual at the training points need not mean a small error between them, and optimisation can stall on solutions that satisfy the equations poorly (Krishnapriyan et al., 2021). A surrogate must therefore be compared with an independent solver on held-out parameter settings and time windows, and checked for mass and positivity, which a network does not respect by construction. Here the equation is itself a proposed model of how borrowers move, so the name physics-informed should not be read as physical validation of anything about disclosure. The engine does not train a surrogate, and the extension plans one only if the cost of the finite-volume solver ever justifies it.

20.11 The baseline the transport model must beat

Every model in this chapter has a simpler competitor. Many of the movements Tomás wanted to forecast are between categories: a borrower moves from one lender to another, from one size class to the next, from performing to non-performing. A continuous-time Markov chain on a graph of such states describes these moves without any geometry, and credit-risk work has long used it for movements between rating grades (Jarrow et al., 1997). If $c_i(t)$ is the mass at node i and Q is a generator, with nonnegative rates Q_{ij} off the diagonal and rows summing to zero, then

$$\dot{c} = Q^\top c, \qquad c(t) = \exp(tQ^\top)\, c(0),$$

and node risks g_i give the model risk $r = g^\top c / \mathbf{1}^\top c$, the graph version of (20.3). The nonnegative rates keep masses nonnegative and the zero row sums keep total mass constant, the same two properties Proposition 20.3 proves for the transport solver. Chapter 19 develops this baseline and its estimation from observed transitions. In the evidence run the engine recovered a four-node generator from 400 simulated paths with a largest rate error of 0.016, and the propagated masses still summed to one.

Figure 20.3 sets out the order of comparison the extension specifies. Each model on the ladder adds assumptions to the one before. A model earns its place only by forecasting

held-out periods better than the model below it, or by carrying a structural constraint that is independently known to hold.

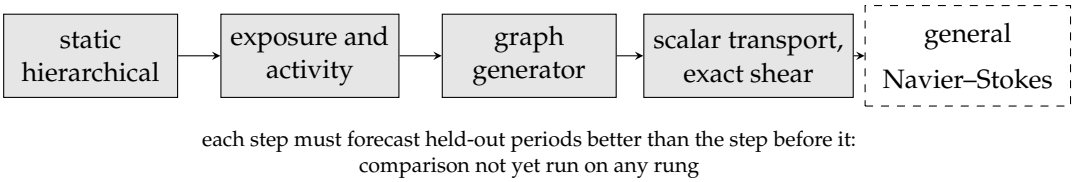


FIGURE 20.3: The ladder of dynamic models, from fewest assumptions (left) to most (right). Models built and tested in the engine have solid borders and grey fill; the general Navier–Stokes solver, not built, has a dashed border and hatching. The arrows mark the comparisons the extension requires; none has been run on held-out data.

The comparison has not been run on any rung. The engine has the static model of the earlier chapters, the exposure and activity panel, the graph generator and the scalar transport solver under an exact shear, each with tests of its own properties. It has no experiment in which any of them forecasts held-out vintages of a register and is scored against another. Until that experiment exists, nothing in this book supports the claim that a transport model, with or without the Navier–Stokes closure, forecasts disclosure risk better than a table of counts.

The committee met to consider Tomás's proposal with Figure 20.3 on the table. The chair asked which rung the proposal sat on, and Naledi pointed to the last one. The committee recorded the model as research, approved Tomás's use of synthetic data to run the comparison against the graph baseline, and stated that no forecast from it would enter a release decision until it had beaten that baseline on held-out vintages. Grace, reading the record later, found the condition written in a form she could check.

20.12 The answer

The chapter asked what a dynamic model of the target population would have to show before its forecast could enter a release decision. It has to show five things, and each has a test. Its density must obey a stated balance law, and its solver must preserve mass and positivity without clipping, which Proposition 20.3 guarantees under a bound the engine enforces. Its link to disclosure must run through an attack response fitted to measured outcomes, as in (20.3). Its parameters must be identified by the data, shown by the singular values of the sensitivity matrix and not by the narrowness of a posterior on a coarse grid. Its validation must use data that the model did not generate. And it must forecast held-out periods better than the graph baseline and the simpler models below it. The engine meets the first and supplies the tools for the second and third. The fourth and fifth have not been done. A model drawn from physics brings structure, and structure is a set of assumptions; until the comparison is run, the Bank's release decisions rest on audits of attacks, as the rest of this book describes.

IN PRACTICE: A PROPOSAL FOR A DYNAMIC RISK MODEL

Ask the proposer to name, in the register's own terms, the feature space, the density and the attack response, and to state where the response was fitted. Ask for the balance law and for the treatment of entry and exit. Require a solver check against an exact solution, with mass

and minimum reported at every run, and refuse any solver that clips. Require the identifiability spectrum beside every posterior, and treat a matched-model recovery as a pipeline check only. Record the model's place on the ladder of Figure 20.3 and the held-out comparison it must win before any forecast enters a decision record. Until then, record it as research.

In the software. The solver, the exact diffusion solution and the model risk integral are in `privacy_risk.transport` (`transport`, `exact_diffusion`, `risk_from_density`; the last uses a declared toy response unless a fitted one is passed). The finite-prior posterior, the transport forward model, the predictive mixture and the spectrum are in `privacy_risk.bayes` (`grid_posterior`, `transport_forward`, `predictive_pmf`, `sensitivity_spectrum`). The graph baseline is in `privacy_risk.graph`. The tests in `tests/test_ext_transport.py` check mass, positivity, refinement, refusal of unstable steps, normalisation of the posterior, Example 20.6 and the spectrum. Algorithm 20.1 sets out one run of the solver with its stability check, and its checks replay Example 20.4 and the refinement study of Figure 20.1.

Algorithm 20.1. One run of the transport solver.

Input. A grid of $n \times n$ cells; a horizon T; a dispersion D_c; a drift amplitude A and smoothing ν; an initial density c_0 (by default (20.8) at $t = 0$, sampled at cell centres); and, optionally, a number of steps M.

Output. The density c_M at time T, with its mass error, its smallest value, the number of steps, the step Δt and the stability number.

1. Refuse the inputs unless n is a whole number of at least 8, T, D_c, A and ν are finite, and T, D_c and ν are not negative. Refuse an initial density that is not $n \times n$, finite and nonnegative.
2. Set $h = 1/n$ and the rate bound $R = |A|/h + 4D_c/h^2$, the bracket of (20.10).
3. If M is not given, take the least $M \geq 1$ with $TR/M \leq 0.8$. If it is given, refuse $M < 1$.
4. Set $\Delta t = T/M$. If $\Delta t\, R > 0.8$, refuse the request and report the stability number $\Delta t\, R$; never clip.
5. For $m = 0, 1, \ldots, M-1$: set the drift in row j to $u_j = Ae^{-\nu(2\pi)^2 m\Delta t} \sin(2\pi y_j)$, from (20.7), with y_j the height of the row's centre, and update every cell by (20.9), with indices taken modulo n.
6. Return c_M, the mass error $\bar{c}_M - \bar{c}_0$ where the bar is the mean over cells, $\min c_M$, M, Δt and $\Delta t\, R$. By Proposition 20.3 the mass error is zero up to rounding and the minimum does not fall.

Checks, each run against the software by the test suite:

- One step of $\Delta t = 0.0667$ from all the mass in one cell at height 0.3125 on an 8×8 grid, with $D_c = 0.03$, $A = 0.2$, $\nu = 0.05$ (Example 20.4), share the cell keeps: 0.3891.
- The same, share of the cell downstream: 0.2267.
- The same, share of the cell upstream: 0.1281.

- The same, stability number: 0.619.
- No drift, a 32×32 grid, horizon 0.2 and $D_c = 0.03$, steps left to the solver, number of steps: 31.
- The same, stability number: 0.793.
- The same, mass error: 0.0000.
- A horizon of zero, number of steps: 1.
- One step of $\Delta t = 0.2$ from the single cell of Example 20.4, stability number 1.856: refused.
- Two steps over a horizon of 0.2 on a 32×32 grid with $A = 0.2$ and $D_c = 0.03$, as in the evidence run: refused.
- A grid of 4 cells a side: refused.

20.13 Proofs

Proof of Proposition 20.1. Take the inner product of (20.4) with u and integrate over the periodic domain. The time derivative gives $\int u \cdot \partial_t u \, dx = \frac{1}{2} \frac{d}{dt} \|u\|_2^2$. For the nonlinear term, $u \cdot (u \cdot \nabla) u = \sum_j u_j \partial_j |u|^2 / 2$, and integration by parts on the periodic domain, where boundary terms vanish, gives $\int u \cdot (u \cdot \nabla) u \, dx = -\int (\nabla \cdot u) |u|^2 / 2 \, dx = 0$ by (20.5). In the same way $\int u \cdot \nabla \pi \, dx = -\int \pi \nabla \cdot u \, dx = 0$. For the smoothing term, integration by parts in each component gives $\nu \int u \cdot \Delta u \, dx = -\nu \sum_{i,j} \int (\partial_j u_i)^2 dx = -\nu \|\nabla u\|_2^2$. The forcing gives $\langle f, u \rangle$. Collecting terms yields (20.6). $\square$

Proof of Proposition 20.2. For part 1, write $u_1 = A e^{-\nu k^2 t} \sin(ky)$ and $u_2 = 0$. Then $\nabla \cdot u = \partial_x u_1 = 0$, since u_1 does not depend on x, so (20.5) holds. The convective term is $(u \cdot \nabla) u = u_1 \partial_x u = 0$. The time derivative is $\partial_t u_1 = -\nu k^2 u_1$ and the Laplacian is $\Delta u_1 = \partial_{yy} u_1 = -k^2 u_1$, so $\partial_t u_1 = \nu \Delta u_1$. With $\pi = 0$ and $f = 0$ both components of (20.4) hold. For the energy identity, the mean of $\sin^2(ky)$ over the unit square is $\frac{1}{2}$, so $\|u\|_2^2 = \frac{1}{2} A^2 e^{-2\nu k^2 t}$ and $\frac{1}{2} \frac{d}{dt} \|u\|_2^2 = -\frac{1}{2} \nu k^2 A^2 e^{-2\nu k^2 t}$; and $\|\nabla u\|_2^2 = \|\partial_y u_1\|_2^2 = \frac{1}{2} k^2 A^2 e^{-2\nu k^2 t}$. The two sides of (20.6) then sum to zero, which is $\langle f, u \rangle$.

For part 2, with $u = 0$ and $s = 0$ the equation is $\partial_t c = D_c \Delta c$. Each of $\cos(kx)$ and $\sin(ky)$ is an eigenfunction of the Laplacian with eigenvalue $-k^2$, so $e^{-D_c k^2 t}$ times either solves the equation, as does the constant 1; by linearity so does (20.8). The two trigonometric terms have mean zero over the unit square, so the mass is one. Finally $c \geq 1 - e^{-D_c k^2 t} (0.35 + 0.20) \geq 1 - 0.55 > 0$. $\square$

Proof of Proposition 20.3. Fix a row j and write $a = \Delta t / h$, $d = D_c \Delta t / h^2$, $u = u_j$. Substituting the fluxes into (20.9),

$$c_{i,j}^{m+1} = c_{i,j} - a\big(u^+ c_{i,j} + u^- c_{i+1,j} - u^+ c_{i-1,j} - u^- c_{i,j}\big) + d\big(c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1} - 4c_{i,j}\big),$$

where all values on the right are at step m. Since $u^+ - u^- = |u|$, collecting terms gives

$$c_{i,j}^{m+1} = \big(1 - a|u| - 4d\big) c_{i,j} + \big(a u^+ + d\big) c_{i-1,j} + \big(d - a u^-\big) c_{i+1,j} + d\, c_{i,j+1} + d\, c_{i,j-1}.$$

The last four weights are nonnegative because $u^+ \geq 0$, $u^- \leq 0$ and $d \geq 0$. The first is at least $1 - \Delta t(|A|/h + 4D_c/h^2) \geq 0.2$ by $|u| \leq |A|$ and (20.10). The weights sum to $1 - a|u| - 4d + au^+ - au^- + 4d = 1$. So $c_{i,j}^{m+1}$ is a weighted average of five values at step m, and it is at least their minimum; hence $\min c^{m+1} \geq \min c^m$, and nonnegativity follows by induction.

For mass, sum (20.9) over all cells. Within each row the flux differences $F_{i+1/2,j} - F_{i-1/2,j}$ telescope to zero around the periodic index i. The diffusion term sums to $\sum_{i,j}(c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1}) - 4\sum_{i,j} c_{i,j} = 0$, because each shifted sum runs over all cells once. Hence $\sum_{i,j} c_{i,j}^{m+1} = \sum_{i,j} c_{i,j}^m$. □

Proof of Proposition 20.5. Write $R = r_a(\eta)$ with η drawn from the posterior, so that $K \mid R$ is Binomial(B, R). By the law of total variance,

$$\operatorname{Var}(K) = \mathbb{E}[\operatorname{Var}(K \mid R)] + \operatorname{Var}(\mathbb{E}[K \mid R]) = \mathbb{E}[BR(1-R)] + \operatorname{Var}(BR).$$

The first term is $B\bar{r} - B\,\mathbb{E}[R^2] = B\bar{r} - B(\operatorname{Var} R + \bar{r}^2) = B\bar{r}(1-\bar{r}) - B\operatorname{Var} R$, and the second is $B^2 \operatorname{Var} R$. Their sum is $B\bar{r}(1-\bar{r}) + B(B-1)\operatorname{Var} R$. The added term is nonnegative, and it is zero exactly when $B = 1$ or $\operatorname{Var} R = 0$, that is, when R takes one value with posterior probability one. □

Chapter 21

Chaos, changing adversaries and quantum capability

Every few months a headline announces that quantum computers will soon break the encryption that protects banks, governments and hospitals. A reader who follows these reports may reach a firm conclusion about disclosure control. The methods in this book assume an adversary of a certain strength. A quantum computer is an adversary of a different order: it can factor the numbers on which public-key encryption rests, and it can search large spaces faster than any classical machine. On this view the floors, rounding and audits of the earlier chapters were calibrated against yesterday's computers, and they will become obsolete when the new machines arrive. The remedy is to move to stronger encryption now, before the data are exposed. The view has real support. Shor's algorithms factor integers and compute discrete logarithms in polynomial time on a quantum computer (Shor, 1997), and the public-key systems in wide use today rest on the hardness of exactly those problems.

The first fact that breaks the belief comes from the Bank's own register. The adversary who most threatens a loan-level credit file is an insider at a reporting lender, who knows the exposure of every loan his bank has reported. Against him the likelihood attack of Chapter 13 names the right loan in 0.772 of attempts at the smallest floor examined, and still in 0.416 at the largest. That attack compares one number with a short list of published numbers. It needs no computer worth the name, and it reads a file the Bank meant to be read. Encryption, of any strength, protects data that should not be read. A release is data that should be. What an adversary can learn from a release is limited by what the release shows and by what the adversary already knows, and computing power changes that limit only for attacks whose bottleneck was computation.

This chapter asks one question. When the adversary changes, through drift in the world it observes or through a new kind of computer, what has to change in the way the Bank decides, and which tool answers each part? The answer separates three tasks that are easily run together: forecasting how a scenario will develop, stress testing a decision against changes in its inputs, and operating the systems that hold the data securely.

21.1 A memo about quantum computers

The chair of the Bank's statistics release committee received a memo from the Bank's information technology department. It summarised a vendor briefing on quantum computing and recommended that every file released by the statistics department be encrypted with a

quantum-resistant algorithm. The chair forwarded it to Naledi with a short question: did the committee's past decisions still hold?

Naledi's reply began with Pieter. Pieter is a credit analyst at one of the Bank's reporting lenders, the insider of Chapter 13. If the Bank released a loan-level file, he would download it from the Bank's website, open it in a spreadsheet, and look for the rows whose exposures matched loans in his own bank's ledger. Encrypting the file in transit would change nothing in that scene, because Pieter would be an authorised reader. The file would be decrypted for him, as for every user, before he ever looked at it.

21.2 Three roles that must stay apart

The extension that this part of the book draws on uses three bodies of mathematics, and it assigns each one a separate role. A dynamic model, such as the transport model of Chapter 20, may be used to forecast how a population of targets will move. Tools from the study of chaos measure how sensitive a forecast, or a decision built on it, is to small changes in its inputs, and so serve stress testing. Post-quantum cryptography protects the register, its communications and its signatures against a specified class of cryptanalytic attack, and so serves secure operation. These roles answer different questions, and naming the three together creates no guarantee. Each component has its own assumptions and its own tests.

The confusion the roles prevent is concrete. A forecast is not a stress test: a model that predicts the most likely path says nothing about how the decision behaves on the unlikely ones. A stress test is not a security control: knowing that a decision survives perturbed inputs says nothing about whether the keys that sign the decision record are safe. And a security control is not a disclosure guarantee, as the scene with Pieter shows. The remainder of the chapter takes the roles in turn, starting with sensitivity.

21.3 Finite-time growth and the tangent equation

Suppose a scenario model has a state $z(t)$ that evolves by $\dot{z} = F(z)$. The state might hold the share of loans active in each size class, an attacker's measurement error, or a correlation between exposures. Two runs that start a small distance apart will diverge or converge, and the rate at which they do so measures how far the model's forecast can be trusted. To first order, a small difference v between two runs obeys the tangent equation

$$\dot{v} = J_F\big(z(t)\big)\, v, \qquad v(0) = v_0, \tag{21.1}$$

where J_F is the Jacobian of F. The finite-time growth rate along one trajectory is

$$\lambda_T(z_0, v_0) = \frac{1}{T} \log \frac{\lVert v(T) \rVert}{\lVert v_0 \rVert}. \tag{21.2}$$

A positive value shows that small differences grew over that interval, for that trajectory, that starting perturbation and that norm. It does not prove that the system is chaotic, which is a statement about infinite time. The estimate also depends on the scaling of the coordinates, the time step and transients at the start, so each of these must be reported. Over long horizons the tangent vector grows too large to store, and the engine renormalises it at fixed intervals and adds up the logarithms of the growth, as in the standard method of Benettin et al. (1980).

A numerical method for (21.2) should be checked on a system whose answer is known before it is trusted on one whose answer is not. Linear systems supply the check.

Proposition 21.1 (Growth rates of a linear system)**.** *Let $\dot{z} = Az$ with A a constant matrix that is diagonalisable with real eigenvalues $\mu_1 > \mu_2 \geq \cdots \geq \mu_d$ and eigenvectors $w_1, \ldots, w_d$.*

1. *If v_0 is an eigenvector with eigenvalue μ, then $\lambda_T(z_0, v_0) = \mu$ for every $T > 0$ and every z_0.*
2. *If $v_0 = \sum_i a_i w_i$ with $a_1 \neq 0$, then $\lambda_T(z_0, v_0) = \mu_1 + O(1/T)$ as $T \to \infty$.*

The engine's check takes A diagonal with entries 0.5 and -1 and starts the tangent along the first axis, so the true rate is 0.5 by part 1. Over a horizon of 4 the estimate was 0.5 to twelve decimal places (the draft of this book reports thirteen; the repository's code gives 0.499999999999836). The check confirms the integration and the renormalisation. Because the starting direction is already the growing one, it does not test part 2, the slower convergence of a general starting direction towards the fastest-growing one; a second check with a general start would cover that.

Lorenz (1963) gave the classical example of a deterministic system with strong sensitivity to initial conditions:

$$\dot{z}_1 = \sigma(z_2 - z_1), \qquad \dot{z}_2 = z_1(\rho - z_3) - z_2, \qquad \dot{z}_3 = z_1 z_2 - \beta z_3, \tag{21.3}$$

with the familiar values $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$. These are a benchmark, not fitted financial parameters. The engine ran the system for ten time units from $(1, 1, 1)$ to leave the transient behind, then estimated the growth rate over a horizon of 60 with a time step of 0.005, starting the tangent along each of the three axes in turn. The three estimates were 0.882, 0.893 and 0.861, and the repository's code reproduces all three. They are positive and close to one another, as part 2 of Proposition 21.1 leads one to expect once each starting direction has turned towards the fastest-growing one. The spread of about 0.03 between them is the finite-time variation that the definition warns about, and a longer horizon or several starting points would be needed to narrow it.

The Bank's use for this is limited and specific. Naledi's scenario models for the credit file describe how loan activity, exposures and an attacker's knowledge change from one vintage to the next, in the manner of Chapter 19. A growth rate computed on such a model tells her how quickly two scenarios that start almost together will come apart. A chaotic benchmark such as Lorenz is a stress generator: a bounded transform of its trajectory can drive a synthetic scenario parameter through irregular changes. The labels in such a study still come from running the actual attack under each scenario. A chaotic trajectory is never evidence of disclosure, and nothing in this book claims that real disclosure processes are chaotic. Nor is the decaying shear of Chapter 20 chaotic; it was chosen because its solution is known.

21.4 The diagnostic horizon

The chair's reply to Naledi's first note asked a practical question. If a scenario model can drift away from the truth, for how long does a release decision based on it remain valid? The growth rate suggests an answer, and the answer must be handled carefully.

Suppose the attack response g of Chapter 20 changes by at most L_g for each unit of change in the state, and that two states which start ϵ_0 apart are about $e^{\lambda T}\epsilon_0$ apart at time T.

Then the risks computed from them differ by about $L_g e^{\lambda T} \epsilon_0$. Setting this equal to an allowed error $\delta > L_g \epsilon_0$ gives the diagnostic horizon

$$T_{\text{diag}} \approx \frac{1}{\lambda} \log \frac{\delta}{L_g \epsilon_0}. \tag{21.4}$$

With the Lorenz rate of about 0.88 and an allowed error one hundred times the initial error after scaling by L_g, the horizon is a little over five time units ($\log 100 / 0.88 = 5.2$). The logarithm makes the horizon grow slowly with precision: measuring the initial state ten times more accurately adds only $\log(10)/\lambda$ to it.

Example 21.2 (How long a scenario can be trusted)**.** Suppose Naledi's scenario model grows small differences at the Lorenz rate, $\lambda = 0.88$ per unit of time. The other inputs are illustrative. The attack response moves by at most $L_g = 0.5$ per unit of state, the state is known to within $\epsilon_0 = 0.002$, and the committee will tolerate an error of $\delta = 0.02$ in a risk figure. Then $L_g \epsilon_0 = 0.001$, the ratio $\delta / (L_g \epsilon_0)$ is 20, and (21.4) gives $\log(20)/0.88 = 3.4$ units of time. Measuring the state ten times more finely, to 0.0002, raises the ratio to 200 and the horizon to 6.0, an extra $\log(10)/0.88 = 2.6$ units for a tenfold gain in precision. If the model's unit is a month, Naledi would plan to rerun the scenario study after three months. That is a date for a review. It does not end the certificate.

Equation (21.4) is a diagnostic, and it is not an expiry rule for audit certificates. Three things stand in the way of reading it as one. The constant L_g is rarely known. The growth rate is not constant along a trajectory, as the spread of the Lorenz estimates shows. And the decision is not a smooth function of the state: a release either passes the rule or it does not, and the exponential bound says nothing about when a discontinuous decision changes. The engine's function returns an infinite horizon for a nonpositive rate, which says only that this diagnostic sets no limit. What it can do is shorten the period over which a forecast is relied on, and prompt a reassessment when the scenario has had time to drift. The audit's validity rests on the conditions of Chapter 6: the release has not changed and the declared adversary still describes the threat.

21.5 Decisions are discontinuous: perturb the decision

A release rule compares an upper bound with a tolerance, and its output jumps from withhold to release as the bound crosses the line. Near the line, the smallest change in an input can reverse the decision; far from it, large changes leave it alone. A smooth sensitivity measure averages over both situations and so describes neither. Global sensitivity analysis, which divides the variance of a model's output among its inputs (Sobol', 2001; Saltelli et al., 2007), is built for smooth outputs; a decision that jumps needs a measure of its own. The extension measures the decision directly. Perturb the inputs many times, recompute the decision each time, and report the share of perturbed runs in which it changed, the flip rate.

For a single risk estimate b, a tolerance τ and Gaussian perturbations, the flip rate has a closed form.

Proposition 21.3 (Flip rate near a threshold)**.** *Let the decision be $d(z) = \mathbf{1}\{z > \tau\}$, let $b \neq \tau$, and let $Z \sim N(0,1)$. For $s > 0$, the probability that $d(b + sZ) \neq d(b)$ is $\Phi\big(-|\tau - b|/s\big)$, where Φ is the standard normal distribution function.*

The flip rate depends only on the distance to the tolerance measured in units of the perturbation. Figure 21.1 sets the formula beside the engine's simulation. With a tolerance of 0.12, a perturbation standard deviation of 0.01 and 2,000 draws at each point, the repository's run with seed 2026 gave flip rates of 0 at an estimate of 0.05, 0.022 at 0.10, 0.298 at 0.115 and 0.454 at 0.119. The formula gives 0.000, 0.023, 0.309 and 0.460. The draft reports 0.309 and 0.470 at the last two points, from a different random stream; with 2,000 draws the standard error at these points is about 0.010 to 0.011, so the simulated rates differ from the formula by at most about one standard error in every case.

Example 21.4 (One estimate, two widths of doubt)**.** An upper bound of 0.110 sits 0.010 below the tolerance of 0.12. If the inputs behind it are uncertain by a standard deviation of 0.01 in their effect on the bound, the distance is one width, and Proposition 21.3 gives a flip rate of $\Phi(-1) = 0.159$. The engine, perturbing the bound 2,000 times with seed 2026, counted 290 flips, a rate of 0.145, within one standard error (0.008) of the formula. If the uncertainty doubles to 0.02, the distance is half a width and the formula gives $\Phi(-0.5) = 0.309$; the engine counted 595 flips, 0.298, again within one standard error. (The draft prints 293 and 624 from its own random stream.) The bound did not move. Only the committee's doubt about its inputs did, and that alone doubled the share of nearby worlds in which the decision would go the other way.

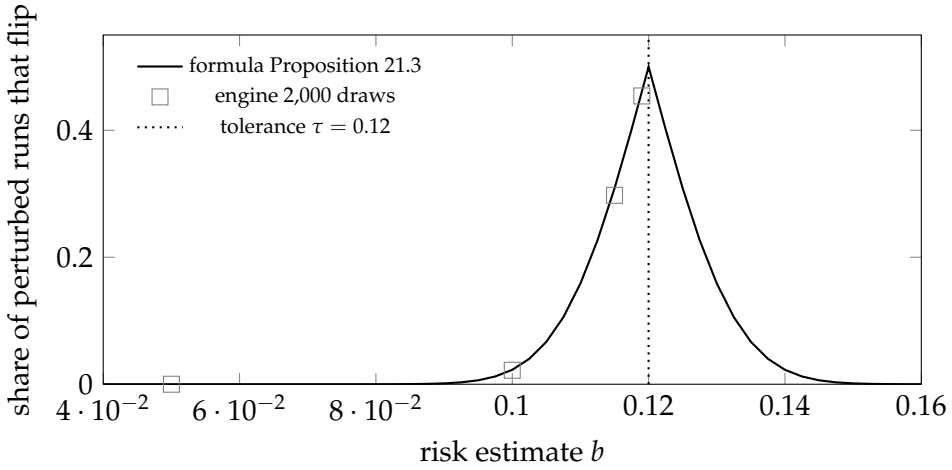


FIGURE 21.1: Flip rate of a threshold decision against the risk estimate, with Gaussian perturbations of standard deviation 0.01 and tolerance 0.12. The formula of Proposition 21.3 (solid black line) is set beside the engine's simulated rates (open grey squares); the dotted line marks the tolerance. Synthetic study.

The figure carries the lesson of the section. An estimate of 0.10, which a committee would read as comfortably inside a tolerance of 0.12, flips in about one perturbed run in fifty. An estimate of 0.115 flips in almost one in three. A model that forecasts risk well on average can be unsafe for exactly the decisions near the line, and those are the ones a committee most needs to get right. Stress effort belongs there: on configurations close to the tolerance and on cases where the ranking of candidate settings reverses.

In practice the perturbed quantities are the inputs of the attack: the attacker's measurement error, correlations between exposures, which loans are active, when the release happens and what the attacker knows. The decision to perturb is the one the release rule makes, recomputed with the attack run again under each perturbed input. For Pieter the input that

moves is the precision of his reading. A lender that improves its ledger, or a merger that brings a second lender's ledger into his hands, changes the adversary without changing the release. The flip rate shows how close the committee's decision sits to the point where such a change would reverse it.

21.6 A stress gate over a preregistered set

A flip rate describes the decision. To act on it, the release rule itself must take account of the stress scenarios. Let $\mathcal{S}$ be a finite set of stress scenarios, written down before any audit is run, and let $R_{a,s}(\theta)$ be an upper confidence bound on the success of attack a under scenario s for protection setting θ. The gate releases θ only if

$$\max_{s \in \mathcal{S}} R_{a,s}(\theta) \le \tau_a \qquad \text{for every required attack } a. \tag{21.5}$$

The engine calls this a robust gate. Its guarantee is exactly as wide as the set.

Proposition 21.5 (Coverage of the stress gate)**.** *Let the bounds $R_{a,s}(\theta)$ be simultaneous at level $1-\alpha$ over every candidate setting θ, required attack a and scenario $s \in \mathcal{S}$: with probability at least $1-\alpha$, $r_{a,s}(\theta) \le R_{a,s}(\theta)$ for all of them at once. Then with probability at least $1-\alpha$, every setting that passes* (21.5) *has true risk at most τ_a for every required attack and every scenario in $\mathcal{S}$, whatever setting is chosen among those that pass.*

The proposition needs the error allocation to cover every coordinate the gate looks at. Adding scenarios to $\mathcal{S}$ therefore costs audit budget, in the way that adding candidate settings did in Chapter 5. It says nothing about scenarios outside the set. A finite grid of stress scenarios does not bound a continuum of them unless a separate argument, a covering or a regularity condition, is supplied. And a scenario with no bound counts as a failure: the gate cannot pass an attack it has not audited.

The evidence run illustrates the last point with the credit-register figures of the earlier chapters, through the release gate of Chapter 18. With the pooled history attack and the insider both required at a tolerance of 0.12, the gate withheld the file: at a floor of 75 the insider had not been audited, and at a floor of 150 its bound of 0.42 failed. With only the outside attack required, the gate released at a floor of 75. The choice of which adversaries are required is a declaration the committee makes, as Chapter 2 insists, and the gate makes the consequence of that declaration visible.

Example 21.6 (Three stress scenarios for one attack)**.** Take the floor of 75 and the external attack at a tolerance of 0.12, with three preregistered scenarios: the base case with a bound of 0.104, a case in which the adversary's readings are twice as precise with a bound of 0.112, and a case in which a lender merger gives the adversary a second ledger, with a bound of 0.131. The base bound is the run's figure; the other two are illustrative. The stress gate takes the worst scenario, 0.131, and fails the floor, naming the merger as the reason. Drop the merger from the set and the gate passes, because its guarantee covers only the set. Add the insider as a required attack with no audit and the gate fails again, on the missing coordinate. Each scenario also costs attacks: with two candidates and three scenarios the 0.05 is split six ways, and the zero-success audit at 0.12 needs 38 attacks per coordinate instead of 24 ($\lceil \log(0.05/6)/\log(0.88) \rceil = 38$ against $\lceil \log(0.05)/\log(0.88) \rceil = 24$).

Naledi's draft reply to the chair now had two parts. The first explained that the committee's decisions depended on which adversaries it declared, and that the insider row, Pieter's

row, was the one that withheld the loan-level file. The second part was about the memo, and for that she needed to separate what a quantum computer could do to the Bank's systems from what it could do with the Bank's releases.

21.7 What quantum capability changes

A quantum threat model has to say what algorithm the attacker runs, how the input is represented, what access the attacker has to the system under attack, what resources the attack needs and what counts as success. Two algorithms anchor most discussion. Shor's algorithms factor integers and compute discrete logarithms in polynomial time (Shor, 1997), which breaks the public-key systems based on those problems once a large enough fault-tolerant machine exists. Grover's algorithm searches an unstructured space of N items in about $\sqrt{N}$ queries to an oracle, where a classical search needs about N (Grover, 1996). The first is a large speed-up on a narrow class of problems. The second is a modest speed-up on a broad one, and it assumes an oracle that can be built and run coherently. Neither result says that a quantum computer can identify every record in a database or solve every learning problem. Loading data, building a reversible oracle, correcting errors and repeating measurements all bear on what a concrete attack would cost.

Figure 21.2 separates the two threats that the memo had merged. The first is an attack on the cryptography that protects the register, the channels that carry its data, the keys and the signatures on decision records. Shor's algorithms are the reason this threat is real, and an adversary may record encrypted traffic today in the hope of decrypting it later. The second is an attack that uses stronger computation to infer identities or attributes from an authorised or public release. These threats call for different defences, run by different parts of an institution. Stronger encryption does not remove information that a release shows to its readers, and a good disclosure audit does not protect a weak transport protocol or a stolen signing key.

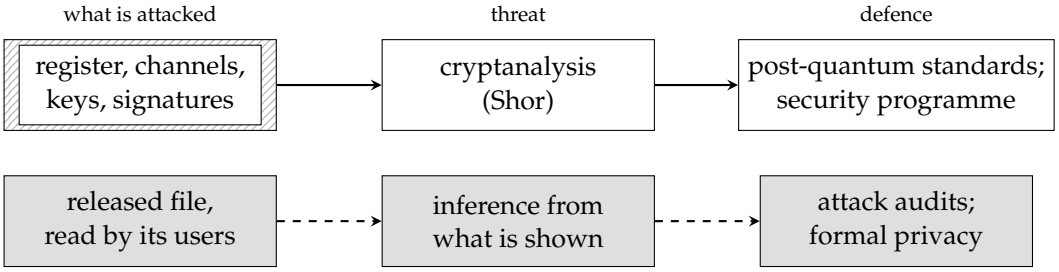


FIGURE 21.2: Two threats from a quantum-capable adversary and their separate defences. The upper row (hatched box, solid arrows) is the attack on cryptography, answered by the security programme. The lower row (grey box, dashed arrows) is inference from a release, answered by attack audits and, where it applies, formal privacy. Each row's defence does nothing against the other row's threat.

For the second threat, the useful question is whether the attack was ever limited by computation. Some attacks are. The reconstruction of census microdata from published tables solved a very large system of constraints, and computing power was part of what made it possible (Garfinkel et al., 2019; Abowd et al., 2022). A faster solver could reach further along that road. Other attacks are limited by information. A linkage attack against a cell of a few hundred loans compares a reading with a few hundred numbers, and no speed-up is needed. What stops it, when anything does, is that the release does not distinguish the

target from the other candidates. No amount of computation changes that.

Proposition 21.7 (Computation cannot separate what the release does not)**.** *Let the target be one of the candidates in a released file, and let Z be everything the adversary holds: the release and its own auxiliary information. Suppose that, given Z, the target is equally likely to be each of $r \geq 1$ candidates and has probability zero of being any other. Let $d(Z, U)$ be any procedure that names a candidate, where U is randomness independent of the target given Z. Then $\mathbb{P}\{d(Z, U) = \text{target}\} \leq 1/r$.*

The procedure d is unrestricted. It may run for any length of time, and it may be a quantum computation that starts from a fixed state, takes Z as classical input and is measured at the end, because the outcome of such a computation is a random function of Z with randomness that does not depend on which candidate is the target. The bound is the plain maximum of the posterior from Chapter 2, and it applies to every attack at once. Chapter 13 gave the same ceiling of $1/r$ for Pieter's likelihood attack when r candidates publish his target's value; the present result says that a cleverer attack, run on a stronger computer, could not do better. The condition is about information. If the adversary gains new information, Z changes and the ceiling may rise. A stronger computer gives no new information.

The same reasoning shows the limit of the proposition. When the posterior is not flat, when the release does separate the target, a classical attack that fails to find the most probable candidate leaves room that a stronger computer might fill. The audits in this book measure the attacks they implement, and a classical red-team exercise cannot certify all quantum attacks by giving them a larger budget label. A quantum attack study needs an implemented attack or a defensible resource analysis. Until one exists, the engine records quantum attack coverage on released data as unresolved, neither zero nor covered.

The chair read this section of Naledi's note twice. Her question at the committee was whether the Bank could say that its releases were safe against quantum computers. Naledi said the Bank could say three narrower things: that its released files were audited against the attacks it had declared, that for files where the target cannot be separated from others no computer could do better than chance among them, and that quantum attacks on releases were recorded as not yet studied. The committee minuted all three.

21.8 Post-quantum standards belong in the security programme

The memo's recommendation was not wrong about the first threat. The register, the channels that carry supervisory data to the Bank and the signatures on its records do rest on public-key systems that a large quantum computer could break. The answer to that threat is published. NIST has standardised ML-KEM for establishing keys (NIST, 2024a), and ML-DSA and SLH-DSA for digital signatures (NIST, 2024b,c). These are designed to resist quantum cryptanalysis under their stated assumptions. They do not guarantee security against every future discovery or against a faulty implementation.

The standards are components, and their use has conditions. ML-KEM establishes shared key material; it does not by itself encrypt bulk data, and it must sit inside an authenticated protocol with key derivation, authenticated encryption and key management. A signature shows that a record came from the holder of a key and has not changed since. It does not show that the counts in the record are true or that the operator followed the audit procedure. The trust and independence requirements on the audit registry, set out in Chapter 18, stay in force whatever signature scheme is used.

The engine follows a simple division of labour. It contains no cryptography of its own. Its decision records carry SHA-256 digests, which detect change but say nothing about who made a record; signing records, and protecting the register and its transport against quantum-capable attackers, belong to the institution's security programme, using maintained implementations of the three standards through supported protocols. The engine's documentation also forbids two tempting shortcuts. It forbids bespoke ciphers, including the chaotic ciphers that a reader of the earlier sections might be drawn to and that cryptographers have found weak in many published designs (Alvarez and Li, 2006), and it forbids deriving keys or randomness from model trajectories. A Lorenz trajectory is unpredictable in the sense of Section 21.3: small errors in its start grow fast. It is fully determined by that start, and an adversary who learns the start and the parameters can reproduce it exactly. Numerical unpredictability is not cryptographic entropy.

The security work the memo called for is therefore real, and it has a sequence: list the public-key dependencies, establish how long each kind of data must stay confidential and compare that with the time a migration will take (Mosca, 2018), identify stored ciphertext that must survive future cryptanalysis, choose maintained implementations through the security process, and test interoperability and downgrade behaviour. The chair sent the memo back to the information technology department with a request for that plan, and a note that encrypting released files was not part of it.

21.9 A guarantee that does not depend on the attacker's computer

There is one disclosure guarantee in this book that holds whatever computer the attacker uses. Chapter 10 set out differential privacy: a randomised mechanism M is (ε, δ)-differentially private when, for neighbouring datasets D and D' and every set S of outputs,

$$\mathbb{P}\{M(D) \in S\} \le e^{\varepsilon}\, \mathbb{P}\{M(D') \in S\} + \delta. \tag{21.6}$$

The guarantee is a property of the mechanism's output distributions, so it does not depend on the attacker's running time (Dwork and Roth, 2014). What it does not say directly is how it applies when the classical output is fed into a quantum computation. The extension supplies the step for one restricted setting.

Proposition 21.8 (Postprocessing by a data-independent quantum computation)**.** *Let M satisfy* (21.6)*. Let a later computation take the classical output y of M, start from a fixed state that does not depend on the dataset, and end in an event B with probability $h_B(y) \in [0,1]$. Then*

$$\mathbb{E}\, h_B\big(M(D)\big) \le e^{\varepsilon}\, \mathbb{E}\, h_B\big(M(D')\big) + \delta. \tag{21.7}$$

The left side is the probability that the whole procedure, mechanism followed by quantum computation, ends in B when the data are D, and the right side bounds it in terms of the same procedure on D'. So the classical guarantee carries over, with the same parameters, to anything such a computation does with the output. The proof applies (21.6) to the sets on which h_B exceeds each level and integrates over the levels.

The engine checks the inequality numerically. For randomised response with $\varepsilon = \log 3$, which reports a respondent's true bit with probability $3/4$, the output distributions on neighbouring datasets are $(0.75, 0.25)$ and $(0.25, 0.75)$. Figure 21.3 shows the two sides of (21.7) for three choices of h_B. When h_B is the indicator of the first output, the left side is 0.75 and the right side is $3 \times 0.25 = 0.75$, so the inequality holds with equality: this is the privacy

condition itself on its tightest set. For $h_B = (0.3, 0.9)$ the sides are 0.45 and 2.25, and for the constant $h_B = (0.6, 0.6)$ they are 0.60 and 1.80.

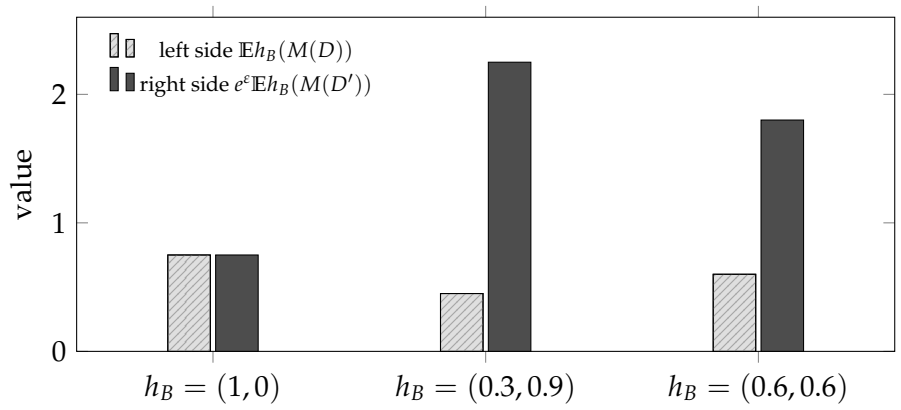


FIGURE 21.3: The two sides of the postprocessing inequality (21.7) for randomised response with $\varepsilon = \log 3$ and three later computations h_B. Left sides are light and hatched; right sides are dark and plain. The first pair is equal, the tight case; in the other two the bound holds with room. Synthetic check.

The proposition has firm limits, and they should travel with it. It assumes the quantum computation starts from a state that does not depend on the confidential data. If the attacker holds quantum side information correlated with the dataset, or can query the private mechanism coherently, the argument does not apply, and a guarantee for that setting needs its own definition and proof. It is a statement about a distributional inequality, not a claim that any system defeats a quantum computer. And everything Chapter 10 said about the meaning of ε still holds: the guarantee limits the change caused by one person's contribution, it does not hide population facts, and against an adversary who arrives with a strong prior it can permit a high posterior. Repeated releases need composition accounting, and any published model, selected setting or audit statistic is an output that must be assessed too.

Pieter, at his desk at the lender, has an ordinary laptop. The committee's decision on the loan-level file did not turn on what that laptop could compute. It turned on what the file would show him, and that would be the same on any machine he could buy or rent.

21.10 The answer

The chapter asked what has to change in the way the Bank decides when the adversary changes, and which tool answers each part. When the world the adversary observes drifts, the Bank should measure how sensitive its decision is: by finite-time growth rates checked first on a system with a known answer, by flip rates of the decision itself near the tolerance, and by a gate that requires every preregistered stress scenario to pass under a simultaneous error allocation. The diagnostic horizon prompts reassessment and does not replace it. When the adversary acquires a new kind of computer, two separate things change. The cryptography that protects the register and its records must move to the published post-quantum standards through the security programme, with no bespoke or chaotic ciphers. The disclosure risk of a release changes only where an attack was limited by computation; where the release cannot separate a target from its neighbours, no computation can, and where it can, the Bank must record quantum attacks as unstudied until a study exists. Differential privacy

is the one guarantee whose strength does not depend on the attacker's computer, and it carries over to data-independent quantum postprocessing. Quantum computers do not make disclosure control obsolete, and stronger encryption does not replace it, because disclosure control governs what the Bank chooses to show.

IN PRACTICE: CHANGING ADVERSARIES AND NEW COMPUTERS

Declare the stress scenarios before the audit, allocate the error budget across every candidate, attack and scenario, and treat a scenario with no bound as a failure. Report the flip rate of the decision for configurations within a few perturbation widths of the tolerance. Use growth rates and the diagnostic horizon to schedule reassessment, never as an expiry date. Keep two registers of threat: send attacks on cryptography to the security programme, with a migration plan built on FIPS 203, 204 and 205 and maintained implementations, and keep attacks on releases in the disclosure record, with quantum attack coverage marked as unresolved until a study exists. Where a release carries a differential privacy guarantee, state that it extends to data-independent quantum postprocessing and does not extend to correlated quantum side information.

In the software. Growth rates, the Lorenz benchmark, the diagnostic horizon, flip rates and the stress gate are in `privacy_risk.sensitivity` (`finite_time_growth`, `lorenz`, `diagnostic_horizon`, `decision_flip_rate`, `robust_gate`). The postprocessing check is `privacy_risk.dp.postprocessing_bound`. The tests are in `tests/test_ext_sensitivity.py`: the growth rate on a linear system and then Lorenz, the flip rate with the stress gate, and the postprocessing inequality for randomised response. The cryptography policy is in `SECURITY.md`. Algorithm 21.1 gives the flip rate, which is a direct count, and Algorithm 21.2 the stress gate; their checks replay Examples 21.4 and 21.6.

Algorithm 21.1. Decision flip rate.

Input. A decision rule d; base inputs $b = (b_1, \dots, b_m)$; perturbation standard deviations $s = (s_1, \dots, s_m)$; a number of draws N; a seeded random source.

Output. The reference decision $d(b)$, the number of flips, N, and the flip rate.

1. Refuse the inputs unless s has one entry for each input, every $s_i \geq 0$, and $N \geq 1$.
2. Compute the reference decision $d(b)$.
3. For each of N draws, draw independent standard normal $Z_1, \dots, Z_m$, compute $d(b_1 + s_1 Z_1, \dots, b_m + s_m Z_m)$, and count a flip if it differs from $d(b)$.
4. Return the number of flips F and the flip rate F/N. For one estimate and the rule $d(z) = \mathbf{1}\{z > \tau\}$, the rate estimates $\Phi(-|\tau - b|/s)$ of Proposition 21.3.

Checks (seed 2026, the repository's random stream; the draft's counts of 293 and 624 came from another):

- An upper bound of 0.110 against a tolerance of 0.12, standard deviation 0.01, 2,000 draws (Example 21.4), number of flips: 290.
- The same with standard deviation 0.02: 595.
- The same, flip rate: 0.298 (to three places, 0.2975).
- A standard deviation of zero, flip rate: 0.000.

- A negative standard deviation: refused.
- Two standard deviations for one input: refused.

Algorithm 21.2. The stress gate over a preregistered set.

Input. The required attacks a, each with a tolerance τ_a; for each attack, the upper bounds $R_{a,s}$ under the preregistered scenarios $s \in \mathcal{S}$, computed with an error allocation that covers every candidate, attack and scenario.

Output. Pass or fail, with each failure named by its attack and its worst scenario, or marked missing.

1. For each required attack a: if it has no bound under any scenario, record it as missing.
2. Otherwise find the scenario s with the largest bound; if $R_{a,s} > \tau_a$, record a as failed under s, with that bound.
3. Pass if and only if nothing was recorded, which is (21.5). The pass says nothing about scenarios outside $\mathcal{S}$ (Proposition 21.5).

The gate checks no input beyond the presence of a tolerance, so its only refusal is an empty list of required attacks.

Checks:

- The external attack at 0.12 with base, precise-reading and merger scenarios at 0.104, 0.112 and 0.131 (Example 21.6), scenario named in the failure: merger.
- The same without the merger, number of failures: 0.
- The same with the insider also required and no insider audit, reason for the failure: missing.

21.11 Proofs

Proof of Proposition 21.1. For a linear system the Jacobian is A at every state, so the tangent equation (21.1) is $\dot{v} = Av$ with solution $v(T) = e^{AT}v_0$, whatever z_0. For part 1, $Av_0 = \mu v_0$ gives $v(T) = e^{\mu T}v_0$, so $\|v(T)\|/\|v_0\| = e^{\mu T}$ and $\lambda_T = \mu$.

For part 2, $v(T) = \sum_i a_i e^{\mu_i T} w_i = e^{\mu_1 T} q(T)$, where $q(T) = a_1 w_1 + \sum_{i \geq 2} a_i e^{(\mu_i - \mu_1)T} w_i$. Since $\mu_i - \mu_1 < 0$ for $i \geq 2$, $q(T) \to a_1 w_1 \neq 0$, so for T large $\|q(T)\|$ lies between two positive constants m and m'. Then $\lambda_T = \mu_1 + \big(\log\|q(T)\| - \log\|v_0\|\big)/T$, and the second term is at most C/T in absolute value, with $C = \max(|\log m|, |\log m'|) + |\log\|v_0\||$. Hence $\lambda_T = \mu_1 + O(1/T)$. □

Proof of Proposition 21.3. If $b < \tau$ then $d(b) = 0$, and a flip occurs when $b + sZ > \tau$, that is when $Z > (\tau - b)/s$, an event of probability $1 - \Phi((\tau - b)/s) = \Phi(-(\tau - b)/s)$ by the symmetry of the normal law. If $b > \tau$ then $d(b) = 1$, and a flip occurs when $b + sZ \leq \tau$, that is when $Z \leq -(b - \tau)/s$, an event of probability $\Phi(-(b - \tau)/s)$. In both cases the probability is $\Phi(-|\tau - b|/s)$. □

Proof of Proposition 21.5. Let E be the event that $r_{a,s}(\theta) \leq R_{a,s}(\theta)$ for every candidate θ, required attack a and scenario $s \in \mathcal{S}$. By assumption $\mathbb{P}(E) \geq 1 - \alpha$. On E, take any θ that

passes (21.5). For each required a and each $s \in \mathcal{S}$, $r_{a,s}(\theta) \le R_{a,s}(\theta) \le \max_{s'} R_{a,s'}(\theta) \le \tau_a$. Because E covers all candidates at once, the conclusion holds for the setting chosen, even when the choice depends on the bounds. □

Proof of Proposition 21.7. Write T for the index of the target. By assumption $\mathbb{P}(T = j \mid Z) \le 1/r$ for every candidate j. Because U is independent of T given Z,

$$\mathbb{P}\{d(Z,U) = T\} = \mathbb{E}\Big[\sum_j \mathbb{P}\{d(Z,U) = j \mid Z\}\mathbb{P}(T = j \mid Z)\Big] \le \mathbb{E}\Big[\frac{1}{r}\sum_j \mathbb{P}\{d(Z,U) = j \mid Z\}\Big] \le \frac{1}{r},$$

since the probabilities of naming each candidate sum to at most one. For a quantum procedure that starts from a fixed state, applies operations chosen from Z and measures, the probability of each outcome is a function of Z alone, so the procedure is of the form $d(Z,U)$ with U the measurement randomness, independent of T given Z. □

Proof of Proposition 21.8. For a classical input y, the computation starts from a fixed state ρ_0, applies a quantum operation Φ_y chosen by y, and measures; with E_B the measurement operator for the event B, the probability of B is $h_B(y) = \operatorname{tr}\big(E_B \Phi_y(\rho_0)\big)$, which lies in $[0,1]$. Because ρ_0 does not depend on the dataset, h_B is the same function under D and D', and the probability of B is $\mathbb{E}[h_B(M(D))]$ and $\mathbb{E}[h_B(M(D'))]$ respectively. For any random variable X with values in $[0,1]$, $\mathbb{E}X = \int_0^1 \mathbb{P}\{X > t\}\,dt$. Apply this to $X = h_B(M(D))$ and use (21.6) on the sets $S_t = \{y : h_B(y) > t\}$, assumed measurable:

$$\mathbb{E}h_B(M(D)) = \int_0^1 \mathbb{P}\{M(D) \in S_t\}\,dt \le \int_0^1 \big(e^{\varepsilon}\mathbb{P}\{M(D') \in S_t\} + \delta\big)dt = e^{\varepsilon}\,\mathbb{E}h_B(M(D')) + \delta.$$

If ρ_0 depended on the dataset, h_B would differ between D and D' and the first equality would fail for one of them; this is why correlated side information lies outside the result. □

Chapter 22

Learning to plan audits

Planned chapter, not yet drafted.

The question this chapter will answer is what a learned planner may do in the audit process and what it may not. The 304-page draft trains a planner on counts, separates four authorities, shows why a credible interval is not an audit certificate, lets a Bayesian forecast set the bets of a betting audit so that it changes the cost and not the error, and ends with what can be claimed today and a claim–evidence ledger. Chapter 18 already states the gate and the budget the planner works inside. The chapter is not yet rebuilt in this copy.

Part VIII

Conclusion

Chapter 23

What remains open

A reader who has followed the book this far may hold a belief that the earlier chapters have encouraged. The method has propositions with full proofs, a pipeline that regenerates every number from one configuration, falsification tests that can refuse a decision, a protocol with roles and a filled record. On this view the scientific question is settled, and what remains is adoption: an institution has to install the software, train its staff and fill in the worksheets. The view is understandable, because the parts of the method fit together and each part has been checked.

The view fails on one table. Every floor this book reports for the credit register depends on how precisely the adversary can read a loan's exposure, and no institution has measured that precision. When the standard deviation of the reading on the log scale is 0.6, the programme floor against the full outside adversary has a median of 150 over twenty draws of the synthetic register. When it is 0.3, the median is 300, and in three of the twenty draws no floor up to 300 passes at all. When it is 1, the median is 75. The decision rule is settled. The number it produces for a real register is not, because the inputs that decide it are declarations about the world outside the institution, and the evidence for them has not yet been gathered on any real register.

This chapter asks which of the book's claims would change if the method were run on a real register by a real institution, and which claims the method cannot make at all. It closes by answering the question that Chapter 1 put at the start of the book.

23.1 Real institutional registers

The credit register of Chapter 13 is synthetic. It shares the structure of a supervisory collection such as AnaCredit (Israël et al., 2017; ECB, 2016): firms as borrowers, lenders with unequal shares, sector, region and size class as the key, monthly vintages. Its parameters were chosen, not estimated, and three of the choices bear directly on the floors. Exposures do not change between vintages. Whether a loan is live in a vintage is drawn independently of its size and its key. And the adversary's reading error is a declared normal error on the log scale. The register was rebuilt once after referees found that its loan panel gave the history adversary a signal no real register has: every loan had been forced to be live in the first month, and a third vanished and returned in a fixed pattern. The generator now has originations during the panel, a monthly exit hazard of 0.04 with no return, and a stable stock of about 27,000 loans. That correction moved the results, which is a measure of how much they depend on the generator.

The repository states what an institution would have to supply to replace the synthetic evidence with its own. An adapter, written inside the institution and outside the package, would emit only aggregates: de-identified key-class counts, held-out predicted match probabilities with their outcomes for the calibration test, external frequency benchmarks for the transport test, and aligned attack-success indicators for every release. No direct identifier, auxiliary register or row-level confidential data would leave the secure environment. The demographic path of the code has been run this way on two public files (Chapter 12). The credit path has not been run on any real register.

Three replications would tell an institution most of what the synthetic study cannot. The first is the reading precision: the institution has to argue, from what it knows of the sources outsiders hold, for the error with which an outsider reads exposure, and it should report the floor across a range of precisions, as the replication table of the long form does, and not at one value. The second is the composition gap on its own programme: the ratio of the pooled risk to the bound the marginals imply, which on the simulated population ranged from 1.57 to 2.05 across register sizes from 50,000 to 400,000 units and reached 1.86 on the public Census extract. The third is what a floor keeps of the credit volume. On the synthetic register the floor of 150 keeps 20 per cent of loans and 4 per cent of exposure; on a register with more lender concentration and regional specialisation, both of which the referees asked for and the generator does not yet have, the share could be very different.

The book's prose has to carry this limit wherever the credit floors appear, and the chapters that report them say so. The floors of 75, 150 and none are properties of a synthetic register under declared parameters. The decision rule that produced them is the transferable result.

At the Bank, the chair asked Naledi in the first committee meeting of the year which of the book's numbers she would put in front of the board. Her answer was none of the floors and all of the method. The floors described a register the Bank did not hold. What she could present was the record for Tomás's request, filled with the Bank's own counts inside the secure environment, and the range of floors it gave across the reading precisions the reviewer had accepted as plausible.

23.2 Time-varying exposures and dependent loan activity

Proposition 9.3 explains a result that surprised the referees: an adversary who pools six vintages without knowing the target's history gains nothing over the first vintage, and the floor stays at 1. The proposition rests on two conditions that a real register will not meet exactly. Each loan's published key and exposure are the same in every vintage in which it appears, and whether a loan is live in a vintage is independent of its key, its exposure and the adversary's information.

Both conditions fail in ordinary credit data. Exposures amortise, are drawn down and are restructured, so a loan's published exposure differs from month to month. An adversary who holds a reading of the target in each vintage then compares each published value with a different reading, and the later vintages are no longer copies of the first. Loan activity also depends on size: large loans to large firms may persist longer, and small credit lines may open and close more often. When activity depends on the key or the exposure, the pattern of appearances across vintages carries information about which candidate is the target, even for an adversary who does not know the target's own pattern. In both cases the pooled adversary without history would gain something, and the zero gain of the synthetic register

is a lower limit.

The referees listed amortising exposure among the open items, together with the programme analysis of monthly aggregate tables and the harm that a lender-insider causes to other lenders' relationships. None of these needs new theory. Each needs a generator, or better a real register, with the feature in it, and a rerun of the joint attack. The proposition says what to look for: the gain from pooling without history is exactly the information that changing values and dependent activity add, and on a register without them it is zero.

23.3 Attack coverage

Every certificate in this book covers the attacks that were declared. The admissible release is one whose declared attacks succeed at a rate below the tolerance with stated confidence, and the method says nothing about attacks outside the declared set. Two documented cases show what an undeclared attack looks like. Narayanan and Shmatikov (2008) showed that subscribers in a film-rating data set released for a prediction contest could be identified from a handful of ratings with approximate dates, using a public rating site as the auxiliary source; the release had been reviewed without an adversary who held such a source. The United States Census Bureau's experiments on the tables of the 2010 census showed that published tables could be inverted to reconstruct records (Garfinkel et al., 2019), and the Bureau adopted a formal privacy mechanism for 2020 (Abowd et al., 2022). In each case the attack was not new in principle. It was new to the declaration.

The declared attack can also be weaker than the declared knowledge allows. The referees found that the history adversary, who knows when the target's loan is live, also knows when it is absent, and that the attack first used only half of that knowledge. Using both halves raised the risk at the old floor of 40 from 0.105 to 0.179. The attack now requires a candidate's pattern of appearances to match the target's in both directions, but it does so through a large penalty in the score and not through exact exclusion, and whether the maximum-posterior rule is the best attack for each declared information set has not been proved. Simulation can show what a declared rule achieves. It cannot deliver the supremum over rules that a formal guarantee delivers by construction.

The insider is the clearest case of what a floor cannot do, and Proposition 23.1 states it without simulation.

Proposition 23.1 (An exact insider is not governed by a floor)**.** *Suppose a release publishes, for each record it keeps, a continuous attribute whose values are distinct across records with probability one, and that the protection mechanism removes records, by any rule including a minimum cell count, without altering the published values of the records it keeps. An adversary who knows the target's published value exactly identifies the target with probability one whenever the target is released. Its membership-conditional risk is therefore one at every setting of the mechanism, and its unconditional risk equals the probability that the target is released.*

The proposition explains the insider row of the decision record. An insider who reads exposure with a log error of 0.01 is close to exact, and on the synthetic register its success falls only from 0.772 with no suppression to 0.416 at a floor of 300. What lowers it is the removal of records, not their protection. Against such an adversary the levers are the access route, the precision of what is published, or a formal mechanism. Rounding exposure to one significant figure halves the insider's success on the synthetic register.

The public-file experiments add a measured view of coverage. On thirty state samples of the Census extract, a search guided by a neural network confirmed 31.57 of 32 high-risk settings on average, and a simple domain heuristic confirmed all 32. A quasi-random Sobol search confirmed only 22.03, but it covered 13.83 of 24 scenario cells, against 10.07 for the neural search and 11.97 for the heuristic. A search that targets what it already believes is risky finds risky settings and covers less of the space. The cells are fixed bins of declared configuration choices, not new attack families, so even the widest search here explores a space the analyst wrote down. No method in the repository proposes an attack outside the declared space and then measures it.

When Grace asked what the Bank's records could not show, Naledi's list began with attacks. The record showed what each declared adversary achieved and how confident the Bank could be about it. It could not show that the declared adversaries were the ones that would come, and she wrote that sentence into the record's limits section.

23.4 Eliciting the harm weight

Chapter 8 showed that a stated weight on harm relative to lost utility implies a tolerance, through the point of the risk–utility frontier that maximises utility minus the weighted risk bound. The result is useful only if a committee can state the weight, and no committee currently does. The repository reports the implied tolerance at six weights on the simulated population: 0.464 at a weight of 0.25, 0.345 at 0.5, 0.145 at 1 and at 2, and 0.081 at 5 and at 10.

The figures contain a practical point. Because the frontier is a finite set of settings, the implied tolerance is a step function of the weight: every weight between 1 and 2 implies the same setting, and so does every weight between 5 and 10. A committee does not need to state its weight to two decimal places. It needs to place the weight in an interval wide enough to contain the steps, and it can do so by choosing between pairs of settings shown with what each keeps and what each risks. Whether committees can make such choices consistently, and whether their past release decisions identify a weight, are empirical questions. The long form proposes estimating the weight from revealed institutional behaviour, since agencies have made these decisions for decades and their past choices constrain it under the framework. That study has not been done.

The harm weight also hides a scope limit. The framework represents harm by a scalar within a declared harm class. Harms that fall on groups and not on individuals, and harms that arise from correct inference about a population and not from the identification of a person, are outside it (Skinner, 2012; Dwork and Roth, 2014).

23.5 The model-proposed attack line

Chapter 14 reported a pipeline in which a model proposes scenarios and bets, a frozen plan records them, and an independent audit decides. Its results are mostly negative, and the remaining questions are best stated with those results beside them. On the synthetic benchmark, trained to convergence over ten seeds, the logistic and deep proposal models reach held-out losses that cannot be told apart. On the public Census file, one- and three-layer networks improved on plain logistic regression in three overlapping state splits, and a logistic model with explicit pairwise products and squares did better than both. The evidence supports a role for nonlinearity, not for depth.

The language-model arm is where the gap between what is built and what is measured is widest. A hosted pilot was prepared, with prompts saved and hashed before collection, and its first request returned an error with no proposal; collection stopped without retrying, and no hosted model response exists. A second run used a small open-weight model with four billion parameters on the author's own machine. It produced a valid list of proposals for 20 of 30 states within three attempts, and 50 of its 70 attempts were rejected, most for repeated or off-catalogue proposals. On those 20 states its proposals covered 6.05 scenario cells on average, against 13.65 for the Sobol search, and confirmed 27.60 of 32 settings. In 19 of the 20 states every proposal used the finest geography, in the catalogue's own order. The model copied the catalogue. Its run settles that this small model, under this prompt, adds no coverage over quasi-random or neural search. It does not settle what a larger model, a prompt that names the coverage goal, or a worked example would do, and any of those is a new experiment to be declared before it is run.

There is therefore no genuine comparison, at a fixed evaluation budget, of a capable language model against the strongest baseline that uses no language model, which is the primary contrast of the confirmatory protocol in the repository. The infrastructure for it exists: a replay contract that binds each response to the hash of the exact prompt, a schema that refuses counts and code, and a ledger that records the order of proposal, review, measurement, training, freeze and audit. Until the comparison is run, the claim the line can support is architectural. A model can propose scenarios into an audit without being able to change its error statement, and a poor model costs attacks, not validity.

23.6 Claims the method cannot make

Some limits are not gaps that more work will close. They follow from what the method is, and a release record should state them.

The method cannot establish legal compliance. Data protection law asks whether identification is reasonably likely by the means an adversary could use, and gives no number (European Parliament and Council, 2016). The statistical confidentiality duty of the European System of Central Banks likewise names no probability (Council of the European Union, 1998). A tolerance is the institution's written reading of that phrase for a declared adversary and harm. The method makes the reading explicit, owned and dated, and leaves to lawyers the question whether it is a reasonable reading.

The method cannot protect against adversaries who were not declared. Its certificate covers registered attacks. An adversary with better auxiliary information, a new linkage method or a different notion of success lies outside the claim, and Proposition 23.1 shows that one such adversary, an insider who knows a published value exactly, defeats every floor. The falsification tests and the expiry date narrow this gap; they do not close it. The method gives no differential privacy guarantee either, and where a guarantee over all attacks is required, a formal mechanism with an accountant is the tool (Chapter 10).

The method cannot enforce its own controls. The implementation is a library and not a security boundary, the registry service that a deployment needs is specified and not built, and the hashes a run writes detect change but do not prove who made it. Separation of duties, external time stamps and audit inside the secure environment are institutional controls, and the method depends on them.

The method also cannot certify what it has not shown in the open. The papers describe the pipeline as open, and a reader's ability to rerun a number is part of the evidence for it. At

the time of writing the repository is private, and either it is made public or that description changes before the papers or this book circulate.

23.7 Open statistical questions

Four extensions look tractable, and the long form names them. The selected point of the risk–utility frontier is chosen by maximising an estimate, so it is biased in the way any such choice is, and the frontier needs a proper inferential treatment. The risk budget, which allocates a programme tolerance across releases, controls only the union event; a joint accountant that admits dependence between releases would need a composition theorem for scenario-conditional risk under a stated class of dependence. The implied tolerance could be estimated from revealed behaviour, as above. And the calibration test of Chapter 15 could be developed into a formal specification test with a distribution under the null, which would replace the analyst's band with a level. The sequential audit has open items of its own: a truncated Beta mixture in place of the capped bet, exact group-sequential boundaries, and a bound on the expected length of the rule actually used.

23.8 How an institution should turn a measured risk into a release decision

Chapter 1 began with a gap. Institutions have many ways to protect a release and few rules that turn a measured risk into a decision to release or to withhold, a rule that someone else could apply to the same evidence and reach the same answer. The book's answer can now be stated plainly.

An institution should first say what it is measuring. Disclosure risk is the probability that a declared adversary, holding declared information, names the right record for a declared set of targets. A figure without that declaration has no referent, and the familiar measures are answers to different declarations (Chapters 2 and 3).

It should then treat the figure as an estimate. The decision compares an exact upper confidence bound with the tolerance, not a point estimate, because a point estimate at the tolerance passes about half the time. The number of simulated attacks is fixed in advance from the tolerance and the error level, a protection setting is chosen by testing a fixed sequence from the most protective down and confirmed on fresh targets, and an audit that is watched as it runs uses a sequence that stays valid under watching (Chapters 4 to 6).

The tolerance has to belong to someone. It is written before the risk is measured, with a named owner, an expiry date, a harm class, an error level, the legal basis and the parties it protects, the access route and, for a programme, a horizon. It can be stated directly or read off the risk–utility frontier from a stated weight on harm (Chapters 7 and 8).

The object assessed is the programme, not the release. When records keep a pseudonym across releases, no function of the per-release risks bounds an adversary who pools them, so the joint attack is simulated directly and compared with a programme tolerance over the declared horizon (Chapter 9).

The model behind the figure is tested before the figure is used, against bands fixed in advance, and a failed test makes the assessment invalid. The figure is not adjusted (Chapter 15). The whole sequence is carried out in a fixed order by separate people, recorded in

a registry that holds every audit, and filed as a decision record with a row for every harm, including those for which no tolerance has been set (Chapter 16).

The release goes out when the upper bound for every declared scenario, pooled ones included, lies at or below its tolerance, the model has passed its tests, and the owner has signed. Otherwise the record names the remedy, and the changed product is assessed again.

That rule is reproducible: two analysts with the same declarations and the same simulation output reach the same decision, and an auditor can check afterwards that the decision followed from the evidence and that the evidence was gathered in the stated order. The rule does not remove judgement. It places the judgement in the declarations and the tolerance, where it can be seen, disagreed with and revised, and it leaves the probability calculation open to being proved wrong. What remains open, as this chapter has set out, is the evidence from real registers that would show how large the numbers are, and the declarations that no method can make on an institution's behalf.

IN PRACTICE: STATING THE LIMITS IN THE RECORD

Add a limits section to every decision record. State that the certificate covers the declared attacks only, name any adversary that was considered and not declared, and say why. Report floors across a range of the parameters that decide them, the adversary's reading precision above all, and not at one value. State whether exposures change between vintages and whether loan activity depends on size, since the no-history result needs both to fail. Name any insider separately and govern it through access. Record that the tolerance is the institution's reading of its legal duty, not a finding of compliance.

In the software. The floors across reading precisions and register draws come from `scripts/programme_robustness.py` and `scripts/robustness_table.py`. The public-file search and the language-model arm are `privacy_risk.injector_search`, `privacy_risk.llm_pilot_local` and `scripts/public_injector_search.py`. The adapter for a real register is not in the package; `privacy_risk.realdata` is the pattern for writing one.

23.9 Proofs

Proof of Proposition 23.1. Let v_j be the published value of the continuous attribute for each released record j, and let t be a released target with value v_t. The mechanism keeps v_t unaltered, so the adversary, who knows v_t exactly, can form the set $\{j : v_j = v_t\}$ of released records sharing it. With probability one the values are distinct across records, so the set is $\{t\}$, and naming its only member identifies the target. The membership-conditional risk is therefore one for every rule that removes records, at every setting. The unconditional risk is the product of the probability that the target is released and the membership-conditional risk, by the factorisation of Chapter 2, and so equals the probability of release. □

Appendix A

The software

Every number in this book comes from one Python package, `privacy_risk`, in the repository that accompanies it, and from the scripts beside it. This appendix is a reader's map of that code. It says what each module computes, names its main functions, gives the chapters that use it and the test file that checks it, and then explains how to reproduce the evidence. It ends with the gaps a review of the code found, which the book reports and does not repair, since the book branch does not change code.

The map is written for two readers. A release committee member who wants to know where a figure in a decision record came from can follow a module name to a chapter and back. A researcher who wants to change an assumption and rerun a result can follow a chapter to the function that computes it and the test that pins it down.

A.1 Layout and installation

The repository has five parts that matter to a reader of this book: the package in `src/privacy_risk/`, the scripts in `scripts/`, the tests in `tests/`, the configurations in `configs/` and the recorded runs in the directories whose names begin with `artifacts-`. The manuscripts in `paper/` read their tables from `paper/generated/`, which a script copies from a recorded run, so a table in a paper cannot disagree with the run behind it once that step has been taken.

The package needs Python 3.10 to 3.12 and installs with its development extras from the repository root. It depends on NumPy, pandas, SciPy and Matplotlib, and on version 0.5.0 of the `wooldridge` package, which supplies the two public microdata files of Chapter 12. No confidential data and no network access are needed, and no module reads real confidential data. The installation defines four commands: `privacy-risk-run` runs the paper's experiments and the governance gate, `privacy-risk-neural` runs the neural release pipeline, `privacy-risk-neural-demo` runs a small synthetic example of the proposal-and-audit method, and `sentry-r` drives the proposal model of Chapter 14.

A.2 The modules

The package has 27 modules besides its initialiser. They fall into three groups, shown in Tables A.1, A.2 and A.3: the core of the method, which implements the definitions and propositions of Parts I to IV; the case studies and the run machinery, which produce the

evidence of Part V; and the model-proposed attack line with its audit, which Chapter 14 reports.

The core of the method

Table A.1 lists the eleven modules that implement the method itself. The first four build the objects of Chapter 2. `synthetic` generates the simulated register of 200,000 units, the sampling design and the protection mechanisms, and `keys` computes the sample and population key frequencies. `linkage` holds the posterior linkage rule and the Monte Carlo attack of the declared scenario, with the adversary's error model as a small class whose `scaled` and `misspecified` methods produce the stronger and the mistaken adversaries of Chapter 15. `risk` computes the prosecutor, journalist and marketer measures of Chapter 3 and fits the negative binomial model for population cell sizes.

The next three implement the evidence of Part II. `uncertainty` gives the exact one-sided Clopper–Pearson bounds, the number of attacks needed to certify a tolerance with no success, the selection-corrected bound on the worst target and the split of variance between simulation and design. `sequential` gives the beta–binomial e-process and the confidence sequence it induces, the posterior threshold at which a paid adversary acts, and the transport bound used for the temporal test. `governance` holds the threshold record, the rule `decide`, the implied tolerance and the programme risk budget, and `frontier` finds the non-dominated settings and fits the value curve from which a harm weight implies a tolerance.

The last three serve Part IV and Part VI. `composition` computes the sharp bounds that marginal risks imply and runs the pooled intersection attack of Chapter 9; `dp` states what a privacy budget permits as a bound on posterior movement; and `validation` holds the falsification tests of Chapter 15.

Case studies and the run

Table A.2 lists the six modules that produce the evidence of Part V and assemble a run. `credit` builds the synthetic credit register of 20,000 firms and 60 lenders over six monthly vintages and measures its five harms: borrower identification, the predictive value of an alleged lending relationship, attribute disclosure of arrears status, reconstruction of a lender's position from published totals, and differencing across vintages. `evaluation` attacks the same register as a security evaluation would: seven protection policies against several attacks, scored by precision, recall, worst subgroup and calibration under a wrong error model, with the search for the floor a tolerance implies, per release and pooled, and the share of loans a rotated pseudonym still allows to be relinked. `realdata` runs the demographic path on the two public files.

`experiments` calls the others and returns each number the long form reports, keeping four kinds of work apart: proof checks, simulation checks, falsification tests and the gate. `pipeline` runs the experiments, applies the falsification tests and the gate, and writes the evidence; it is the entry point of `privacy-risk-run`. `reporting` draws the figures, writes the LaTeX table fragments and the file of numbers, and writes the manifest of SHA-256 hashes.

Model-proposed attacks and the registered audit

Table A.3 lists the ten modules of the line that Chapter 14 reports and Chapter 23 assesses. `adaptive_audit` is the registered sequential gate of the companion journal article: coordi-

TABLE A.1: Modules that implement the method. Chapters are those that report results from the module; the test file is in `tests/`.

Module	What it computes	Main functions	Chapters	Test file
`synthetic`	Simulated register, sampling design, mechanisms, small-cell suppression	`make_register`, `apply_mechanism`, `suppress_small_cells`	2, 11	`test_synthetic`
`keys`	Sample and population key frequencies	`key_frequencies`, `sample_population_frequencies`	2	`test_keys`
`linkage`	Posterior linkage and the Monte Carlo attack; error models	`posterior_linkage`, `attack`, `per_target_risk`, `ErrorModel`	2, 11, 15	`test_linkage`
`risk`	Prosecutor, journalist and marketer measures; frequency model	`prosecutor_risk`, `fit_frequency_model`, `journalist_from_fit`	3, 15	`test_risk`
`uncertainty`	Exact bounds, certification cost, selection-corrected worst target, nested variance	`binomial_upper`, `certification_replicates`, `selected_maximum_bound`, `nested_variance`	4, 5	`test_uncertainty`
`sequential`	E-process, confidence sequence, adversary's acting threshold, transport bound	`anytime_upper`, `attack_threshold`, `transport_bound_kl`, `monitoring_study`	6, 15	`test_sequential`
`governance`	Threshold record, decision rule, implied tolerance, risk budget	`GovernanceThreshold`, `decide`, `implied_tolerance`, `risk_budget`	7, 8, 16	`test_governance`
`frontier`	Nondominated settings, fitted value curve	`nondominated`, `choose_max_utility`, `fit_value_curve`	8	`test_frontier`
`composition`	Marginal bounds and the pooled intersection attack	`marginal_bounds`, `independent_union`, `pooled_attack`, `pooled_attack_from_sets`	9	`test_composition`
`dp`	Posterior bound a privacy budget permits; basic composition	`posterior_bound`, `basic_composition`	10	`test_dp`
`validation`	The falsification tests	`calibration_table`, `stress_ratios`, `composition_test`, `full_validation_report`	15	`test_validation`

nates with their own error shares, release chains that audit floors from the strictest down, the zero-success budget, and the controls on the audit process that Chapter 16 describes, with targets derived from a committed secret and a beacon and bets recorded in a hash-chained ledger before each outcome. `neural_pipeline` is the package's main entry point for the proposal line: a network fitted to development attacks proposes the bets and starting floor, a plan is frozen with a digest, and the release chain decides on fresh targets. `neural_trust` is an earlier proposal-and-audit procedure that supplies no input to the book's tables. `sentry_r` holds the scenario schema, the intake of language-model proposals, the review and evidence join, the deep and logistic models, the shortlist freeze and the final audit. `provenance` is the keyed, hash-chained ledger that records every stage of that line in order and debits each audit against a programme error budget. `multiscale` adds a feature dictionary, exact sensitivities of the network and an audit bound under a declared distribution shift.

The remaining four are experiments around the line. `injector_search` runs the budget-

TABLE A.2: Modules that produce the case-study evidence and assemble a run.

Module	What it computes	Main functions	Chapters	Test file
`credit`	Synthetic credit register and its five harms	`make_credit_register`, `borrower_identification`, `relationship_disclosure`, `reconstruction_from_totals`, `differencing_risk`	13, 16	`test_credit`
`evaluation`	Seven policies against the credit attacks; floor search; pooled attack; relinking	`evaluate`, `identification_attack`, `programme_attack`, `relink_share`	13, 9, 16	`test_evaluation`
`realdata`	Census 2000 extract and mortgage file as releases; pooled attack on real keys	`load_census2000`, `build_release`, `attack`, `composition`	12	`test_realdata`
`experiments`	Every experiment of the long form	`run_all`, `programme`, `programme_scaling`, `credit_register`	all of Part V	`test_experiments`
`pipeline`	Falsification, gate and evidence for one run	`validate`, `gate`, `run`	15, 16	`test_pipeline`
`reporting`	Figures, table fragments, numbers, SHA-256 manifest	`figures`, `paper_tables`, `paper_numbers`, `write_evidence`	this appendix	`test_reporting`

matched search on public Census states and confirms the selected settings on fresh draws. `llm_pilot` collects proposals from a hosted model under a fixed cost cap and never invents a response; `llm_pilot_local` does the same through a local open-weight model. `redteam` turns penetration-test findings into declared adversary capabilities and audits the release against the stronger view they grant.

Three test files check code outside the package: `test_linkage_injector` checks the executable linkage experiment in `scripts/`, and `test_colab_setup` checks the helper that clones and synchronises the repository from a notebook. The initialiser itself only sets a version string.

A.3 The scripts

The package computes; the scripts run studies, write tables and check manuscripts. Table A.4 groups the 33 scripts in `scripts/` by what they do, leaving aside the superseded scripts in `scripts/archive`. The first group writes the evidence the book reports, and each of its scripts names its output file at the top of its source. The second group checks that the manuscripts agree with that evidence. The third covers the public-file and model-proposed attack studies, and the fourth is housekeeping.

A.4 Reproducing the evidence

The evidence for Parts II to V comes from one configuration, `configs/paper.json`, which fixes a seed, a register of 200,000 units with a 5 per cent sample, 45 mechanism settings, the credit register, and the bands of the falsification tests. A run with it takes a few minutes on one core. A smaller configuration, `configs/base.json`, finishes in seconds and is what

TABLE A.3: Modules of the model-proposed attack line and the registered audit.

Module	What it computes	Main functions	Chapters	Test file
adaptive_audit	Registered sequential gate; release chains; committed targets; ledger	AdaptiveAudit, register_release_chain, zero_success_budget, target_seed, AuditLedger	6, 16, 14	test_adaptive_audit
neural_pipeline	Network proposes bets and start floor; frozen plan; release chain	run, fit, release_chain	14	test_neural_pipeline
neural_trust	Earlier proposal and simultaneous audit	certify, propose_and_audit, zero_success_budget	14	test_neural_trust
sentry_r	Schema, language-model intake, review, models, freeze, final audit	stage_llm, review_proposals, DeepRiskNet, shortlist, final_audit	14	test_sentry_r, test_injector_integrity
provenance	Keyed hash-chained ledger of every stage; error budget	ProvenanceLedger, record_audit, verify	14, 16	test_provenance, test_injector_integrity
multiscale	Feature dictionary, network sensitivities, shifted audit	MultiscaleFeatures, shifted_audit, zero_success_plan	14	test_multiscale
injector_search	Budget-matched search on public states; fresh confirmation	select_and_measure, confirm, paired_bootstrap	14, 23	test_injector_search
llm_pilot	Hosted-model proposal collection with a cost cap	collect, request_payload	23	test_llm_pilot
llm_pilot_local	Local open-weight proposal collection	collect, response_schema	23	test_llm_pilot_local
redteam	Penetration-test findings as adversary capabilities; audit	adversary_view, PenTestPlan, run	16	test_redteam

continuous integration runs. The sequence below installs the package, reruns the paper's configuration, publishes the tables, checks the manuscript against the run and runs the test suite.

```
python -m pip install -e ".[dev]"
privacy-risk-run --config configs/paper.json --output artifacts-paper
python scripts/rebuild_evidence.py --artifacts artifacts-paper \
    --publish paper/generated
python scripts/check_paper_numbers.py --artifacts artifacts-paper \
    --paper paper/bis_ifc_full.tex
python scripts/programme_robustness.py
python scripts/robustness_table.py
python scripts/audit_process_attacks.py
pytest
```

A run writes six things into its output directory: the configuration copied verbatim, `results.json` with every experiment's output, `decision.json` with the governance record, the figures, the LaTeX table fragments with a file of named numbers, and `manifest.json`, which holds the SHA-256 hash of every other file. Figures are written without creation dates and the results are strict JSON, so two runs of one configuration produce identical files and an identical manifest. `rebuild_evidence` rewrites the presentation from a stored run with-

out simulating again, which is the step to use when only a table's layout changes.

The manifest proves less than it may appear to. A hash shows that a file has changed since the run that wrote the manifest. It carries no key and no trusted time stamp, so anyone who can rewrite the files can rewrite the manifest as well; for evidence against the operator the manifest's own hash has to be published or time-stamped outside the operator's control. The script `write_manifest` says this in its documentation.

The test suite has 195 tests in 30 files. On 1 November 2026, run in the Linux environment used to build this book, 194 passed and one failed. The failure, `test_shipped_manifest_matches_the_files_it_lists`, is an artefact of line endings: the repository now asks Git to store JSON with Unix line endings, but the Windows working copy still holds files with Windows line endings, so `artifacts-paper/config.json` hashes differently from the committed file, which matches the manifest. Checking the files out again, or renormalising them once and committing, removes the failure. The suite's configuration also requires 85 per cent line coverage of the package. Continuous integration runs the linter, the tests, the notebook, the fast pipeline, the public-file scripts and both number checks on Python 3.10, 3.11 and 3.12, and a second workflow reruns the paper's configuration on each push to the main branch that touches code or configuration, publishes the tables and commits the refreshed evidence.

The repository holds an older count of its own tests. Its validation note records 178 and later 183 passing tests for earlier snapshots, and says that a claimed 194 was not reproduced at the time. The count of 194 passing tests reported here is from the current branch.

A.5 Modules added for Chapters 18 and 19

Six modules implement the learning-system chapters. Table A.5 lists them. They were written from the chapters' specifications and are tested against the checks the chapters print, not against an earlier engine, so a figure the book takes from the extension's own run is reproduced in direction and size and not digit for digit. `scripts/extension_evidence.py` regenerates the pooled-attack table of Chapter 19.

A.6 Gaps in the code

A review of the repository before the book was written, and the work for the chapters of Part VI, found the following gaps. None affects a number the book reports, since each number is checked against the stored run, but several affect what the code can be trusted to do on its own, and an institution that adopts the code should close them.

The governance record is thinner than the threshold. `governance.GovernanceThreshold` records six of the nine elements of the threshold: tolerance, owner, expiry date, adversary, harm class and error level. The legal basis with the protected parties, the access route and the programme horizon are missing, so the record the code writes is thinner than the one Chapter 7 describes and Table 16.2 shows. `decide` also does not check that the bound it receives was computed at the threshold's error level.

One tolerance for the programme. Validation and the gate now judge the pooled bound against one declared value, `programme.pooled_tolerance` (0.12 in the paper's configuration,

defaulting to the release tolerance), and `decision.json` records it once. The entry `tau_programme` (0.5) is the total of the union risk budget and is recorded as `union_risk_budget`. An earlier version judged the same bound against both values.

Falsification bands. The calibration band in the configuration is now 0.03, down from 0.06, which a model with a misstated error scale met with a gap of 0.049. The pipeline also requires the band to reject that misspecified model before the calibration test counts as valid (check `calibration_band_has_power`), which test configurations can switch off with `require_calibration_power`. The frequency test checks the mean population cell size, which the fitted law recovers to 0.055 cells, and not the uniqueness probability that the journalist measure uses, whose error reaches 25 per cent. The composition and temporal tests are recorded beside the validity flag and in the expiry check, as the long form intends.

Counts rebuilt from rates. `uncertainty.selected_maximum_bound` used to rebuild success counts by rounding rates times replicates. It now takes the integer counts directly through its `successes` argument, and keeps the rounding only as a fallback for callers that hold rates alone. A test checks that the two agree when the rates are exact counts and that non-integer or out-of-range counts are refused.

An unstated normality assumption. `uncertainty.nested_variance` gives a variance split from which a chi-square interval for the share of variance due to the design is built. That interval assumes the outer estimates are roughly normal. The docstring now says so, and adds that the assumption is poor when the risk is near zero or the outer replicates are few.

Version strings. The package metadata and the package initialiser now both say version 0.7.0. Archives of the repository named 0.7.1 are not matched by any tag. A reader who cites a version should cite the commit.

Differential privacy. `dp` covered the posterior bound and basic composition only. It now also has `attack_ceiling`, the ceiling $\min(1, e^{\varepsilon}\kappa + \delta)$ of Chapter 10, `advanced_composition`, which is never worse than the basic total, and `laplace_epsilon`, which reports ε for a given noise scale and sensitivity. Noise of scale $1/\varepsilon$ on a count gives ε when a record is added or removed and 2ε when one is replaced, and the guarantee is per record: a borrower with several loans needs a sensitivity that counts all of them. The code still does not implement a mechanism.

Line endings. The one failing test, described above, and the long list of files that show as modified in the Windows working copy, have the same cause and the same remedy.

What is not built. The registry service that a deployment needs, with durable per-programme budgets, signing keys and external time stamps, is specified and only partly built: `registry.Registry` keeps budgets and plan hashes in a local SQLite file, and has no signing keys or external time stamps. The adapter for a real register is described and not written. Dependencies other than `wooldridge` are not pinned, so a fixed seed does not promise bit-for-bit equality across library versions. The pooled attack in `composition.pooled_attack`, by contrast, was checked against Proposition 9.2: a target absent from a

release contributes zero to that release's marginal risk and does not narrow the intersection, as the proposition and the paper state.

Availability. The papers describe the pipeline as open. At the time of writing the repository is private. Either it is made public before the papers or this book circulate, or the description changes.

Gaps and the claims they touch. The gaps above and the fit of Chapter 8 bear on specific claims. The governance record that omits three elements limits the completeness claims of Chapters 7 and 16; the work to close it is to fill those fields from a signed form and test the strict check on the record. The value-curve fit returns no standard errors, which limits the implied tolerance of Chapter 8; the work is to return the covariance and bootstrap over the frontier points.

TABLE A.4: The scripts in `scripts/`, grouped by purpose, with the chapters that report what they produce.

Script	What it does	Chapters
Evidence for the book		
`rebuild_evidence`	Rewrites figures, tables, numbers and manifest from a stored run without re-simulating; `-publish` copies tables and figures to the manuscripts	all
`programme_robustness`	Floors by adversary knowledge, tolerance, register draw and reading noise; writes `artifacts-robustness/robustness.json`	13, 23
`robustness_table`	Tables of floors from that file	13, 23
`audit_process_attacks`	Seed grinding, shadow audits and post-hoc bets, with and without controls	16
`adaptive_audit_chain`	The release chain on the credit register	6, 14
`adaptive_audit_comparison`	Operating characteristics of the gate on Bernoulli streams	6
`estimand_check`	Checks that two identification tables share one estimand	13
`worked_examples`	Recomputes every number in the worked examples	16
`neural_results`	Figure and table of the network's predictions	14
Checks on the manuscripts		
`check_paper_numbers`	Every run value appears in the long form; every quoted decimal in the article has a source	all
`check_float_refs`	Every table, figure and algorithm is labelled and cited	–
`check_realdata_paper`	Archived public-data claims checked separately	12
Public files and model-proposed attacks		
`realdata_replication`, `realdata_neural_pilot`, `realdata_neural_comparison`, `plot_realdata_neural`	Public-file replication with source hashes; state-held-out neural pilot and comparison; their plot	12, 14
`public_injector_search`, `report_public_search`	Budget-matched search on 30 Census states and its tables	14, 23
`collect_llm_pilot`, `collect_llm_pilot_local`, `report_llm_local`	Language-model proposal collection, hosted and local, and the local arm's table	23
`injector_experiment`, `linkage_injector_experiment`	Synthetic injector ablation; executable nearest-record linkage pilot	14
`sentry_r_example`, `sentry_r_benchmark`	Synthetic inputs and the logistic against deep benchmark	14
`multiscale_demo`	Training, sensitivity and shifted-audit demonstration	14
`pentest_demo`	Penetration-test findings on a public Census release	16
Housekeeping		
`write_manifest`	SHA-256 manifest of a list of files	this appendix
`build_diagrams`, `build_notebook`, `validate_notebook`	Editable diagrams; the Colab notebook and its check	–
`make_acmart`, `sync`	ACM version of the earlier manuscript; repository synchronisation	–

TABLE A.5: Modules added for Chapters 18 and 19. The test files are in `tests/`.

Module	What it computes	Chapter	Test file
`gate`	The release gate: candidates, required attacks, bounds from audits only	18	`test_ext_learning`
`betting`	Predictable bets and the betting audit	6, 18	`test_ext_learning`
`evidence`	Measurement records as integer counts; refusal of the final-audit split	18	`test_ext_learning`
`registry`	Programme budget, frozen plans and their hashes in SQLite	18	`test_ext_learning`
`dynamics`	Changing and frozen panels, and the pooled attack using presence and absence	19	`test_ext_models`
`graph`	Generator checks, propagation, node risk and the maximum-likelihood generator	19	`test_ext_models`

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